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WAVES AND INSTABILITIES IN PLASMAS

by Liu Chen

Retypesetted and modified version

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Preface

These notes have been given over the past several years in an one-semester course to second-year graduate students in the Plasma Physics Section of the Department of Astrophysical Sciences of Princeton University. The students generally have (or are assumed to have) some basic knowledge of plasma dynamics in terms of single-particle and fluid descriptions. The topics covered in these notes, therefore, are selective and tend to emphasize more on kinetic-theoretical approaches to waves and instabilities in both uniform and nonuniform plasmas.

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Retypesetting Description

All the figures are re-drawn with colors, and many modifications are made to the original text —— most of these modifications attempt to make the text more readable or mathematically accurate. CAUTION: there might be some typos and\or mistakes, and the original text should be referred when necessary. All the footnotes and appendices are added by the retypesetter. The retypesetter will be responsible for almost all the potential *physical* mistakes.

Conventions and Symbols

- Throughout this book, CGS Gaussian unit system is used, and the Boltzmann constant k_{B} is set to unity.
- The abbreviation "R.H.S." means "right-hand side" and "L.H.S." means "left-hand side".
- The statement $A \equiv a$ means that A and a are identified. This is used when we want to define a new quantity, or to use a more convenient symbol for an already defined quantity; for the former case, the statement $A := a$ is also used, which means that A is defined to be a.
- Let X be a point set with a limit point x_0, $|f(x)| = \alpha(x)|g(x)|$. As $x \to x_0$, the statement $f \sim g$ means $\alpha \to 1$, $f = \mathcal{o}(g)$ means $\alpha \to 0$, and $f = \mathcal{O}(g)$ means that α is bounded; the $\mathcal{O}$ symbol can also apply to the whole set X and not just as $x \to x_0$. Note that in the context of asymptotic analysis, the symbols $\sim$ and $\mathcal{O}$ do not simply mean that f and g are "of the same order"; instead, they have more precise meanings. For more on asymptotic expansion, see [1], [2].
- The number Pi $= 4\sum_{n=0}^{+\infty}(-1)^n/(2n+1)$ is denoted by the upright π. Euler's number (natural constant) $\lim_{n\to+\infty}(1+1/n)^n = \sum_{n=0}^{+\infty} 1/n!$ is denoted by the the upright e; please distinguish between e and e, with the latter denotes the elementary charge. The pure imaginary number $\sqrt{-1}$ is denoted by the upright i.
- Complex conjugation of a complex number D is denoted by an asterisk, D^*, and the real and imaginary parts of D are denoted by $(\mathrm{Re}\ D) \equiv D_{\mathrm{r}}$ and $(\mathrm{Im}\ D) \equiv D_{\mathrm{i}}$, respectively.
- Vectors in $\mathbb{R}^3$ are denoted by boldface letters, e.g., $\boldsymbol{a}$, and their contravariant/covariant components are denoted by the corresponding normal-face letters with upper/lower indices, e.g., $\boldsymbol{a} = a^i \boldsymbol{e}_i$ where $\boldsymbol{e}_i := \partial\boldsymbol{r}/\partial x^i$ is the i-th coordinate basis vector. Repeated upper and lower indices are meant to sum over all possible values, e.g. $a^i b_i \equiv \sum_i a^i b_i$. Unit vectors are usually denoted by $\hat{\boldsymbol{e}}_{\dots}$, and are always emphasized by a hat, e.g., $\hat{\boldsymbol{k}} = \boldsymbol{k}/|\boldsymbol{k}|$.

  Cartesian tensors are denoted by normal-face letters with a tilde, e.g., $\tilde{T}$, and their contravariant/covariant components are denoted by the corresponding normal-face letters with upper/lower indices, e.g., $\tilde{T} = T_{ij}\boldsymbol{e}^i\boldsymbol{e}^j$. Matrix/tensor transposition of $\tilde{T}$ is denoted by a superscript "T", e.g., $\tilde{T}^{\mathrm{T}} = T_{ji}\boldsymbol{e}^i\boldsymbol{e}^j$; a transposition followed by complex conjugation is denoted by a dagger, i.e., $\tilde{T}^\dagger \equiv \left(\tilde{T}^{\mathrm{T}}\right)^* \equiv \left(\tilde{T}^*\right)^{\mathrm{T}} = T_{ji}^*\boldsymbol{e}^i\boldsymbol{e}^j$. For more on tensor analysis, see [3].

Formulae

- Maxwell equations:

$$\begin{aligned} \nabla \cdot \boldsymbol{E} &= 4\pi\rho, \quad \nabla \times \boldsymbol{E} + \frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{0}, \\ \nabla \cdot \boldsymbol{B} &= 0, \qquad \nabla \times \boldsymbol{B} - \frac{1}{c}\frac{\partial \boldsymbol{E}}{\partial t} = \frac{4\pi}{c}\boldsymbol{J}. \end{aligned} \tag{0.0.1}$$

  - Poynting's theorem:

$$\frac{\partial}{\partial t}\left(\frac{1}{8\pi}(|\boldsymbol{E}|^2 + |\boldsymbol{B}|^2)\right) + \nabla \cdot \left(\frac{c}{4\pi}\boldsymbol{E} \times \boldsymbol{B}\right) = -\boldsymbol{J} \cdot \boldsymbol{E}. \tag{0.0.2}$$

  - Potentials:

$$\boldsymbol{E} = -\nabla\varphi - \frac{1}{c}\frac{\partial \boldsymbol{A}}{\partial t}, \quad \boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{0.0.3}$$

  - Wave equations under the Coulomb gauge condition $\nabla \cdot \boldsymbol{A} = 0$:

$$\begin{aligned} -\nabla^2\varphi &= 4\pi\rho, \\ -\nabla^2\boldsymbol{A} + \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}}{\partial t^2} + \frac{1}{c}\frac{\partial}{\partial t}\nabla\varphi &= \frac{4\pi}{c}\boldsymbol{J}. \end{aligned} \tag{0.0.4}$$

- Material derivative:

$$\begin{gathered} \frac{\mathrm{d}f(t,\boldsymbol{x}(t,\boldsymbol{\xi}))}{\mathrm{d}t} = \frac{\partial f(t,\boldsymbol{x}(t,\boldsymbol{\xi}))}{\partial t} + \frac{\partial \boldsymbol{x}(t,\boldsymbol{\xi})}{\partial t} \cdot \nabla f(t,\boldsymbol{x}(t,\boldsymbol{\xi})), \\ \text{i.e.,} \quad \frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \boldsymbol{v} \cdot \nabla. \end{gathered} \tag{0.0.5}$$

- Dot products:

  Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$, $\tilde{T}, \tilde{P}, \tilde{Q}$ be second-order Cartesian tensors and $\boldsymbol{e}_i := \partial\boldsymbol{r}/\partial q^i$ be the i-th coordinate basis vector, then

$$\begin{aligned} \boldsymbol{a} \cdot \boldsymbol{b} &= a^i b_i, \qquad \tilde{T} \cdot \boldsymbol{a} = T^{ij} a_j \boldsymbol{e}_i, \\ \tilde{P} : \tilde{Q} &= P^{ij} Q_{ij}, \quad \tilde{P} \cdot\cdot\, \tilde{Q} = P^{ij} Q_{ji}. \end{aligned} \tag{0.0.6}$$

- Permutation identity:

$$\varepsilon^{ijk}\varepsilon_{klm} = \delta^i_l \delta^j_m - \delta^i_m \delta^j_l. \tag{0.0.7}$$

- Multi-dimensional Taylor expansion:

$$f(\boldsymbol{x} + \boldsymbol{a}) = f(\boldsymbol{x}) + a^i \nabla_i f(\boldsymbol{x}) + \frac{1}{2!} a^i a^j \nabla_i \nabla_j f(\boldsymbol{x}) + \mathcal{O}(a^3). \tag{0.0.8}$$

- Properties of the Jacobian [4]:

  Let f, g, h, k be functions of two indenpendent variables x and y, and denote the Jacobian as

$$\frac{\partial(f,g)}{\partial(x,y)} \equiv \det \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} = \frac{\partial f}{\partial x}\frac{\partial g}{\partial y} - \frac{\partial f}{\partial y}\frac{\partial g}{\partial x}. \tag{0.0.9}$$

Then

$$\begin{aligned}
\frac{\partial(f,g)}{\partial(h,k)} &= \frac{\partial(f,g)}{\partial(x,y)}\Big/\frac{\partial(h,k)}{\partial(x,y)},\\
\frac{\partial(f,g)}{\partial(x,y)} &= 1\Big/\frac{\partial(x,y)}{\partial(f,g)},\\
\left(\frac{\partial f}{\partial g}\right)_h &= \frac{\partial(f,h)}{\partial(g,h)}.
\end{aligned} \tag{0.0.10}$$

- Fourier\Laplace transform:
  - Fourier transform:

$$\mathcal{F}\{f(x)\}(k) \equiv \mathcal{F}(k)\{f(x)\} := \int_{\mathbb{R}} \mathrm{d}x\, f(x)\mathrm{e}^{-\mathrm{i}kx}. \tag{0.0.11}$$

  - Unilateral Laplace transform:

$$\mathcal{L}\{f(t)\}(\omega) \equiv \mathcal{L}(\omega)\{f(t)\} := \int_0^{+\infty} \mathrm{d}t\, f(t)\mathrm{e}^{\mathrm{i}\omega t}, \quad \omega \in \mathrm{ROC}, \tag{0.0.12}$$

where "ROC" means "Range of Convergence". The Laplace transform is originally defined only in ROC, although it may be analytically continued to the whole complex ω plane.

  - Bilateral Laplace transform:

$$\mathcal{L}\{f(t)\}(\omega) \equiv \mathcal{L}(\omega)\{f(t)\} := \int_{\mathbb{R}} \mathrm{d}t\, f(t)\mathrm{e}^{\mathrm{i}\omega t}, \quad \omega \in \mathrm{ROC}. \tag{0.0.13}$$

- Integration formula:

$$\int_0^{+\infty} \mathrm{d}u\, \frac{\cos(xu)}{1+u^2} = \frac{\pi}{2}\mathrm{e}^{-|x|}. \tag{0.0.14}$$

- Gamma function:
  - Definition:

$$\Gamma(z) := \int_0^{+\infty} \mathrm{d}x\, x^{z-1}\mathrm{e}^{-x}. \tag{0.0.15}$$

  - Recurrence relation:

$$\Gamma(z+1) = z\Gamma(z) \equiv z!. \tag{0.0.16}$$

  - Special values:

$$\Gamma(1) = 1, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \tag{0.0.17}$$

- Asymptotic behavior of the plasma dispersion function:

$$Z(\zeta) = -2\zeta\left(1 - \frac{2}{3}\zeta^2 + \frac{4}{15}\zeta^4 - \frac{8}{105}\zeta^6 + \mathcal{O}(\zeta^8)\right) + \mathrm{i}\sqrt{\pi}\mathrm{e}^{-\zeta^2} \text{ for } \zeta \to 0 \tag{0.0.18}$$

$$= -\frac{1}{\zeta}\left(1+\frac{1}{2\zeta^2}+\frac{3}{4\zeta^4}+\frac{15}{8\zeta^6}+\mathcal{O}\left(\frac{1}{\zeta^8}\right)\right)+\mathrm{i}\sigma\sqrt{\pi}\mathrm{e}^{-\zeta^2} \quad \text{for } \zeta\to\infty, \tag{0.0.18}$$

where

$$\sigma = \begin{cases} 0 \text{ for } \mathrm{Im}\ \zeta > 0 \\ 1 \text{ for } \mathrm{Im}\ \zeta = 0. \\ 2 \text{ for } \mathrm{Im}\ \zeta < 0 \end{cases} \tag{0.0.19}$$

- Properties of Bessel functions of integer order:
  - Asmptotic behaviors:

$$\begin{aligned} J_n(x) &= \frac{(x/2)^n}{n!} - \frac{(x/2)^{n+2}}{(n+1)!} + \frac{(x/2)^{n+4}}{2(n+2)!} + \mathcal{O}(x^{n+6}) && \text{for } |x| \ll 1, \\ &\sim \sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{n\pi}{2}-\frac{\pi}{4}\right) && \text{for } |x|\to\infty, |\arg x|<\pi; \\ I_n(x) &\equiv \mathrm{e}^{-\mathrm{i}n\pi/2}J_n\left(\mathrm{e}^{\mathrm{i}\pi/2}x\right) \\ &= \frac{(x/2)^n}{n!} + \frac{(x/2)^{n+2}}{(n+1)!} + \frac{(x/2)^{n+4}}{2(n+2)!} + \mathcal{O}(x^{n+6}) && \text{for } |x| \ll 1 \\ &= \frac{\mathrm{e}^x}{\sqrt{2\pi x}}\left(1-\frac{n-1}{8x}+\frac{(n-1)(n-9)}{128x^2}+\mathcal{O}\left(\frac{1}{x^3}\right)\right) && \text{for } x \gg 1. \end{aligned} \tag{0.0.20}$$

  - Recurrence relations:

$$\begin{aligned} J_{n-1}(x)+J_{n+1}(x) &= \frac{2n}{x}J_n(x), \\ J_{n-1}(x)-J_{n+1}(x) &= 2J_n'(x). \end{aligned} \tag{0.0.21}$$

  - Symmetry properties:

$$\begin{aligned} J_n(-x) = J_{-n}(x) &= (-1)^n J_n(x), \\ I_{-n}(x) &= I_n(x). \end{aligned} \tag{0.0.22}$$

  - Generating function:

$$\begin{aligned} \exp\left(\frac{x}{2}\left(t-\frac{1}{t}\right)\right) &= \sum_{n=-\infty}^{+\infty} J_n(x)\,t^n, \quad 0<|t|<+\infty \\ \Rightarrow \mathrm{e}^{\pm\mathrm{i}x\sin\theta} &= \sum_{n=-\infty}^{+\infty} J_n(x)\,\mathrm{e}^{\pm\mathrm{i}n\theta}. \end{aligned} \tag{0.0.23}$$

  - Summation formulae:

$$\begin{aligned} &\sum_{n=-\infty}^{+\infty} J_n^2(x) = 1, \quad \sum_{n=-\infty}^{+\infty} J_n(x)J_n'(x) = 0, \\ &\sum_{n=-\infty}^{\infty} n^2 J_n^2(x) = \frac{x^2}{2}, \quad \sum_{n=-\infty}^{\infty} \left(J_n'(x)\right)^2 = \frac{1}{2}. \end{aligned} \tag{0.0.24}$$

- Weber integral:

$$\int_0^{+\infty} \mathrm{d}x\, \mathrm{e}^{-\rho^2 x^2} x J_n(\alpha x) J_n(\beta x) = \frac{1}{2\rho^2} \exp\left(-\frac{\alpha^2+\beta^2}{4\rho^2}\right) I_n\left(\frac{\alpha\beta}{2\rho^2}\right)$$

$$\Rightarrow \int_0^{+\infty} \mathrm{d}x\, \mathrm{e}^{-\rho^2 x^2} x^2 J_n(\alpha x) J_n'(\beta x) = \frac{1}{4\rho^4} \exp\left(-\frac{\alpha^2+\beta^2}{4\rho^2}\right) (\alpha I_n' - \beta I_n)\left(\frac{\alpha\beta}{2\rho^2}\right) \tag{0.0.25}$$

$$\Rightarrow \int_0^{+\infty} \mathrm{d}x\, \mathrm{e}^{-\rho^2 x^2} x^3 J_n'(\alpha x) J_n'(\beta x) = \frac{1}{8\rho^6} \exp\left(-\frac{\alpha^2+\beta^2}{4\rho^2}\right) (\alpha\beta I_n'' + (2\rho^2 - \alpha^2 - \beta^2) I_n' + \alpha\beta I_n)\left(\frac{\alpha\beta}{2\rho^2}\right).$$

From the front matter of [5].

Vector Formulas

$$
\begin{aligned}
\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) &= \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})\\
\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) &= (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}\\
(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d}) &= (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})\\
\nabla\times\nabla\psi &= 0\\
\nabla\cdot(\nabla\times\mathbf{a}) &= 0\\
\nabla\times(\nabla\times\mathbf{a}) &= \nabla(\nabla\cdot\mathbf{a}) - \nabla^2\mathbf{a}\\
\nabla\cdot(\psi\mathbf{a}) &= \mathbf{a}\cdot\nabla\psi + \psi\nabla\cdot\mathbf{a}\\
\nabla\times(\psi\mathbf{a}) &= \nabla\psi\times\mathbf{a} + \psi\nabla\times\mathbf{a}\\
\nabla(\mathbf{a}\cdot\mathbf{b}) &= (\mathbf{a}\cdot\nabla)\mathbf{b} + (\mathbf{b}\cdot\nabla)\mathbf{a} + \mathbf{a}\times(\nabla\times\mathbf{b}) + \mathbf{b}\times(\nabla\times\mathbf{a})\\
\nabla\cdot(\mathbf{a}\times\mathbf{b}) &= \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})\\
\nabla\times(\mathbf{a}\times\mathbf{b}) &= \mathbf{a}(\nabla\cdot\mathbf{b}) - \mathbf{b}(\nabla\cdot\mathbf{a}) + (\mathbf{b}\cdot\nabla)\mathbf{a} - (\mathbf{a}\cdot\nabla)\mathbf{b}
\end{aligned}
$$

If $\mathbf{x}$ is the coordinate of a point with respect to some origin, with magnitude $r = |\mathbf{x}|$, $\mathbf{n} = \mathbf{x}/r$ is a unit radial vector, and $f(r)$ is a well-behaved function of r, then

$$
\nabla\cdot\mathbf{x} = 3 \qquad \nabla\times\mathbf{x} = 0
$$

$$
\nabla\cdot[\mathbf{n}f(r)] = \frac{2}{r}f + \frac{\partial f}{\partial r} \qquad \nabla\times[\mathbf{n}f(r)] = 0
$$

$$
(\mathbf{a}\cdot\nabla)\mathbf{n}f(r) = \frac{f(r)}{r}[\mathbf{a} - \mathbf{n}(\mathbf{a}\cdot\mathbf{n})] + \mathbf{n}(\mathbf{a}\cdot\mathbf{n})\frac{\partial f}{\partial r}
$$

$$
\nabla(\mathbf{x}\cdot\mathbf{a}) = \mathbf{a} + \mathbf{x}(\nabla\cdot\mathbf{a}) + i(\mathbf{L}\times\mathbf{a})
$$

where $\mathbf{L} = \frac{1}{i}(\mathbf{x}\times\nabla)$ is the angular-momentum operator.

Theorems from Vector Calculus

In the following ϕ, ψ, and $\mathbf{A}$ are well-behaved scalar or vector functions, V is a three-dimensional volume with volume element d^3x, S is a closed two-dimensional surface bounding V, with area element da and unit outward normal $\mathbf{n}$ at da.

$$
\int_V \nabla\cdot\mathbf{A}\, d^3x = \int_S \mathbf{A}\cdot\mathbf{n}\, da \qquad \text{(Divergence theorem)}
$$

$$
\int_V \nabla\psi\, d^3x = \int_S \psi\mathbf{n}\, da
$$

$$
\int_V \nabla\times\mathbf{A}\, d^3x = \int_S \mathbf{n}\times\mathbf{A}\, da
$$

$$
\int_V (\phi\nabla^2\psi + \nabla\phi\cdot\nabla\psi)\, d^3x = \int_S \phi\mathbf{n}\cdot\nabla\psi\, da \qquad \text{(Green's first identity)}
$$

$$
\int_V (\phi\nabla^2\psi - \psi\nabla^2\phi)\, d^3x = \int_S (\phi\nabla\psi - \psi\nabla\phi)\cdot\mathbf{n}\, da \qquad \text{(Green's theorem)}
$$

In the following S is an open surface and C is the contour bounding it, with line element $d\mathbf{l}$. The normal $\mathbf{n}$ to S is defined by the right-hand-screw rule in relation to the sense of the line integral around C.

$$
\int_S (\nabla\times\mathbf{A})\cdot\mathbf{n}\, da = \oint_C \mathbf{A}\cdot d\mathbf{l} \qquad \text{(Stokes's theorem)}
$$

$$
\int_S \mathbf{n}\times\nabla\psi\, da = \oint_C \psi\, d\mathbf{l}
$$

From the back matter of [5].

Explicit Forms of Vector Operations

Let $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ be orthogonal unit vectors associated with the coordinate directions specified in the headings on the left, and A_1, A_2, A_3 be the corresponding components of $\mathbf{A}$. Then

Cartesian $(x_1, x_2, x_3 = x, y, z)$

$$\nabla\psi = \mathbf{e}_1 \frac{\partial\psi}{\partial x_1} + \mathbf{e}_2 \frac{\partial\psi}{\partial x_2} + \mathbf{e}_3 \frac{\partial\psi}{\partial x_3}$$

$$\nabla\cdot\mathbf{A} = \frac{\partial A_1}{\partial x_1} + \frac{\partial A_2}{\partial x_2} + \frac{\partial A_3}{\partial x_3}$$

$$\nabla\times\mathbf{A} = \mathbf{e}_1\left(\frac{\partial A_3}{\partial x_2} - \frac{\partial A_2}{\partial x_3}\right) + \mathbf{e}_2\left(\frac{\partial A_1}{\partial x_3} - \frac{\partial A_3}{\partial x_1}\right) + \mathbf{e}_3\left(\frac{\partial A_2}{\partial x_1} - \frac{\partial A_1}{\partial x_2}\right)$$

$$\nabla^2\psi = \frac{\partial^2\psi}{\partial x_1^2} + \frac{\partial^2\psi}{\partial x_2^2} + \frac{\partial^2\psi}{\partial x_3^2}$$

Cylindrical (ρ, ϕ, z)

$$\nabla\psi = \mathbf{e}_1 \frac{\partial\psi}{\partial\rho} + \mathbf{e}_2 \frac{1}{\rho}\frac{\partial\psi}{\partial\phi} + \mathbf{e}_3 \frac{\partial\psi}{\partial z}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_1) + \frac{1}{\rho}\frac{\partial A_2}{\partial\phi} + \frac{\partial A_3}{\partial z}$$

$$\nabla\times\mathbf{A} = \mathbf{e}_1\left(\frac{1}{\rho}\frac{\partial A_3}{\partial\phi} - \frac{\partial A_2}{\partial z}\right) + \mathbf{e}_2\left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial\rho}\right) + \mathbf{e}_3\frac{1}{\rho}\left(\frac{\partial}{\partial\rho}(\rho A_2) - \frac{\partial A_1}{\partial\phi}\right)$$

$$\nabla^2\psi = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\phi^2} + \frac{\partial^2\psi}{\partial z^2}$$

Spherical (r, θ, ϕ)

$$\nabla\psi = \mathbf{e}_1 \frac{\partial\psi}{\partial r} + \mathbf{e}_2 \frac{1}{r}\frac{\partial\psi}{\partial\theta} + \mathbf{e}_3 \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_1) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta A_2) + \frac{1}{r\sin\theta}\frac{\partial A_3}{\partial\phi}$$

$$\nabla\times\mathbf{A} = \mathbf{e}_1\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_3) - \frac{\partial A_2}{\partial\phi}\right] + \mathbf{e}_2\left[\frac{1}{r\sin\theta}\frac{\partial A_1}{\partial\phi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_3)\right] + \mathbf{e}_3\frac{1}{r}\left[\frac{\partial}{\partial r}(rA_2) - \frac{\partial A_1}{\partial\theta}\right]$$

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}$$

$$\left[\text{Note that } \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) \equiv \frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi).\right]$$

Physical Constants [6]

Name	Symbol	Value (SI)
Speed of light in vacuum	c	2.99792458×10^{8} m/s (exact)
Newtonian constant of gravitation	G	$6.67430(15) \times 10^{-11}$ $\mathrm{m}^3 \cdot \mathrm{kg}^{-1} \cdot \mathrm{s}^{-2}$
Planck constant	h	$6.62607015 \times 10^{-34}$ J · s (exact)
Reduced Planck constant $h/2\pi$	$\hbar$	$1.054571817... \times 10^{-34}$ J · s (exact)
Elementary charge	e	$1.602176634 \times 10^{-19}$ C (exact)
Vacuum permittivity	ε_0	$8.8541878188(14) \times 10^{-12}$ F/m
Vacuum permeability $1/c^2\varepsilon_0$	μ_0	$1.25663706127(20) \times 10^{-7}$ H/m
Electron mass	m_e	$9.109383(28) \times 10^{-31}$ kg
Proton mass	m_p	$1.67262192595(52) \times 10^{-27}$ kg
Proton-electron mass ratio	m_p/m_e	1836.152673426(32)
Boltzmann constant	k_B	1.380649×10^{-23} J/K (exact)
Avogadro constant	N_A	$6.02214076 \times 10^{23}$ mol^{-1} (exact)
Electron volt (e/C) J	eV	$1.602176634 \times 10^{-19}$ J (exact)
(Unified) atomic mass unit $m(^{12}\mathrm{C})/12$	u, m_u	$1.66053906892(52) \times 10^{-27}$ kg
Temperature 1 eV	–	$1.160451812... \times 10^{4}$ K (exact)

Plasma Parameters

Name	**Symbol**	**Definition**
(Kinetic) temperature	T	$m\langle\lvert\boldsymbol{v}-\langle\boldsymbol{v}\rangle\rvert^2\rangle/3$
(Kinetic) pressure	p	nT
Thermal speed	$v_\mathrm{t}, v_\mathrm{T}, v_\mathrm{th}$	$\sqrt{2T/m}$ or $\sqrt{T/m}$
Ion acoustic speed	c_s	$\sqrt{(T_e+3T_i)/m_i}$
Alfvén speed	$v_\mathrm{A}, c_\mathrm{A}$	$B/\sqrt{4\pi\rho_\mathrm{m}}$
Plasma frequency	ω_p	$\sqrt{4\pi nq^2/m}$
Gyrofrequency\cyclotron frequency	$\Omega, \omega_\mathrm{c}, \omega_\mathrm{g}$	qB/mc
Transit frequency	ω_t	v_t/L
Lower-hybrid frequency	$\omega_\mathrm{LH}, \omega_\mathrm{lh}$	$\lvert\Omega_e\rvert\omega_{\mathrm{p}i}/\sqrt{\Omega_e^2+\omega_\mathrm{pe}^2}$
Upper-hybrid frequency	$\omega_\mathrm{UH}, \omega_\mathrm{uh}$	$\sqrt{\omega_\mathrm{pe}^2+\Omega_e^2}$
de Broglie wavelength	λ_de	h/mv_t
Inertial length\skin depth	d, λ	c/ω_p
Debye length	$\lambda_\mathrm{D}, \lambda_\mathrm{De}$	$\sqrt{T/4\pi nq^2}$
(Thermal) gyroradius	$\rho_\mathrm{t}, \rho, r_\mathrm{c}, r_\mathrm{g}, \cdots$	v_t/Ω
Plasma parameter	Λ	$4\pi n\lambda_\mathrm{D}^3$ or $4\pi n\lambda_\mathrm{D}^3/3$
Plasma beta	β	$8\pi p/B^2$
$E\times B$ drift\electric drift	$\boldsymbol{v}_E$	$c\boldsymbol{E}\times\boldsymbol{B}/B^2$
Magnetic drift\gradient drift	$\boldsymbol{v}_{\nabla B}, \boldsymbol{v}_\mathrm{grad}$	$(v_\perp^2/2\Omega)\hat{\boldsymbol{b}}\times\nabla\ln B$
Curvature drift	$\boldsymbol{v}_\mathrm{curv}$	$\left(v_\parallel^2/\Omega\right)\hat{\boldsymbol{b}}\times\left(\hat{\boldsymbol{b}}\cdot\nabla\hat{\boldsymbol{b}}\right)$
Polarization drift	$\boldsymbol{v}_\mathrm{pola}$	$(1/\Omega)\hat{\boldsymbol{b}}\times(\mathrm{d}\boldsymbol{v}_E/\mathrm{d}t)$
First-order parallel drift\Baños drift	–	$(v_\perp^2/2\Omega)\hat{\boldsymbol{b}}\hat{\boldsymbol{b}}\cdot\left(\nabla\times\hat{\boldsymbol{b}}\right)$
Diamagnetic drift	$\boldsymbol{v}_\mathrm{diam}$	$(1/nm\Omega)\hat{\boldsymbol{b}}\times\nabla p$

Contents

1 General Properties of Electromagnetic Waves in Dielectric Media

In this chapter, we shall discuss some general properties pertinent to linear electromagnetic waves in plasmas; in fact, these properties are sufficiently general and, hence, fundamental that they are applicable to other dielectric media.

1.1 Plasma as a dielectric medium

Let's assume that the plasma, or, generally, the dielectric medium is temporally stationary and spatially homogeneous. The perturbed (wave) electric field[1], $\boldsymbol{E}_1$, can then be taken to assume the *plane wave* form

$$\boldsymbol{E}_1(t, \boldsymbol{x}) = \hat{\boldsymbol{E}}_1 \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}, \tag{1.1.1}$$

where $\boldsymbol{k}$ is the wavevector and ω the angular frequency. Whether $(\omega, \boldsymbol{k})$ is real or complex depends on the detailed nature of the problem. For example, $\omega = \omega_0$ is real if we are interested in wave properties driven by an external source oscillating at angular-frequency ω_0; $\boldsymbol{k}$, however, can become complex as in the case of evanescent waves. On the other hand, if we are interested in an initial value problem (e.g., stability analysis), then ω is complex and $\boldsymbol{k}$ is real. This can be realized by interpreting $\hat{\boldsymbol{E}}_1(\omega, \boldsymbol{k})$ as the Laplace-in-time and Fourier-in-space transform of $\boldsymbol{E}_1(t, \boldsymbol{x})$; note that $\mathrm{Im}\ \omega$ has to be positive to ensure *causality*[2], although $\hat{\boldsymbol{E}}_1$ may then be analytically continued to the whole complex ω plane.

In response to $\boldsymbol{E}_1$, charged particles will jiggle and acquire velocities, $\boldsymbol{V}_1$, and, hence, there will be perturbed current densities, $\boldsymbol{J}_1$. Now since the background plasma is taken to be both time-stationary and spatially homogeneous, to lowest order it responses to $\boldsymbol{E}_1$ linearly as[3]

$$\begin{aligned} \boldsymbol{J}_1(t, \boldsymbol{x}) &= \int_{\mathbb{R}} \mathrm{d}t' \int_{\mathbb{R}^3} \mathrm{d}^3x'\ \tilde{\sigma}(t - t', \boldsymbol{x} - \boldsymbol{x}') \cdot \boldsymbol{E}_1(t', \boldsymbol{x}') \\ \Rightarrow \hat{\boldsymbol{J}}_1(\omega, \boldsymbol{k}) &= \tilde{\sigma}(\omega, \boldsymbol{k}) \cdot \hat{\boldsymbol{E}}_1(\omega, \boldsymbol{k}), \end{aligned} \tag{1.1.2}$$

where $\tilde{\sigma}(t, \boldsymbol{x})$ is the real-space *conductivity tensor* and

[1] The original book uses the notation δA for the perturbed quantity, here we use the symbol A_1 instead.

[2] With $\mathrm{Im}\ \omega > 0$, that $|\boldsymbol{E}_1(t)| \to +\infty$ as $t \to +\infty$ is unimportant since it cannot affect what is observed at finite times t; with $\mathrm{Im}\ \omega < 0$, however, $|\boldsymbol{E}_1(t)| \to +\infty$ as $t \to -\infty$, which means that the field would have been large in the past and the approximation of linearization is invalidated. [7]

[3] For a time-stationary system, the real-space response function, generally denoted $\chi(t, t'; \boldsymbol{x}, \boldsymbol{x}')$, must be invariant under time translation $t \mapsto t + a$, so that it only depends on the time delay $\chi(t, t'; \boldsymbol{x}, \boldsymbol{x}') \equiv \chi(t - t'; \boldsymbol{x}, \boldsymbol{x}')$; similarly, for spatially homogeneous systems, we have $\chi(t, t'; \boldsymbol{x}, \boldsymbol{x}') \equiv \chi(t, t'; \boldsymbol{x} - \boldsymbol{x}')$. [8] Here, the conductivity tensor serves as the response function.

$$\tilde{\sigma}(\omega, \boldsymbol{k}) = \int_{\mathbb{R}} \mathrm{d}t \int_{\mathbb{R}^3} \mathrm{d}^3x\, \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} \tilde{\sigma}(t, \boldsymbol{x}) \tag{1.1.3}$$

the $(\omega, \boldsymbol{k})$ space one. Eq. (1.1.2) is the so-called *Ohm's law.*

We cannot affect the past; this statement of *causality* [8], [9] implies that

$$\begin{aligned} &\tilde{\sigma}(t) = 0 \text{ for } t < 0 \\ \Rightarrow\ &\tilde{\sigma}(\omega) = \int_0^{+\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \tilde{\sigma}(t). \end{aligned} \tag{1.1.4}$$

From Eq. (1.1.4) we know that $\tilde{\sigma}(\omega)$ will not blow up for $\mathrm{Im}\ \omega > 0$ and, hence, it is *holomorphic* (complex *analytic*) in the upper half ω plane.

For the present discussion, we shall simply treat $\tilde{\sigma}$ as given; it is in the next chapters that we will derive specific expressions for $\tilde{\sigma}$ and discuss the results.

Substituting Eq. (1.1.1) and Eq. (1.1.2) into the following first-order Maxwell equations

$$\begin{aligned} \nabla \times \boldsymbol{E}_1 + \frac{1}{c}\frac{\partial \boldsymbol{B}_1}{\partial t} &= \boldsymbol{0}, \\ \nabla \times \boldsymbol{B}_1 - \frac{1}{c}\frac{\partial \boldsymbol{E}_1}{\partial t} &= \frac{4\pi}{c}\boldsymbol{J}_1, \end{aligned} \tag{1.1.5}$$

we find

$$\begin{aligned} \boldsymbol{k} \times \hat{\boldsymbol{E}}_1 &= \frac{\omega}{c}\hat{\boldsymbol{B}}_1, \\ \boldsymbol{k} \times \hat{\boldsymbol{B}}_1 &= -\frac{\omega}{c}\hat{\boldsymbol{E}}_1 - \mathrm{i}\frac{4\pi}{c}\hat{\boldsymbol{J}}_1 \\ &= -\frac{\omega}{c}\left(\tilde{g} + \mathrm{i}\frac{4\pi\tilde{\sigma}}{\omega}\right)\cdot \hat{\boldsymbol{E}}_1 \equiv -\frac{\omega}{c}\tilde{\epsilon}\cdot\hat{\boldsymbol{E}}_1, \end{aligned} \tag{1.1.6}$$

where $\tilde{g}$ is the *metric tensor*[4] and $\tilde{\epsilon} \equiv \tilde{g} + \mathrm{i}4\pi\tilde{\sigma}/\omega$ the *dielectric tensor*[5].

Eliminating $\hat{\boldsymbol{B}}_1$ using the vector identity $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b})$, we obtain the wave equation[6]

[4]The metric tensor is often referred to as the "unit tensor", in that its dot product with any vector is equal to the vector itself

$$\tilde{g}\cdot\boldsymbol{a} = \boldsymbol{a} \text{ for } \boldsymbol{a} \in \mathbb{R}^3. \tag{1.1.7}$$

[5]Unlike in electrostatics, the charged particles in plasmas are not bounded and frequently undergo oscillations. So it is fruitless to distinguish between the free charges and the polarization charges; instead, it is customary [9] to regard all the charges as polarization ones, i.e., to define the "polarization" $\boldsymbol{P}_1$ such that

$$\rho_1 =: -\nabla\cdot\boldsymbol{P}_1, \quad \boldsymbol{J}_1 =: \partial\boldsymbol{P}_1/\partial t. \tag{1.1.8}$$

Then $\hat{\boldsymbol{J}}_1 = -\mathrm{i}\omega\hat{\boldsymbol{P}}_1 \Rightarrow \hat{\boldsymbol{P}}_1 = \mathrm{i}(\tilde{\sigma}/\omega)\cdot\hat{\boldsymbol{E}}_1$, and the "dielectric tensor" is defined in the same manner as in electrostatics

$$\hat{\boldsymbol{D}}_1 := \hat{\boldsymbol{E}}_1 + 4\pi\hat{\boldsymbol{P}}_1 = \left(\tilde{g} + \mathrm{i}\frac{4\pi\tilde{\sigma}}{\omega}\right)\cdot\hat{\boldsymbol{E}}_1 \equiv \tilde{\epsilon}\cdot\hat{\boldsymbol{E}}_1; \tag{1.1.9}$$

$\tilde{\chi} \equiv \mathrm{i}4\pi\tilde{\sigma}/\omega$ is the *susceptibility tensor.*

[6]Sometimes $\tilde{D}$ is referred to as the electromagnetic/dispersion tensor, and D the dispersion function.

$$\left(\boldsymbol{kk} - k^2\tilde{g} + \frac{\omega^2}{c^2}\tilde{\epsilon}\right) \cdot \hat{\boldsymbol{E}}_1 \equiv \tilde{D} \cdot \hat{\boldsymbol{E}}_1 = \boldsymbol{0}. \tag{1.1.10}$$

In order that Eq. (1.1.10) has non-trival solution of $\hat{\boldsymbol{E}}_1$, the determinant of $\tilde{D}$ must vanish

$$D(\omega, \boldsymbol{k}) \equiv \det \tilde{D}(\omega, \boldsymbol{k}) = 0. \tag{1.1.11}$$

Eq. (1.1.11) is the so-called linear *dipsersion ralation*, from which we can solve for a relation between ω and $\boldsymbol{k}$, $\omega = \omega(\boldsymbol{k}) \in \mathbb{C}$ and, thus, plot the $\omega - \boldsymbol{k}$ dispersion curves.

In general, for a fixed $\boldsymbol{k}$, the dispersion relation can have multiple (or, sometimes, infinite) number of ω roots. Indexing these ω branches by j, we then have

$$\omega = \omega_j(\boldsymbol{k}), \quad j = 1, 2, \cdots. \tag{1.1.12}$$

In plasma physics literature, the various ω branches are sometimes called *normal modes*. Substituting Eq. (1.1.12) into Eq. (1.1.10), we can then determine the eigenvector $\hat{\boldsymbol{E}}_{1j}$ corresponding to the eigenvalue ω_j, which reveals the *polarization*[7] of the perturbed electric (or magnetic) field for the j-th branch (normal mode).

Hereafter the subscript 1 indicating perturbation will be dropped for simplicity (e.g., $\hat{\boldsymbol{E}}_{1j}$ will be written simply as $\hat{\boldsymbol{E}}_j$).

Homework #1

(1) Illustrate with an example (e.g., ideal magnetohydrodynamic waves) the above mentioned concepts: conductivity tensor $\tilde{\sigma}$, dispersion relation D, ω branches (or normal modes) $\{\omega_j\}$, eigenvectors $\hat{\boldsymbol{E}}_j$ and wave polarizations.

(2) Consider the following inverse Laplace transform

$$\phi(t) = \int_C \frac{\mathrm{d}\omega}{2\pi} \frac{\mathrm{e}^{-\mathrm{i}\omega t}}{D(\omega)}. \tag{H.1.1}$$

How will you choose the integration contour C while ensuring causality if

(i) $D(\omega)$ has no zeros in the upper half ω plane (Im $\omega > 0$)?

(ii) $D(\omega)$ has finite number of zeros in the upper half ω plane (Im $\omega > 0$)?

(3) (i) What is the condition that a function $\phi(t, x)$ is Fourier transformable in x, i.e., $\phi(t, k) = \int_{\mathbb{R}} \mathrm{d}x\, \mathrm{e}^{-ikx}\phi(t, x)$ exists?

(ii) Suppose that $\phi(t = 0, x)$ is sufficiently smooth and of finite spatial extent, i.e., $\phi(t = 0, x) = 0$ for $|x| \geq d$. Is $\phi(t, x)$ Fourier transformable for any <u>finite</u> t?

[7] Loosely speaking, polarization is defined as the relative orientation of the perturbed electric (or magnetic) field with respect to some particular direction, e.g., with respect to $\boldsymbol{k}$ or to the equilibrium (background) magnetic field $\boldsymbol{B}_0$.

1.2 Analysis with two time scales

In plasmas, due to the presence of weak dissipation or source, waves tend to either grow or decay slowly with time. In many cases of interest, the dissipation or source is so weak that the characteristic growth or decay rate is much smaller than the typical oscillation frequency ω_0. The wave process is then said to possess *two time scales*, and we may assume the wave electric field $\boldsymbol{E}$ to be of the following form

$$\boldsymbol{E}(t,\boldsymbol{x}) = \hat{\boldsymbol{E}}(\varepsilon t)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega_0 t)}, \quad \omega_0 \in \mathbb{R}. \tag{1.2.1}$$

Here for simplicity of discussions, we have neglected the spatial dependence of the amplitude (this is discussed in the next section) and will not write the spatial phase factor $\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x})$ when unnecessary; $\varepsilon \ll 1$ is a small parameter such that

$$\frac{\left|\partial\hat{\boldsymbol{E}}/\partial t\right|}{\left|\omega_0\hat{\boldsymbol{E}}\right|} = \mathcal{O}(\varepsilon) \ll 1. \tag{1.2.2}$$

We shall explore some implications of the two time scales.

First, we show that given the relation

$$A(\omega) = B(\omega)E(\omega) \tag{1.2.3}$$

where ω is the Laplace-transform variable, we have

$$\hat{A}(\varepsilon t) = B\left(\omega_0 + \mathrm{i}\frac{\partial}{\partial t}\right)\hat{E}(\varepsilon t). \tag{1.2.4}$$

Proof of Eq. (1.2.4):

The Laplace transform of $E(t)$ is

$$E(\omega) = \int_0^{+\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}E(t) = \int_0^{+\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\omega_0)t}\hat{E}(\varepsilon t) \equiv \hat{E}(\omega-\omega_0), \tag{1.2.5}$$

then

$$\begin{aligned}
A(t) &= \int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega}{2\pi}\mathrm{e}^{-\mathrm{i}\omega t}A(\omega) = \int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega}{2\pi}\mathrm{e}^{-\mathrm{i}\omega t}B(\omega)\hat{E}(\omega-\omega_0) \\
&\overset{\omega'=\omega-\omega_0}{=\!=\!=} \int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega'}{2\pi}\mathrm{e}^{-\mathrm{i}(\omega_0+\omega')t}B(\omega_0+\omega')\hat{E}(\omega') \\
&\underset{\text{about } \omega'=0}{\overset{\text{expand } B(\omega_0+\omega')}{=\!=\!=\!=\!=}} \mathrm{e}^{-\mathrm{i}\omega_0 t}\int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega'}{2\pi}\mathrm{e}^{-\mathrm{i}\omega' t}\left(B(\omega_0) + \frac{\partial B(\omega_0)}{\partial\omega_0}\omega'\right)\hat{E}(\omega') \\
&= \mathrm{e}^{-\mathrm{i}\omega_0 t}\int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega'}{2\pi}\left(B(\omega_0) + \frac{\partial B(\omega_0)}{\partial\omega_0}\left(\mathrm{i}\frac{\partial}{\partial t}\right)\right)\hat{E}(\omega')\mathrm{e}^{-\mathrm{i}\omega' t} \\
&= \mathrm{e}^{-\mathrm{i}\omega_0 t}\left(B(\omega_0) + \frac{\partial B(\omega_0)}{\partial\omega_0}\left(\mathrm{i}\frac{\partial}{\partial t}\right)\right)\hat{E}(\varepsilon t);
\end{aligned} \tag{1.2.6}$$

that is,

$$\hat{A}(\varepsilon t) = \left(B(\omega_0) + \frac{\partial B(\omega_0)}{\partial \omega_0} \left(\mathrm{i} \frac{\partial}{\partial t} \right) \right) \hat{E}(\varepsilon t) \approx B\left(\omega_0 + \mathrm{i} \frac{\partial}{\partial t} \right) \hat{E}(\varepsilon t). \tag{1.2.7}$$

The expansion of $B(\omega_0 + \omega')$ about $\omega' = 0$ is justified by the fact that $\hat{E}(\varepsilon t)$ changes only slowly with time, which implies that $\hat{E}(\omega')$ is significant only when $\omega' \approx 0$.

Alternatively, note that

$$\mathcal{L}^{-1}\{-\mathrm{i}\omega E(\omega)\}(t) = \frac{\partial E(t)}{\partial t} = -\mathrm{i}\mathrm{e}^{-\mathrm{i}\omega_0 t} \left(\omega_0 + \mathrm{i} \frac{\partial}{\partial t} \right) \hat{E}(\varepsilon t). \tag{1.2.8}$$

Treating ω on the L.H.S. as an operator, Eq. (1.2.3) inversely transforms to

$$A(t) = \mathrm{e}^{-\mathrm{i}\omega_0 t} B\left(\omega_0 + \mathrm{i} \frac{\partial}{\partial t} \right) \hat{E}(\varepsilon t) \tag{1.2.9}$$

or

$$\hat{A}(\varepsilon t) = B\left(\omega_0 + \mathrm{i} \frac{\partial}{\partial t} \right) \hat{E}(\varepsilon t). \tag{1.2.10}$$

■

Next, we note that *the existence of two disparate time scales presents an unique opportunity of separating the dynamics on these scales.* For example, we may average out the fast oscillations in order to examine only the slow timescale phenomena. This concept of averaging out the fast oscillations is essentially the same as the one used in deriving the guiding-center motion [10] where the cyclotron motion is the fast oscillation and is averaged out. For the present case, the method of averaging is simple.

Let[8]

$$\begin{aligned} A(t) &= \mathrm{Re}\left(\hat{A}(\varepsilon t) \mathrm{e}^{-\mathrm{i}\omega_0 t} \right) = \frac{1}{2}\left(\hat{A}(\varepsilon t) \mathrm{e}^{-\mathrm{i}\omega_0 t} + \text{c.c.} \right), \\ B(t) &= \mathrm{Re}\left(\hat{B}(\varepsilon t) \mathrm{e}^{-\mathrm{i}\omega_0 t} \right) = \frac{1}{2}\left(\hat{B}(\varepsilon t) \mathrm{e}^{-\mathrm{i}\omega_0 t} + \text{c.c.} \right), \end{aligned} \tag{1.2.11}$$

then

$$\langle A(t) B(t) \rangle = \frac{1}{T} \int_0^T \mathrm{d}t\, A(t) B(t) = \frac{1}{4}\left(\hat{A}^*(\varepsilon t) \hat{B}(\varepsilon t) + \hat{A}(\varepsilon t) \hat{B}^*(\varepsilon t) \right). \tag{1.2.12}$$

Here, T is a time scale that is much longer than the fast oscillation period but much shorter than the slow timescale

$$1 \ll \omega_0 T \ll \frac{1}{\varepsilon}; \tag{1.2.13}$$

[8]When dealing with multiplications, we need to adopt a real representation for a real quantity.

i.e., for the T time scale, $\hat{A}(\varepsilon t)$ is frozen (adiabatic) and can be taken outside the integral, whereas the fast oscillation $\mathrm{e}^{\pm \mathrm{i}2\omega_0 t}$ is averaged out.

We now apply the above results, Eq. (1.2.4) and Eq. (1.2.12), to the Maxwell equations. The Poynting's theorem for the wave quantities writes

$$\frac{1}{4\pi}\left(\boldsymbol{B}\cdot\frac{\partial\boldsymbol{B}}{\partial t}+\boldsymbol{E}\cdot\left(4\pi\boldsymbol{J}+\frac{\partial\boldsymbol{E}}{\partial t}\right)\right)+\frac{c}{4\pi}\nabla\cdot(\boldsymbol{E}\times\boldsymbol{B})=\boldsymbol{0}. \tag{1.2.14}$$

Let

$$\boldsymbol{E}(t)=\mathrm{Re}\Big(\hat{\boldsymbol{E}}(\varepsilon t)\mathrm{e}^{-\mathrm{i}\omega_0 t}\Big)=\frac{1}{2}\Big(\hat{\boldsymbol{E}}(\varepsilon t)\mathrm{e}^{-\mathrm{i}\omega_0 t}+\text{c.c.}\Big) \tag{1.2.15}$$

and similarly for $\boldsymbol{B}$, then

$$\boldsymbol{E}(\omega)=\frac{1}{2}\Big(\hat{\boldsymbol{E}}(\omega-\omega_0)+\hat{\boldsymbol{E}}^*(\omega+\omega_0)\Big) \tag{1.2.16}$$

where $\hat{\boldsymbol{E}}^*(\omega)\equiv\mathcal{L}\Big\{\hat{\boldsymbol{E}}^*(\varepsilon t)\Big\}(\omega)\neq\Big(\hat{\boldsymbol{E}}(\omega)\Big)^*$.
Since the spatial dependences of $\hat{\boldsymbol{E}}$ and $\hat{\boldsymbol{B}}$ are neglected, after phase-averaging the Poynting vector has no contribution.
Note that

$$\mathcal{L}\left\{4\pi\boldsymbol{J}(t)+\frac{\partial\boldsymbol{E}(t)}{\partial t}\right\}(\omega)=4\pi\boldsymbol{J}(\omega)-\mathrm{i}\omega\boldsymbol{E}(\omega)=-\mathrm{i}\omega\tilde{\epsilon}(\omega)\cdot\boldsymbol{E}(\omega), \tag{1.2.17}$$

using Eq. (1.2.4) we find

$$\begin{aligned}4\pi\boldsymbol{J}+\frac{\partial\boldsymbol{E}}{\partial t}&=\frac{1}{2}\mathrm{e}^{-\mathrm{i}\omega_0 t}\left(-\mathrm{i}\left(\omega_0+\mathrm{i}\frac{\partial}{\partial t}\right)\tilde{\epsilon}\left(\omega_0+\mathrm{i}\frac{\partial}{\partial t}\right)\cdot\hat{\boldsymbol{E}}\right)\\&\quad+\frac{1}{2}\mathrm{e}^{\mathrm{i}\omega_0 t}\left(-\mathrm{i}\left(-\omega_0+\mathrm{i}\frac{\partial}{\partial t}\right)\tilde{\epsilon}\left(-\omega_0+\mathrm{i}\frac{\partial}{\partial t}\right)\cdot\hat{\boldsymbol{E}}^*\right)\\&\approx\frac{1}{2}\mathrm{e}^{-\mathrm{i}\omega_0 t}\left(-\mathrm{i}\omega_0\tilde{\epsilon}(\omega_0)+\frac{\partial(\omega_0\tilde{\epsilon}(\omega_0))}{\partial\omega_0}\frac{\partial}{\partial t}\right)\cdot\hat{\boldsymbol{E}}\\&\quad+\frac{1}{2}\mathrm{e}^{i\omega_0 t}\left(\mathrm{i}\omega_0\tilde{\epsilon}(-\omega_0)+\frac{\partial(\omega_0\tilde{\epsilon}(-\omega_0))}{\partial\omega_0}\frac{\partial}{\partial t}\right)\cdot\hat{\boldsymbol{E}}^*;\end{aligned} \tag{1.2.18}$$

since $4\pi\boldsymbol{J}+\partial\boldsymbol{E}/\partial t$ are real physical quantities, the dielectric tensor must satisfy the following realizability condition

$$\tilde{\epsilon}^*(\omega)=\tilde{\epsilon}(-\omega)\ \text{for}\ \omega\in\mathbb{R}. \tag{1.2.19}$$

Substituting Eq. (1.2.18) into Eq. (1.2.14), and averaging out the fast oscillations using Eq. (1.2.12), we find (denoting $\tilde{\epsilon}(\omega_0)\equiv\tilde{\epsilon}$)

$$\left\langle\boldsymbol{E}\cdot\left(4\pi\boldsymbol{j}+\frac{\partial\boldsymbol{E}}{\partial t}\right)\right\rangle=\frac{1}{4}\Bigg(-\mathrm{i}\omega_0\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}\cdot\hat{\boldsymbol{E}}+\mathrm{i}\omega_0\hat{\boldsymbol{E}}\cdot\tilde{\epsilon}^*\cdot\hat{\boldsymbol{E}}^* \tag{1.2.20}$$

$$
\begin{aligned}
&+\hat{\boldsymbol{E}}^* \cdot \frac{\partial(\omega_0\tilde{\epsilon})}{\partial\omega_0} \cdot \frac{\partial\hat{\boldsymbol{E}}}{\partial t} + \hat{\boldsymbol{E}} \cdot \frac{\partial(\omega_0\tilde{\epsilon}^*)}{\partial\omega_0} \cdot \frac{\partial\hat{\boldsymbol{E}}^*}{\partial t}\Bigg) \\
&\approx \frac{1}{4}\left(2\omega_0\hat{\boldsymbol{E}}^* \cdot \tilde{\epsilon}^{\mathrm{a}} \cdot \hat{\boldsymbol{E}} + \frac{\partial}{\partial t}\left(\hat{\boldsymbol{E}}^* \cdot \frac{\partial\left(\omega_0\tilde{\epsilon}^{\mathrm{h}}\right)}{\partial\omega_0} \cdot \hat{\boldsymbol{E}}\right)\right),
\end{aligned}
\tag{1.2.20}
$$

where we have decomposed the dielectric tensor into its Hermitian and anti-Hermitian parts

$$
\tilde{\epsilon} \equiv \tilde{\epsilon}^{\mathrm{h}} + \mathrm{i}\tilde{\epsilon}^{\mathrm{a}}, \quad \tilde{\epsilon}^{\mathrm{h}} \equiv \frac{1}{2}\left(\tilde{\epsilon} + \tilde{\epsilon}^{\dagger}\right), \quad \tilde{\epsilon}^{\mathrm{a}} \equiv \frac{1}{2\mathrm{i}}\left(\tilde{\epsilon} - \tilde{\epsilon}^{\dagger}\right)
\tag{1.2.21}
$$

and assumed that $\left\|\tilde{\epsilon^{\mathrm{a}}}\right\| / \left\|\tilde{\epsilon^{\mathrm{h}}}\right\| = \mathcal{O}(\varepsilon) \ll 1$ or more precisely

$$
\frac{\left|\hat{\boldsymbol{E}}^* \cdot \partial\left(\omega_0\tilde{\epsilon}^{\mathrm{a}}\right)/\partial\omega_0 \cdot \hat{\boldsymbol{E}}\right|}{\left|\hat{\boldsymbol{E}}^* \cdot \partial\left(\omega_0\tilde{\epsilon}^{\mathrm{h}}\right)/\partial\omega_0 \cdot \hat{\boldsymbol{E}}\right|} = \mathcal{O}(\varepsilon) \ll 1.
\tag{1.2.22}
$$

All together, the phase-averaged Poynting's theorem writes

$$
\frac{\partial W}{\partial t} + \frac{\omega_0}{8\pi}\hat{\boldsymbol{E}}^* \cdot \tilde{\epsilon}^{\mathrm{a}} \cdot \hat{\boldsymbol{E}} = 0,
\tag{1.2.23}
$$

where

$$
W \equiv \frac{1}{16\pi}\left(\left|\hat{\boldsymbol{B}}\right|^2 + \hat{\boldsymbol{E}}^* \cdot \frac{\partial\left(\omega_0\tilde{\epsilon}^{\mathrm{h}}\right)}{\partial\omega_0} \cdot \hat{\boldsymbol{E}}\right)
\tag{1.2.24}
$$

is the *wave energy*. Note that W consists of three components: the magnetic field energy $\left|\hat{\boldsymbol{B}}\right|^2/16\pi$, the electric field energy $\left|\hat{\boldsymbol{E}}\right|^2/16\pi$, and the coherent particle kinetic (mechanical) energy

$$
\frac{1}{16\pi}\hat{\boldsymbol{E}}^* \cdot \frac{\partial\left(\omega_0\tilde{\epsilon}^{\mathrm{h}}\right)}{\partial\omega_0} \cdot \hat{\boldsymbol{E}} - \frac{1}{16\pi}\left|\hat{\boldsymbol{E}}\right|^2 \overset{\tilde{\epsilon}=\tilde{g}+\mathrm{i}4\pi\tilde{\sigma}/\omega_0}{=\!=\!=\!=} -\frac{1}{4}\hat{\boldsymbol{E}}^* \cdot \frac{\partial\left(\omega_0\sigma^{\mathrm{a}}\right)}{\partial\omega_0} \cdot \hat{\boldsymbol{E}};
\tag{1.2.25}
$$

since $\tilde{\sigma}$ is due to particle coherent dynamics, the meaning of particle kinetic (mechanical) energy (which could be negative) is transparent.

Eq. (1.2.23) clearly shows that in plasmas, wave energy may either decrease or increase depending on the anti-Hermitian part of $\tilde{\epsilon}$. As a simple example, let's take $\boldsymbol{k} = \boldsymbol{0}$ and $\tilde{\epsilon} = \epsilon\tilde{g}$, then $\hat{\boldsymbol{B}} = \boldsymbol{0}$, and (noting that $\tilde{\epsilon}^{\mathrm{h}} = \epsilon_{\mathrm{r}}\tilde{g}, \tilde{\epsilon}^{\mathrm{a}} = \epsilon_{\mathrm{i}}\tilde{g}$)

$$
\begin{aligned}
&\frac{\partial}{\partial t}\left(\frac{\partial(\omega_0\epsilon_{\mathrm{r}})}{\partial\omega_0}\left|\hat{\boldsymbol{E}}\right|^2\right) + 2\omega_0\epsilon_{\mathrm{i}}\left|\hat{\boldsymbol{E}}\right|^2 = 0 \\
&\Rightarrow \left|\hat{\boldsymbol{E}}(\varepsilon t)\right|^2 = \left|\hat{\boldsymbol{E}}(t=0)\right|^2 \mathrm{e}^{2\gamma t}
\end{aligned}
\tag{1.2.26}
$$

where

$$
\gamma = -\frac{\omega_0\epsilon_{\mathrm{i}}}{\partial(\omega_0\epsilon_{\mathrm{r}})/\partial\omega_0}.
\tag{1.2.27}
$$

Thus, a wave with positive energy, $\partial(\omega_0\epsilon_{\rm r})/\partial\omega_0 > 0$, would grow (decay) if the dissipation $\omega_0\epsilon_{\rm i}$ is negative (positive); the converse is true for negative-energy waves. Note that the consistency of the two-time-scale analysis requires the previously stated assumption

$$\frac{|\epsilon_{\rm i}|}{|\epsilon_{\rm r}|} = \mathcal{O}(\varepsilon) \ll 1. \tag{1.2.28}$$

Another example is the electrostatic wave, for which $\nabla\times\boldsymbol{E} = \boldsymbol{0} \Rightarrow \boldsymbol{k}\times\hat{\boldsymbol{E}} = \boldsymbol{0}$. The linear responses of the plasma to the electrostatic waves are encoded in the longitudinal dielectric function (which is, a scalar)[9]

$$\epsilon_{ll} \equiv \hat{\boldsymbol{k}}\cdot\tilde{\epsilon}\cdot\hat{\boldsymbol{k}}. \tag{1.2.31}$$

The remaining analyses are the same as the above example.

1.3 Analysis with two time and space scales

In plasmas, spatial dependence plays a crucial role. In particular, it gives rise to the wave dispersion; i.e., ω depends on $\boldsymbol{k}$. As in the case of temporal variations, *wave spatial variations also often exhibits two scales*: one rapid and one slow. Such two-space-scale phenomena can occur due to various reasons, e.g., dissipation or source, propagation of wave packet, and weak nonuniformity, etc.

Thus, in general, we would allow fast and slow variations both in time and space. The wave electric field $\boldsymbol{E}$ can then expressed in the following form

$$\boldsymbol{E}(t,\boldsymbol{x}) = \hat{\boldsymbol{E}}(\varepsilon t,\varepsilon\boldsymbol{x})\mathrm{e}^{\mathrm{i}(\boldsymbol{k}_0\cdot\boldsymbol{x}-\omega_0 t)}, \quad \boldsymbol{k}_0\in\mathbb{R}^3, \omega_0\in\mathbb{R}. \tag{1.3.1}$$

We can carry out analyses similar to those in Section 1.2. For example, given the following relation in $(\omega,\boldsymbol{k})$ space (Laplace-Fourier space)

$$\boldsymbol{A}(\omega,\boldsymbol{k}) = \tilde{B}(\omega,\boldsymbol{k})\cdot\boldsymbol{E}(\omega,\boldsymbol{k})), \tag{1.3.2}$$

we find, in $(t,\boldsymbol{x})$ space (real space), that

$$\hat{\boldsymbol{A}}(\varepsilon t,\varepsilon\boldsymbol{x}) = \tilde{B}\left(\omega_0 + \mathrm{i}\frac{\partial}{\partial t}, \boldsymbol{k}_0 - \mathrm{i}\nabla\right)\cdot\hat{\boldsymbol{E}}(\varepsilon t,\varepsilon\boldsymbol{x}), \tag{1.3.3}$$

i.e., in indicial tensor representation,

[9]Noting the wave equation for the electrostatic waves

$$\left(\boldsymbol{k}\boldsymbol{k} - k^2\tilde{g} + \frac{\omega^2}{c^2}\tilde{\epsilon}\right)\cdot\hat{\boldsymbol{E}} = \frac{\omega^2}{c^2}\tilde{\epsilon}\cdot\hat{\boldsymbol{k}}\hat{E} = \boldsymbol{0}, \tag{1.2.29}$$

the dispersion relation writes

$$\epsilon_{ll} = 0 \text{ and } \epsilon_{tl} \equiv \hat{\boldsymbol{t}}\cdot\tilde{\epsilon}\cdot\hat{\boldsymbol{k}} = 0 \text{ for } \hat{\boldsymbol{t}}\perp\hat{\boldsymbol{k}}. \tag{1.2.30}$$

The condition $\epsilon_{tl} = 0$ ensures that the wave dispersion is purely electrostatic, see Section 3.3.

$$\hat{A}_i(\varepsilon t, \varepsilon \boldsymbol{x}) = \left(B_{ij}(\omega_0, \boldsymbol{k}_0) + \frac{\partial B_{ij}(\omega_0, \boldsymbol{k}_0)}{\partial \omega_0} \left(\mathrm{i} \frac{\partial}{\partial t} \right) \right. \\ \left. + \left(\nabla^m_{\boldsymbol{k}_0} B_{ij}(\omega_0, \boldsymbol{k}_0) \right) (-\mathrm{i} \nabla_m) \right) \hat{E}^j(\varepsilon t, \varepsilon \boldsymbol{x}). \tag{1.3.4}$$

We can of course carry out a similar phase-averaging both in time and space. This is straightforward and we shall not discuss it.

Instead, we shall substitute Eq. (1.3.1) and Eq. (1.3.4) into the Maxwell equations Eq. (1.1.5), and examine the results order by order. For this purpose, we assume that $\|\tilde{\epsilon^{\mathrm{a}}}\| / \|\tilde{\epsilon^{\mathrm{h}}}\| = \mathcal{O}(\varepsilon) \ll 1$, or more precisely,

$$\frac{\left| \hat{\boldsymbol{E}}^* \cdot \partial(\omega_0 \tilde{\epsilon}^{\mathrm{a}}) / \partial \omega_0 \cdot \hat{\boldsymbol{E}} \right|}{\left| \hat{\boldsymbol{E}}^* \cdot \partial(\omega_0 \tilde{\epsilon}^{\mathrm{h}}) / \partial \omega_0 \cdot \hat{\boldsymbol{E}} \right|} = \mathcal{O}(\varepsilon) \ll 1. \tag{1.3.5}$$

At zeroth order, $\mathcal{O}(1)$, we obtain the following equations (denoting $\tilde{\epsilon}(\omega_0, \boldsymbol{k}_0) \equiv \tilde{\epsilon}$)

$$\boldsymbol{k}_0 \times \hat{\boldsymbol{E}} = \frac{\omega_0}{c} \hat{\boldsymbol{B}}, \quad \boldsymbol{k}_0 \times \hat{\boldsymbol{B}} = -\frac{\omega_0}{c} \tilde{\epsilon}^{\mathrm{h}} \cdot \hat{\boldsymbol{E}}, \tag{1.3.6}$$

$$\left(\boldsymbol{k}_0 \boldsymbol{k}_0 - k_0^2 \tilde{g} + \frac{\omega_0^2}{c^2} \tilde{\epsilon}^{\mathrm{h}} \right) \cdot \hat{\boldsymbol{E}} \equiv \tilde{D}^{\mathrm{h}} \cdot \hat{\boldsymbol{E}} = \boldsymbol{0}. \tag{1.3.7}$$

The wave equation Eq. (1.3.7) gives the zeroth-order Hermitian dipsersion relation

$$D^{\mathrm{h}}(\omega_0, \boldsymbol{k}_0) \equiv \det \tilde{D}^{\mathrm{h}}(\omega_0, \boldsymbol{k}_0) = 0 \\ \Rightarrow \omega_0 = \omega_{0j}(\boldsymbol{k}_0), \quad j = 1, 2, \cdots; \tag{1.3.8}$$

an alternative expression for the dipsersion relation can be obtained by dotting $\hat{\boldsymbol{E}}^*$ into Eq. (1.3.7), the result is

$$\left| \boldsymbol{k}_0 \times \hat{\boldsymbol{E}} \right|^2 = \frac{\omega_0^2}{c^2} \hat{\boldsymbol{E}}^* \cdot \tilde{\epsilon}^{\mathrm{h}} \cdot \hat{\boldsymbol{E}}. \tag{1.3.9}$$

At first order, $\mathcal{O}(\varepsilon)$, we have

$$c \nabla \times \hat{\boldsymbol{E}} = -\frac{\partial \hat{\boldsymbol{B}}}{\partial t}, \tag{1.3.10}$$

$$c \nabla \times \hat{\boldsymbol{B}} = \omega_0 \tilde{\epsilon}^{\mathrm{a}} \cdot \hat{\boldsymbol{E}} + \frac{\partial \left(\omega_0 \tilde{\epsilon}^{\mathrm{h}} \right)}{\partial \omega_0} \cdot \frac{\partial \hat{\boldsymbol{E}}}{\partial t} - \omega_0 \left(\nabla \cdot \nabla_{\boldsymbol{k}_0} \right) \left(\tilde{\epsilon}^{\mathrm{h}} \cdot \hat{\boldsymbol{E}} \right). \tag{1.3.11}$$

Performing the following operations

$$\hat{\boldsymbol{E}}^* \cdot \text{Eq. (1.3.11)} + \hat{\boldsymbol{E}} \cdot \text{Eq. (1.3.11)}^* - \hat{\boldsymbol{B}}^* \cdot \text{Eq. (1.3.10)} - \hat{\boldsymbol{B}} \cdot \text{Eq. (1.3.10)}^*,$$

we obtain

$$-c \nabla \cdot \left(\hat{\boldsymbol{E}}^* \times \hat{\boldsymbol{B}} + \hat{\boldsymbol{E}} \times \hat{\boldsymbol{B}}^* \right) = \frac{\partial}{\partial t} \left(\left| \hat{\boldsymbol{B}} \right|^2 + \hat{\boldsymbol{E}}^* \cdot \frac{\partial \left(\omega_0 \tilde{\epsilon}^{\mathrm{h}} \right)}{\partial \omega_0} \cdot \hat{\boldsymbol{E}} \right) + 2 \omega_0 \hat{\boldsymbol{E}}^* \cdot \tilde{\epsilon}^{\mathrm{a}} \cdot \hat{\boldsymbol{E}} \tag{1.3.12}$$

$$-\omega_0\left(\nabla\cdot\nabla_{\boldsymbol{k}_0}\right)\left(\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}^{\mathrm{h}}\cdot\hat{\boldsymbol{E}}\right). \tag{1.3.12}$$

Divided by 16π, and noting from Eq. (1.3.9) that

$$\left|\hat{\boldsymbol{B}}\right|^2=\frac{c^2}{\omega_0^2}\left|\boldsymbol{k}_0\times\hat{\boldsymbol{E}}\right|^2=\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}^{\mathrm{h}}\cdot\hat{\boldsymbol{E}}, \tag{1.3.13}$$

we see that the Poynting's theorem can be written as

$$\frac{\partial W}{\partial t}+\nabla\cdot(\boldsymbol{P}+\boldsymbol{T})+\frac{\omega_0}{8\pi}\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}^{\mathrm{a}}\cdot\hat{\boldsymbol{E}}=0, \tag{1.3.14}$$

where

$$\begin{aligned}W&\equiv\frac{1}{16\pi}\left(\left|\hat{\boldsymbol{B}}\right|^2+\hat{\boldsymbol{E}}^*\cdot\frac{\partial\left(\omega_0\tilde{\epsilon}^{\mathrm{h}}\right)}{\partial\omega_0}\cdot\hat{\boldsymbol{E}}\right)\\&=\frac{1}{16\pi}\frac{1}{\omega_0}\hat{\boldsymbol{E}}^*\cdot\frac{\partial\left(\omega_0^2\tilde{\epsilon}^{\mathrm{h}}\right)}{\partial\omega_0}\cdot\hat{\boldsymbol{E}}\end{aligned} \tag{1.3.15}$$

is the wave energy,

$$\boldsymbol{P}\equiv\frac{c}{8\pi}\mathrm{Re}\left(\hat{\boldsymbol{E}}^*\times\hat{\boldsymbol{B}}\right) \tag{1.3.16}$$

is the Poynting vector representing the energy flux carried by the electromagnetic fields, and

$$\boldsymbol{T}\equiv-\frac{\omega_0}{16\pi}\nabla_{\boldsymbol{k}_0}\left(\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}^{\mathrm{h}}\cdot\hat{\boldsymbol{E}}\right) \tag{1.3.17}$$

is the so-called *kinetic flux* representing the energy flux carried by the particles themselves. CAUTION: the energy flux is not solely given by the Poynting vector. Let

$$\boldsymbol{v}_{\mathrm{g}}:=\frac{\text{total wave energy flux}}{\text{wave energy}}=\frac{\boldsymbol{P}+\boldsymbol{T}}{W}, \tag{1.3.18}$$

Eq. (1.3.14) can then be written in the conservative form

$$\frac{\partial W}{\partial t}+\nabla\cdot\left(\boldsymbol{v}_{\mathrm{g}}W\right)=2\omega_{\mathrm{i}}W \tag{1.3.19}$$

where

$$\omega_{\mathrm{i}}\equiv-\frac{\omega_0}{16\pi}\frac{\hat{\boldsymbol{E}}^*\cdot\tilde{\epsilon}^{\mathrm{a}}\cdot\hat{\boldsymbol{E}}}{W}. \tag{1.3.20}$$

Since for the solution $\omega_0=\Omega_0(\boldsymbol{k}_0)$ of the zeroth-order Hermitian dipsersion relation Eq. (1.3.8) we have (see Homework #2 (3))

$$\boldsymbol{v}_{\mathrm{g}}=\nabla_{\boldsymbol{k}_0}\Omega_0(\boldsymbol{k}_0), \tag{1.3.21}$$

Eq. (1.3.19) shows that the wave energy is carried by the *wave packet* or, sometimes called, *qusi-particle* which propagates at the *group velocity* $\boldsymbol{v}_{\mathrm{g}}$. If $\omega_{\mathrm{i}}=0$, i.e., if the plasma is Hermitian, then the wave energy remains constant along the propagation path of the wave packet; with $\omega_{\mathrm{i}}\neq0$, however, the wave energy can either increase or decrease along the path.

Homework #2

(1) Using the physical realizability condition

$$\tilde{\epsilon}^*(\omega, \boldsymbol{k}) = \tilde{\epsilon}(-\omega, -\boldsymbol{k}) \text{ for } \omega \in \mathbb{R}, \boldsymbol{k} \in \mathbb{R}^3, \tag{H.2.1}$$

prove that[10]

$$(\tilde{\epsilon}(\omega, \boldsymbol{k}))^* = \tilde{\epsilon}(-\omega^*, -\boldsymbol{k}) \text{ for } \omega \in \mathbb{C}, \boldsymbol{k} \in \mathbb{R}^3. \tag{H.2.2}$$

Please note that $(f(z))^* = f^*(z^*) \neq f^*(z), z \in \mathbb{C}$. Eq. (H.2.1) or Eq. (H.2.2) also implies that

$$\tilde{\epsilon}^{\mathrm{h}}(-\omega, -\boldsymbol{k}) = \tilde{\epsilon}^{\mathrm{h}}(\omega, \boldsymbol{k}) \tag{H.2.3}$$

and

$$\tilde{\epsilon}^{\mathrm{a}}(-\omega, -\boldsymbol{k}) = -\tilde{\epsilon}^{\mathrm{a}}(\omega, \boldsymbol{k}). \tag{H.2.4}$$

(2) The cold plasma Langmuir oscillation with collisional dissaption is governed by the following equations

$$\frac{\partial \boldsymbol{V}}{\partial t} = -\frac{e}{m_e}\boldsymbol{E} - \nu \boldsymbol{V}, \quad \frac{\partial \boldsymbol{E}}{\partial t} = 4\pi e n_e \boldsymbol{V}, \tag{H.2.5}$$

or

$$\frac{\partial^2 \boldsymbol{V}}{\partial t^2} + \nu \frac{\partial \boldsymbol{V}}{\partial t} + \omega_{\mathrm{pe}}^2 \boldsymbol{V} = \boldsymbol{0}. \tag{H.2.6}$$

Here $\omega_{\mathrm{pe}} \equiv (4\pi e^2 n_e)/m_e$ is the electron plasma frequency, and we shall assume $|\nu| \ll \omega_{\mathrm{pe}}$.

(i) Solving for $\boldsymbol{V}$ using the two-time-scale approach.

(ii) Derive the dielectric tensor $\tilde{\epsilon}$ and indicate its Hermitian and anti-Hermitian parts.

(iii) Derive an expression for the wave energy W .

(iv) Calculate the damping or growth rate using W and $\tilde{\epsilon}^{\mathrm{a}}$.

(3) Using the disperison relation Eq. (1.3.9), prove that the expression of $\boldsymbol{v}_{\mathrm{g}}$ given by Eq. (1.3.18) agrees with the more familiar one Eq. (1.3.21).

Hint: differentiate Eq. (1.3.9) with respect to $(\omega_0, \boldsymbol{k}_0)$.

1.4 Instabilities

Plasmas in both laboratories and space are often far from thermal equilibrium. For example, in mirror machines as well as earth's radiation belt, the velocity distribution function (VDF) is non-Maxwellian due to the existence of loss cones. Furthermore, in any confined plasmas (either inertially or magnetically), there is also nonuniformities in macroscopic thermodynamic quantities such as density, temperature, pressure, etc. In a way, these *deviations from thermal*

[10]This amounts to say that the real-space dielectric tensor is real [9]. Since $D(\omega, \boldsymbol{k}) = \det(\boldsymbol{k}\boldsymbol{k} - k^2\tilde{g} + (\omega^2/c^2)\tilde{\epsilon})$, the dispersion function $D(\omega, \boldsymbol{k})$ statisfies the same property $(D(\omega, \boldsymbol{k}))^* = D(-\omega^*, -\boldsymbol{k})$. Thus, if $\omega = \Omega(\boldsymbol{k}) = \Omega_{\mathrm{r}}(\boldsymbol{k}) + \mathrm{i}\Omega_{\mathrm{i}}(\boldsymbol{k})$ is a solution to $D(\omega, \boldsymbol{k}) = 0$, i.e., if $D(\Omega(\boldsymbol{k}), \boldsymbol{k}) = 0$, then $(D(\Omega(\boldsymbol{k}), \boldsymbol{k}))^* = D(-\Omega^*(\boldsymbol{k}), -\boldsymbol{k}) = 0$, which implies that $\omega = -\Omega^*(-\boldsymbol{k}) = -\Omega_{\mathrm{r}}(-\boldsymbol{k}) + \mathrm{i}\Omega_{\mathrm{i}}(-\boldsymbol{k})$ is also a solution. Q: Are these two solutions the same? What are the meanings of negative frequencies ($\omega < 0$)?

equilibrium may be regarded as free energies stored in the plasma. Conventionally, they may be categorized as the following three types:

- Velocity-space free energy: velocity-space anisotropy, beams. Examples are loss-cone and two-stream instabilities.
- Magnetic free energy: current and magnetic inhomogeneities. Examples are kink and tearing instabilities.
- Expansion free energy: density, temperature, pressure inhomogeneities. Examples are drift, ballooning/interchange and trapped-particle instabilities.

Instabilities are collective processes which may be triggered (if certain threshold conditions are satisfied) to release the above-mentioned free energies. That is, via the instabilities, these free energies can be converted into either turbulent plasma motions and/or electromagnetic radiations. In this respect, instabilities may be viewed as anomalous (compared to collisional) processes whereby plasmas can relax toward thermal equilibrium and, therefore, play crucial roles in our understanding of important subjects in both laboratory and space plasmas, such as disruption in tokamaks, anomalous transport processes, collisionless shocks, solar radio bursts, etc.

In an infinite, uniform plasma, we say that an instability exists if and only if for some real $\boldsymbol{k}$ the dispersion relation $D(\omega, \boldsymbol{k}) = 0$ admits a solution with $(\mathrm{Im}\ \omega) \equiv \gamma > 0$. Here, we note our convention is

$$\boldsymbol{E}(t, \boldsymbol{x}) = \hat{\boldsymbol{E}} \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)} \tag{1.4.1}$$

with $(\omega, \boldsymbol{k})$ properly understood to be the Laplace-in-time and Fourier-in-space transform pair. That is, if a linear instability exists, then a wave, which is periodic in space with a wavevector $\boldsymbol{k}$, will exponentiate in time with the growth rate given by γ.

Based on the instability excitation mechanisms, we may divide instabilities into two types: dissipative-type and reactive-type.

Dissipative-type instabilities are excited via (negative or positive) dissipative processes or, more generally speaking, the *anti-Hermitian part of the dielectric tensor*, as we have discussed in Section 1.2 and Section 1.3. Thus, we decompose the dispersion function and the ω solution into real and imaginary parts, respectively[11]

$$D(\omega, \boldsymbol{k}) \equiv D_{\mathrm{r}}(\omega, \boldsymbol{k}) + \mathrm{i} D_{\mathrm{i}}(\omega, \boldsymbol{k}) = 0, \quad \omega \equiv \omega_{\mathrm{r}} + \mathrm{i}\gamma, \tag{1.4.3}$$

and assume that

[11]By definition

$$D_{\mathrm{r}} \equiv \frac{1}{2}(D + D^*), \quad D_{\mathrm{i}} \equiv \frac{1}{2\mathrm{i}}(D - D^*); \tag{1.4.2}$$

note that $D^*(\omega_{\mathrm{r}}, \boldsymbol{k}) = (D(\omega_{\mathrm{r}}, \boldsymbol{k}))^*$ whereas $D^*(\omega, \boldsymbol{k}) \neq (D(\omega, \boldsymbol{k}))^*$ for $\omega \in \mathbb{C}$.

$$\frac{|D_{\rm i}|}{|D_{\rm r}|} = \mathcal{O}(\varepsilon) \ll 1, \quad \left|\frac{\gamma}{\omega_{\rm r}}\right| = \mathcal{O}(\varepsilon) \ll 1. \tag{1.4.4}$$

Then the dipsersion relation becomes, at zeroth order $\mathcal{O}(1)$,

$$\begin{aligned} &D_{\rm r}(\omega_{\rm r}, \boldsymbol{k}) = 0 \\ &\Rightarrow \omega_{\rm r} = \omega_{{\rm r}j}(\boldsymbol{k}), \quad j = 1, 2, \cdots; \end{aligned} \tag{1.4.5}$$

at first order $\mathcal{O}(\varepsilon)$, we have, from the Taylor expansion of $D_{\rm r}$ about $\omega = \omega_{{\rm r}j}(\boldsymbol{k})$,

$$\gamma_j(\boldsymbol{k}) = -\left.\frac{D_{\rm i}(\omega, \boldsymbol{k})}{\partial D_{\rm r}(\omega, \boldsymbol{k})/\partial\omega}\right|_{\omega=\omega_{{\rm r}j}(\boldsymbol{k})}. \tag{1.4.6}$$

Thus, for plasmas with $D_{\rm i} < 0$, a wave with $\partial D_{\rm r}(\omega_{\rm r})/\partial\omega_{\rm r} > 0$ is unstable ($\gamma > 0$); a wave with $\partial D_{\rm r}(\omega_{\rm r})/\partial\omega_{\rm r} < 0$, however, is stable or damped ($\gamma < 0$). Converse conclusions can be made for $D_{\rm i} > 0$. In Homework #3, we shall have an example of a negative-energy wave driven unstable via positive dissipations.

As to reactive-type instabilities, the most well-known example is the electrostatic beam-plasma instability. First, however, we shall illustrate with the following model dispersion relation:

$$\epsilon(\omega, k) = \epsilon_{\rm r}(\omega, k) = 1 - \frac{\omega_{\rm p}}{\omega} + \frac{\omega_{\rm b}}{\omega - kv_0} = 0. \tag{1.4.7}$$

Eq. (1.4.7) is a quadratic equation in ω and, hence, can be readily solved. For simplicity, we shall assume $\omega_{\rm b} \ll \omega_{\rm p}$. Thus

$$\begin{aligned} \omega_{1,2} &= \frac{1}{2}\left(kv_0 + \omega_{\rm p} - \omega_{\rm b} \pm \sqrt{\left(\omega_{\rm p} - kv_0\right)^2 - 2\omega_{\rm b}\left(\omega_{\rm p} + \omega_{\rm b} + kv_0\right)}\right) \\ &\approx \frac{1}{2}\left(kv_0 + \omega_{\rm p} \pm \sqrt{\left(\omega_{\rm p} - kv_0\right)^2 - 2\omega_{\rm b}\left(\omega_{\rm p} + kv_0\right)}\right). \end{aligned} \tag{1.4.8}$$

Thus, there exists an instability if

$$\left|\omega_{\rm p} - kv_0\right| \lesssim 2\sqrt{\omega_{\rm b}\omega_{\rm p}}, \tag{1.4.9}$$

and the maximum growth rate $\gamma_{\rm max} \approx \sqrt{\omega_{\rm b}\omega_{\rm p}}$ occurs at $kv_0 \approx \omega_{\rm p}$. These results are sketched in terms of the $\omega - k$ dispersion curves shown in Figure 1.1.

Since instability occurs at $kv_0 \approx \omega_{\rm p}$ this model reactive-type instability can be understood in terms of coupling between the $\omega_1 \approx \omega_{\rm p}$ and $\omega_2 \approx kv_0$ modes. Now, away from coupling, the dipsersion relations for the ω_1 and ω_2 modes are, respectively,

$$\begin{aligned} &0 = \epsilon_{\rm r1}(\omega, k) = 1 - \frac{\omega_{\rm p}}{\omega} \text{ for } \left|\frac{kv_0}{\omega}\right| \gg 1 \text{ or } \left|\frac{kv_0}{\omega}\right| \ll 1 \\ &\Rightarrow \omega_1 = \omega_{\rm p} \end{aligned} \tag{1.4.10}$$

and

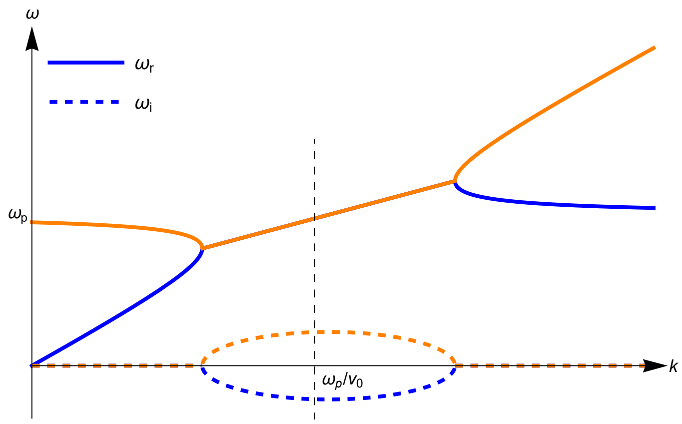


Figure 1.1: Sketch of $\omega - k$ curves for the model dispersion relation of Eq. (1.4.7).

$$0 = \epsilon_{r2}(\omega, k) = \begin{cases} 1 + \omega_b/(\omega - kv_0) & \text{for } kv_0 \sim \omega \gg \omega_p \\ -\omega_p/\omega + \omega_b/(\omega - kv_0) & \text{for } kv_0 \sim \omega \ll \omega_p \end{cases}$$
$$\Rightarrow \omega_2 = \begin{cases} -\omega_b + kv_0 & \text{for } kv_0 \gg \omega_p \\ kv_0(1 + \omega_b/\omega_p) & \text{for } kv_0 \ll \omega_p \end{cases}. \tag{1.4.11}$$

Since $\partial(\omega_1\epsilon_{r1})/\partial\omega_1 = 1 > 0$, the ω_1 mode has positive energy; on the other hand, $\partial(\omega_2\epsilon_{r2})/\partial\omega_2 = 1 - \omega_b kv_0/(\omega_2 - kv_0)^2 < 0$, i.e., the ω_2 mode has negative energy. Thus, the above model reactive-type instability can be explained in terms of the *coupling between a negative-energy and a positive-energy wave*. It is easy to see that if both modes are positive-energy waves, then the system is stable. For example, the following model dispersion relation

$$\epsilon(\omega, k) = \epsilon_r(\omega, k) = 1 - \frac{\omega_p}{\omega} - \frac{\omega_b}{\omega - kv_0} = 0 \tag{1.4.12}$$

has no unstable solutions, as is shown in Figure 1.2. Similarly, we can show that the coupling between two negative-energy waves introduces no instabilities.

We now briefly examine the beam-plasma instability. Here, we let a cold electron beam streaming through a background cold plasma. Let the beam velocity be v_b and density n_b. For

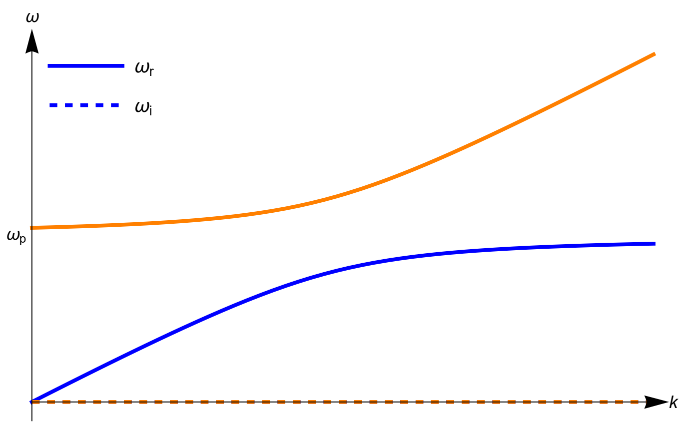


Figure 1.2: Sketch of $\omega - k$ curves for the model dispersion relation of Eq. (1.4.12).

the background plasma, we have density n_0 and $m_i \to +\infty$, i.e., immobile ions. For simplicity we assume the beam is weak, $n_b \ll n_0$. The corresponding electrostatic dispersion relation can then be shown to be

$$\epsilon(\omega, k) = \epsilon_{\mathrm{r}}(\omega, k) = 1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2} - \frac{\omega_{\mathrm{p}b}^2}{(\omega - kv_b)^2} = 0. \tag{1.4.13}$$

Similar to the model problem, away from coupling, the dipsersion relations for the ω_1 and ω_2 modes are, respectively,

$$\begin{aligned} 0 = \epsilon_{\mathrm{r}1}(\omega, k) = 1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2} \text{ for } \left|\frac{kv_b}{\omega}\right| \gg 1 \text{ or } \left|\frac{kv_b}{\omega}\right| \ll 1 \\ \Rightarrow \omega_1 = \pm\omega_{\mathrm{pe}} \end{aligned} \tag{1.4.14}$$

and

$$\begin{aligned} 0 = \epsilon_{\mathrm{r}2}(\omega, k) &= \begin{cases} 1 - \omega_{\mathrm{p}b}^2/(\omega - kv_b)^2 & \text{for } kv_b \sim \omega \gg \omega_{\mathrm{pe}} \\ -\omega_{\mathrm{pe}}^2/\omega^2 - \omega_{\mathrm{p}b}^2/(\omega - kv_b)^2 & \text{for } kv_b \sim \omega \ll \omega_{\mathrm{pe}} \end{cases} \\ \Rightarrow \omega_2 &= \begin{cases} \pm\omega_{\mathrm{p}b} + kv_b & \text{for } kv_b \gg \omega_{\mathrm{pe}} \\ kv_b(1 \pm \mathrm{i}\omega_{\mathrm{p}b}/\omega_{\mathrm{pe}}) & \text{for } kv_b \ll \omega_{\mathrm{pe}} \end{cases}. \end{aligned} \tag{1.4.15}$$

The corresponding $\omega - k$ dispersion curves are sketched in Figure 1.3. One can prove that the coupling between the positive-energy wave $\omega_1 = \omega_{\mathrm{pe}}$ and the negative-energy wave $\omega_2 = -\omega_{\mathrm{p}b} + kv_b$, which leads to the beam-plasma instability, happens at $k \le k_{\mathrm{c}}$ where $k_{\mathrm{c}} v_b = \left(1 + (n_b/n_0)^{1/3}\right)^{3/2} \omega_{\mathrm{pe}}$, and that the maximum growth rate $\gamma_{\max} \approx \left(\sqrt{3}/4^{4/3}\right)\left(\omega_{\mathrm{pe}}\omega_{\mathrm{p}b}^2\right)^{1/3}$ occurs at $kv_b \approx \omega_{\mathrm{pe}}$.

1.5 Nyquist technique for stability analysis

In many practical applications, the dispersion relation $D(\omega, \boldsymbol{k}) = 0$ is generally too complicated to solve analytically for instability growth rates. In fact, the first thing one would like to know

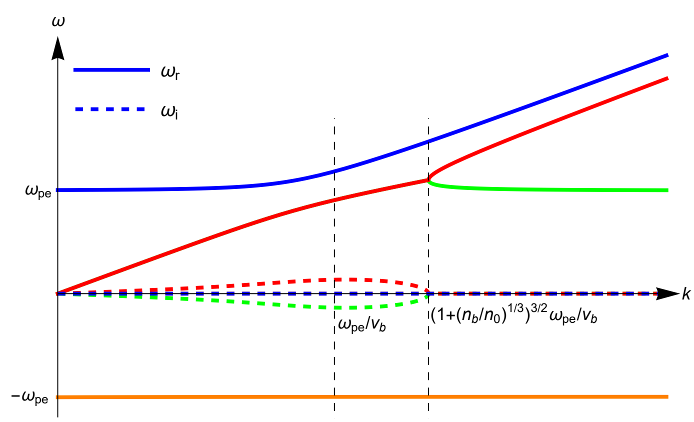


Figure 1.3: Sketch of $\omega - k$ curves for the beam-plasma dispersion relation of Eq. (1.4.13).

is if the dispersion relation admits solutions of ω with $\mathrm{Im}\ \omega > 0$. For this purpose, there exists the powerful *Nyquist technique*.

Theorem (argument principle [11]):
Assume that the function $f : \mathbb{C} \to \mathbb{C}$ is holomorphic inside a contour $C \subset \mathbb{C}$ apart from a finite number of poles $\{b_j \in \mathbb{C} \mid j = 1, 2, \cdots, P\}$, with $\{a_i \in \mathbb{C} \mid i = 1, 2, \cdots, N\}$ the zeros (finite number) of f inside C. If f has neither zeros nor poles on C, then

$$\begin{aligned} \frac{1}{2\pi\mathrm{i}} \oint_C \mathrm{d}z\, \frac{f'(z)}{f(z)} &= \sum_{i=1}^{N} n_i - \sum_{j=1}^{P} p_j \\ &= \#(\text{zeros of } f \text{ inside } C) - \#(\text{poles of } f \text{ inside } C), \end{aligned} \tag{1.5.1}$$

where n_i and p_j are the multiplicities of the zero a_i and the pole b_j, respectively, and $\#$ is the counting measure. Note that the zeros and poles are counted with their multiplicities.

In particular, if f has no poles inside C, i.e., if f is holomorphic inside C, then

$$\frac{1}{2\pi\mathrm{i}} \oint_C \mathrm{d}z\, \frac{f'(z)}{f(z)} = \sum_{i=1}^{N} n_i = \#(\text{zeros of } f \text{ inside } C). \tag{1.5.2}$$

Let $w = f(z), \Gamma = f(C)$, then the quantity

$$W := \frac{1}{2\pi\mathrm{i}} \oint_C \mathrm{d}z\, \frac{f'(z)}{f(z)} = \frac{1}{2\pi\mathrm{i}} \oint_\Gamma \frac{\mathrm{d}w}{w} = \frac{1}{2\pi} \Delta_\Gamma \mathrm{Arg}\ w \tag{1.5.3}$$

counts how many times the image $\Gamma = f(C)$ encircles the origin $w = 0$ in the counterclockwise direction; in other words, W is the *winding number*[12] of the map f (with respect to the origin).

Proof of the argument principle:
Let c_k be a zero or pole of f inside C with multiplicity m_k. Then we can write

$$f(z) = (z - c_k)^{m_k} g_k(z), \quad z \in V_k, \tag{1.5.4}$$

where V_k is a neighborhood of c_k which does not contain any other zeros or poles of f, and g_k is holomorphic and non-zero in V_k. Thus

$$\frac{f'(z)}{f(z)} = \frac{m_k}{z - c_k} + \frac{g_k'(z)}{g_k(z)}, \quad z \in V_k. \tag{1.5.5}$$

Note that $g_k'(z)/g_k(z)$ is holomorphic in V_k,

$$\begin{aligned} \frac{1}{2\pi\mathrm{i}} \oint_C \mathrm{d}z\, \frac{f'(z)}{f(z)} &\overset{\text{Cauchy's theorem}}{=\!=\!=\!=} \frac{1}{2\pi\mathrm{i}} \sum_k \oint_{\partial V_k} \mathrm{d}z\, \frac{f'(z)}{f(z)} \\ &\overset{\text{residue theorem}}{=\!=\!=\!=} \sum_k m_k = \sum_{i=1}^{N} n_i - \sum_{j=1}^{P} p_j. \end{aligned} \tag{1.5.6}$$

■

[12] Note that the winding number is a topological invariant in that it does not change under continuous deformation of the contour C as long as C encloses the same number of zeros.

Since the dielectric tensor is holomorphic in the upper half ω plane, the dispersion relation $D(\omega, \boldsymbol{k})$ is also holomorphic there. We now apply the argument principle to $D(\omega, \boldsymbol{k})$ to determine, for a fixed $\boldsymbol{k}$, whether $D(\omega, \boldsymbol{k})$ has zeros in the upper half ω plane, i.e., whether there exists instabilities. For this purpose, we choose the contour C in the upper half ω plane to be $C = C_\delta + C_R$, where

$$\begin{aligned} C_\delta &= \{\omega_\mathrm{r} + \mathrm{i}\delta \mid \omega_\mathrm{r} \in [-R, R]\}, \\ C_R &= \left\{\mathrm{i}\delta + R\mathrm{e}^{\mathrm{i}\theta} \mid \theta \in [0, \pi]\right\}; \end{aligned} \tag{1.5.7}$$

see Figure 1.4. Here, δ and R are two positive real numbers. In order that C encloses all the unstable roots, we need to take δ arbitrarily small ($\delta = 0^+$ is ok) and R sufficiently large ($R = +\infty$ is ok); if $\tilde{\epsilon}$ (and thus D) has no singularity on the real ω axis, we may simple take $\delta = 0$. From the argument principle Eq. (1.5.2), for a fixed $\boldsymbol{k}$, the number of zeros of $D(\omega, \boldsymbol{k})$ in the upper half ω plane is given by

$$W(\boldsymbol{k}) = \frac{1}{2\pi\mathrm{i}} \oint_C \mathrm{d}\omega \, \frac{\partial D(\omega, \boldsymbol{k})/\partial\omega}{D(\omega, \boldsymbol{k})}; \tag{1.5.8}$$

thus, for the chosoen $\boldsymbol{k}$, there exists instabilities if and only if $W(\boldsymbol{k})$ is a positive integer.[13] This is the, sometimes called, Nyquist instability theorem. On the other hand, noting the geometric interpretation Eq. (1.5.3), we have now reduced the stability analysis to the problem of mapping the contour C to the D complex plane and counting the winding number; the image $D(C, \boldsymbol{k})$ is called the *Nyquist diagram.*

Let's illustrate this technique with the beam-plasma instability discussed in Section 1.4. The corresponding linear dispersion relation is

$$\epsilon(\omega, k) = \epsilon_\mathrm{r}(\omega, k) = 1 - \frac{\omega_\mathrm{pe}^2}{\omega^2} - \frac{\omega_{\mathrm{p}b}^2}{(\omega - kv_b)^2} = 0 \tag{1.5.9}$$

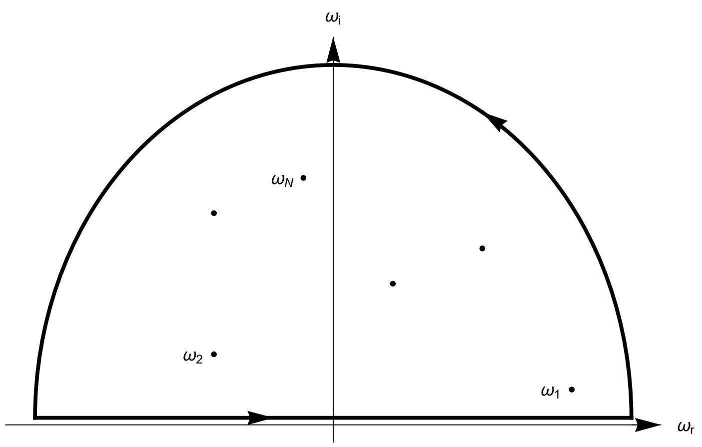
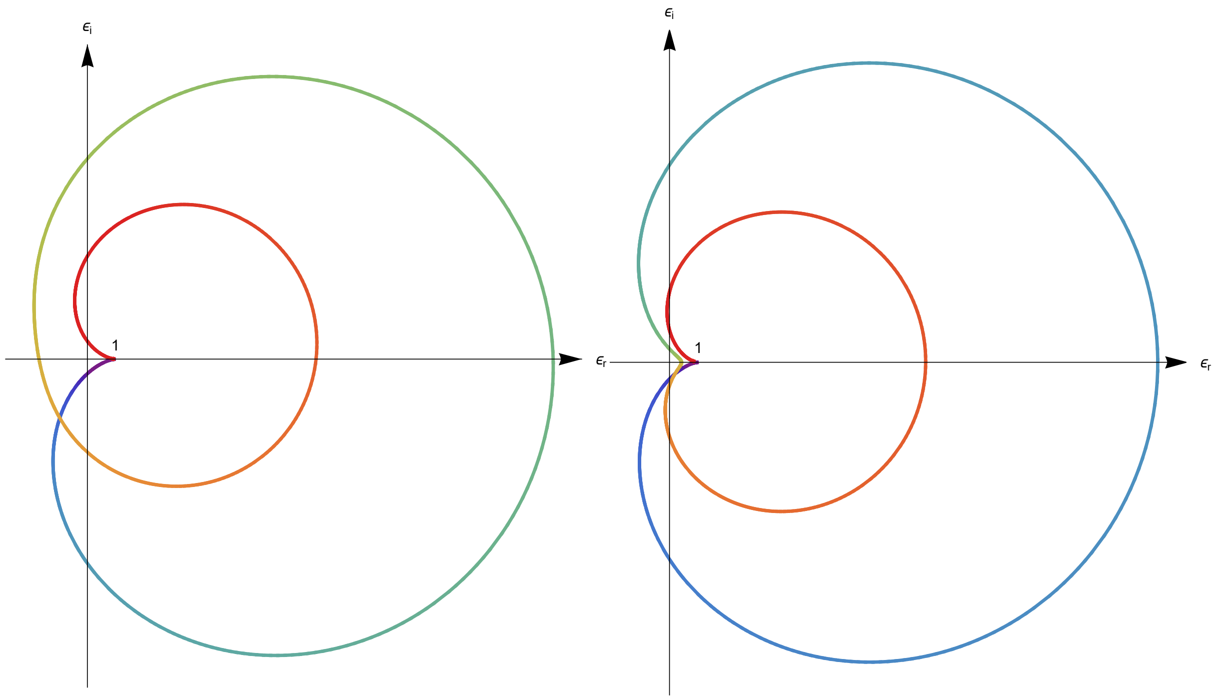


Figure 1.4: Integration contour in the complex ω plane for Nyquist stability analysis.

[13]Note that the winding number $W(\boldsymbol{k})$ can be calculated numerically for a given contour C. For numerical applications of the Nyquist technique, see [12].



(a) (b)

Figure 1.5: Nyquist diagram for the beam-plasma instability with (a) $kv_b \ll \omega_{\rm pe}$ and (b) $kv_b \gg \omega_{\rm pe}$. The large semicircle C_R is mapped to $\epsilon = 1 \pm {\rm i}0$, and the image of the straight line C_δ is colored with a rainbow gradient, where purple corresponds to $\omega_{\rm r} \to -\infty$ and red corresponds to $\omega_{\rm r} \to +\infty$.

with $\omega_{\rm pe} \gg \omega_{{\rm p}b}$ assumed for simplicity. Since ϵ has three poles on the real ω axis, we have to take the distance between C_δ and the real ω axis, δ, to be arbitrarily small but still positive. For the $kv_b \ll \omega_{\rm pe}$ limit which we know is unstable, the image $\epsilon(C)$ (shown in Figure 1.5 (a)) encircles the origin $\epsilon = 0$ once in the counterclockwise direction, i.e., $W = 1$, which agrees with the result obtained in Section 1.4. On the other hand, for the $kv_b \gg \omega_{\rm pe}$ limit which we know is stable, the image $\epsilon(C)$ (shown in Figure 1.5 (b)) does not encircle the origin and we have $W = 0$, which means there is no instability (as it should).

The Nyquist technique for stability analyses is not limited to simple dispersion relation. As an example, we shall apply it to a stability analysis governed by a differential equation. This application, which is relevant to the eigenmode stability problems to be discussed later, is presented here to further demonstrate the Nyquist technique. Consider the following model ordinary differential equation (ODE)

$$\left(\frac{{\rm d}^2}{{\rm d}x^2} + \omega^2 + {\rm i}\nu\omega - \omega_0^2 - x^2\right) y(x) = 0, \quad \nu > 0 \tag{1.5.10}$$

with the boundary conditions

$$y(x) \to 0 \text{ as } |x| \to +\infty. \tag{1.5.11}$$

As a physical motivation, Eq. (1.5.10) may be regarded as modelling the wave equation for the Bohm-Gross (warm electron plasma) waves in a Gaussian density cavity. In order to apply the Nyquist technique, we need to obtain an equivalent dispersion relation in ω. For that purpose, we apply the operation $\int_{\mathbb{R}} \mathrm{d}x\, y^*(x)$ to Eq. (1.5.10) and make use of the boundary condition Eq. (1.5.11) to find

$$0 = D(\omega) \equiv \omega^2 + \mathrm{i}\nu\omega - \omega_0^2 - \frac{\langle |xy|^2 \rangle + \left\langle |\mathrm{d}y/\mathrm{d}x|^2 \right\rangle}{\langle |y|^2 \rangle}, \tag{1.5.12}$$

where $\langle A \rangle \equiv \int_{\mathbb{R}} \mathrm{d}x\, A(x)$.

Now since $D(\omega) \to \infty$ as $\omega \to \infty$, we shall choose the radius of the semicircle, R, to be sufficiently large but still finite. The image $D(C)$ is shown in Figure 1.6, from which we conclude that Eq. (1.5.10) predicts no instability. In fact, we may choose C_δ to be infinitesimally below the real ω axis and prove that no marginally stable solution exists either.

Finally, we remark that, while in this model example the above conclusions can also be obtained by either noting that Eq. (1.5.10) is a Weber equation[14] or solving Eq. (1.5.12) which is a

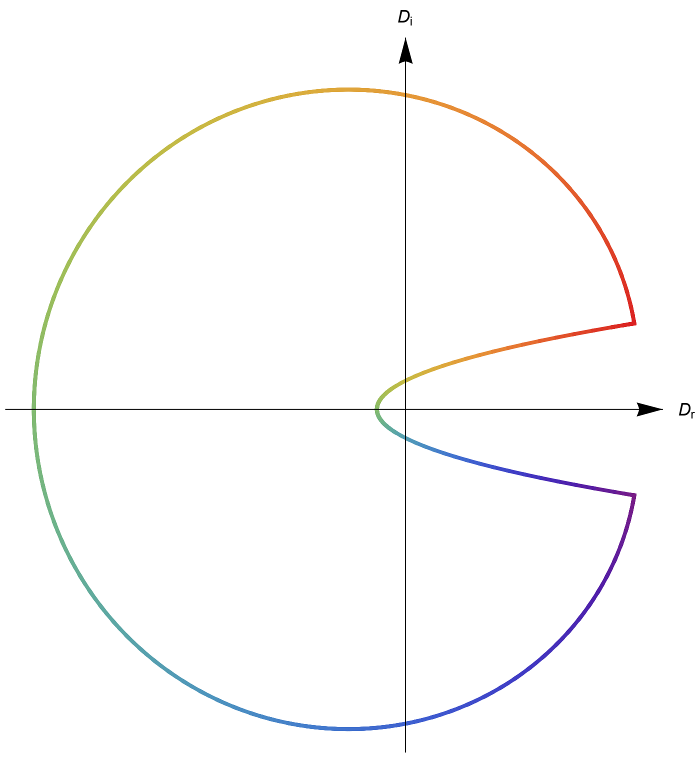


Figure 1.6: Nyquist diagram for the model ODE Eq. (1.5.10), which looks like the Pac-Man. The large semicircle C_R and the image of the straight line C_δ are all colored with a rainbow gradient, where purple corresponds to $\theta \to \pi$ or $\omega_\mathrm{r} \to -\infty$ and red corresponds to $\theta \to 0$ or $\omega_\mathrm{r} \to +\infty$.

[14]The general form of the Weber equation is $\phi''(z) + (\mathcal{E} - z^2)\phi(z) = 0$, where $\mathcal{E}$ is a constant; bounded solutions exist only for $\mathcal{E} = 2n+1, n \in \mathbb{Z}_{\geq 0}$. [11] Recall that the time-independent Schrödinger equation for the harmonic oscillator is a Weber equation.

quadratic equation in ω, the Nyquist technique is more powerful and applicable to a broader range of problems.

Homework #3

(1) Consider the beam-plasma model, where cold electrons streaming through background immobile ions with velocity v_b. The corresponding electrostatic dispersion relation is

$$\epsilon(\omega, k) = 1 - \frac{\omega_{pb}^2}{(\omega - kv_b)^2} = 0. \tag{H.3.1}$$

Suppose there is an electric field of the plane wave form

$$E(t, x) = \frac{1}{2}\left(\hat{E}e^{i(kx-\omega t)} + \text{c.c.}\right) \tag{H.3.2}$$

where $\hat{E}$ is independent of t and x. Calculate the phase-averaged particle kinetic energy density for both the slow and fast modes.

(2) The following dispersion relation describes beam-plasma interaction including collisional effects on the background electrons.

$$1 - \frac{\omega_{pe}^2}{\omega(\omega + i\nu)} - \frac{\omega_{pb}^2}{(\omega - kv_b)^2} = 0, \quad \nu > 0. \tag{H.3.3}$$

Show that in the $kv_b \gg \omega_{pe}$ limit which is stable in the reactive limit $\nu \to 0$, the negative-energy wave (i.e., the slow beam mode) is unstable due to the positive collisional dissipation.

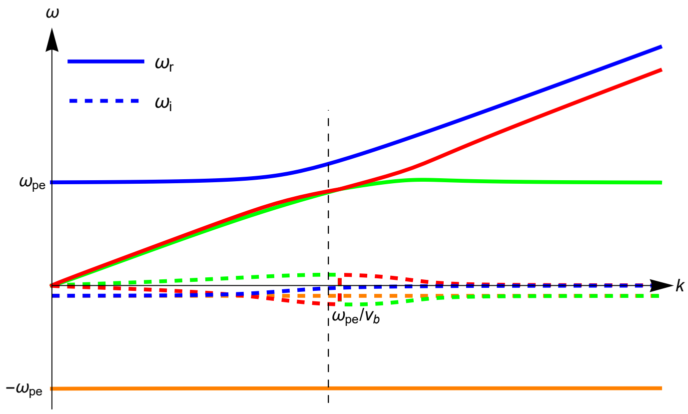


Figure 1.7: Negative-energy slow beam mode becomes unstable in the $kv_b \gg \omega_{pe}$ limit due to positive collisional dissipation.

(3) Using Nyquist technique to prove that, as discussed in Section 1.4, the model dispersion relation Eq. (1.4.7), $1 - \omega_p/\omega + \omega_b/(\omega - kv_0) = 0$, predicts

(i) stability for $kv_0 \ll \omega_p$; (ii) stability for $kv_0 \gg \omega_p$; (iii) instability for $kv_0 = \omega_p$.

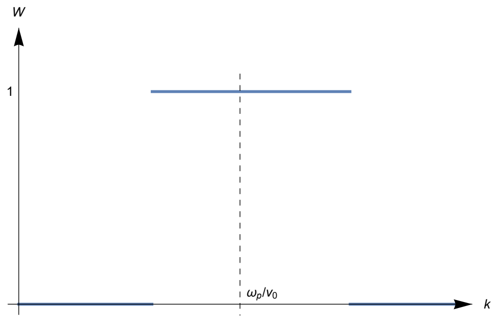


Figure 1.8: Winding number W as a function of k for the model dispersion relation Eq. (1.4.7).

1.6 Absolute and convective instabilities

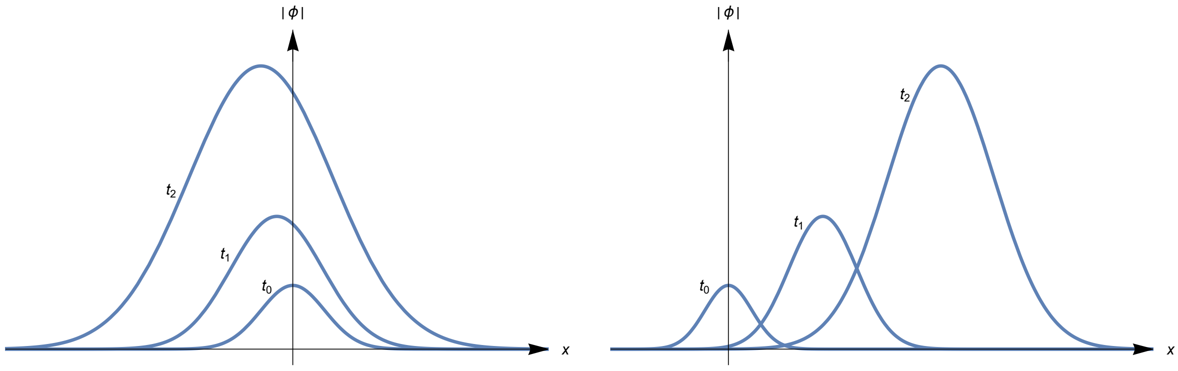
The reference of this section is Chapter 2 of [13].

Up to now, our definition of instabilities (see Section 1.4) is based on perturbations which have plane wave spatial dependence, i.e., $\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x})$. In other words, the perturbations are taken to be periodic in $\boldsymbol{x}$ and, hence, have infinite spatial extent. In reality, however, the perturbations are generally of finite spatial extent. Treating as an initial value problem, there then exists two types of time-asymptotic ($t \to +\infty$) behaviors at a fixed finite spatial point. One is the absolute instability characterized by perturbations which become unbounded as $t \to +\infty$ at every fixed finite point in space. The other is the convective instability characterized by perturbations which propagate and grow along the system such that, at any fixed finite point in space, the perturbations van ish t ime-asymptotically. Sketches of the time-asymptotic behaviors of these two types are shown in Figure 1.9.

We emphasize that the distinguish bewteen these two types of instabilities is not simply an academic exercise but carries important practical implications. This is because (i) instabilities are grown out of low-level thermal fluctuations and (ii) the unstable region is, in practice, of finite spatial extent. For convective instabilities, the perturbations are spatially amplified inside the unstable region. Thus, their amplitudes as well as their nonlinear consequences can be estimated. For example, if the unstable region is small, then the convertive instabilities may be practically insignificant. On the other hand, absolute instabilities can grow to large amplitudes such that their saturation can only be due to nonlinear effects.

In terms of operators, let's assume that the perturbation, $\phi(t,x)$, is governed by the following initial value problem (we will only consider the one-dimensional case, generalization to higher dimensions is straightforward)

$$D\left(\mathrm{i}\frac{\partial}{\partial t}, -\mathrm{i}\frac{\partial}{\partial x}\right)\phi(t,x) = 0, \quad \phi(t=0,x) = \phi_0(x). \tag{1.6.1}$$



(a) (b)

Figure 1.9: Sketches of the time-asymptotic behaviors of (a) an absolute instability for which $|\phi(x,t)| \to +\infty$ as $t \to +\infty$ for a fixed finite x and (b) a convective instability for which $|\phi(x,t)| \to 0$ as $t \to +\infty$ for a fixed finite x.

Here, we reiterate that the system is linearly unstable, i.e., the linear dispersion relation $D(\omega, k) = 0$ has unstable (Im $\omega > 0$) roots for real k.

Laplace transforming Eq. (1.6.1) in time, we obtain an ODE

$$D\left(\omega, -\mathrm{i}\frac{\partial}{\partial x}\right)\phi(\omega, x) = s(x), \tag{1.6.2}$$

where $s(x)$ represents the effective initial perturbations. The *Green's function* of the above ODE, $G(\omega, x)$, satisfies

$$D\left(\omega, -\mathrm{i}\frac{\partial}{\partial x}\right)G(\omega, x) = \delta(x). \tag{1.6.3}$$

Since the solution $\phi(\omega, x)$ is determined by $G(\omega, x)$, i.e.,

$$\phi(\omega, x) = \int_{\mathbb{R}} \mathrm{d}x\, s(x')G(\omega, x - x'), \tag{1.6.4}$$

in order to determine the time-asymptotic behavior of $\phi(t, x)$ it suffices to investigate

$$G(t, x) = \int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega}{2\pi} G(\omega, x)\mathrm{e}^{-\mathrm{i}\omega t}. \tag{1.6.5}$$

Fourier transforming Eq. (1.6.3) in space, we obtain[15]

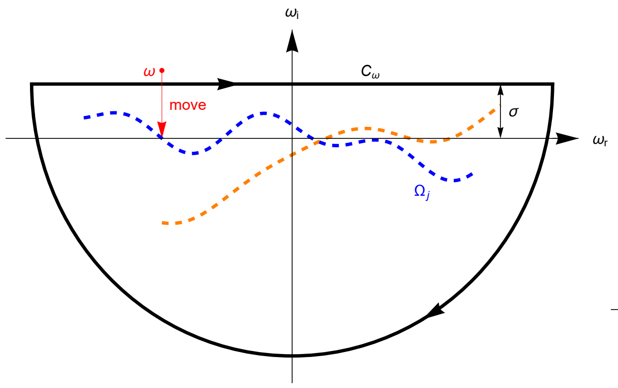
$$D(\omega, k)G(\omega, k) = 1 \Rightarrow G(\omega, k) = \frac{1}{D(\omega, k)}, \tag{1.6.6}$$

thus

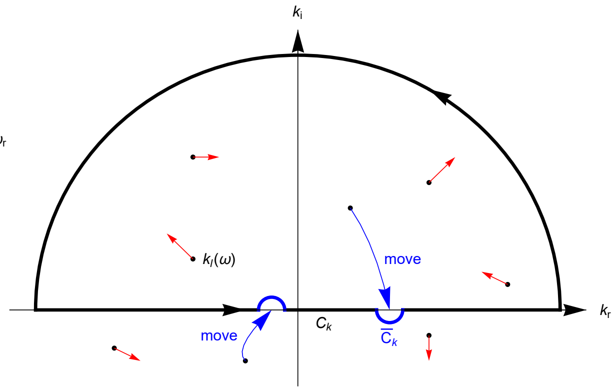
$$G(\omega, x) = \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \frac{\mathrm{e}^{\mathrm{i}kx}}{D(\omega, k)}. \tag{1.6.7}$$

Causility requires that

[15]Question: $G(\omega, k) + = C\delta(D(\omega, k))$?



(a)



(b)

Figure 1.10: Sketches of integration contours for (a) Eq. (1.6.5) and (b) Eq. (1.6.7)

$$\sigma > \sup\{\text{Im } \omega_j(k) \mid k \in \mathbb{R}, j = 1, 2, \cdots\}, \tag{1.6.8}$$

where $\mathbb{R} + \mathrm{i}\sigma$ is the inverse Laplace transform contour, and $\omega = \omega_j(k) \in \mathbb{C}, j = 1, 2, \cdots$ are the solutions of $D(\omega, k) = 0$ for fixed $k \in \mathbb{R}$. That Im $\omega_j(k) > 0$ for some (j, k), of course, is consistent with the fact that the system is unstable. Furthermore, as being physically resonable, we assume that $D(\omega, k)$ is sufficiently well-hebaved as ω or k approaches the infinity of the ω or k complex plane, respectively. The ω and k integration contours for, respectively, $t > 0$ and $x > 0$, can then be closed by sufficiently large semicircles in the lower half ω plane and upper half k plane, respectively, and the resultant contours are denoted by $C_\omega \equiv (\mathbb{R} + \mathrm{i}\sigma) + C_R^-$ and $C_k \equiv \mathbb{R} + C_R^+$, respectively; see the black lines of Figure 1.10.

Since $G(\omega, x)$ is holomorphic only for Im $\omega \geq \sigma$, we need to analytically continuate it as we close the ω integration contour in the lower half ω plane (Im $\omega < 0$). Let's take one point on (or above) $\mathbb{R} + \mathrm{i}\sigma$, ω, and continuate it downward; see the red line in Figure 1.10(a).

Let $k = k_l(\omega) \in \mathbb{C}, l = 1, 2, \cdots$ be the solutions of $D(\omega, k) = 0$ for fixed $\omega \in \mathbb{C}$. Since $k_l(\omega)$ depends on ω, the poles of $1/D(\omega, k)$ in the k plane will move around as ω moves downward; see the red lines in Figure 1.10(b). In particular, we konw that as $\omega \to \zeta_j \in \Omega_j$ where $\Omega_j \subset \mathbb{C}$ is the image of the map $\omega_j : \mathbb{R} \to \mathbb{C}, k \mapsto \omega_j(k)$ (see the dashed lines in Figure 1.10(a)), some pole(s) in k plane must move toward the real k axis; this is because ω_j is defined by $D\big(\omega_j(k), k\big) = 0$ for $k \in \mathbb{R}$. As, say, $k_m(\omega)$ becomes real, $G(\omega, x)$ becomes non-holomorphic since $k_m\big(\zeta_j\big)$ now lies on the k integration contour C_k. In this respect, $\big\{\Omega_j \mid j = 1, 2, \cdots\big\}$ are the branch cuts of $G(\omega, x)$ in the ω plane. In order to analytically continuate $G(\omega, x)$ below Ω_j, it is then necessary to deform C_k around those poles which approach or cross the real k axis, in a way that the <u>identities</u> of those poles which contribute to the integral do <u>not</u> change. Denote the new contour as $\overline{C}_k$ (see the blue lines in Figure 1.10), the analytic continuation of $G(\omega, x)$ is then given by

$$\overline{G}(\omega, x) = \oint_{\overline{C}_k} \frac{\mathrm{d}k}{2\pi} \frac{\mathrm{e}^{\mathrm{i}kx}}{D(\omega, k)} \xlongequal[\text{assume simple poles}]{\text{residue theorem}} \mathrm{i}\sum_l{}' \left.\frac{\mathrm{e}^{\mathrm{i}kx}}{\partial D(\omega, k)/\partial k}\right|_{k=k_l(\omega)} \tag{1.6.9}$$

with the understanding that only those k_l for which

$$\text{Im } k_l(\omega) > 0 \text{ for } \omega \text{ on (or above) } \mathbb{R} + \mathrm{i}\sigma \tag{1.6.10}$$

are included in the primed summation.

Substituting Eq. (1.6.9) into Eq. (1.6.5), it is clear that the poles of $\overline{G}(\omega, x)$ will contribute to the ω integral. In particular, in the time-asymptotic limit $t \to +\infty$, the pole with maximum ω_{i} dominates. For each l in the primed summation, we can solve for several ω from the equation

$$\left.\frac{\partial D(\omega, k)}{\partial k}\right|_{k=k_l(\omega)} = 0 \Rightarrow \omega = \omega_{nl} \in \mathbb{C}, n = 1, 2, \cdots. \tag{1.6.11}$$

Let $\omega_{\mathrm{M}} \in \mathbb{C}$ be the solution with maximum ω_{i}, i.e.,

$$\text{Im } \omega_{\mathrm{M}} \geq \text{Im } \omega_{nl} \text{ for all } n, l, \tag{1.6.12}$$

and denote the particular branch k_l from which ω_{M} is solved as k_M, with $k_{\mathrm{M}} \equiv k_M(\omega_{\mathrm{M}}) \in \mathbb{C}$. Thus, $(\omega_{\mathrm{M}}, k_{\mathrm{M}})$ satisfies

$$D(\omega_{\mathrm{M}}, k_{\mathrm{M}}) = 0, \quad \frac{\partial D(\omega_{\mathrm{M}}, k_{\mathrm{M}})}{\partial k_{\mathrm{M}}} = 0, \tag{1.6.13}$$

with ω_{M} being the most unstable solution. This implies that $D(\omega_{\mathrm{M}}, k) = 0$ has a double root at $k = k_{\mathrm{M}}$; in other words, as $\omega \to \omega_{\mathrm{M}}$, two of the k plane poles coalesce at $k = k_{\mathrm{M}}$. Expanding $D(\omega, k)$ about $(\omega_{\mathrm{M}}, k_{\mathrm{M}})$, we find

$$\begin{aligned} D(\omega, k) &\approx \frac{\partial D(\omega_{\mathrm{M}}, k_{\mathrm{M}})}{\partial \omega_{\mathrm{M}}}(\omega - \omega_{\mathrm{M}}) + \frac{1}{2}\frac{\partial^2 D(\omega_{\mathrm{M}}, k_{\mathrm{M}})}{\partial k_{\mathrm{M}}^2}(k - k_{\mathrm{M}})^2 \\ &\equiv A(\omega - \omega_{\mathrm{M}}) + \frac{1}{2}B(k - k_{\mathrm{M}})^2. \end{aligned} \tag{1.6.14}$$

We now Substitute Eq. (1.6.14) into Eq. (1.6.9). Note that[16] $\overline{G}(\omega_{\mathrm{M}}, x)$ is singular (regular) if the two converging poles lie in the opposite (same) side of $\overline{C}_k$, we need only to deal with the singular case for which $k_{\mathrm{M}} \in \overline{C}_k$. Near $k = k_{\mathrm{M}}$, $\overline{C}_k$ looks like a straight line, and we can write, for $\omega \approx \omega_{\mathrm{M}}$ and finite x,

$$\begin{aligned} \overline{G}(\omega, x) &\approx \frac{2}{B}\mathrm{e}^{\mathrm{i}k_{\mathrm{M}}x}\int_{-\varepsilon+k_{\mathrm{M}}}^{\varepsilon+k_{\mathrm{M}}} \frac{\mathrm{d}k}{2\pi}\frac{\mathrm{e}^{\mathrm{i}(k-k_{\mathrm{M}})x}}{2A(\omega-\omega_{\mathrm{M}})/B + (k-k_{\mathrm{M}})^2} \\ &\xlongequal[\alpha=\sqrt{2A(\omega-\omega_{\mathrm{M}})/B}]{k'=k-k_{\mathrm{M}}} \frac{2}{B}\mathrm{e}^{\mathrm{i}k_{\mathrm{M}}x}\int_{-\varepsilon}^{\varepsilon}\frac{\mathrm{d}k'}{2\pi}\frac{\cos(k'x)}{\alpha^2 + k'^2} \\ &\xlongequal[\varepsilon/\alpha\to\infty]{k'=\alpha u} \pm\frac{1}{B}\mathrm{e}^{\mathrm{i}k_{\mathrm{M}}x}\frac{1}{\alpha}\frac{1}{\pi}\int_{\mathbb{R}}\mathrm{d}u\,\frac{\cos(\alpha x u)}{1+u^2} \\ &\xlongequal{\exp(-|\alpha x|)\to 1} \pm(2AB)^{-1/2}(\omega-\omega_{\mathrm{M}})^{-1/2}\mathrm{e}^{\mathrm{i}k_{\mathrm{M}}x}. \end{aligned} \tag{1.6.15}$$

[16] If the two converging poles lie in the same side of $\overline{C}_k$, then we may continue to deform $\overline{C}_k$ to avoid singularity; if, however, they lie in the opposite side, then the contour $\overline{C}_k$ is clamped and the singularity is inevitable.

Here we have used the integration formula [14]

$$\int_0^{+\infty} \mathrm{d}u\, \frac{\cos(xu)}{1+u^2} = \frac{\pi}{2}\mathrm{e}^{-|x|}. \tag{1.6.16}$$

Finally,

$$G(t,x) = \oint_{C_\omega} \frac{\mathrm{d}\omega}{2\pi} \overline{G}(\omega,x)\mathrm{e}^{-\mathrm{i}\omega t} \approx \pm(2AB)^{-1/2}\mathrm{e}^{\mathrm{i}(k_{\mathrm{M}}x-\omega_{\mathrm{M}}t)} \oint_{C_\omega} \frac{\mathrm{d}\omega}{2\pi} \frac{\mathrm{e}^{-\mathrm{i}(\omega-\omega_{\mathrm{M}})t}}{\sqrt{\omega-\omega_{\mathrm{M}}}}. \tag{1.6.17}$$

Noting that the function $\sqrt{\omega-\omega_{\mathrm{M}}}$ is multivalued, we cannot complete the integration contour with C_R^- since it intersects the branch cut $\{\omega_{\mathrm{M}} - \mathrm{i}\rho \mid \rho \in [0,+\infty)\}$; instead, we choose an integration contour C'_ω which consists of three parts and is consistent with the branch cut, see Figure 1.11. By Cauchy's theorem, the contour integral along C'_ω equals the integral along the two sides of the branch cut[17]

$$\begin{aligned}\int_{C'_\omega} \frac{\mathrm{d}\omega}{2\pi} \frac{\mathrm{e}^{-\mathrm{i}(\omega-\omega_{\mathrm{M}})t}}{\sqrt{\omega-\omega_{\mathrm{M}}}} &= \frac{1}{2\pi}\left(\frac{\mathrm{e}^{\mathrm{i}(2n\pi-\pi/2)}}{\mathrm{e}^{\mathrm{i}(n\pi-\pi/4)}} - \frac{\mathrm{e}^{\mathrm{i}(2n\pi+3\pi/2)}}{\mathrm{e}^{\mathrm{i}(n\pi+3\pi/4)}}\right)\int_0^{+\infty} \mathrm{d}\rho\, \rho^{-\frac{1}{2}}\mathrm{e}^{-\rho t} \\ &= \frac{\mathrm{e}^{\mathrm{i}(n\pi-\pi/4)}}{\pi}\frac{1}{\sqrt{t}}\Gamma\left(\frac{1}{2}\right) = (-1)^n\frac{1-\mathrm{i}}{\sqrt{2\pi}}\frac{1}{\sqrt{t}}.\end{aligned} \tag{1.6.18}$$

Here, we have choosen $\arg(\omega-\omega_{\mathrm{M}})|_{\text{right}} = 2n\pi - \pi/2$, then $\arg(\omega-\omega_{\mathrm{M}})|_{\text{left}} = 2n\pi + (3\pi)/2$ is not arbitrary.

Thus, our final result for the time-asymptotic behavior of the Green's function is

$$|G(t,x)| \propto \frac{1}{\sqrt{t}}\mathrm{e}^{(\mathrm{Im}\ \omega_{\mathrm{M}})t-(\mathrm{Im}\ k_{\mathrm{M}})x} \text{ as } t\to+\infty \text{ for finite } x. \tag{1.6.19}$$

Eq. (1.6.19) clearly shows that if $\mathrm{Im}\ \omega_{\mathrm{M}} > 0$, then we have $|G(t,x)| \to +\infty$ as $t \to +\infty$ for finite x, i.e., an absolute instability; otherwise, for $\mathrm{Im}\ \omega_{\mathrm{M}} \le 0$, we have $|G(t,x)| \to 0$ as $t \to +\infty$ for finite x, i.e., a convective instability.

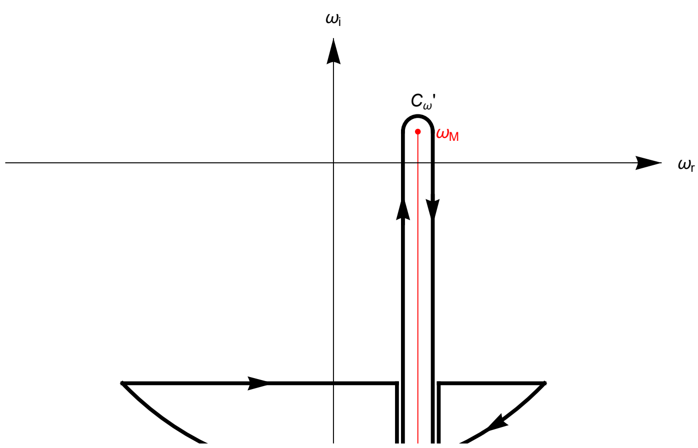


Figure 1.11: ω integration contour which is consistent with the branch cut of $\sqrt{\omega-\omega_{\mathrm{M}}}$.

[17] The integral on the small semicircle around ω_{M} vanishes as the radius of the semicircle goes to zero because $\sqrt{\omega-\omega_{\mathrm{M}}}\mathrm{e}^{-\mathrm{i}(\omega-\omega_{\mathrm{M}})t} \rightrightarrows 0$ as $\omega \to \omega_{\mathrm{M}}$; see lemma 3.4 of [15].

Let's summarize the reuslts obtained in this section. First, we assume that the system is linearly unstable, i.e., the linear dispersion relation $D(\omega, k) = 0$ has unstable ($\mathrm{Im}\ \omega > 0$) roots for real k. We then show that, if

(1) there exists two k solutions of $D(\omega, k) = 0$ which lie, respectively, in the upper and lower half of the k plane for ω on (or above) $\mathbb{R} + \mathrm{i}\sigma$, and

(2) these two k solutions coalesce ($\partial D(\omega, k)/\partial k = 0$) at $\omega = \omega_{\mathrm{M}}$ with $\mathrm{Im}\ \omega_{\mathrm{M}} > 0$,

then we have an absolute instability; otherwise, the system only exhibits convective instabilities. We remark that the condition $\partial D(\omega_{\mathrm{M}}, k_{\mathrm{M}})/\partial k_{\mathrm{M}} = 0$ is equivalent to the condition that the group velocity vanishes, $\partial \omega_{\mathrm{M}}/\partial k_{\mathrm{M}} = 0$. Thus, the above requirements for absolute instabilities are also consistens with our physical intuition that absolute instabilities correspond to standing-still wave packets which, while growing in time, also tend to spread out (in both positive and negative directions) in space. Finally, we emphasize that the nature of the instability depends crucially on the reference frame. This is intuitively expected, because a convective instability in the laboratory frame will appear to be an absolute one to the co-moving observer who travels at the group velocity of the wave packet.

Let us illustrate the application of the above stability criteria with the following three examples:

(1) Consider the model dispersion relation:

$$D(\omega, k) = 1 + \frac{k_0^2}{(k - \omega/v_1)(k - \omega/v_2)} = 0, \quad v_1 v_2 > 0. \tag{1.6.20}$$

By considering the $k \to 0$ limit, it is clear that Eq. (1.6.20) predicts instability. Meanwhile, in the limit $\mathrm{Im}\ \omega \to +\infty$ we find $k_{1,2} \to \omega/v_{1,2}$, which implies that the two k plane poles lie on the same side of the k integration contour $\overline{C}_k$ for ω on (or above) $\mathbb{R} + \mathrm{i}\sigma$ and, hence, the instability is convective.

(2) Consider another model dispersion relation:

$$D(\omega, k) = 1 - \frac{k_0^2}{(k - \omega/v_1)(k - \omega/v_2)} = 0, \quad v_1 v_2 > 0. \tag{1.6.21}$$

Similarly, by considering the $k \to 0$ limit it is clear that Eq. (1.6.21) predicts instability. Meanwhile, in the limit $\mathrm{Im}\ \omega \to +\infty$ we find $k_1 \to \omega/v_1$ and $k_2 \to -\omega/v_2$, which implies that the two k plane poles lie on the opposite side of $\overline{C}_k$. We now calculate ω_{M} where k_1 and k_2 coalesce:

$$\begin{aligned} \frac{\partial D(\omega_{\mathrm{M}}, k_{\mathrm{M}})}{\partial k_{\mathrm{M}}} &= \frac{k_0^2}{(k_{\mathrm{M}} - \omega_{\mathrm{M}}/v_1)^2 (k_{\mathrm{M}} - \omega_{\mathrm{M}}/v_2)^2} \left(2k_{\mathrm{M}} + \frac{\omega_{\mathrm{M}}}{v_2} - \frac{\omega_{\mathrm{M}}}{v_1} \right) = 0 \\ &\Rightarrow k_{\mathrm{M}} = \frac{\omega_{\mathrm{M}}}{2} \left(\frac{1}{v_1} - \frac{1}{v_2} \right), \end{aligned} \tag{1.6.22}$$

then

$$D(\omega_{\mathrm{M}}, k_{\mathrm{M}}) = 0 \Rightarrow \omega_{\mathrm{M}} = \mathrm{i}|k_0\overline{v}|, \quad \frac{2}{\overline{v}} \equiv \frac{1}{v_1} + \frac{1}{v_2}. \tag{1.6.23}$$

Thus, Im $\omega_{\mathrm{M}} > 0$ and we have an absolute instability. Note that k_{M} is not, in general, a real number, i.e., the two k plane poles coalesce on $\overline{C}_k$ instead of on C_k.

(3) Consider the beam-plasma instability

$$\epsilon(\omega, k) = 1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2} - \frac{\omega_{\mathrm{p}b}^2}{(\omega - kv_b)^2} = 0. \tag{1.6.24}$$

Now, as in Example (1), we have, in the limit Im $\omega \to +\infty$, $k_{1,2} \to \omega/v_b$ and, thus, this instability is convective.

2 Linear Waves and Instabilities in Uniform Unmagnetized Plasmas

In the previous chapter, we have discussed some general wave properties in a dielectric medium, assuming that the dielectric tensor, $\tilde{\epsilon}$, is given. In this and the next chapters, we shall derive specific expressions for $\tilde{\epsilon}$ and discuss the results. Since only high-temperature plasmas are of interest here, we assume, unless otherwise stated, that the plasmas are collisionless. Thus, the plasma dynamics obey the following Vlasov equation[18]

$$\left(\frac{\partial}{\partial t}+\boldsymbol{v}\cdot\nabla+\frac{q}{m}\left(\boldsymbol{E}+\frac{\boldsymbol{v}}{c}\times\boldsymbol{B}\right)\cdot\nabla_{\boldsymbol{v}}\right)f(t,\boldsymbol{x},\boldsymbol{v})=0 \tag{2.0.1}$$

plus the Maxwell equations. Here, $f=f(t,\boldsymbol{x},\boldsymbol{v})$ is the ensemble-averaged one-particle [16] phase-space density (PSD), i.e., $f(t,\boldsymbol{x},\boldsymbol{v})\,\mathrm{d}^3x\,\mathrm{d}^3v$ gives the number of particles inside the phase-spcae volume element $\mathrm{d}^3x\,\mathrm{d}^3v$ at time t; the normalization of f is given by

$$\int_{\mathbb{R}^3}\mathrm{d}^3x\int_{\mathbb{R}^3}\mathrm{d}^3v\,f(t,\boldsymbol{x},\boldsymbol{v})=N,\quad n(t,\boldsymbol{x})=\int_{\mathbb{R}^3}\mathrm{d}^3v\,f(t,\boldsymbol{x},\boldsymbol{v}), \tag{2.0.2}$$

where N is the total number of particles and n the local number density.

2.1 Solution of the linearized Vlasov equation in the electroasttic limit

Let

$$f=f_0(\boldsymbol{x},\boldsymbol{v})+\delta f(t,\boldsymbol{x},\boldsymbol{v}), \tag{2.1.1}$$

where f_0 and δf are, respectively, the equilibrium and perturbed phase-space densities. Now with $\boldsymbol{E}_0=\boldsymbol{B}_0=\boldsymbol{0}$, we find $\nabla f_0=\boldsymbol{0}$, i.e.,

$$\begin{aligned} f_0&=f_0(\boldsymbol{v})\\ \Rightarrow n_0&=\text{constant}. \end{aligned} \tag{2.1.2}$$

As for δf, we shall assume, for simplicity of discussion, that the wave perturbations[19] are <u>electrostatic</u>, i.e., $\boldsymbol{B}\propto\nabla\times\boldsymbol{E}=\boldsymbol{0}$. Thus, we have

$$\left(\frac{\partial}{\partial t}+\boldsymbol{v}\cdot\nabla\right)\delta f=-\frac{q}{m}\boldsymbol{E}\cdot\nabla_{\boldsymbol{v}}(f_0+\delta f). \tag{2.1.3}$$

Linearizing Eq. (2.1.3) or, specifically, assuming

[18]The essence of the Vlasov equation is to neglect collisional effects or *correlations* between particles [16], [17]; in other words, Vlasov theory is a self-consistent *mean-field theory*.

[19]Remember that the subscript 1 indicating perturbation is omitted.

$$\frac{|\nabla_{\boldsymbol{v}}\delta f|}{|\nabla_{\boldsymbol{v}}f_0|} \ll 1, \tag{2.1.4}$$

we arrive at the linearized Vlasov equation

$$\mathcal{L}_0 f_1 \equiv \left(\frac{\partial}{\partial t} + \boldsymbol{v}\cdot\nabla\right) f_1 = -\frac{q}{m}\boldsymbol{E}\cdot\nabla_{\boldsymbol{v}} f_0. \tag{2.1.5}$$

Here we use a different symbol f_1 for δf in the linearized equation just to distinguish between the exact δf in Eq. (2.1.3) and the $\delta f \equiv f_1$ in Eq. (2.1.5). Eq. (2.1.5) is a first-order linear partial differential equation (PDE) and, hence, can be readily solved by integrating along its *characteristics* or, in this case, the *zeroth-order (unperturbed) particle orbits* determined by the following ODEs

$$\frac{\mathrm{d}\boldsymbol{x}'(t')}{\mathrm{d}t'} = \boldsymbol{v}'(t'), \quad \frac{\mathrm{d}\boldsymbol{v}'(t')}{\mathrm{d}t'} = \boldsymbol{0} \tag{2.1.6}$$

with the boundary conditions

$$\boldsymbol{x}'(t'=t) = \boldsymbol{x}, \quad v'(t'=t) = \boldsymbol{v}. \tag{2.1.7}$$

The solution to Eq. (2.1.5) is then

$$\begin{aligned} f_1(t,\boldsymbol{x},\boldsymbol{v}) &= f_{10}(\boldsymbol{x}_0,\boldsymbol{v}_0) - \frac{q}{m}\int_0^t \mathrm{d}t'\,\boldsymbol{E}(t',\boldsymbol{x}'(t'))\cdot\nabla_{\boldsymbol{v}'} f_0(\boldsymbol{v}'(t')) \\ &= f_{10}(\boldsymbol{x}-\boldsymbol{v}t,\boldsymbol{v}) - \frac{q}{m}\nabla_{\boldsymbol{v}} f_0\cdot\int_0^t \mathrm{d}t'\,\boldsymbol{E}(t',\boldsymbol{x}-\boldsymbol{v}(t-t')), \end{aligned} \tag{2.1.8}$$

where $\boldsymbol{x}_0 \equiv \boldsymbol{x}'(t'=0) = \boldsymbol{x}-\boldsymbol{v}t$, $\boldsymbol{v}_0 \equiv \boldsymbol{v}'(t'=0) = \boldsymbol{v}$ and $f_{10}(\boldsymbol{x},\boldsymbol{v}) \equiv f_1(t=0,\boldsymbol{x},\boldsymbol{v})$. Substituting Eq. (2.1.8) into the Gauss's law (which is conventionally referred to as the Poisson's equation in plasma physics literature)

$$\nabla\cdot\boldsymbol{E} = 4\pi\sum_s q_s\int_{\mathbb{R}^3}\mathrm{d}^3v\, f_{1s}, \quad s = \text{species}, \tag{2.1.9}$$

we then have the linear dynamics completely determined.

We now examine the linerazation from a different point of view. Noting that the full perturbed Vlasov equation can also be written as

$$\left(\frac{\partial}{\partial t} + \boldsymbol{v}\cdot\nabla + \frac{q}{m}\boldsymbol{E}\cdot\nabla_{\boldsymbol{v}}\right)\delta f = -\frac{q}{m}\boldsymbol{E}\cdot\nabla_{\boldsymbol{v}} f_0, \tag{2.1.10}$$

δf, thus, can be formally solved as

$$\delta f(t,\boldsymbol{x},\boldsymbol{v}) = \delta f_0(\boldsymbol{X}_0,\boldsymbol{V}_0) - \frac{q}{m}\int_0^t \mathrm{d}t'\,\boldsymbol{E}(t',\boldsymbol{X}'(t'))\cdot\nabla_{\boldsymbol{V}'} f_0(\boldsymbol{V}'(t')), \tag{2.1.11}$$

where $(\boldsymbol{X}'(t'),\boldsymbol{V}'(t'))$ is, however, particle's <u>exact</u> phase-space trajectory determined by the following ODEs

$$\frac{\mathrm{d}\boldsymbol{X}'(t')}{\mathrm{d}t'} = \boldsymbol{V}'(t'), \quad \frac{\mathrm{d}\boldsymbol{V}'(t')}{\mathrm{d}t'} = \frac{q}{m}\boldsymbol{E}(t', \boldsymbol{X}'(t')) \tag{2.1.12}$$

with the boundary conditions

$$\boldsymbol{X}'(t' = t) = \boldsymbol{x}, \quad v'(t' = t) = \boldsymbol{v}. \tag{2.1.13}$$

Let

$$\boldsymbol{X}' = \boldsymbol{x}' + \delta\boldsymbol{x}', \quad \boldsymbol{V}' = \boldsymbol{v}' + \delta\boldsymbol{v}', \tag{2.1.14}$$

we then have

$$\frac{\mathrm{d}\delta\boldsymbol{x}'(t')}{\mathrm{d}t'} = \delta\boldsymbol{v}'(t'), \quad \frac{\mathrm{d}\delta\boldsymbol{v}'(t')}{\mathrm{d}t'} = \frac{q}{m}\boldsymbol{E}(t', \boldsymbol{X}'(t')). \tag{2.1.15}$$

Substituting Eq. (2.1.14) in to Eq. (2.1.11) and expanding $\boldsymbol{X}'$ and $\boldsymbol{V}'$ about, respectively, $\boldsymbol{x}'$ and $\boldsymbol{v}'$, it is then clear by comparing the results with Eq. (2.1.8) that linearization corresponds to assuming

$$\frac{\left|\int_0^t \mathrm{d}t'\, \delta\boldsymbol{x}'(t') \cdot \nabla_{\boldsymbol{x}'}\boldsymbol{E}(t', \boldsymbol{x}'(t'))\right|}{\left|\int_0^t \mathrm{d}t'\, \boldsymbol{E}(t', \boldsymbol{x}'(t'))\right|} \ll 1 \tag{2.1.16}$$

and

$$\frac{\left|\nabla_{\boldsymbol{v}}\nabla_{\boldsymbol{v}} f_0(\boldsymbol{v}) : \int_0^t \mathrm{d}t'\, \boldsymbol{E}(t', \boldsymbol{x}'(t'))\, \delta\boldsymbol{v}'(t')\right|}{\left|\nabla_{\boldsymbol{v}} f_0(\boldsymbol{v}) \cdot \int_0^t \mathrm{d}t'\, \boldsymbol{E}(t', \boldsymbol{x}'(t'))\right|} \ll 1. \tag{2.1.17}$$

Having established the criteria for linearization, either Eq. (2.1.4) or Eq. (2.1.16) and Eq. (2.1.17), let's go back to Eq. (2.1.8). To go further, we would perform Laplace-in-time and Fourier-in-space transforms,

$$\begin{aligned} \hat{f}_1(\omega, \boldsymbol{k}, \boldsymbol{v}) &= \int_0^{+\infty} \mathrm{d}t \int_{\mathbb{R}^3} \mathrm{d}^3x\, f_1(t, \boldsymbol{x}, \boldsymbol{v}) \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)} \equiv \mathcal{L}(\omega)\mathcal{F}(\boldsymbol{k})\{f_1(t, \boldsymbol{x}, \boldsymbol{v})\}, \\ f_1(t, \boldsymbol{x}, \boldsymbol{v}) &= \int_{\mathbb{R}+\mathrm{i}\sigma} \frac{\mathrm{d}\omega}{2\pi} \int_{\mathbb{R}^3} \frac{\mathrm{d}^3k}{(2\pi)^3} \hat{f}_1(\omega, \boldsymbol{k}, \boldsymbol{v}) \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)} \end{aligned} \tag{2.1.18}$$

where the inverse Laplace transform ω integral contour $\mathbb{R} + i\sigma$ lies above all the poles of $\hat{f}_1$. Equation Eq. (2.1.5) then becomes

$$-\mathrm{i}(\omega - \boldsymbol{k}\cdot\boldsymbol{v})\hat{f}_1(\omega, \boldsymbol{k}, \boldsymbol{v}) = \hat{f}_{10}(\boldsymbol{k}, \boldsymbol{v}) - \frac{q}{m}\hat{\boldsymbol{E}}(\omega, \boldsymbol{k}) \cdot \nabla_{\boldsymbol{v}} f_0(\boldsymbol{v}), \tag{2.1.19}$$

where $\hat{f}_{10}(\boldsymbol{k}, \boldsymbol{v}) = \mathcal{F}\{f_{10}(\boldsymbol{x}, \boldsymbol{v})\}(\boldsymbol{k})$. In the electrostatic limit, we have

$$\begin{aligned} &\boldsymbol{E}(t, \boldsymbol{x}) = -\nabla\varphi(t, \boldsymbol{x}) \\ \Rightarrow\ &\hat{\boldsymbol{E}}(\omega, \boldsymbol{k}) = -\mathrm{i}\boldsymbol{k}\hat{\varphi}(\omega, \boldsymbol{k}). \end{aligned} \tag{2.1.20}$$

We, thus, obtain

$$\hat{f}_1 = \frac{\mathrm{i}\hat{f}_{10} - \frac{q}{m}\hat{\varphi}\boldsymbol{k}\cdot\nabla_v f_0}{\omega - \boldsymbol{k}\cdot\boldsymbol{v}}. \tag{2.1.21}$$

Here, the dependences on $\omega, \boldsymbol{k}$ and $\boldsymbol{v}$ have been surpressed. Noting that in Eq. (2.1.21), only the component of $\boldsymbol{v}$ in the $\boldsymbol{k}$ direction matters, we shall let

$$\begin{aligned} \boldsymbol{k} &= k\hat{\boldsymbol{k}}, \quad k > 0, \\ \boldsymbol{v} &= u\hat{\boldsymbol{k}} + \boldsymbol{v}_\perp \end{aligned} \tag{2.1.22}$$

and

$$(F_1, F_0) \equiv \int_{\mathbb{R}^2} \mathrm{d}^2 v_\perp \, (f_1, f_0). \tag{2.1.23}$$

Thus, from Eq. (2.1.21), we have

$$\hat{F}_1 = \frac{\mathrm{i}\hat{F}_{10} - \frac{q}{m}\hat{\varphi}kF_0'}{\omega - ku}. \tag{2.1.24}$$

Substituting Eq. (2.1.24) into Possion's equation

$$k^2\hat{\varphi} = 4\pi \sum_s q_s \int_{\mathbb{R}} \mathrm{d}u \, \hat{F}_{1s}, \tag{2.1.25}$$

we obtain

$$\hat{\varphi}(\omega, \boldsymbol{k}) = \mathrm{i}\frac{4\pi}{k^2 D(\omega, \boldsymbol{k})} \sum_s q_s \int_{\mathbb{R}} \mathrm{d}u \, \frac{\hat{F}_{10s}(\boldsymbol{k}, u)}{\omega - ku}, \tag{2.1.26}$$

where

$$D(\omega, \boldsymbol{k}) = 1 + \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2} \int_{\mathbb{R}} \mathrm{d}u \, \frac{kF_{0s}'(u)}{\omega - ku}, \quad \omega_{\mathrm{p}s} \equiv \sqrt{\frac{4\pi n_{0s} q_s^2}{m_s}} \tag{2.1.27}$$

is a Cauchy-type integral and, hence, is holomorphic in the upper half complex ω plane (Im $\omega >$ 0) as long as $F_0 \in C^1(\mathbb{R})$. [20] Noting that

$$-\mathrm{i}\boldsymbol{k}\cdot\hat{\boldsymbol{P}} = \hat{\rho} = \sum_s q_s \int_{\mathbb{R}^3} \mathrm{d}^3 v \, \hat{f}_{1s} \overset{\hat{f}_{10s}=0}{=\!=\!=} -\hat{\varphi} \sum_s \frac{q_s^2}{m_s} \int_{\mathbb{R}^3} \mathrm{d}^3 v \, \frac{\boldsymbol{k}\cdot\nabla_{\boldsymbol{v}} f_{0s}}{\omega - \boldsymbol{k}\cdot\boldsymbol{v}}, \tag{2.1.28}$$

from the definition of the susceptibility tensor $\hat{\boldsymbol{P}} = (\tilde{\chi}/4\pi)\cdot\hat{\boldsymbol{E}}$, we find the longitudinal susceptibility

$$\chi \equiv \hat{\boldsymbol{k}}\cdot\tilde{\chi}\cdot\hat{\boldsymbol{k}} = \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2} \int_{\mathbb{R}^3} \mathrm{d}^2 v \, \frac{\boldsymbol{k}\cdot\nabla_{\boldsymbol{v}} f_{0s}}{\omega - \boldsymbol{k}\cdot\boldsymbol{v}} = \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2} \int_{\mathbb{R}} \mathrm{d}u \, \frac{kF_{0s}'}{\omega - ku}; \tag{2.1.29}$$

hence, $D = 1 + \chi$ is the longitudinal dielectric function.

[20] $D(\omega, \boldsymbol{k})$, as a mathematical function, is also holomorphic in the lower half ω plane (Im $\omega < 0$), although it is originally defined only in the upper half ω plane. Note, however, that this integral is singular along the real ω axis (Im $\omega = 0$); see next section.

2.2 Landau damping and phase mixing of ballistic modes

Thus,

$$\begin{aligned}\varphi(t,\boldsymbol{k}) &= \int_{\mathbb{R}+\mathrm{i}\sigma}\frac{\mathrm{d}\omega}{2\pi}\hat{\varphi}(\omega,\boldsymbol{k})\mathrm{e}^{-\mathrm{i}\omega t}\\ &= \mathrm{i}\frac{4\pi}{k^2}\sum_s q_s\int_{\mathbb{R}}\mathrm{d}u\,\hat{F}_{10s}(\boldsymbol{k},u)\int_{\mathbb{R}+\mathrm{i}\sigma}\frac{\mathrm{d}\omega}{2\pi}\frac{\mathrm{e}^{-\mathrm{i}\omega t}}{(\omega-ku)D(\omega,\boldsymbol{k})}\end{aligned} \tag{2.2.1}$$

with $\sigma>0$. For $t>0$, we close the ω integration contour at the lower half plane $C_\omega\equiv(\mathbb{R}+\mathrm{i}\sigma)+C_R^-$. We can, thus, pick up the pole contribution from $\omega=ku$ and $D(\omega,\boldsymbol{k})=0$. The $\omega=ku$ pole gives a time response of the form $\mathrm{e}^{-\mathrm{i}kut}$; this is called the *ballistic mode*, which we shall discuss in more details later. The equation $D(\omega,\boldsymbol{k})=0$ yields the normal modes of the system. Since, however, $D(\omega,\boldsymbol{k})$ is defined only for $\operatorname{Im}\omega>0$ (where it is holomorphic), we need to analytically continuate it below $\mathbb{R}+\mathrm{i}0^+$. Referring to Eq. (2.1.27), it is clear that as $\operatorname{Im}\omega\to 0$, $D(\omega,\boldsymbol{k})$ becomes singular since the pole at ω/k approaches the integration contour, i.e., the real u axis. Thus, the real ω axis is the branch cut of $D(\omega,\boldsymbol{k})$. To analytically continuate $D(\omega,\boldsymbol{k})$ to $\operatorname{Im}\omega\le 0$, we need to deform the u integration contour *below* the ω/k pole. This deformed contour, shown as u_{L} in Figure 2.1, was first prescribed by Л. Д. Landau in his classical paper [18] where he resolved the singularity encountered in the u integration via the initial-value approach. The *analytic continuation*[21] of $D(\omega,\boldsymbol{k})$ is thus given by

$$\overline{D}(\omega,\boldsymbol{k})=1+\sum_s\frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{u_{\mathrm{L}}}\mathrm{d}u\,\frac{kF_{0s}'(u)}{\omega-ku}; \tag{2.2.3}$$

noting the Plemelj formula [20]

$$\left(x-a\mp\mathrm{i}0^+\right)^{-1}=\mathcal{P}(x-a)^{-1}\pm\mathrm{i}\pi\delta(x-a),\quad x,a\in\mathbb{R} \tag{2.2.4}$$

where $\mathcal{P}$ stands for the principal value, an equivalent description is to replace $(u-\omega/k)^{-1}$ in Eq. (2.1.27) with

[21]Here we present the definition of the concept of analytic continuation.

Definition (analytic continuation [19]):
Let f be a holomorphic function on a region $U\subset\mathbb{C}, G$ be a region that contains U and F be a holomorphic function on G. If $f(z)=F(z)$ for $z\in U$, then we say that F is the analytic continuation of f to $G\setminus U$.
By the theorem of the uniqueness of holomorphic functions, if such an F exists on G, then it is <u>unique</u>: two different methods of analytically continuating f would have to agree, and we are free to choose whichever method that is convenient.

Similarly, let f_1, f_2 be holomorphic functions on regions U_1, U_2, respectively, where $U_1\cap U_2\neq\varnothing$ and $f_1(z)=f_2(z)$ for $z\in U_1\cap U_2$. Then

$$f(z):=\begin{cases}f_1(z) \text{ for } z\in U_1\\ f_2(z) \text{ for } z\in U_2\end{cases} \tag{2.2.2}$$

is holomorphic on $U_1\cup U_2$ and f_1, f_2 are called analytical continuations of each other.

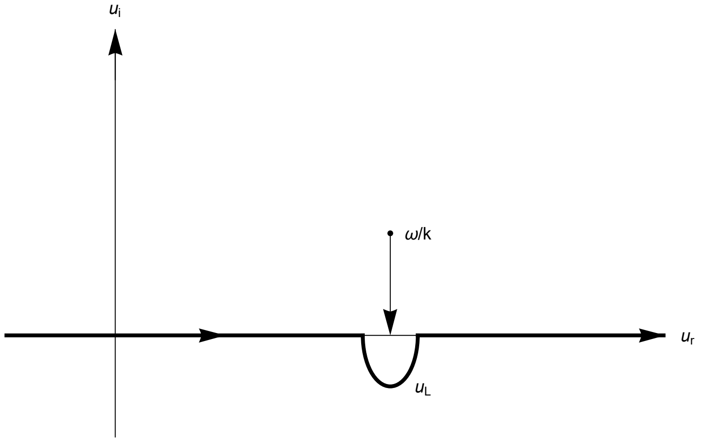


Figure 2.1: Sketch of the Landau contour u_{L} in the u complex plane.

$$\overline{\left(u-\frac{\omega}{k}\right)^{-1}} = \begin{cases} \left(u-\frac{\omega}{k}\right)^{-1} & \text{for Im } \frac{\omega}{k} > 0 \\ \mathcal{P}\left(u-\frac{\omega}{k}\right)^{-1} + \mathrm{i}\pi\delta\left(u-\frac{\omega}{k}\right) & \text{for Im } \frac{\omega}{k} = 0. \\ \left(u-\frac{\omega}{k}\right)^{-1} + \mathrm{i}2\pi\delta\left(u-\frac{\omega}{k}\right) & \text{for Im } \frac{\omega}{k} < 0 \end{cases} \tag{2.2.5}$$

For simplicity of natations, we shall still use the original symbol D to denote its analytic continuation $\overline{D}$.

We now examine qualitatively the ω poles given by $D(\omega, \boldsymbol{k}) = 0$. From Eq. (2.2.3) or Eq. (2.2.5), it is clear that

$$D = D_{\mathrm{r}} + \mathrm{i}D_{\mathrm{i}}, \tag{2.2.6}$$

where D_{i} is due to the presence of the Landau pole $u = \omega/k$. Since the Landau pole only involves the resonant particles with $u = \omega/k$, we can, in general, assume $|D_{\mathrm{i}}| \ll |D_{\mathrm{r}}|$ and correspondingly $\omega = \omega_{\mathrm{r}} + \mathrm{i}\gamma$ with $|\gamma| \ll |\omega_{\mathrm{r}}|$. At the zeroth order, we have

$$\begin{aligned} &D_{\mathrm{r}}(\omega_{\mathrm{r}}, \boldsymbol{k}) = 0 \\ &\Rightarrow \omega_{\mathrm{r}} = \omega_{\mathrm{r}j}(\boldsymbol{k}), \quad j = 1, 2, \cdots; \end{aligned} \tag{2.2.7}$$

at the first order, we find

$$\gamma_j(\boldsymbol{k}) = -\left.\frac{D_{\mathrm{i}}(\omega, \boldsymbol{k})}{\partial D_{\mathrm{r}}(\omega, \boldsymbol{k})/\partial\omega}\right|_{\omega=\omega_{\mathrm{r}j}(\boldsymbol{k})}. \tag{2.2.8}$$

Specifically, using Eq. (2.2.5), the functional form of D_{r} and D_{i} are given by, respectively,

$$D_{\mathrm{r}}(\omega_{\mathrm{r}}, \boldsymbol{k}) = 1 + \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\mathcal{P}\int_{\mathbb{R}} \mathrm{d}u\, \frac{kF_{0s}'(u)}{\omega_{\mathrm{r}} - ku} \tag{2.2.9}$$

and

$$D_{\mathrm{i}}(\omega_{\mathrm{r}}, \boldsymbol{k}) = -\pi \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2} F_{0s}'\left(\frac{\omega_{\mathrm{r}}}{k}\right). \tag{2.2.10}$$

From Eq. (2.2.10), we see that the *resonant wave-particle interaction can play the role of dissipation in collisionless plasmas.* According to discussions in Section 1.4, waves can either grow or decay depending on the signs of $\partial D_\mathrm{r}/\partial\omega_\mathrm{r}$ and D_i. This type of wave damping (growth) is called *Landau damping (growth)*, named after, obviously, Л. Д. Landau.[22] It is interesting to note that the sign of D_i depends on the slope of the VDF at the phase velocity $F_0'(\omega_\mathrm{r}/k)$. This has a simple physical explanation because, for a plane wave, particles travelling faster (slowlier) than the wave phase velocity are, on average, decelerated (accelerated) by the wave and, hence, feed (extract) energy to (from) the wave; see Chapter 8 of [21].

Having found the zeros of D, we can go back to Eq. (2.2.1). Since we are interested in time-*asymptotic* rather than transient responses, only the $\omega = ku$ pole and the normal mode with the maximun γ, denoted as $\omega = \omega_\mathrm{M}(\boldsymbol{k})$, need to be kept. Hence, as $t \to +\infty$, the perturbed fields can be approximated by

$$\begin{aligned}\varphi(t,\boldsymbol{k}) = \mathrm{i}\frac{4\pi}{k^2}\sum_s q_s\int_{u_\mathrm{L}}\mathrm{d}u\,\hat{F}_{10s}(\boldsymbol{k},u)\\ \left(\frac{\mathrm{e}^{-\mathrm{i}kut}}{D(ku,\boldsymbol{k})}+\frac{\mathrm{e}^{-\mathrm{i}\omega_\mathrm{M}(\boldsymbol{k})t}}{(\omega_\mathrm{M}(\boldsymbol{k})-ku)\partial D(\omega,\boldsymbol{k})/\partial\omega|_{\omega=\omega_\mathrm{M}(\boldsymbol{k})}}\right)\end{aligned} \tag{2.2.11}$$

where $\omega = \omega_\mathrm{M}(\boldsymbol{k})$ is assumed to be a simple zero of D. Now the first term corresponds to contributions from the ballistic modes. As $t \to +\infty$, $\mathrm{e}^{-\mathrm{i}kut}$ becomes highly oscillatory in u. Thus, given $\hat{F}_{10}$ sufficiently smooth in u, we expect that the ballistic-mode contribution decay in time[23] at a rate $k\Delta u$, where Δu is the shortest scale of velocity-varivation of $\hat{F}_{10}$ [22]. This is called *phase mixing* of the ballistic modes. Hence, if the system is unstable or marginally stable, i.e., if $\mathrm{Im}\ \omega_\mathrm{M}(\boldsymbol{k}) \geq 0$, then the time-asymptotic response is determined by the most unstable normal mode $\omega = \omega_\mathrm{M}(\boldsymbol{k})$; if, however, the normal modes are all damped, i.e., if $\gamma_j(\boldsymbol{k}) < 0$ for all j, then the time-asymptotic responses will depend on how fast the ballistic modes decay in time. In order to answer this question, let's consider the following two cases:

(1)
$$\hat{F}_{10}(u) = A_0\exp\left(-\frac{(u-u_0)^2}{(\Delta u)^2}\right) \tag{2.2.12}$$

where A_0 is a small but finite constant in u. To further simplify the analysis, we assume $|u_0| \gg \max\{|\Delta u|, |\omega_\mathrm{p}/k|, |v_\mathrm{d}|, v_\mathrm{t}\}$ where v_d is the mean of the distribution F_0/n_0 and $v_\mathrm{t}^2/2$ the variance. Then $D(ku,\boldsymbol{k}) \approx 1$, and

$$\int_\mathbb{R}\mathrm{d}u\,\hat{F}_{10}(u)\mathrm{e}^{-\mathrm{i}kut} = \sqrt{\pi}A_0\Delta u\,\mathrm{e}^{-\mathrm{i}ku_0t}\exp\left(-\frac{(k\Delta u)^2}{4}t^2\right) \tag{2.2.13}$$

[22]Liu Chen 2026-05-18: Landau damping is a consequence of ensemble averaging.
[23]The rigorous version of this assertion is given by the Riemann-Lebesgue lemma [20].

decays much more rapidly than the exponentially damped normal modes. Thus, in this case, the least damped normal mode determines the $t \to +\infty$ response.

(2)
$$\hat{F}_{10}(u) = A_0 \begin{cases} 1 \text{ for } |u - u_0| \leq \Delta u \\ 0 \text{ for } |u - u_0| > \Delta u \end{cases}, \tag{2.2.14}$$

where we also assume $|u_0| \gg \max\{|\Delta u|, |\omega_\mathrm{p}/k|, |v_\mathrm{d}|, v_\mathrm{t}\}$. In this case,

$$\int_{\mathbb{R}} \mathrm{d}u\, \hat{F}_{10}(u) \mathrm{e}^{-\mathrm{i}kut} = \frac{2A_0}{kt} \sin(k\Delta ut) \mathrm{e}^{-\mathrm{i}ku_0 t} \tag{2.2.15}$$

decays only algebraically in t, dominating over the exponentially damped normal modes in the time-asmptotic limit.

Now, in real physical situations, $\hat{F}_{10}$ must be a smooth function in u and has a sufficiently board Δu. It is in this perspective that we expect the normal modes to determine the time-asymptotic behaviours of the system.

It is worthwhile to note that, while ballistic modes do not contribute to field perturbations, they do carry information about the initial phase-space perturbations. To see this point, note that, from

$$\left(\frac{\partial}{\partial t} + \mathrm{i}ku\right) F_1(t, \boldsymbol{k}, u) = \frac{q}{m}(\mathrm{i}k\varphi(t, \boldsymbol{k}))F_0'(u), \tag{2.2.16}$$

we have

$$F_1(t, \boldsymbol{k}, u) = \mathrm{e}^{-\mathrm{i}kut}\left(\hat{F}_{10}(\boldsymbol{k}, u) + \mathrm{i}\frac{q}{m}kF_0'(u)\int_0^t \mathrm{d}t'\, \varphi(t', \boldsymbol{k})\mathrm{e}^{\mathrm{i}kut'}\right). \tag{2.2.17}$$

Eq. (2.2.17) clearly indicates that initial phase-spcae perturbations, $\hat{F}_{10}(\boldsymbol{k}, u)$, propagate without any damping even though the perturbed fields decay away. This *conservation of information* is, of course, due to the fact that the Vlasov plasma conserves (fine-grained) entropy. We remark that that $F_1(t, \boldsymbol{k}, u)$ remains undamped is the basis for the interesting nonlinear phenomenon known as plasma echoes.

Finally, let me suggest two advanced readings on Landau damping [23], [24]. Both papers requires some mathematical inclination. The ballistic modes are also called *Van Kampen modes* or *Case-Van Kampen modes*, which can be attributed to the continuous spectrum. This concept of continuum is, surprisingly, often very useful in plasma physics research with regard to resonance heating, ideal MHD waves in nonuniform plasmas, etc.

Homework #4

(1) Consider a cold plasma with two equal-density contour-streaming electron beams with velocity $\pm v_b$, respectively. In the electrostatic limit, show that

(i) the linear dispersion relation is given by

$$1 - \frac{\omega_{\mathrm{p}b}^2}{(\omega - kv_b)^2} - \frac{\omega_{\mathrm{p}b}^2}{(\omega + kv_b)^2} = 0, \tag{H.4.1}$$

where $\omega_{\mathrm{p}b} \equiv \sqrt{4\pi n_b e^2/m_e}$ and $n_b = n_0/2$ with n_0 being the electron density. Note that this can be solved analytically by hand.

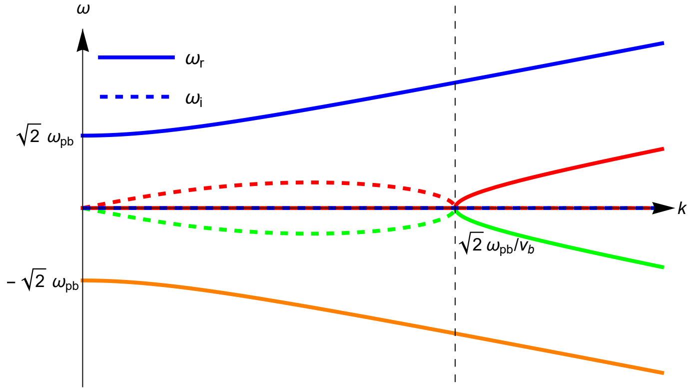


Figure 2.2: Dispersion curves for the equal-density contour-streaming electron instability.

(ii) the system is linearly unstable and plot the $\omega - k$ dispersion curves.

(iii) the instability is absolute.

(iv) in a moving reference frame with velocity v_b, the instability becomes convective.

(2) Consider the weak beam-plasma instability with the dispersion relation

$$1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2} - \frac{\omega_{\mathrm{p}b}^2}{(\omega - kv_b)^2} = 0, \quad |\omega_{\mathrm{p}b}| \ll |\omega_{\mathrm{pe}}|. \tag{H.4.2}$$

Given a steady-state driving source with $\omega = \omega_0$ at $x = 0$ and letting $k = k_{\mathrm{r}} + \mathrm{i}k_{\mathrm{i}}$, calculate

(i) the range of k_{r} that the source will be spatially amplified, and

(ii) the corresponding spatial amplification factor $|k_{\mathrm{i}}x|$.

(3) Using the Nyquist technique to prove the Newcomb-Gardner theorem: the electron Langmuir (Bohm-Gross) wave is stable if $F_{0e}(u)$ has only one extrema (which is the global maximum).

2.3 Particle trapping and the breakdown of linearization

So far we have considered linear wave dynamics via the linearized Vlasov equation. While the validity conditions for the linear theory are formally expressed in Eq. (2.1.4) or Eq. (2.1.16) and Eq. (2.1.17), we would like to further quantify them for the simple case of a monochromatic travelling wave, i.e.,

$$E(t, x) = \hat{E}_0 \mathrm{e}^{\gamma t} \sin(\omega_{\mathrm{r}} t - kx). \tag{2.3.1}$$

Now from discussions of Section 2.1, we know that linearization is equivalent to approximate the exact particle orbits with the unperturbed ones, i.e., linear theory assumes[24]

$$\left|(\omega - ku)\hat{F}_1\right| = \left|\frac{q}{m}\hat{E}F_0'\right| \gg \left|\frac{q}{m}\hat{E}\frac{\partial \hat{F}_1}{\partial u}\right|. \tag{2.3.2}$$

Here $\omega = \omega_r + i\gamma$ and $\hat{E} = \hat{E}_0 e^{\gamma t}$. It is, thus, clear that linearization will first breakdown for those particles with minimal $|\omega - ku|$, i.e., resonant particles with $u = \omega_r/k$. Equation Eq. (2.3.2) then becomse

$$\left|F_0'\left(\frac{\omega_r}{k}\right)\right| \gg \left|\left.\frac{\partial \hat{F}_1}{\partial u}\right|_{u=\frac{\omega_r}{k}}\right|. \tag{2.3.3}$$

Noting that $\hat{F}_1 = -i(q/m)\hat{E}F_0'/(\omega - ku)$, Eq. (2.3.3) can be readily shown to yield

$$\gamma^2 \gg \left|\frac{q}{m}k\hat{E}\right| \equiv \omega_b^2. \tag{2.3.4}$$

Here, ω_b corresponds to the bounce frequency of a charged particle deeply trapped in an electrostatic wave with amplitude $\hat{E}$ and wavenumber k.

Let's consider the trapping process in more details. Furthermore, we assume $\gamma = 0$ and, hence, E in Eq. (2.3.1) has a constant amplitude. The corresponding equation of motion of a charged particle is

$$\frac{d^2x}{dt^2} = -\frac{q}{m}\hat{E}_0 \sin(kx - \omega_r t). \tag{2.3.5}$$

Changing to a frame moving with the phase velocity of the wave, i.e., letting $\delta x = x - (\omega_r/k)t$, Eq. (2.3.5) becomes

$$\frac{d^2\delta x}{dt^2} = -\frac{q}{m}\hat{E}_0 \sin(k\delta x). \tag{2.3.6}$$

Eq. (2.3.6) can be solved exactly in terms of the elliptic integrals. For the present purpose, we consider those deeply trapped particles, for which $|k\delta x| \ll 1$. Eq. (2.3.6) then becomes

$$\frac{d^2\delta x}{dt^2} = -\omega_{b0}^2 \delta x, \tag{2.3.7}$$

which clearly exhibits the bounce motion of particles trapped in the travelling electrostatic wave. The phase-space $(\delta x, \delta v)$ plot for Eq. (2.3.6) is sketched in Figure 2.3, where $\delta v = v - \omega_r/k$, and $\Delta v = 2\omega_{b0}/k$ is sometimes called the trapping width.

From the particle trapping picture, it is then straightforward to develop a physical explanation for the validity condition givey by Eq. (2.3.4). Noting that Landau damping is due to the presence of more faster, with respect to the wave, resonant particles than slowlier ones. On the other hand, the resonant particles change their relative velocities with respect to the wave

[24]Since the time-asymptotic behaviours of the system are dtermined by normal modes, we ignore the initial perturbation $\hat{F}_{10}$.

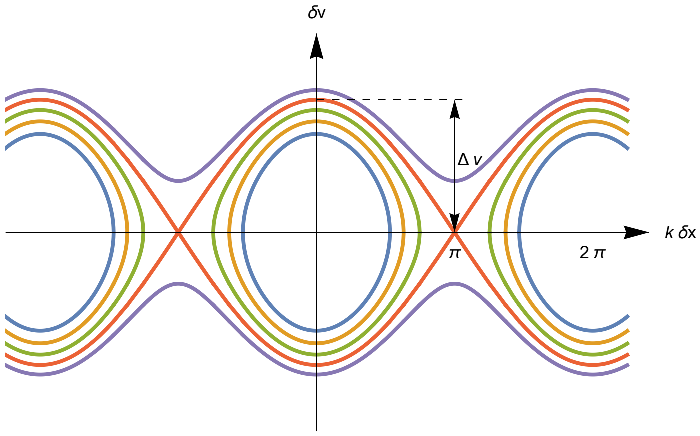


Figure 2.3: Sketch of the $(\delta x, \delta v)$ phase-space plot for Eq. (2.3.6), i.e., the corresponding energy contours for different energy values $\mathcal{E} = (m/2)(\delta v)^2 + (m\omega_{\mathrm{b0}}^2/2k^2)\cos(k\delta x)$, where different energy values are indicated by different colors.

on the characteristic time scale of ω_{b}^{-1}. Thus, the *time-asymptotic linear theory is meaningful only if the wave can decay for several Landau-damping time, γ^{-1}, before particle can execute a bounce motion*. Similar conclusions can, of course, be drawn for the case with Landau growth. In the case of damping, it is clear that, if Eq. (2.3.4) holds initially, it will be valid for all time; in the case of instability, since $\hat{E}$ and hence, ω_{b} increase with time, we know that Eq. (2.3.4) will be violated sooner or later and, thereby, nonlinear treatment is called for.

Let's now consisder the opposite limit $|\gamma| \gg \omega_{\mathrm{b0}}$. In this limit, time-asymptotic linear theory becomes meaningless, so we consider the initial-value problem. Further more, we let $t \ll |\gamma|^{-1}$ so that E in Eq. (2.3.1) has a constant amplitude. From the Possion's equation, we know that $E(t=0,x) = -\hat{E}_0 \sin(kx)$ is consistent with an initial perturbed VDF of the form

$$F_1(x,u,t=0) = F_{10}(u)\cos(kx), \quad F_{10}(u) \equiv F_1(x=0,u,t=0). \tag{2.3.8}$$

From Eq. (2.1.8) we then find

$$\begin{aligned} F_1(x,u,t) &= F_{10}(u)\cos(k(x-ut)) + \frac{q}{m}\hat{E}_0 F_0'(u)\int_0^t \mathrm{d}t' \sin(k(x-ut)+(ku-\omega_{\mathrm{r}})t') \\ &= F_{10}(u)\cos(k(x-ut)) + \frac{q}{m}\hat{E}_0 F_0'(u)\frac{\cos(kx-\omega_{\mathrm{r}}t)-\cos(kx-kut)}{\omega_{\mathrm{r}}-ku}; \end{aligned} \tag{2.3.9}$$

note that $u = \omega_{\mathrm{r}}/k$ is a removable singularity. Taking the velocity differentiation, we find

$$\begin{aligned} \left.\frac{\partial F_1(x,u,t)}{\partial u}\right|_{u=\frac{\omega_{\mathrm{r}}}{k}} &= \left(F_{10}\left(\frac{\omega_{\mathrm{r}}}{k}\right)\frac{k}{\omega_{\mathrm{b0}}} + F_0''\left(\frac{\omega_{\mathrm{r}}}{k}\right)\frac{\omega_{\mathrm{b0}}}{k}\right)\omega_{\mathrm{b0}}t\sin(kx-\omega_{\mathrm{r}}t) \\ &+\left(F_{10}'\left(\frac{\omega_{\mathrm{r}}}{k}\right) - F_0'\left(\frac{\omega_{\mathrm{r}}}{k}\right)(\omega_{\mathrm{b0}}t)^2\right)\cos(kx-\omega_{\mathrm{r}}t). \end{aligned} \tag{2.3.10}$$

Eq. (2.3.10), thus, becomes secular in time and the linear theory breaks down for $t \gtrsim \omega_{\mathrm{b0}}^{-1}$. For more detailed analysis of particle-trapping effects on Landau damping, we refer to [25].

2.4 Normal modes

Our discussions here are standard and, hence, will be brief. The reader are referred to typical textbooks such as [26] for detailed analysis. These normal modes, of course, correspond to $D = 0$ with D given by Eq. (2.2.3) or, equivalently, Eq. (2.2.9) and Eq. (2.2.10).

First, in the high-frequency electrostatic regime, we have the Bohm-Gross (Langmuir, or *space charge*) wave.[25] Here, $|\omega_{\rm r}| \approx \omega_{\rm pe} \gg \max\{\omega_{\rm p\it i}, |k|v_{\rm te}, |k|v_{\rm t\it i}\}$. More precisely, we have

$$\omega_{\rm r}^2 = \omega_{\rm pe}^2 + \frac{3}{2}k^2 v_{\rm te}^2 = \omega_{\rm pe}^2(1 + 3k^2\lambda_{\rm De}^2) \tag{2.4.1}$$

where $v_{\rm te}^2/2 \equiv \int_{\mathbb{R}} {\rm d}u\, u^2 F_{0e}(u)/n_{0e} =: T_e/m_e$ and $\lambda_{\rm De}^2 \equiv v_{\rm te}^2/2\omega_{\rm pe}^2$. For a monotonically decreasing F_{0e} such as a Maxwellian one $F_{0e}(u) = (n_{0e}/\sqrt{\pi}v_{\rm te})\exp(-u^2/v_{\rm te}^2)$, there is weak Landau damping.

Next, there is the low-frequency electrostatic ion acoustic wave (IAW), which has no counterpart in the cold plasma model.[26] This wave has an intermediate phase velocity, i.e., $v_{{\rm t}i} \ll |\omega_{\rm r}/k| \ll v_{\rm te}$. We further take F_{0e} to be Maxwellian, then

$$\begin{aligned} D_{\rm r} &= 1 - \frac{\omega_{{\rm p}i}^2}{\omega_{\rm r}^2}\left(1 + \frac{3}{2}\frac{k^2 v_{{\rm t}i}^2}{\omega_{\rm r}^2}\right) + \frac{1}{k^2\lambda_{\rm De}^2} \\ \Rightarrow \omega_{\rm r}^2 &= \frac{k^2 c_{\rm s}^2}{1 + k^2\lambda_{\rm De}^2} + \frac{3}{2}k^2 v_{{\rm t}i}^2. \end{aligned} \tag{2.4.2}$$

Here $c_{\rm s}^2 \equiv \omega_{{\rm p}i}^2\lambda_{\rm De}^2 = T_e/m_i$. We note that, to satisfy the $|\omega_{\rm r}/k| \gg v_{{\rm t}i}$ condition, $T_e \gg T_i$ is demanded. For $T_e \approx T_i$, the IAW is heavily ion Landau damped and, therefore, is not a normal mode in its proper sense — it is referred to as a *virtual* mode, see Section 5.3; for $T_e \gg T_i$, it suffers mainly weak electron Landau damping.

Now it is interesting to calculate the wave energies for both the Bohm-Gross mode and the IAW in the $k^2\lambda_{\rm D\it e}^2 \ll 1$ limit. For the Bohm-Gross wave,

$$\widehat{W}_{\rm BG} = \frac{|\hat{\boldsymbol{E}}|^2}{16\pi}(1+1) = \frac{|\hat{\boldsymbol{E}}|^2}{8\pi}; \tag{2.4.3}$$

for the IAW,

$$\widehat{W}_{\rm IAW} = \frac{|\hat{\boldsymbol{E}}|^2}{16\pi}\left(1 + \frac{1}{k^2\lambda_{\rm De}^2} + \left(1 + \frac{1}{k^2\lambda_{\rm De}^2}\right)\right) \approx \frac{|\hat{\boldsymbol{E}}|^2}{8\pi}\frac{1}{k^2\lambda_{\rm De}^2}. \tag{2.4.4}$$

[25] The cold plasma oscillation is due to the separation of space charges, and becomes propagatable driven by the electron pressure.

[26] The electrons form a quasi-neutral background (Boltzmann distribution), and the wave becomes propagatable driven by the electron\ion pressure.

Thus, in the case of IAW, we have particle kinetic energy dominates over the electric field energy, which is, of course, related to the fact that the wave has so long wavelength that the qusi-neutrality condition $k^2\lambda_{\mathrm{D}e}^2 \ll 1$ (see below) is maintained.

At this juncture, let me briefly introduce the *plasma dispersion function* or, sometimes called, the Z dunction, defined as the Hilbert transform of a Gaussian function

$$
\begin{aligned}
Z(\zeta) &= \frac{1}{\sqrt{\pi}} \int_{u_{\mathrm{L}}} \mathrm{d}x\, \frac{\mathrm{e}^{-x^2}}{x-\zeta} \\
&= \begin{cases} \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \mathrm{d}x\, \frac{\exp(-x^2)}{x-\zeta} & \text{for Im } \zeta > 0 \\ \frac{1}{\sqrt{\pi}} \mathcal{P} \int_{\mathbb{R}} \mathrm{d}x\, \frac{\exp(-x^2)}{x-\zeta} + \mathrm{i}\sqrt{\pi}\exp(-\zeta^2) & \text{for Im } \zeta = 0. \\ \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \mathrm{d}x\, \frac{\exp(-x^2)}{x-\zeta} + \mathrm{i}2\sqrt{\pi}\exp(-\zeta^2) & \text{for Im } \zeta < 0 \end{cases}
\end{aligned}
\tag{2.4.5}
$$

This function is useful for Maxwellian distribution functions. It has been tabulated [27] and there exists standard computer subroutine. The asymptotic behaviors of $Z(\zeta)$ is given by

$$
\begin{aligned}
Z(\zeta) &= -2\zeta\left(1 - \frac{2}{3}\zeta^2 + \frac{4}{15}\zeta^4 - \frac{8}{105}\zeta^6 + \mathcal{O}(\zeta^8)\right) + \mathrm{i}\sqrt{\pi}\mathrm{e}^{-\zeta^2} \quad \text{for } \zeta \to 0 \\
&= -\frac{1}{\zeta}\left(1 + \frac{1}{2\zeta^2} + \frac{3}{4\zeta^4} + \frac{15}{8\zeta^6} + \mathcal{O}\left(\frac{1}{\zeta^8}\right)\right) + \mathrm{i}\sigma\sqrt{\pi}\mathrm{e}^{-\zeta^2} \quad \text{for } \zeta \to \infty,
\end{aligned}
\tag{2.4.6}
$$

where

$$
\sigma = \begin{cases} 0 \text{ for Im } \zeta > 0 \\ 1 \text{ for Im } \zeta = 0. \\ 2 \text{ for Im } \zeta < 0 \end{cases}
\tag{2.4.7}
$$

For a drifting Maxwellian distribution $f_{0s}(\boldsymbol{v}) = \left(n_{0s}/\pi^{3/2}v_{\mathrm{t}s}^3\right)\exp\left(-|\boldsymbol{v} - \boldsymbol{v}_{\mathrm{d}s}|^2/v_{\mathrm{t}s}^2\right)$, the longitudinal susceptibility is given by

$$
\chi_s = \frac{1}{k^2\lambda_{\mathrm{D}s}^2}(1 + \zeta_s Z(\zeta_s)), \quad \zeta_s \equiv \frac{\omega - \boldsymbol{k}\cdot\boldsymbol{v}_{\mathrm{d}s}}{kv_{\mathrm{t}s}}.
\tag{2.4.8}
$$

- A semi-quantitative criterion for linear Landau resonance [28]:
  Let $f(\zeta) = \zeta\mathrm{e}^{-\zeta^2}$, then $f'(\zeta) = \mathrm{e}^{-\zeta^2}(1 - 2\zeta^2) = 0 \Rightarrow \zeta = 1/\sqrt{2}$. Thus, a resonance species s would satisfy $|\zeta_s| = \mathcal{O}(1)$, and the complementary condition for non-resonance is then $|\zeta_s| \gg 1$ or $|\zeta_s| \to 0$.

At long wavelength limit where $k^2\lambda_{\mathrm{D}s}^2 \ll 1 \Rightarrow \chi_s \gg 1$, the dispersion relation $1 + \sum_s \chi_s = 0$ becomes, approximately, $\sum_s \chi_s \approx 0$ or

$$
\sum_s \hat{\rho}_s \approx 0.
\tag{2.4.9}
$$

Here, $\hat{\rho}(\omega, \boldsymbol{k}) = \mathcal{L}(\omega)\mathcal{F}(\boldsymbol{k})\{\rho(t, \boldsymbol{x})\}$ is the Laplace-in-time and Fourier-in-space transforms of the charge density. Eq. (2.4.9) is often referred to as the *quasi-neutrality condition*.

Finally, for isotropic zeroth-order VDF, i.e., $f_0 = \tilde{f}_0(v^2)$, we have the electromagnetic normal mode

$$\omega^2 = \sum_s \omega_{ps}^2 + k^2c^2 \approx \omega_{pe}^2 + k^2c^2, \tag{2.4.10}$$

where c is the speed of light. Note, since $\omega/k > c$, that this mode has no collisionlesss wave-particle resonance; it can, thus, be damped only by collisional dissipations.

2.5 Instabilities

(1) Warm beam-plasma instability.

For Simplicity, we shall assume cold background electrons but a warm beam. The corresponding electrostatic dispersion relation is then

$$1 - \frac{\omega_{pe}^2}{\omega^2} - \frac{\omega_{pb}^2}{n_b k^2}\int_{u_L} du \frac{F_b'(u)}{u - \omega/k} = 0, \tag{2.5.1}$$

where $\omega_{pb} = \sqrt{4\pi n_b q_b^2/m_b}$ is the beam plasma frequency. To further make Eq. (2.5.1) analytically tractable, we shall take F_b to be Lorentzian [29], i.e.,

$$F_b(u) = n_b \frac{v_{tb}}{\pi}\frac{1}{(u - v_b)^2 + v_{tb}^2}, \quad 0 < v_{tb} < v_b. \tag{2.5.2}$$

Eq. (2.5.1) then becomes[27]

$$1 - \frac{\omega_{pe}^2}{\omega^2} - \frac{\omega_{pb}^2}{(\omega - kv_b + ikv_{tb})^2} = 0, \tag{2.5.3}$$

which can be readily analyzed in the various limits of the parameter $|\omega_{pe}/kv_b|$.

First, for $|kv_b/\omega_{pe}| \gg 1$, we find, for the electron plasma wave $(\omega \sim \omega_{pe})$,

$$\omega_1 = \omega_{pe} + \delta\omega_1, \quad \frac{2\delta\omega_1}{\omega_{pe}} \approx \left(\frac{\omega_{pb}}{kv_b}\right)^2\left(1 + i\frac{2v_{tb}}{v_b}\right). \tag{2.5.4}$$

Thus, in contrast to the cold beam results, the electron plasma wave is untable in this regime, which can be understood in terms of destabilization of a positive-energy wave by negative collisionless dissipation. Meanwhile, the two beam modes $(\omega \sim kv_b)$

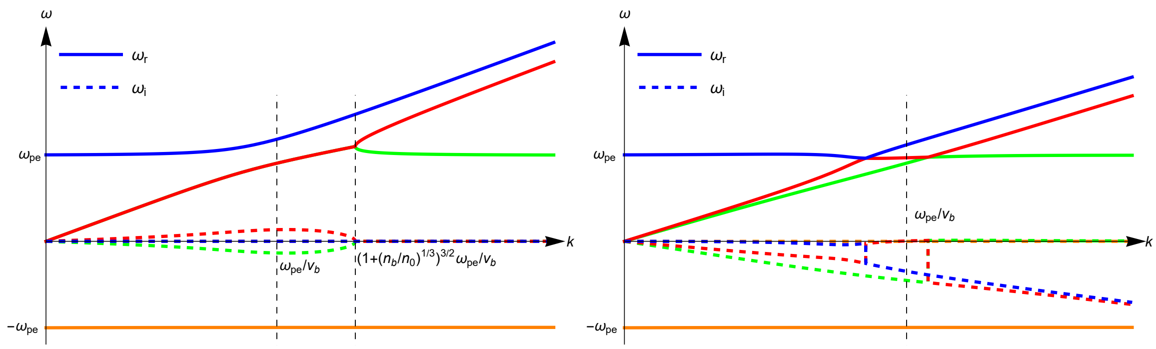
$$\omega_2 \approx kv_b \pm \omega_{pb} - ikv_{tb} \tag{2.5.5}$$

are stable, which may be regarded as the effect of negative (positive) dissipation on negative(positive)-energy waves.

In the opposite limit $|kv_b/\omega_{pe}| \ll 1$, we have, for the electron plasma wave,

$$\omega_1 = \omega_{pe} + \delta\omega_1, \quad \frac{2\delta\omega_1}{\omega_{pe}} \approx \left(\frac{\omega_{pb}}{\omega_{pe}}\right)^2\left(1 - i\frac{2kv_{tb}}{\omega_{pe}}\right). \tag{2.5.6}$$

[27] This result is easily calculated for Im $\omega/k > 0$, and then analytically continuated to the whole ω plane.



(a) (b)

Figure 2.4: Sketches of the $\omega - k$ dispersion curves for Eq. (2.5.1) in the limits of (a) cold beam and (b) warm beam.

Thus, it is stable due to the effect of positive dissipation on a positive-energy wave. As to the beam modes, we find

$$\omega_2 \approx (kv_b - \mathrm{i}kv_{tb})\left(1 \pm \mathrm{i}\frac{\omega_{pb}}{\omega_{pe}}\right) = kv_b \pm \frac{\omega_{pb}}{\omega_{pe}}kv_{tb} - \mathrm{i}kv_{tb} \pm \mathrm{i}\frac{\omega_{pb}}{\omega_{pe}}kv_b. \tag{2.5.7}$$

Hence, for $v_{tb}/v_b > \omega_{pb}/\omega_{pe} = \sqrt{n_b/n_0}$, all the modes are stable in this long wavelength limit.

The dispersion curves for the cold and warm beam cases are sketched in Figure 2.4; note that there exists a critical beam temperature (critical distribution spread $v_{tbc} = v_b(n_b/2n_e)^{1/3}$) signifying the *transition* from the cold beam fluid-like instability to the warm beam kinetic (micro) instability.

(2) Electromagnetic instability due to velocity-space anisotropy (Weibel instability).

The polarization of the wave considered here is

$$\boldsymbol{k} = k\hat{\boldsymbol{e}}_z, \quad \hat{\boldsymbol{E}} \parallel \hat{\boldsymbol{e}}_x, \quad \hat{\boldsymbol{B}} \parallel \hat{\boldsymbol{e}}_y. \tag{2.5.8}$$

The linearized Vlasov equation

$$\mathrm{i}(\omega - kv_z)\hat{f}_1 = \frac{q}{m}\left(\hat{\boldsymbol{E}} + \frac{\boldsymbol{v}}{c} \times \hat{\boldsymbol{B}}\right) \cdot \nabla_{\boldsymbol{v}} f_0 \tag{2.5.9}$$

then yields

$$\hat{f}_1 = -\mathrm{i}\frac{q}{m}\frac{\hat{E}}{\omega - kv_z}\left(\left(1 - \frac{kv_z}{\omega}\right)\frac{\partial}{\partial v_x} + \frac{kv_x}{\omega}\frac{\partial}{\partial v_z}\right) f_0. \tag{2.5.10}$$

With $f_0 = \tilde{f}_0(v_x^2, v_y^2, v_z^2)$, it is clear from Eq. (2.5.10) that $\boldsymbol{J} = J_x\hat{\boldsymbol{e}}_x$ and the dispersion relation is given by

$$-k^2c^2 + \omega^2\epsilon_{xx} = 0 \tag{2.5.11}$$

where

$$\epsilon_{xx} = 1 + \mathrm{i}\frac{4\pi}{\omega}\sigma_{xx} = 1 + \sum_s \frac{\omega_{\mathrm{p}s}^2}{\omega^2}\left(\frac{1}{n_{0s}}\int_{\mathbb{R}^3} \mathrm{d}^3v\, \frac{kv_x^2\, \partial f_{0s}/\partial v_z}{\omega - kv_z} - 1\right). \tag{2.5.12}$$

We further assume $|\omega/kv_{\mathrm{t}z}| \gg 1$ so that we can neglect the resonant wave-particle interaction, and define

$$\frac{1}{n_{0s}}\int_{\mathbb{R}^3} \mathrm{d}^3v\, v_x^2 f_{0s} \equiv \frac{v_{\mathrm{t}sx}^2}{2}, \quad v_{\mathrm{t}x}^2 \equiv \sum_s \frac{\omega_{\mathrm{p}s}^2}{\omega_{\mathrm{p}}^2} v_{\mathrm{t}sx}^2, \tag{2.5.13}$$

then the dispersion relation Eq. (2.5.11) becomes

$$\begin{aligned} &\omega^4 - (\omega_{\mathrm{p}}^2 + k^2c^2)\omega^2 - \frac{1}{2}\omega_{\mathrm{p}}^2 k^2 v_{\mathrm{t}x}^2 = 0 \\ \Rightarrow\ &\omega^2 = \frac{1}{2}\left(\omega_{\mathrm{p}}^2 + k^2c^2 \pm \sqrt{\left(\omega_{\mathrm{p}}^2 + k^2c^2\right)^2 + 2\omega_{\mathrm{p}}^2 k^2 v_{\mathrm{t}x}^2}\right). \end{aligned} \tag{2.5.14}$$

The minus sign gives $\omega^2 < 0$ and, hence, a standing-still purely growing instability. The growth rate is sketched in Figure 2.5.

We conclude this discussion by developing a physical picture for the Weibel instability. For this purpose, we may take the electrons to be current sheets with the following equilibrium distribution

$$f_{0e}(v_x, v_y, v_z) = \frac{n_{0e}}{2}(\delta(v_x - v_0) + \delta(v_x + v_0))\delta(v_y)\delta(v_z). \tag{2.5.15}$$

Now, at $t = 0$, let's give the system an initial perturbation of the form

$$\boldsymbol{B}(z, t = 0) = \hat{\boldsymbol{e}}_y \hat{B} \sin kz. \tag{2.5.16}$$

In response to B_y, the electrons will move according to the equation of motion

$$\frac{\mathrm{d}v_z}{\mathrm{d}t} = -\frac{e}{m_e c} v_x B_y. \tag{2.5.17}$$

In a time setp Δt, we then have

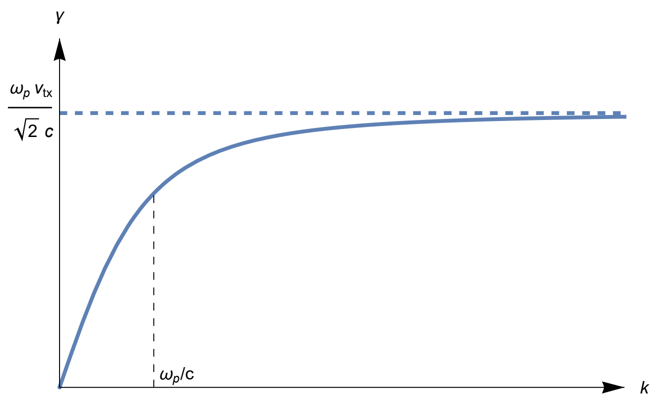


Figure 2.5: Sketch of the growth rate, γ, of the Weibel instability as a function of the wavenumber, k.

$$(\Delta v_z)_\pm = \mp \frac{ev_0}{m_e c} \hat{B} \Delta t \sin kz \tag{2.5.18}$$

where the $\pm$ sign corresponds to $v_x = \pm v_0$. It is clear that the current sheets will start *pinching*, i.e., there will be more positive current sheets at center regions in space (e.g., $kz = \pi$) and more negative current sheets at other regions (e.g., $kz = 0$). We, therefore, expect perturbations in J_x and B_y, which we denote as ΔJ_x and ΔB_y, respectively. If ΔB_y is in phase with the initial perturbation B_y, we then have a positive feedback, i.e., an instability.

To see it more specifically, we use the continuity equation to calculate the density perturbations in the current sheets. We find

$$(\Delta n_e)_\pm = -\Delta t \frac{n_{0e}}{2} \frac{\partial (\Delta v_z)_\pm}{\partial z} = \pm (\Delta t)^2 \frac{n_{0e}}{2} \frac{ev_0}{m_e c} k \hat{B} \cos kz. \tag{2.5.19}$$

The current density perturbation, ΔJ_x, is then given by

$$\Delta J_x = -ev_0 \Big((\Delta n_e)_+ - (\Delta n_e)_- \Big) = -\frac{n_{0e} e^2 v_0^2 k}{m_e c} (\Delta t)^2 \hat{B} \cos kz. \tag{2.5.20}$$

From Amperé's law $\nabla \times \Delta \boldsymbol{B} = (4\pi/c) \Delta \boldsymbol{J}$ it is easy to show that

$$\Delta B_y = \left(\frac{\omega_{\rm pe} v_0}{c} \right)^2 (\Delta t)^2 B_y. \tag{2.5.21}$$

Thus, ΔB_y reinforces B_y and the initial perturbation grows at a rate $\sim \omega_{\rm pe} v_0 / c$.

Homework #5

(1) Consider the warm beam-plasma instability discussed in class.

(i) For a beam with a Lorentzian distribution

$$F_b(u) = n_b \frac{v_{\rm tb}}{\pi} \frac{1}{(u - v_b)^2 + v_{\rm tb}^2}, \quad 0 < v_{\rm tb} < v_b, \tag{H.5.1}$$

show that the instability makes a transition from the cold beam-plasma instability to the warm beam one for $v_{\rm tb}/v_b > (n_b/2n_0)^{1/3}$.

(ii) Show that this conclusion also holds for a beam with a drifting Maxwellian distribution

$$F_b(u) = \frac{n_b}{\sqrt{\pi} v_{\rm tb}} \exp\left(-\frac{(u - v_b)^2}{v_{\rm tb}^2} \right). \tag{H.5.2}$$

(2) Show that in an unmagnetized plasma, the dispersion tensor, $\tilde{D}$, with $\boldsymbol{k} = k \hat{\boldsymbol{e}}_z$, is given by

$$\tilde{D} \cdot \big[\hat{\boldsymbol{e}}_x |\ \hat{\boldsymbol{e}}_y |\ \hat{\boldsymbol{e}}_z \big] = \big[\hat{\boldsymbol{e}}_x |\ \hat{\boldsymbol{e}}_y |\ \hat{\boldsymbol{e}}_z \big] \cdot \mathrm{diag}\big\{ D_{xx}, D_{yy}, D_{zz} \big\}, \tag{H.5.3}$$

where

$$
\begin{aligned}
D_{\left\{\substack{xx \\ yy}\right.} &= -k^2 + \frac{\omega^2}{c^2}\left(1 - \sum_s \frac{\omega_{\mathrm{p}s}^2}{\omega^2} + \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}\omega^2}\int_{\mathbb{R}^3} \mathrm{d}^3v\, \frac{k v_{\left\{\substack{x \\ y}\right.}^2\, \partial f_{0s}/\partial v_z}{\omega - k v_z}\right), \\
D_{zz} &= \frac{\omega^2}{c^2}\left(1 - \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{\mathbb{R}^3} \mathrm{d}^3v\, \frac{\partial f_{0s}/\partial v_z}{v_z - \omega/k}\right).
\end{aligned}
\tag{H.5.4}
$$

(3) Consider the Weibel instability where $\boldsymbol{k} = k\hat{\boldsymbol{e}}_z$.

(i) Keeping the order $\mathcal{O}\big((kv_{\mathrm{t}z}/\omega)^2\big)$ correction, show that finite $v_{\mathrm{t}z}$ is stabilizing.

(ii) Demonstrate the $v_{\mathrm{t}z}$ stabilization effect by examining the instability in the $|\omega/kv_{\mathrm{t}z}| \ll 1$ limit. For this purpose, assuming

$$
f_{0e}(v_x, v_y, v_z) = n_{0e}\tilde{f}_{0\perp}(v_x, v_y)\tilde{f}_{0z}(v_z) \tag{H.5.5}
$$

with

$$
\tilde{f}_{0z}(v_z) = \frac{1}{\sqrt{\pi}v_{\mathrm{t}z}} \exp\left(-\frac{v_z^2}{v_{\mathrm{t}z}^2}\right). \tag{H.5.6}
$$

(iii) Given f_{0e} bi-Maxwellian, i.e., referring to Eq. (H.5.5), Eq. (H.5.6) and

$$
\tilde{f}_{0\perp}(v_x, v_y) = \frac{1}{\pi v_{\mathrm{t}\perp}^2} \exp\left(-\frac{v_x^2 + v_y^2}{v_{\mathrm{t}\perp}^2}\right), \tag{H.5.7}
$$

obtain, via the Nyquist technique, the critical value of $(v_{\mathrm{t}\perp}/v_{\mathrm{t}z})^2$ for the instability.

(4) Assuming that the electron distribution is a drifting Maxwellian

$$
F_{0e}(u) = \frac{n_{0e}}{\sqrt{\pi}v_{\mathrm{t}e}} \exp\left(-\frac{(u - v_{\mathrm{d}})^2}{v_{\mathrm{t}e}^2}\right), \tag{H.5.8}
$$

(i) for cold ions $T_i = 0$, what is the critical value of v_{d} for the IAW to be unstable (current-driven ion acoustic instability)?

(ii) for $T_i = T_e$, is there any electrostatic instability? Substantiate your answer.

3 Linear Waves and Instabilities in Uniform Magnetized Plasmas

3.1 Magnetized Vlasov equation with the guiding-center transformation

In a magnetized plasma, the corresponding Vlasov equation is given by

$$\mathcal{L}_0 f \equiv \left(\frac{\partial}{\partial t} + \boldsymbol{v}\cdot\nabla + \frac{q}{mc}(\boldsymbol{v}\times\boldsymbol{B}_0)\cdot\nabla_{\boldsymbol{v}}\right) f = -\frac{q}{m}\left(\boldsymbol{E} + \frac{\boldsymbol{v}}{c}\times\boldsymbol{B}\right)\cdot\nabla_{\boldsymbol{v}} f. \qquad (3.1.1)$$

Here, we have assumed that there is no equilibrium electric field $\boldsymbol{E}_0 = \boldsymbol{0}$; generalization to $\boldsymbol{E}_0 \neq \boldsymbol{0}$ is not difficult[28]. Letting

$$f = f_0(\boldsymbol{v}) + f_1(t, \boldsymbol{x}, \boldsymbol{v}), \qquad (3.1.2)$$

we then have, for the equilibrium distribution function f_0,

$$\mathcal{L}_0 f_0 = 0, \qquad (3.1.3)$$

and, for the linearly perturbed distribution function f_1, [29]

$$\mathcal{L}_0 f_1 = -\frac{q}{m}\left(\boldsymbol{E} + \frac{\boldsymbol{v}}{c}\times\boldsymbol{B}\right)\cdot\nabla_{\boldsymbol{v}} f_0 \qquad (3.1.4)$$

or, formally,

$$f_1 = \mathcal{L}_0^{-1}\left\{-\frac{q}{m}\left(\boldsymbol{E} + \frac{\boldsymbol{v}}{c}\times\boldsymbol{B}\right)\cdot\nabla_{\boldsymbol{v}} f_0\right\}. \qquad (3.1.5)$$

Here $\mathcal{L}_0^{-1}$ should be interpreted as integration along the particle's unperturbed orbit in the $(\boldsymbol{x}, \boldsymbol{v})$ phase space. Meanwhile, f_0, as given by Eq. (3.1.3), must be a function of particle's unperturbed *constants of motion*. Thus, the solution of both f_0 and f_1 require detailed knowledge of the particle's unperturbed motion. It, furthermore, indicates that, if we can transform to coordinates where the unperturbed motion is simpler, the procedure involved in solving f_0 and f_1 will also be greatly simplified.

Based on the above intuitions and knowing that, in magnetized plasmas particle dynamics is simpler with the *guiding-center* picture [10], it is, therefore, suggestive to make a transformation from the particle phase space, $(\boldsymbol{x}, \boldsymbol{v})$, to the *guiding-center phase space*, $(\boldsymbol{X}, \boldsymbol{V})$, defined by

[28]For example, in [30], the guiding-center "energy" coordinate is defined by $\mathcal{E} = v^2/2 + q\varphi_0(\boldsymbol{x})/m$, where $\varphi_0(\boldsymbol{x})$ is the equilibrium (macroscopic) scalar potential.

[29]Remember that the subscript 1 indicating perturbation is omitted.

$$\begin{aligned}\boldsymbol{X} &:= \boldsymbol{x} + \frac{1}{\Omega}\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel,\\ \boldsymbol{V} &:= (\mathcal{E},\mu,\alpha,\sigma),\end{aligned} \tag{3.1.6}$$

where $\hat{\boldsymbol{e}}_\parallel = \boldsymbol{B}_0/B_0, \Omega = qB_0/mc, \mathcal{E} = v^2/2, \mu = v_\perp^2/2B_0\,(v_\perp > 0), \sigma = \mathrm{sgn}\ v_\parallel$ and α is the velocity *gyrophase* angle of the particle in the plane perpendicular to $\boldsymbol{B}_0$. Note that $\mathcal{E}$ and μ (or $v_\parallel$ and μ) are guiding-center's unperturbed constants of motion in a uniform magnetic field. The inverse transformation is given by[30]

$$\begin{aligned}\boldsymbol{v} &\equiv \hat{\boldsymbol{e}}_\parallel v_\parallel + \boldsymbol{v}_\perp\\ &= \hat{\boldsymbol{e}}_\parallel\sigma\sqrt{2(\mathcal{E}-\mu B_0)} + v_\perp(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha),\\ \boldsymbol{x} &= \boldsymbol{X} - \frac{1}{\Omega}\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel,\end{aligned} \tag{3.1.9}$$

where $\left\{\hat{\boldsymbol{e}}_{\perp 1}, \hat{\boldsymbol{e}}_{\perp 2}, \hat{\boldsymbol{e}}_\parallel\right\}$ is an orthonormal right-handed basis of $\mathbb{R}^3$. Different choices of $\hat{\boldsymbol{e}}_{\perp 1}$ (or $\hat{\boldsymbol{e}}_{\perp 2}$) give rise to values of α that differ by constants, which do not affect the dynamics.

From Eq. (3.1.6), we find

$$\begin{aligned}\nabla_{\boldsymbol{x}} &\mapsto \nabla_{\boldsymbol{X}},\\ \nabla_{\boldsymbol{v}} &\mapsto \nabla_{\boldsymbol{V}} + \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}}\\ &\quad = \boldsymbol{v}\frac{\partial}{\partial\mathcal{E}} + \frac{\boldsymbol{v}_\perp}{B_0}\frac{\partial}{\partial\mu} + \frac{\hat{\boldsymbol{e}}_\alpha}{v_\perp}\frac{\partial}{\partial\alpha} + \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}},\end{aligned} \tag{3.1.10}$$

and[31]

$$\int_{\mathbb{R}^3}\mathrm{d}^3v \mapsto \sum_\sigma\int_0^{+\infty}\mathrm{d}\mathcal{E}\int_0^{+\infty}\mathrm{d}\mu\int_0^{2\pi}\mathrm{d}\alpha\,\frac{B_0}{|v_\parallel|} \equiv \sum_\sigma\iiint\mathrm{d}\mathcal{E}\,\mathrm{d}\mu\,\mathrm{d}\alpha\,\frac{B_0}{|v_\parallel|}. \tag{3.1.12}$$

Here $\hat{\boldsymbol{e}}_\alpha = \hat{\boldsymbol{e}}_\parallel\times\boldsymbol{v}_\perp/v_\perp$. The unperturbed Vlasov operator $\mathcal{L}_0$ then becomes, term by term

$$\begin{aligned}\frac{\partial}{\partial t} &\mapsto \frac{\partial}{\partial t},\\ \boldsymbol{v}\cdot\nabla_{\boldsymbol{x}} &\mapsto \boldsymbol{v}\cdot\nabla_{\boldsymbol{X}}\end{aligned} \tag{3.1.13}$$

---

[30]Defining

$$\hat{\mathrm{e}}_\pm := \frac{1}{\sqrt{2}}(\hat{\mathrm{e}}_{\perp 1} \pm \mathrm{i}\hat{\mathrm{e}}_{\perp 2}), \tag{3.1.7}$$

the perpendicular velocity $\boldsymbol{v}_\perp = v_\perp(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha)$ can also be represented as

$$\boldsymbol{v}_\perp = \frac{v_\perp}{\sqrt{2}}\sum_{\lambda=\pm}\hat{\mathrm{e}}_\lambda\mathrm{e}^{-\mathrm{i}\lambda\alpha}. \tag{3.1.8}$$

[31]It is straightforward to show that

$$\mathrm{d}\mathcal{E}\wedge\mathrm{d}\mu\wedge\mathrm{d}\alpha = \frac{v_\parallel}{B_0}\mathrm{d}v_1\wedge\mathrm{d}v_2\wedge\mathrm{d}v_\parallel\,, \tag{3.1.11}$$

where $\wedge$ is the wedge product of differential forms.

$$
\begin{aligned}
&= v_\parallel \frac{\partial}{\partial X_\parallel} + \boldsymbol{v}_\perp \cdot \nabla_{\boldsymbol{X}_\perp}, \\
\left(\frac{q}{mc}\boldsymbol{v}\times\boldsymbol{B}_0\right)\cdot\nabla_{\boldsymbol{v}} \mapsto \Omega\big(\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel\big)\cdot &\left(\boldsymbol{v}\frac{\partial}{\partial\mathcal{E}} + \frac{\boldsymbol{v}_\perp}{B_0}\frac{\partial}{\partial\mu} + \frac{\hat{\boldsymbol{e}}_\alpha}{v_\perp}\frac{\partial}{\partial\alpha} + \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}}\right) \\
&= \Omega\big(\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel\big)\cdot\left(\frac{\hat{\boldsymbol{e}}_\alpha}{v_\perp}\frac{\partial}{\partial\alpha} + \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}}\right) \\
&= -\Omega\frac{\partial}{\partial\alpha} - \boldsymbol{v}_\perp\cdot\nabla_{\boldsymbol{X}_\perp};
\end{aligned} \tag{3.1.13}
$$

that is,

$$
\mathcal{L}_0 \mapsto \mathcal{L}_\mathrm{g} \equiv \frac{\partial}{\partial t} + v_\parallel\frac{\partial}{\partial X_\parallel} - \Omega\frac{\partial}{\partial\alpha}. \tag{3.1.14}
$$

The equilibrium equation, Eq. (3.1.3), then becomes

$$
\mathcal{L}_\mathrm{g} f_{0\mathrm{g}} = -\Omega\frac{\partial f_{0\mathrm{g}}}{\partial\alpha} = 0. \tag{3.1.15}
$$

Here, subscript g denotes a quantity that is a function of the $(\boldsymbol{X},\boldsymbol{V})$ variables; note that for any quantity A, although $A_\mathrm{g}(\boldsymbol{X},\boldsymbol{V})$ is equal to $A(\boldsymbol{x},\boldsymbol{v})$, A_g and A are different functions (thus denoted by different symbols). Hence, for a uniform magnetized plasma, we have

$$
f_{0\mathrm{g}} = f_{0\mathrm{g}}(\mathcal{E},\mu,\sigma) \tag{3.1.16}
$$

or

$$
f_0 = f_0\big(v_\parallel, v_\perp\big). \tag{3.1.17}
$$

3.2 Solution of the linearized Vlasov equation in the electrostatic limit

In the electrostatic limit, the linearized Vlasov equation, Eq. (3.1.4), transforms to

$$
\mathcal{L}_\mathrm{g} f_{1\mathrm{g}} = \frac{q}{m}(\nabla_{\boldsymbol{X}}\varphi_\mathrm{g})\cdot\left(\hat{\boldsymbol{e}}_\parallel v_\parallel\frac{\partial}{\partial\mathcal{E}} + \boldsymbol{v}_\perp\left(\frac{\partial}{\partial\mathcal{E}} + \frac{1}{B_0}\frac{\partial}{\partial\mu}\right)\right) f_{0\mathrm{g}}. \tag{3.2.1}
$$

Note that for any field quantity (function of t and $\boldsymbol{x}$, not $\boldsymbol{v}$) , e.g., $\varphi = \varphi(t,\boldsymbol{x})$, we have

$$
\nabla_{\boldsymbol{v}}\varphi = \boldsymbol{0}, \tag{3.2.2}
$$

i.e.,

$$
\left(\boldsymbol{v}\frac{\partial}{\partial\mathcal{E}} + \frac{\boldsymbol{v}_\perp}{B_0}\frac{\partial}{\partial\mu} + \frac{\hat{\boldsymbol{e}}_\alpha}{v_\perp}\frac{\partial}{\partial\alpha} + \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}}\right)\varphi_\mathrm{g} = \boldsymbol{0}, \tag{3.2.3}
$$

whose $\hat{\boldsymbol{e}}_\alpha$ component gives

$$\boldsymbol{v}_\perp \cdot \nabla_{\boldsymbol{X}} \varphi_{\mathrm{g}} = -\Omega \frac{\partial \varphi_{\mathrm{g}}}{\partial \alpha}. \tag{3.2.4}$$

Thus Eq. (3.2.1) becomes

$$\mathcal{L}_{\mathrm{g}} f_{1\mathrm{g}} = \frac{q}{m}\left(v_\parallel \frac{\partial \varphi_{\mathrm{g}}}{\partial X_\parallel}\frac{\partial}{\partial \mathcal{E}} - \Omega \frac{\partial \varphi_{\mathrm{g}}}{\partial \alpha}\left(\frac{\partial}{\partial \mathcal{E}} + \frac{1}{B_0}\frac{\partial}{\partial \mu}\right)\right) f_{0\mathrm{g}}. \tag{3.2.5}$$

In order to make contact with the low-frequency limit to be discussed in Section 3.7, we shall remove the $\partial/\partial\alpha$ terms in Eq. (3.2.5). Let

$$f_{1\mathrm{g}} = \frac{q}{m}\varphi_{\mathrm{g}}\left(\frac{\partial}{\partial \mathcal{E}} + \frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}} + G_{\mathrm{g}}, \tag{3.2.6}$$

Eq. (3.2.5) then reduces to

$$\begin{aligned}\mathcal{L}_{\mathrm{g}} G_{\mathrm{g}} &= \mathcal{L}_{\mathrm{g}} f_{1\mathrm{g}} - \left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel} - \Omega \frac{\partial}{\partial \alpha}\right)\left(\frac{q}{m}\varphi_{\mathrm{g}}\left(\frac{\partial}{\partial \mathcal{E}} + \frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}}\right)\\ &= -\frac{q}{m}\left(\frac{\partial \varphi_{\mathrm{g}}}{\partial t}\frac{\partial}{\partial \mathcal{E}} + \left(\left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel}\right)\varphi_{\mathrm{g}}\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}},\end{aligned} \tag{3.2.7}$$

i.e.,

$$\left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel} - \Omega \frac{\partial}{\partial \alpha}\right) G_{\mathrm{g}} = -\frac{q}{m}\left(\frac{\partial \varphi_{\mathrm{g}}}{\partial t}\frac{\partial}{\partial \mathcal{E}} + \left(\left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel}\right)\varphi_{\mathrm{g}}\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}}. \tag{3.2.8}$$

Noting that G_{g} (and $f_{1\mathrm{g}}, \varphi_{\mathrm{g}}$, etc.) is a periodic function in α, i.e.,

$$G_{\mathrm{g}}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma, \alpha + 2\pi) = G_{\mathrm{g}}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma, \alpha), \tag{3.2.9}$$

we can Fourier expand it as

$$G_{\mathrm{g}} = \sum_{n=-\infty}^{+\infty} \left\langle G_{\mathrm{g}}\right\rangle_n \mathrm{e}^{-\mathrm{i}n\alpha}, \tag{3.2.10}$$

where

$$\left\langle G_{\mathrm{g}}\right\rangle_n = \left\langle G_{\mathrm{g}}\right\rangle_n(\boldsymbol{X}, \mathcal{E}, \mu, \sigma) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\alpha\, \mathrm{e}^{\mathrm{i}n\alpha} G_{\mathrm{g}} \tag{3.2.11}$$

is the n-th Fourier coefficient of G_{g}. Substituting Eq. (3.2.10) into Eq. (3.2.8), we find the following equation for $\left\langle G_{\mathrm{g}}\right\rangle_n$

$$\begin{aligned}\mathcal{L}_{\mathrm{g}n}\left\langle G_{\mathrm{g}}\right\rangle_n &\equiv \left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel} + \mathrm{i}n\Omega\right)\left\langle G_{\mathrm{g}}\right\rangle_n\\ &= -\frac{q}{m}\left(\frac{\partial \left\langle \varphi_{\mathrm{g}}\right\rangle_n}{\partial t}\frac{\partial}{\partial \mathcal{E}} + \left(\left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial X_\parallel}\right)\left\langle \varphi_{\mathrm{g}}\right\rangle_n\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}}.\end{aligned} \tag{3.2.12}$$

Eq. (3.2.12) is similar to the unmagnetized case and, can be readily solved by Laplace-in-t and Fourier-in-$\boldsymbol{X}$ transforms. In this chapter we will focus on the normal modes, ignoring the initial values of related quantities. Denoting

$$\hat{A}_{\mathrm{g}}(\omega, \boldsymbol{k}) = \mathcal{L}(\omega)\mathcal{F}(\boldsymbol{k})\big\{A_{\mathrm{g}}(t, \boldsymbol{X})\big\}, \tag{3.2.13}$$

Eq. (3.2.12) yields

$$\left\langle\hat{G}_{\mathrm{g}}\right\rangle_n = -\frac{q}{m}\left\langle\hat{\varphi}_{\mathrm{g}}\right\rangle_n \frac{\omega\,\partial/\partial\mathcal{E} + \big(\big(\omega - k_\parallel v_\parallel\big)/B_0\big)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega} f_{0\mathrm{g}}. \tag{3.2.14}$$

Now, again, in order to make contact with later discussions on nonuniform plasmas where $v_\parallel$ is not a constant of motion due to varying $\boldsymbol{B}_0$, we would like to remove the parallel (to $\boldsymbol{B}_0$) operator $\partial/\partial X_\parallel$ term from the R.H.S. of Eq. (3.2.8). Let

$$f_{1\mathrm{g}} = \frac{q}{m}\varphi_{\mathrm{g}}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + h_{\mathrm{g}}, \tag{3.2.15}$$

then

$$\begin{aligned}\left\langle\hat{h}_{\mathrm{g}}\right\rangle_n &= \left\langle G_{\mathrm{g}}\right\rangle_n + \frac{q}{m}\left\langle\varphi_{\mathrm{g}}\right\rangle_n \frac{1}{B_0}\frac{\partial f_{0\mathrm{g}}}{\partial\mu} \\ &= -\frac{q}{m}\left\langle\hat{\varphi}_{\mathrm{g}}\right\rangle_n \frac{\omega\,\partial/\partial\mathcal{E} + (n\Omega/B_0)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega} f_{0\mathrm{g}}.\end{aligned} \tag{3.2.16}$$

Summarizing the results obtained so far, we have

$$\hat{f}_{1\mathrm{g}} = \frac{q}{m}\hat{\varphi}_{\mathrm{g}}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + \sum_{n=-\infty}^{+\infty}\left\langle\hat{h}_{\mathrm{g}}\right\rangle_n \mathrm{e}^{-\mathrm{i}n\alpha} \tag{3.2.17}$$

with $\left\langle\hat{h}_{\mathrm{g}}\right\rangle_n$ given by Eq. (3.2.16).

Since the field equations are in the $(t, \boldsymbol{x})$ space rather than the $(t, \boldsymbol{X})$ space, we need to establish relations between $\hat{A}_{\mathrm{g}}$ and $\hat{A}$. Noting that $A_{\mathrm{g}}(t, \boldsymbol{X}) = A(t, \boldsymbol{x})$, we have

$$\begin{aligned}\hat{A} &= \int_0^{+\infty}\mathrm{d}t\int_{\mathbb{R}^3}\mathrm{d}^3x\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}A \\ &= \int_0^{+\infty}\mathrm{d}t\int_{\mathbb{R}^3}\mathrm{d}^3X\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{X} - \omega t) + \mathrm{i}\boldsymbol{k}\cdot\left(\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel\right)/\Omega}A_{\mathrm{g}} \\ &= \hat{A}_{\mathrm{g}}\mathrm{e}^{\mathrm{i}L_{\boldsymbol{k}}},\end{aligned} \tag{3.2.18}$$

where

$$L_{\boldsymbol{k}} \equiv \boldsymbol{k}\cdot\frac{\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel}{\Omega}. \tag{3.2.19}$$

Without loss of generality, we may take $\boldsymbol{k} = \hat{\boldsymbol{e}}_1 k_\perp + \hat{\boldsymbol{e}}_\parallel k_\parallel$, thus $L_{\boldsymbol{k}} = \lambda\sin\alpha$ with $\lambda \equiv k_\perp v_\perp/\Omega$. Furthermore, we note the following identity

$$\mathrm{e}^{\pm \mathrm{i}\lambda \sin\alpha} = \sum_{n=-\infty}^{+\infty} J_n(\lambda)\, \mathrm{e}^{\pm \mathrm{i} n\alpha} \tag{3.2.20}$$

where J_n is the n-th order Bessel function (of the first kind). Hence,

$$\hat{A}_{\mathrm{g}} = \hat{A}\,\mathrm{e}^{-\mathrm{i}\lambda\sin\alpha} = \sum_{n=-\infty}^{+\infty} J_n(\lambda)\,\hat{A}\,\mathrm{e}^{-\mathrm{i}n\alpha} \tag{3.2.21}$$

$$\Rightarrow \left\langle \hat{A}_{\mathrm{g}} \right\rangle_n = J_n(\lambda)\,\hat{A}. \tag{3.2.22}$$

From Eq. (3.2.16), Eq. (3.2.17), and Eq. (3.2.21) and Eq. (3.2.22), we obtain

$$\hat{f}_1 = \frac{q}{m}\hat{\varphi}\left(\frac{\partial}{\partial\mathcal{E}} - \sum_{n,m=-\infty}^{+\infty} J_m(\lambda)J_n(\lambda)\frac{\omega\,\partial/\partial\mathcal{E} + (n\Omega/B_0)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega}\mathrm{e}^{\mathrm{i}(m-n)\alpha}\right) f_{0\mathrm{g}} \tag{3.2.23}$$

$$\begin{aligned}
\Rightarrow \left\langle \hat{f}_1 \right\rangle_0 &= \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\alpha\, \hat{f}_1 \\
&= \frac{q}{m}\hat{\varphi}\left(\frac{\partial}{\partial\mathcal{E}} - \sum_{n=-\infty}^{+\infty} J_n^2(\lambda)\frac{\omega\,\partial/\partial\mathcal{E} + (n\Omega/B_0)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega}\right) f_{0\mathrm{g}} \qquad (3.2.24)\\
&= -\frac{q}{m}\sum_{n=-\infty}^{+\infty} J_n^2(\lambda)\frac{\left(k_\parallel v_\parallel + n\Omega\right)\partial/\partial\mathcal{E} + (n\Omega/B_0)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega} f_{0\mathrm{g}}.
\end{aligned}$$

Substituting Eq. (3.2.24) into the Possion's equation

$$\begin{aligned}
k^2\hat{\varphi} &= 4\pi\sum_s q_s \int_{\mathbb{R}^3} \mathrm{d}^3 v\, \hat{f}_{1s} \\
&\mapsto 4\pi\sum_s q_s \sum_\sigma \iint \mathrm{d}\mathcal{E}\,\mathrm{d}\mu\, \frac{B_0}{\left|v_\parallel\right|} 2\pi\left\langle \hat{f}_{1s}\right\rangle_0,
\end{aligned} \tag{3.2.25}$$

we find the following electrostatic dispersion relation

$$D = 1 + \sum_s \chi_s = 0, \tag{3.2.26}$$

where

$$\begin{aligned}
\chi_s = 2\pi\frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\sum_\sigma \iint \mathrm{d}\mathcal{E}\,\mathrm{d}\mu\,\frac{B_0}{\left|v_\parallel\right|}\sum_{n=-\infty}^{+\infty} J_n^2(\lambda_s) \\
\frac{\left(k_\parallel v_\parallel + n\Omega_s\right)\partial/\partial\mathcal{E} + (n\Omega_s/B_0)\partial/\partial\mu}{\omega - k_\parallel v_\parallel - n\Omega_s} f_{0\mathrm{g}s}
\end{aligned} \tag{3.2.27}$$

is the longitudinal susceptibility of species s. χ_s can be weitten in the more conventional form by transforming from the $(\mathcal{E}, \mu, \sigma)$ variables back to the $\left(v_\parallel, v_\perp\right)$ varibales. Noting that[32]

[32] In order to avoid confusion, we have adopted the thermodynamic convention of denoting the independent variables (held constant) in the differentiations as subscripts outside the parentheses; e.g., $(\partial/\partial\mathcal{E})_{\mu,\sigma}$ denotes the partial derivative with respect to $\mathcal{E}$ while holding μ and σ constant.

$$\begin{aligned}\left(\frac{\partial}{\partial\mathcal{E}}\right)_{\mu,\sigma} &\mapsto \frac{1}{v_\parallel}\left(\frac{\partial}{\partial v_\parallel}\right)_{v_\perp}, \\ \frac{1}{B_0}\left(\frac{\partial}{\partial\mu}\right)_{\mathcal{E},\sigma} &\mapsto \frac{1}{v_\perp}\left(\frac{\partial}{\partial v_\perp}\right)_{v_\parallel} - \frac{1}{v_\parallel}\left(\frac{\partial}{\partial v_\parallel}\right)_{v_\perp},\end{aligned} \tag{3.2.28}$$

we have

$$\chi_s = 2\pi\frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{u_\mathrm{L}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \sum_{n=-\infty}^{+\infty} J_n^2(\lambda_s)\frac{k_\parallel\,\partial/\partial v_\parallel + (n\Omega_s/v_\perp)\partial/\partial v_\perp}{\omega - k_\parallel v_\parallel - n\Omega_s} f_{0s}. \tag{3.2.29}$$

Discussions [17]:

(1) For a parallelly drifting Maxwellian distribution with $\boldsymbol{v}_\mathrm{d} = v_\mathrm{d}\hat{\boldsymbol{e}}_\parallel$

$$f_{0s}(\boldsymbol{v}) = \frac{n_{0s}}{\pi^{\frac{3}{2}}v_{\mathrm{t}s}^3}\exp\left(-\frac{\left(v_\parallel - v_{\mathrm{d}s}\right)^2}{v_{\mathrm{t}s}^2}\right)\exp\left(-\frac{v_\perp^2}{v_{\mathrm{t}s}^2}\right), \tag{3.2.30}$$

the longitudinal susceptibility becomes

$$\chi_s = \frac{1}{k^2\lambda_{\mathrm{D}s}^2}\left(1 + \zeta_{0s}\sum_{n=-\infty}^{+\infty} Z(\zeta_{ns})A_n(b_s)\right), \tag{3.2.31}$$

where

$$\zeta_{ns} \equiv \frac{\omega - k_\parallel v_{\mathrm{d}s} - n\Omega_s}{k_\parallel v_{\mathrm{t}s}}, \quad b_s \equiv \frac{k_\perp^2 v_{\mathrm{t}s}^2}{2\Omega_s^2} = \frac{1}{2}k_\perp^2\rho_{\mathrm{t}s}^2, \quad A_n(x) \equiv \mathrm{e}^{-x}I_n(x). \tag{3.2.32}$$

Here, I_n is the n-th order modified Bessel function of the first kind, and Z is the plasma dispersion function as defined in Eq. (2.4.5). The FLR parameter b_s is retated to the *polarization density* [30], [31] of species s.

(2) For a bi-Maxwellian distribution

$$f_{0s}(\boldsymbol{v}) = n_{0s}\frac{1}{\sqrt{\pi}v_{\mathrm{t}\parallel s}}\frac{1}{\pi v_{\mathrm{t}\perp s}}\exp\left(-\frac{v_\parallel^2}{v_{\mathrm{t}\parallel s}^2}\right)\exp\left(-\frac{v_\perp^2}{v_{\mathrm{t}\perp s}^2}\right), \tag{3.2.33}$$

we have

$$\chi_s = \frac{1}{k^2\lambda_{\mathrm{D}\parallel s}^2}\left(1 + \sum_{n=-\infty}^{+\infty}\left(1 + \frac{T_{\parallel s}}{T_{\perp s}}\frac{n\Omega_s}{\omega - n\Omega_s}\right)\zeta_{ns}Z(\zeta_{ns})A_n(b_s)\right), \tag{3.2.34}$$

where

$$\lambda_{\mathrm{D}\parallel s}^2 \equiv \frac{v_{\mathrm{t}\parallel s}^2}{2\omega_{\mathrm{p}s}^2}, \quad \zeta_{ns} \equiv \frac{\omega - n\Omega_s}{k_\parallel v_{\mathrm{t}\parallel s}}, \quad b_s \equiv \frac{k_\perp^2 v_{\mathrm{t}\perp s}^2}{2\Omega_s^2} = \frac{1}{2}k_\perp^2\rho_{\mathrm{t}\perp s}^2. \tag{3.2.35}$$

(3) In the strong magnetic field limit where $B_0 \to +\infty$, we have $\Omega \to \infty, \lambda \to 0$, and

$$\begin{aligned}\chi_s &\to 2\pi \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{u_{\mathrm{L}}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \frac{k_\parallel\, \partial f_{0s}/\partial v_\parallel}{\omega - k_\parallel v_\parallel} \\ &= \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{\mathbb{R}^3} \mathrm{d}^3 v\, \frac{k_\parallel\, \partial f_{0s}/\partial v_\parallel}{\omega - k_\parallel v_\parallel};\end{aligned} \tag{3.2.36}$$

note that in this limit $\chi_s = 0$ for $k_\parallel = 0$. CAUTION: the strong magnetic field limit is not equivalent to the strongly magnetized limit where $b \ll 1$, because b also involves $k_\perp$ and v_{t}; in other words, for a species to be strongly magnetized, the magnetic field not necessarily needs to be strong.

3.3 Quasi-electrostatic and quasi-transverse approximations

In magnetized plasmas, the electrostatic (longitudinal, $\hat{\boldsymbol{E}} \parallel \boldsymbol{k}$) and electromagnetic (transverse, $\hat{\boldsymbol{E}} \perp \boldsymbol{k}$) modes are usually coupled except in rather special limits, such as the exactly parallel propagation. The analysis, therefore, can be significantly simplified if we can assume the mode of interest is either essentially electrostatic or essentially electromagnetic. These so-called *quasi-electrostatic* and *quasi-transverse* approximations are often adopted in plasma physics literature. The validity conditions quoted for the approximations are, however, usually qualitative. In this section, we will try to establish a formal procedure whereby one can quantitatively evaluate the validity limits of the two approximations.

Recall that the wave equation is given by Eq. (1.1.10)

$$\tilde{D}\cdot\hat{\boldsymbol{E}} = \left(\boldsymbol{k}\boldsymbol{k} - k^2\tilde{g} + \frac{\omega^2}{c^2}\tilde{\epsilon}\right)\cdot\hat{\boldsymbol{E}} = \boldsymbol{0}. \tag{3.3.1}$$

Now $\hat{\boldsymbol{E}}$ can be decomposed into its longitudinal and transverse parts

$$\hat{\boldsymbol{E}} = \hat{E}_l\hat{\boldsymbol{k}} + \hat{E}_t\hat{\boldsymbol{t}}, \quad \hat{\boldsymbol{t}} \perp \hat{\boldsymbol{k}}. \tag{3.3.2}$$

Denoting

$$\epsilon_{ij} \equiv \hat{\boldsymbol{e}}_i\cdot\tilde{\epsilon}\cdot\hat{\boldsymbol{e}}_j, \quad i,j = l,t, \tag{3.3.3}$$

Eq. (3.3.1) then writes

$$\begin{cases} D_l\hat{E}_l + \epsilon_{lt}\hat{E}_t = 0 \\ \epsilon_{tl}\hat{E}_l + D_t\hat{E}_t = 0 \end{cases} \tag{3.3.4}$$

where $D_l \equiv \epsilon_{ll}$, $D_t \equiv \epsilon_{tt} - n^2$ with $n = ck/\omega$ being the refractive index.

(1) Quasi-electrostatic approximation:

If $\epsilon_{lt} = 0$, then $D_l = 0$ and the mode dispersion is purely electrostatic. If, however, $\epsilon_{lt} \neq 0$ but $\left|\hat{E}_t/\hat{E}_l\right| \ll 1$, we must have $\omega = \omega_l + \delta\omega$ with $|\delta\omega/\omega_l| \ll 1$, where

$$D_l^l \equiv D_l(\omega_l, \boldsymbol{k}) = 0. \tag{3.3.5}$$

Eq. (3.3.4) then becomes

$$\begin{cases} \delta\omega \frac{\partial D_l^l}{\partial \omega_l} \hat{E}_l + \epsilon_{lt}^l \hat{E}_t = 0 \\ \epsilon_{tl}^l \hat{E}_l + D_t^l \hat{E}_t = 0 \end{cases} \Rightarrow \frac{\delta\omega}{\omega_l} = \left(\frac{\epsilon_{lt}\epsilon_{tl}}{D_t \partial(\omega D_l)/\partial\omega} \right)^l, \tag{3.3.6}$$

which implies the validity condition for the quasi-electrostatic approximation

$$\left| \frac{\delta\omega}{\omega_l} \right| = \left| \frac{\epsilon_{lt}\epsilon_{tl}}{D_t \partial(\omega D_l)/\partial\omega} \right|^l \ll 1. \tag{3.3.7}$$

(2) Quasi-transverse approximation:

Similarly, if $\epsilon_{tl} = 0$, then $D_t = 0$ and the mode dispersion is purely electromagnetic. If, however, $\epsilon_{tl} \neq 0$ but $|\hat{E}_l/\hat{E}_t| \ll 1$, we must have $\omega = \omega_t + \delta\omega$ with $|\delta\omega/\omega_t| \ll 1$, where

$$D_t^t \equiv D_t(\omega_t, \boldsymbol{k}) = 0. \tag{3.3.8}$$

Eq. (3.3.4) then becomes

$$\begin{cases} D_l^t \hat{E}_l + \epsilon_{lt}^t \hat{E}_t = 0 \\ \epsilon_{tl}^t \hat{E}_l + \delta\omega \frac{\partial D_t^t}{\partial \omega_t} \hat{E}_t = 0 \end{cases} \Rightarrow \frac{\delta\omega}{\omega_t} = \left(\frac{\epsilon_{lt}\epsilon_{tl}}{D_l \partial(\omega D_t)/\partial\omega} \right)^t, \tag{3.3.9}$$

which implies the validity condition for the quasi-transverse approximation

$$\left| \frac{\delta\omega}{\omega_t} \right| = \left| \frac{\epsilon_{lt}\epsilon_{tl}}{D_l \partial(\omega D_t)/\partial\omega} \right|^t \ll 1. \tag{3.3.10}$$

Noting that $\tilde{\epsilon} = \tilde{g} + \mathrm{i}4\pi\tilde{\sigma}/\omega$, it is clear that, for $i \neq j$,

$$\epsilon_{ij} = \hat{\boldsymbol{e}}_i \cdot \left(\mathrm{i}\frac{4\pi\tilde{\sigma}}{\omega} \right) \cdot \hat{\boldsymbol{e}}_j = \hat{\boldsymbol{e}}_i \cdot \left(\mathrm{i}\frac{4\pi}{\omega} \right) \tilde{\sigma} \cdot \frac{\hat{\boldsymbol{E}}_j}{\hat{E}_j}. \tag{3.3.11}$$

Thus, a physical explanation for the *coupling between electrostatic (longitudinal) and electromagnetic (transverse) modes* is that the current perturbation associated with one mode has a nonvanishing component when projected onto the polarization of the other mode. For example, if $\epsilon_{lt} \neq 0$, then the current produced by $\hat{\boldsymbol{E}}_t$ has a nonvanishing component along $\hat{\boldsymbol{k}}$; with $\boldsymbol{k} \cdot \hat{\boldsymbol{J}} \neq 0$ there will be charge density perturbations and, hence, space-charge (electrostatic) oscillations. Also, we remark that, since $\tilde{\epsilon} = \tilde{\epsilon}^{\mathrm{h}} + \mathrm{i}\tilde{\epsilon}^{\mathrm{a}}$, this finite coupling can, sometimes, modify the wave growth or damping mechanism.

Let's illustrate the above discussions using the cold-plasma lower-hybrid wave, which is an electrostatic wave with $|k_\parallel/k_\perp| \lesssim \sqrt{m_e/m_i}$ (which will be clear soon) and $\Omega_i \ll |\omega| \approx \omega_{\mathrm{LH}} \equiv |\Omega_e|\omega_{\mathrm{pi}}/\sqrt{\Omega_e^2 + \omega_{\mathrm{pe}}^2} \ll \min\{\omega_{\mathrm{pe}}, |\Omega_e|\}$. The corresponding electrostatic dispersion relation is given by

$$D_l \approx 1 - \frac{\omega_{\mathrm{pi}}^2}{\omega^2} + \frac{\omega_{\mathrm{pe}}^2}{\Omega_e^2} - \frac{\omega_{\mathrm{pe}}^2}{\omega^2}\frac{k_\parallel^2}{k^2} = 0 \tag{3.3.12}$$

$$\Rightarrow \omega_l^2 = \omega_{\mathrm{LH}}^2 \left(1 + \frac{m_i}{m_e}\frac{k_\parallel^2}{k^2}\right). \tag{3.3.12}$$

Here the ions are treated as unmagnetized ($b_i \gg 1$, or $|\omega/\Omega_i| \gg 1$) and nonresonant $|\zeta_i| \gg 1$, while the electrons are strongly magnetized ($b_e \ll 1$, or $|\omega/\Omega_e| \ll 1$). From the argument of associated current perturbations, it is easy to see that couplings to electromagnetic modes with $\hat{\boldsymbol{t}}$ nearly parallel to $\hat{\boldsymbol{e}}_\parallel$ dominate. Since $\hat{\boldsymbol{e}}_\parallel \cdot \tilde{\epsilon} \cdot \hat{\boldsymbol{e}}_\parallel \approx -\omega_{\mathrm{pe}}^2/\omega^2$ and $\hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{e}}_\parallel = k_\parallel/k$, we have

$$\epsilon_{tl} \approx \epsilon_{lt} \approx -\frac{\omega_{\mathrm{pe}}^2}{\omega^2}\frac{k_\parallel}{k}; \tag{3.3.13}$$

meanwhile, we have

$$D_t \approx \frac{c^2k^2 - \omega_{\mathrm{pe}}^2}{\omega^2} \approx \frac{c^2k^2}{\omega^2}, \tag{3.3.14}$$

where we have assumed

$$\frac{ck}{\omega_{\mathrm{pe}}} \gg 1. \tag{3.3.15}$$

The validity criterion for the quasi-electrostatic approximation is then

$$\left|\frac{\delta\omega}{\omega_l}\right| = \frac{\left(\omega_{\mathrm{pe}}k_\parallel/\omega k\right)^4}{2\left(1 + \omega_{\mathrm{pe}}^2/\Omega_e^2\right)}\left(\frac{\omega}{ck_\parallel}\right)^2 \sim n_\parallel^{-2} \ll 1. \tag{3.3.16}$$

Thus, Eq. (3.3.15) and Eq. (3.3.16), which are consistent, constitute the conditions for approximating the lower-hybrid wave as an electrostatic wave.

3.4 Electromagnetic dielectric tensor for magnetized warm plasmas

The discussions in Section 3.1 and Section 3.2 can be straightforwardly generalized to allow full, electromagnetic perturbations. In this case, $f_{1\mathrm{g}}$ satisfies

$$\mathcal{L}_{\mathrm{g}} f_{1\mathrm{g}} = -\frac{q}{m}\boldsymbol{a}_{\mathrm{g}} \cdot \nabla_{\boldsymbol{V}} f_{0\mathrm{g}}, \tag{3.4.1}$$

where

$$\boldsymbol{a}_{\mathrm{g}} = \boldsymbol{E}_{\mathrm{g}} + \frac{\boldsymbol{v}}{c} \times \boldsymbol{B}_{\mathrm{g}} \tag{3.4.2}$$

with $\mathcal{L}_{\mathrm{g}}$ and $\nabla_{\boldsymbol{V}}$ given by, respectively, Eq. (3.1.14) and Eq. (3.1.10).

Again, making the Laplace-in-t and Fourier-in-$\boldsymbol{X}$ transformas, and letting

$$\hat{f}_{1\mathrm{g}} = \sum_{n=-\infty}^{+\infty} \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \mathrm{e}^{-\mathrm{i}n\alpha}, \tag{3.4.3}$$

Eq. (3.4.1) becomes

$$-i\left(\omega - k_\parallel v_\parallel - n\Omega\right)\left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n = -\frac{q}{m}\mathcal{A}_n, \tag{3.4.4}$$

where

$$\begin{aligned}\mathcal{A}_n &\equiv \left\langle \hat{\boldsymbol{a}}_{\mathrm{g}} \cdot \nabla_{\boldsymbol{V}} f_{0\mathrm{g}} \right\rangle_n \\ &= \left\langle \mathrm{i}\frac{v_\perp}{\lambda}\frac{\partial \hat{E}_{\mathrm{g}\perp 1}}{\partial \alpha} P_1(f_{0\mathrm{g}}) + \mathrm{i} v_\perp \frac{\partial \hat{E}_{\mathrm{g}\perp 2}}{\partial \lambda} P_1(f_{0\mathrm{g}}) + v_\parallel \hat{E}_{\mathrm{g}\parallel}\frac{\partial f_{0\mathrm{g}}}{\partial \mathcal{E}} + \mathrm{i}\frac{\Omega}{\omega} v_\parallel \frac{\partial \hat{E}_{\mathrm{g}\parallel}}{\partial \alpha}\frac{1}{B_0}\frac{\partial f_{0\mathrm{g}}}{\partial \mu} \right\rangle_n \\ &= \hat{E}_{\perp 1}\frac{v_\perp n J_n(\lambda)}{\lambda} P_1(f_{0\mathrm{g}}) + \mathrm{i}\hat{E}_{\perp 2} v_\perp J_n'(\lambda) P_1(f_{0\mathrm{g}}) + \hat{E}_\parallel v_\parallel J_n(\lambda) P_2(f_{0\mathrm{g}})\end{aligned} \tag{3.4.5}$$

and

$$P_1 = \frac{\partial}{\partial \mathcal{E}} + \left(1 - \frac{k_\parallel v_\parallel}{\omega}\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}, \quad P_2 = \frac{\partial}{\partial \mathcal{E}} + \frac{n\Omega}{\omega}\frac{1}{B_0}\frac{\partial}{\partial \mu}. \tag{3.4.6}$$

In deriving Eq. (3.4.5), we have used $\hat{\boldsymbol{B}}_{\mathrm{g}} = (c\boldsymbol{k}/\omega) \times \hat{\boldsymbol{E}}_{\mathrm{g}}, \boldsymbol{k}_\perp = k_\perp \hat{\boldsymbol{e}}_1$, Eq. (3.2.21), Eq. (3.2.22), and

$$(\cos\alpha)\hat{E}_{\mathrm{g}\left\{\substack{\perp 1\\ \parallel}\right.} = \frac{\mathrm{i}}{\lambda}\frac{\partial \hat{E}_{\mathrm{g}\left\{\substack{\perp 1\\ \parallel}\right.}}{\partial \alpha}, \quad (\sin\alpha)\hat{E}_{\mathrm{g}\perp 2} = \mathrm{i}\frac{\partial \hat{E}_{\mathrm{g}\perp 2}}{\partial \lambda}. \tag{3.4.7}$$

With $\hat{f}_{1\mathrm{g}}$ (and hence $\hat{f}_1$) determined, we can then calculate the perturbated current density

$$\hat{\boldsymbol{J}} = q\int_{\mathbb{R}^3} \mathrm{d}^3 v\, \boldsymbol{v} f_1 = 2\pi q \int_{u_{\mathrm{L}}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \left\langle \boldsymbol{v}\hat{f}_1 \right\rangle_0 \tag{3.4.8}$$

and obtain the conductivity/susceptibility and dielectric tensors. For this purpose, let's note the following relations:

(1)
$$\begin{aligned}\left\langle (\cos\alpha)\hat{f}_1 \right\rangle_0 &= \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\alpha\, \frac{\mathrm{e}^{\mathrm{i}\alpha} + \mathrm{e}^{-\mathrm{i}\alpha}}{2} \sum_{m,n} J_m(\lambda) \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \mathrm{e}^{\mathrm{i}(m-n)\alpha} \\ &= \sum_{n=-\infty}^{+\infty} \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \frac{J_{n-1}(\lambda) + J_{n+1}(\lambda)}{2} \\ &= \frac{1}{\lambda}\sum_{n=-\infty}^{+\infty} n J_n(\lambda) \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n;\end{aligned} \tag{3.4.9}$$

(2)
$$\begin{aligned}\left\langle (\sin\alpha)\hat{f}_1 \right\rangle_0 &= \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\alpha\, \frac{\mathrm{e}^{\mathrm{i}\alpha} - \mathrm{e}^{-\mathrm{i}\alpha}}{2\mathrm{i}} \sum_{m,n} J_m(\lambda) \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \mathrm{e}^{\mathrm{i}(m-n)\alpha} \\ &= \sum_{n=-\infty}^{+\infty} \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \frac{J_{n-1}(\lambda) - J_{n+1}(\lambda)}{2\mathrm{i}} \\ &= -\mathrm{i}\sum_{n=-\infty}^{+\infty} J_n'(\lambda) \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n;\end{aligned} \tag{3.4.10}$$

(3)
$$\left\langle \hat{f}_1 \right\rangle_0 = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\alpha \sum_{m,n} J_m(\lambda) \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n \mathrm{e}^{\mathrm{i}(m-n)\alpha} \tag{3.4.11}$$

$$= \sum_{n=-\infty}^{+\infty} \left\langle \hat{f}_{1\mathrm{g}} \right\rangle_n J_n(\lambda). \tag{3.4.11}$$

Denoting

$$[\![\cdots]\!]_{n,\left\{\begin{smallmatrix}1\\2\end{smallmatrix}\right.} \equiv \int_{u_\mathrm{L}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp (\cdots) \frac{P_{\left\{\begin{smallmatrix}1\\2\end{smallmatrix}\right.}(f_{0\mathrm{g}})}{\omega - k_\parallel v_\parallel - n\Omega}, \tag{3.4.12}$$

the components of the susceptibility tensor $\tilde{\chi} = \mathrm{i}4\pi\tilde{\sigma}/\omega$ in the basis $\left\{\hat{\boldsymbol{e}}_{\perp 1} \equiv \hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_{\perp 2} \equiv \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_\parallel \equiv \hat{\boldsymbol{e}}_3\right\}$, $\chi_{ij} \equiv \hat{\boldsymbol{e}}_i \cdot \tilde{\chi} \cdot \hat{\boldsymbol{e}}_j$, are then given by

$$\chi_{11} = 2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[\left(\frac{v_\perp n J_n(\lambda)}{\lambda} \right)^2 \right]\!\!\right]_{n,1}, \tag{3.4.13}$$

$$\chi_{12} = \mathrm{i}2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[\frac{v_\perp^2 n J_n(\lambda) J_n'(\lambda)}{\lambda} \right]\!\!\right]_{n,1}, \tag{3.4.14}$$

$$\chi_{13} = 2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[\frac{v_\parallel v_\perp n J_n^2(\lambda)^2}{\lambda} \right]\!\!\right]_{n,2}, \tag{3.4.15}$$

$$\chi_{21} = -\chi_{12}, \tag{3.4.16}$$

$$\chi_{22} = 2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[(v_\perp J_n'(\lambda))^2 \right]\!\!\right]_{n,1}, \tag{3.4.17}$$

$$\chi_{23} = -\mathrm{i}2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[v_\parallel v_\perp J_n(\lambda) J_n'(\lambda) \right]\!\!\right]_{n,2}, \tag{3.4.18}$$

$$\chi_{31} = 2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[\frac{v_\parallel v_\perp n J_n^2(\lambda)}{\lambda} \right]\!\!\right]_{n,1}, \tag{3.4.19}$$

$$\chi_{32} = \mathrm{i}2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[v_\parallel v_\perp J_n(\lambda) J_n'(\lambda) \right]\!\!\right]_{n,1}, \tag{3.4.20}$$

$$\chi_{33} = 2\pi \frac{\omega_\mathrm{p}^2}{\omega} \sum_{n=-\infty}^{+\infty} \left[\!\!\left[\left(v_\parallel J_n(\lambda) \right)^2 \right]\!\!\right]_{n,2}. \tag{3.4.21}$$

We now have the complete electromagnetic dielectric tensor $\tilde{\epsilon} = \tilde{g} + \tilde{\chi}$ for uniform magnetized plasmas including thermal kinetic effects. Finally, we remark that, if $\int_\mathbb{R} \mathrm{d}v_\parallel\, v_\parallel f_0 = 0$, then $\chi_{13} = \chi_{31}, \chi_{32} = -\chi_{23}$.

3.5 Normal modes

In magnetized plasmas, there exists a variety of normal modes and, therefore, our treatment will (and has to) be selective. The emphasis will be on thermal kinetic effects which are outside the scope of the usual cold fluid description. We will, however, first briefly review the cold

leading to the cold-plasma dielectric tensor

$$\boldsymbol{\epsilon}\cdot\mathbf{E} = \begin{pmatrix} S & -iD & 0 \\ iD & S & 0 \\ 0 & 0 & P \end{pmatrix}\begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \tag{18}$$

in which the quantities S (for sum), D (for difference), and P (for plasma) are defined:

$$S = \tfrac{1}{2}(R + L), \quad D = \tfrac{1}{2}(R - L), \tag{19}$$

$$R \equiv 1 + \sum_s \chi_s^- = 1 - \sum_s \frac{\omega_{ps}^2}{\omega(\omega + \Omega_s)}, \tag{20}$$

$$L \equiv 1 + \sum_s \chi_s^+ = 1 - \sum_s \frac{\omega_{ps}^2}{\omega(\omega - \Omega_s)}, \tag{21}$$

$$P \equiv 1 - \sum_s \frac{\omega_{ps}^2}{\omega^2}. \tag{22}$$

Figure 3.1: Cold plasma dielectric tensor [21].

plasma results and the readers are strongly recommended to familiarize themselves with the fluid theory via standard textbooks such as [26]. In the cold fluid theory, the normal modes are characterized by examining the two limiting cases, i.e., parallel and perpendicular propagations. The reason is just simplicity: for a general oblique propagation, the various modes are strongly coupled and discussions can easily become tedious and complicated; the situation becomes even more complicated in the Vlasov kinetic theory.

The cold plasma normal modes with $\boldsymbol{k} \parallel \boldsymbol{B}_0$ is summarized in Figure 3.2, where we have assumed that the density is high $\left(\omega_{\mathrm{pe}} > |\Omega_e|\right)$. In Figure 3.2, R and L denote, respectively, right and left circular polarizations, and

$$\omega_{\left\{\substack{\mathrm{R} \\ \mathrm{L}}\right.} = \frac{1}{2}\left(\sqrt{\Omega_e^2 + 4\omega_{\mathrm{pe}}^2} \pm |\Omega_e|\right); \tag{3.5.1}$$

ECW and ICW correspond to electron and ion cyclotron waves, respectively; WHW denotes whistler waves; CAW and SAW denote compressional and shear Alfvén waves, respectively, with $v_{\mathrm{A}} = B_0/\sqrt{4\pi\rho_{\mathrm{m}}}$ being the Alfvén speed.

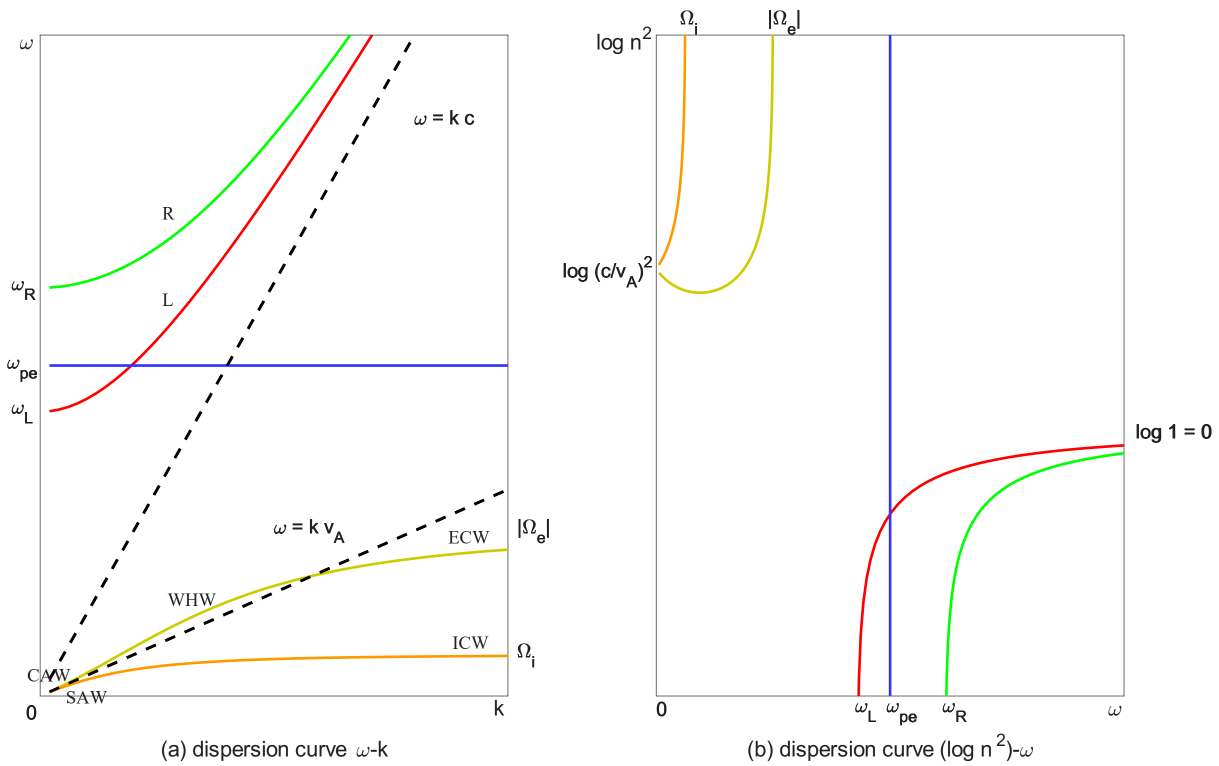


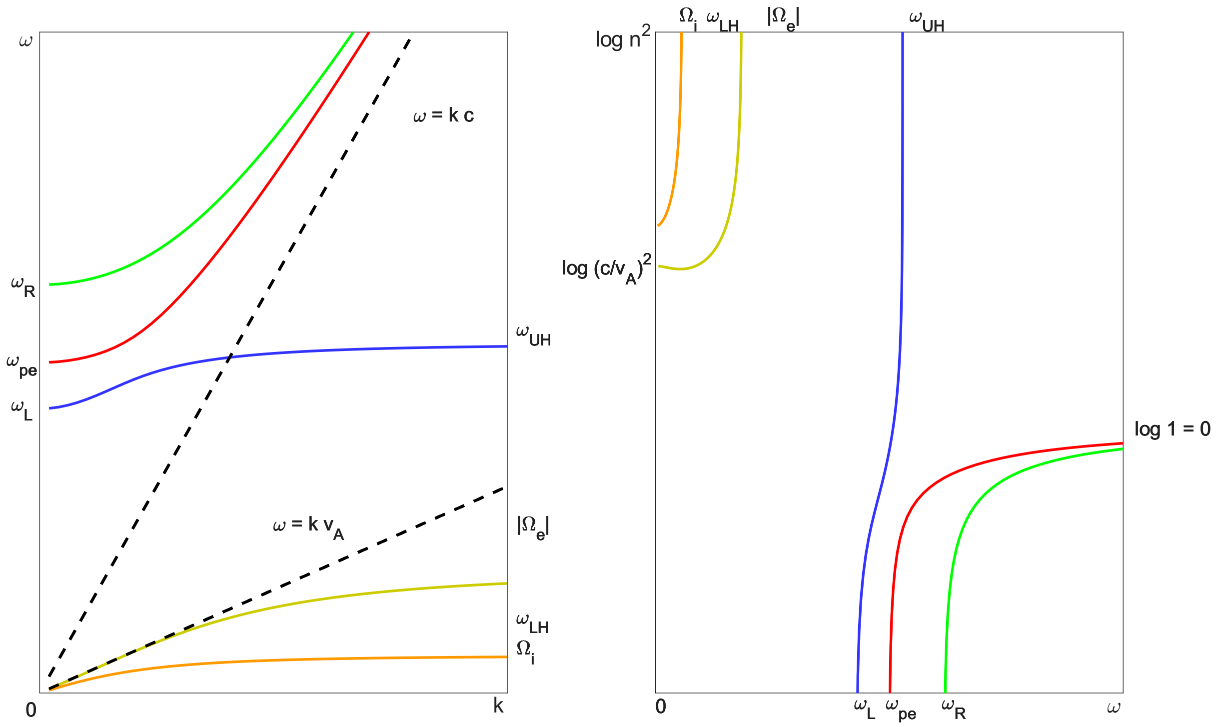
Figure 3.2: Sketch of dispersion curves for parallel waves with $\boldsymbol{k} \parallel \boldsymbol{B}_0$ in a cold, magnetized, high-density $\left(\omega_{\mathrm{pe}} > |\Omega_e|\right)$ plasma; an artificial ion-electron mass ratio $m_i/m_e = 4$ is employed. The cold plasma normal modes (Figure 3.2, Figure 3.3 and Figure 3.4) are ploted using the linear kinetic dispersion relation solver BO (波) [32], [33].

As the wave normal angle (WNA, the angle between $\boldsymbol{k}$ and $\boldsymbol{B}_0$) increases, the L mode couples with the plasma oscillation. The dispersion curves for the obliquely propagating normal modes with WNA $= 45°$ are shown in Figure 3.3.

Figure 3.4 summarizes the cold plasma normal modes with $\boldsymbol{k} \perp \boldsymbol{B}_0$. Here, X and O denote, respectively, extraordinary and ordinary modes; ω_{LH} and ω_{UH} are the lower- and upper-hybrid frequencies, respectively

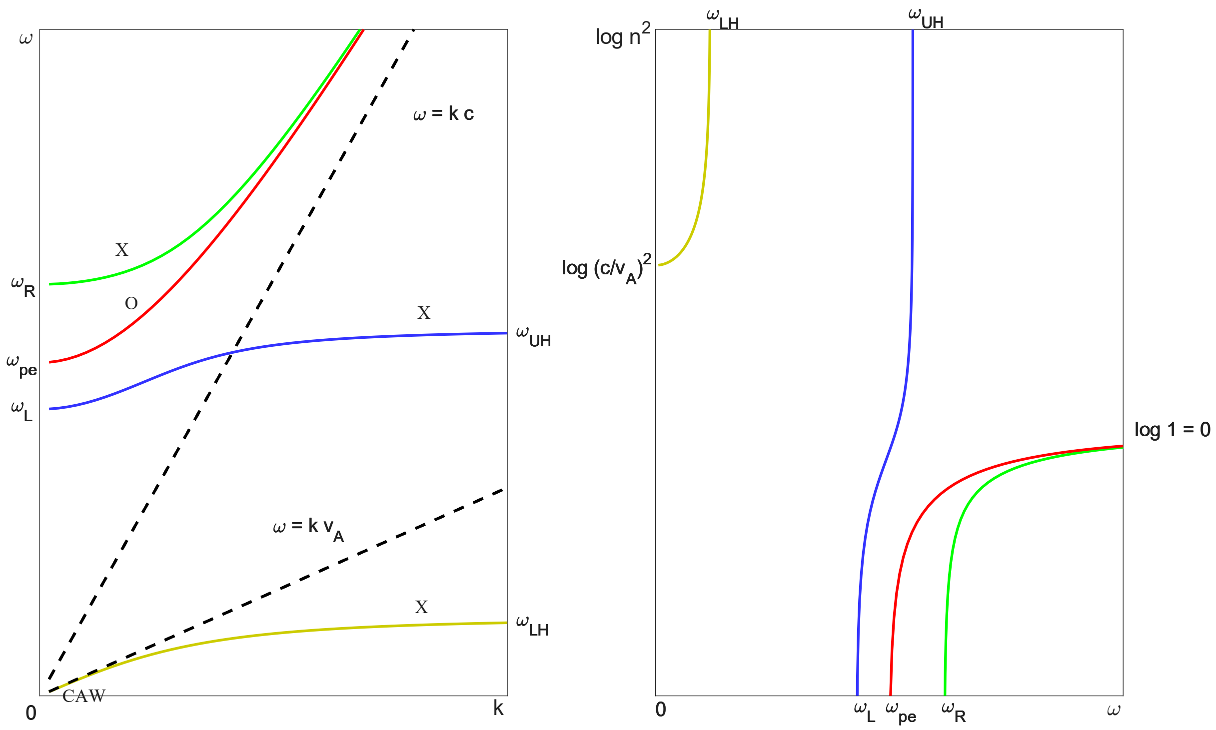
$$\begin{aligned} \omega_{\mathrm{LH}} &= \sqrt{\frac{\omega_{\mathrm{p}i}^2}{1+\frac{\omega_{\mathrm{pe}}^2}{\Omega_e^2}}} = \sqrt{\frac{\Omega_i|\Omega_e|}{1+\frac{\Omega_e^2}{\omega_{\mathrm{pe}}^2}}} \overset{|\Omega_e|/\omega_{\mathrm{pe}} \ll 1}{=\!=\!=} \sqrt{\Omega_i|\Omega_e|}, \\ \omega_{\mathrm{UH}} &= \sqrt{\omega_{\mathrm{pe}}^2+\Omega_e^2}. \end{aligned} \tag{3.5.2}$$

Now, as we add the kinetic effects into the normal-mode description, two dominant features appear. One is the *finite-Larmor-radius* (FLR) effect, which, inaddition to modifying the cold plasma modes, can give rise to new modes near the harmonics of the cyclotron frequencies (e.g., the Bernstein waves). The other feature is the wave-particle resonance, i.e., the *cyclotron resonance* whose resonance condition is $\omega - k_\parallel v_\parallel - n\Omega = 0$. Thus, similar to the unmagnetized case, waves can be damped or excited by tapping the velocity-space free energy. It is worthy



(a) dispersion curve ω-k (b) dispersion curve (log n^2)-ω

Figure 3.3: Sketch of dispersion curves for oblique waves with WNA $= 45°$ in a cold, magnetized, high-density $(\omega_{\mathrm{pe}} > |\Omega_e|)$ plasma; an artificial ion-electron mass ratio $m_i/m_e = 4$ is employed.



(a) dispersion curve ω-k (b) dispersion curve (log n^2)-ω

Figure 3.4: Sketch of dispersion curves for perpendicular waves with $\boldsymbol{k} \perp \boldsymbol{B}_0$ in a cold, magnetized, high-density $(\omega_{\mathrm{pe}} > |\Omega_e|)$ plasma; an artificial ion-electron mass ratio $m_i/m_e = 4$ is employed.

noting that finite plasma pressure p can also give tise to new modes (e.g., the ion acoustic waves) as well as the so-called finite-beta ($\beta = 8\pi p/B_0^2$) effects which are responsible for certain types of instabilities. Since, in most cases, thermal effects can be included via warm fluid description, we shall be concentrating in these lectures on the kinetic effects.

First, we illustrate the FLR physics by examining the ion Bernstein waves (IBWs) which are electrostatic waves with $k_\parallel = 0$. The corresponding dispersion relation is then given by

$$1 + \sum_s \chi_s = 0 \tag{3.5.3}$$

where

$$\begin{aligned}\chi_s &= 2\pi \frac{\omega_{ps}^2}{n_{0s}k^2} \int_{u_L} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \sum_{n=-\infty}^{+\infty} J_n^2(\lambda_s) \frac{n\Omega_s}{\omega - n\Omega_s} \frac{1}{v_\perp} \frac{\partial f_{0s}}{\partial v_\perp} \\ &= 2\pi \frac{\omega_{ps}^2}{n_{0s}k^2} \int_{u_L} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \sum_{n=1}^{+\infty} J_n^2(\lambda_s) \frac{2n^2\Omega_s^2}{\omega^2 - n^2\Omega_s^2} \frac{1}{v_\perp} \frac{\partial f_{0s}}{\partial v_\perp}.\end{aligned} \tag{3.5.4}$$

Taking f_{0s} to be Maxwellian in $v_\perp$, i.e.,

$$f_{0s}\left(v_\parallel, v_\perp\right) = n_{0s} f_{0\parallel s}\left(v_\parallel\right) \frac{1}{\pi v_{ts}^2} \exp\left(-\frac{v_\perp^2}{v_{ts}^2}\right), \tag{3.5.5}$$

and noting the following integral formula (Weber integral) [11], [34]

$$\int_0^{+\infty} \mathrm{d}x\, \mathrm{e}^{-\rho^2 x^2} x J_n(\alpha x) J_n(\beta x) = \frac{1}{2\rho^2} \exp\left(-\frac{\alpha^2+\beta^2}{4\rho^2}\right) I_n\left(\frac{\alpha\beta}{2\rho^2}\right) \tag{3.5.6}$$

where I_n is the n-th order modified Bessel function of the first kind, χ_s reduces to

$$\chi_s = -\frac{1}{k^2\lambda_{Ds}^2} \sum_{n=1}^{+\infty} \frac{2n^2\Omega_s^2}{\omega^2 - n^2\Omega_s^2} A_n(b_s), \tag{3.5.7}$$

where

$$b_s \equiv \frac{k_\perp^2 v_{ts}^2}{2\Omega_s^2} = \frac{1}{2} k_\perp^2 \rho_{ts}^2, \quad A_n(x) \equiv \mathrm{e}^{-x} I_n(x). \tag{3.5.8}$$

For IBWs, we have $|\omega/\Omega_i| = \mathcal{O}(1) \Rightarrow |\omega/\Omega_e| = \mathcal{O}(\Omega_i/|\Omega_e|) \ll 1$ and $|k_\perp \rho_{ti}| = \mathcal{O}(1) \Rightarrow |k_\perp \rho_{te}| = \mathcal{O}(\rho_{te}/\rho_{ti}) \ll 1$, which means that χ_e is dominated by the $n = 1$ term

$$\chi_e \approx -\frac{1}{k^2\lambda_{Ds}^2} \frac{2\Omega_e^2}{\omega^2 - \Omega_e^2} I_1(b_e) \mathrm{e}^{-b_e} \approx \frac{\omega_{pe}^2}{\Omega_e^2}. \tag{3.5.9}$$

The dipsersion relation, Eq. (3.5.3), can then be written as

$$1 + \frac{\omega_{pe}^2}{\Omega_e^2} \approx \frac{1}{k^2\lambda_{Di}^2} \sum_{n=1}^{+\infty} \frac{2n^2\Omega_i^2}{\omega^2 - n^2\Omega_i^2} A_n(b_i) \equiv F_i(\omega). \tag{3.5.10}$$

For a given k , Eq. (3.5.10) can be solved by graphic method, finding the intersection between the two function plots $y = 1 + \omega_{\text{pe}}^2/\Omega_e^2$ and $y = F_i(\omega)$. Noting that $1 + \omega_{\text{pe}}^2/\Omega_e^2 = \mathcal{O}(1)$, we can estimate the solution of Eq. (3.5.10) by examining the limiting forms of $F_i(\omega)$:

- In the $k \to +\infty$ limit, we have $A_n(b_i)/k^2 = \mathcal{O}\Big(\big(k^3\rho_{\text{t}i}\big)^{-1}\Big) \to 0^+$ and, thus,

$$(\omega^2 - n^2\Omega_i) \to 0^+ \text{ for } n = 1, 2, \cdots. \tag{3.5.11}$$

- In the $k \to 0^+$ limit, noting that $A_n(b_i) = \mathcal{O}\big(\big(k^2\rho_{\text{t}i}^2\big)^n\big)$, we find

$$F_i(\omega) \to \frac{\omega_{\text{p}i}^2}{\omega^2 - \Omega_i^2} + \sum_{n=2}^{+\infty} \frac{n^2\Omega_i^2}{\omega^2 - n^2\Omega_i^2}\mathcal{O}\left(k^{2(n-1)}\frac{\rho_{\text{t}i}^{2n}}{\lambda_{\text{D}i}^2}\right). \tag{3.5.12}$$

  Eq. (3.5.10), thus, has no root between $0 < \omega^2 < \Omega_i^2$, and produces roots at $\omega^2 \approx n^2\Omega_i^2$ for $n \geq 2$ as well as at

$$\begin{aligned} \frac{\omega_{\text{p}i}^2}{\omega^2 - \Omega_i^2} &= 1 + \frac{\omega_{\text{pe}}^2}{\Omega_e^2} \\ \Rightarrow \omega^2 &= \Omega_i^2 + \omega_{\text{LH}}^2 \approx \omega_{\text{LH}}^2. \end{aligned} \tag{3.5.13}$$

The dispersion relation is shown in Figure 3.5. Its most important feature is that the frequency bands where the IBWs propagate are separated by gaps, which makes them difficult to be excited [22]. Finally, we remark that the IBWs can propagate across the magnetic field ($\partial\omega/\partial k_\perp \neq 0$) and thus can lead to ion heating, called ion Bernstein wave heating (IBWH) [35].

In the next example, we shall consider the parallel-propagating electron and ion cyclotron waves including wave-particle resonances. From the cold fluid theory, we know that both waves are electromagnetic modes with $\boldsymbol{k} \cdot \hat{\boldsymbol{E}} = 0$. Since their phase speeds are, however, much less than the speed of light[33], we expect effects of wave-particle resonances to be important. To recover these waves from the Vlasov theory, we note that, as $k_\perp \to 0$, $J_0(\lambda) = 1 + o(\lambda)$, $J_{\pm 1}(\lambda) = \pm\lambda/2 + o(\lambda^2)$, $J_{|n|\geq 2}(\lambda) = o(\lambda)$. Thus, for $k_\perp = 0$, we have

$$\begin{aligned} \chi_{11} &= \frac{\pi}{2}\sum_s \frac{\omega_{\text{p}s}^2}{\omega}\sum_{n=\pm 1} [\![v_\perp^2]\!]_{n,1,s} \equiv d_+ + d_-, \\ \chi_{12} &= \mathrm{i}(d_+ - d_-), \\ \chi_{21} &= \mathrm{i}(d_- - d_+), \\ \chi_{22} &= d_+ + d_- \end{aligned} \tag{3.5.14}$$

with

$$\chi_{33} = 2\pi\sum_s \frac{\omega_{\text{p}s}^2}{\omega}\Big[\!\!\Big[v_\parallel^2\Big]\!\!\Big]_{0,2,s} \tag{3.5.15}$$

[33]Refer to footnote[36]. One consequence is that, unlike in vacuum, the spectral energy density of the magnetic field becomes much larger than that of the electric field (which follows from Faraday's law); this excess magnetic field arises from the current carried by the plasma particles.

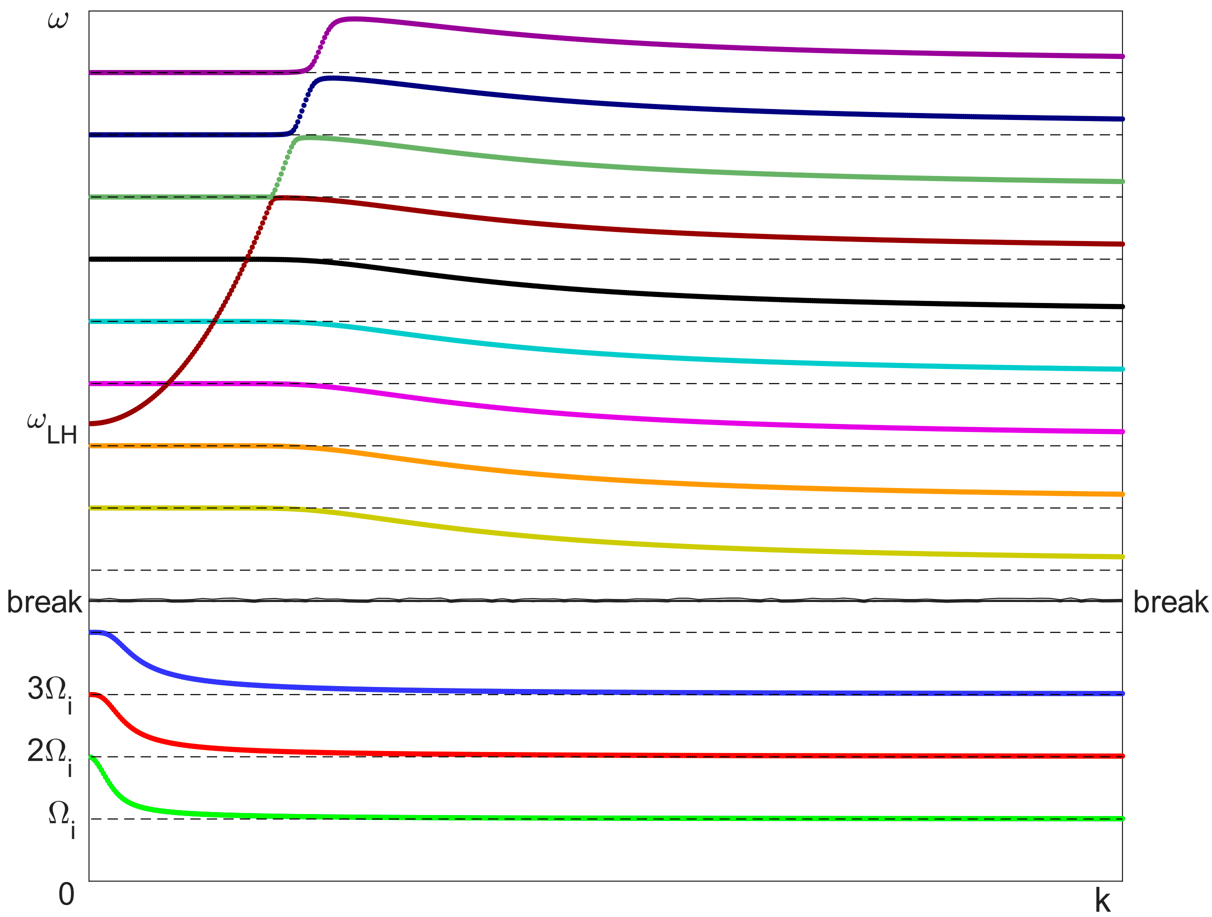


Figure 3.5: Sketch of $\omega - k$ dispersion curves for IBWs. Note that $|\omega| \sim \Omega_i$ and $\lambda \sim \rho_{ti}$.

$$= 2\pi \sum_s \frac{\omega_{ps}^2}{k_\parallel} \int_{u_L} dv_\parallel \int_0^{+\infty} dv_\perp \, v_\perp \frac{\partial f_{0s}/\partial v_\parallel}{\omega - k_\parallel v_\parallel} \quad \text{(electrostatic)} \tag{3.5.15}$$

and $\chi_{13} = \chi_{23} = \chi_{31} = \chi_{32} = 0$. Here

$$\begin{aligned} d_\pm &\equiv \frac{\pi}{2} \sum_s \frac{\omega_{ps}^2}{\omega} [\![v_\perp^2]\!]_{\pm 1,1,s} \\ &= \frac{\pi}{2} \sum_s \frac{\omega_{ps}^2}{\omega^2} \int_{u_L} dv_\parallel \int_0^{+\infty} dv_\perp \, v_\perp^3 \frac{k_\parallel \, \partial/\partial v_\parallel + \big((\omega - k_\parallel v_\parallel)/v_\perp \big) \partial/\partial v_\perp}{\omega - k_\parallel v_\parallel \mp \Omega_s} f_{0s}. \end{aligned} \tag{3.5.16}$$

The electromagnetic-mode wave equation then writes

$$\begin{bmatrix} -n^2 + 1 + d_+ + d_- & \mathrm{i}(d_+ - d_-) \\ \mathrm{i}(d_- - d_+) & -n^2 + 1 + d_+ + d_- \end{bmatrix} \begin{bmatrix} \hat{E}_1 \\ \hat{E}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{3.5.17}$$

which, in the new basis $\big\{ \hat{e}_\mathrm{R} := (\hat{e}_{\perp 1} + \mathrm{i}\hat{e}_{\perp 2})/\sqrt{2}, \hat{e}_\mathrm{L} := (\hat{e}_{\perp 1} - \mathrm{i}\hat{e}_{\perp 2})/\sqrt{2}, \hat{e}_\parallel \big\}$, becomes

$$\begin{bmatrix} D_\mathrm{R} & 0 \\ 0 & D_\mathrm{L} \end{bmatrix} \begin{bmatrix} \hat{E}_\mathrm{R} \\ \hat{E}_\mathrm{L} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{3.5.18}$$

where

$$D_{\mathrm{R}} \equiv 1 + 2d_{-} - n^2,$$
$$D_{\mathrm{L}} \equiv 1 + 2d_{+} - n^2. \tag{3.5.19}$$

Noting that Ω carries sign, it is clear that, for $\omega > 0$, $D_{\mathrm{R}} = 0$ corresponds to right-handed circularly polarized waves[34] and, hence, includes electron cyclotron damping whose resonance condition is[35] $\omega - k_\parallel v_\parallel - |\Omega_e| = 0$; similarly, for $\omega > 0$ and $D_{\mathrm{L}} = 0$, we have left-handed circularly polarized waves which resonant with ions with resonance condition $\omega - k_\parallel v_\parallel - \Omega_i = 0$. As a specific example, it is easy to show that, for the right-handed circularly polarized wave and $|\omega| \gg \Omega_i$, the corresponding dispersion relation is given by (noting that $\omega_{\mathrm{p}i} \ll \omega_{\mathrm{pe}}$)

$$1 - \frac{c^2k^2}{\omega^2} + \frac{\omega_{\mathrm{pe}}^2}{\omega^2}\int_{u_{\mathrm{L}}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp^3 \frac{k_\parallel\, \partial/\partial v_\parallel + \big(\big(\omega - k_\parallel v_\parallel\big)/v_\perp\big)\partial/\partial v_\perp}{\omega - k_\parallel v_\parallel - |\Omega_e|} f_{0e} = 0. \tag{3.5.22}$$

Eq. (3.5.22) will be used in the next section to discuss the electron cyclotron (whistler) instabilities driven by velocity-space anisotropy.

3.6 Instabilities

In this section, we shall consider two instabilities due to anisotropic distributions. One is the whistler instability and the other is the electeostatic ion loss-cone instability.

For the whistler instability, we let $\boldsymbol{k}$ be parallel to $\boldsymbol{B}_0 = B_0\hat{\boldsymbol{e}}_\parallel$, i.e., $\boldsymbol{k} = k\hat{\boldsymbol{e}}_\parallel$. We further simplify the analysis by taking

$$f_{0e}\big(v_\parallel, v_\perp\big) = n_{0e}\delta\big(v_\parallel\big)\frac{1}{\pi v_{\mathrm{t}\perp e}^2}\exp\left(-\frac{v_\perp^2}{v_{\mathrm{t}\perp e}^2}\right). \tag{3.6.1}$$

---

[34]For a mode with $\boldsymbol{E} = \boldsymbol{E}_{\mathrm{R}}(t,\boldsymbol{x}) = \mathrm{Re}\,\left(\hat{\boldsymbol{e}}_{\mathrm{R}}\big|\hat{E}_{\mathrm{R}}\big|\mathrm{e}^{\mathrm{i}\varphi_{\mathrm{R}}}\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}\right)$, we have, at a *fixed space point* $\boldsymbol{x} = \boldsymbol{x}_0$, that

$$\boldsymbol{E}_{\mathrm{R}}(t,\boldsymbol{x}=\boldsymbol{x}_0) = \hat{\boldsymbol{e}}_{\perp 1}\big|\hat{E}_{\mathrm{R}}\big|\cos(\omega t - \varphi_{\mathrm{R}0}) + \hat{\boldsymbol{e}}_{\perp 2}\big|\hat{E}_{\mathrm{R}}\big|\sin(\omega t - \varphi_{\mathrm{R}0}), \quad \varphi_{\mathrm{R}0} \equiv \varphi_{\mathrm{R}} + \boldsymbol{k}\cdot\boldsymbol{x}_0. \tag{3.5.20}$$

If $\omega > 0$, then $\boldsymbol{E}_{\mathrm{R}}(t,\boldsymbol{x}=\boldsymbol{x}_0)$ rotates in the anticlockwise direction in the $1O2$ plane as t increases, i.e., the wave is right-handed circularly polarized —— here polarizations are defined with repsect to the background magnetic field $\boldsymbol{B}_0$ and have nothing to do with the wave propagation directions; if $\omega < 0$, however, $\boldsymbol{E}_{\mathrm{R}}(t,\boldsymbol{x}=\boldsymbol{x}_0)$ rotates in the clockwise direction and the wave is left-handed circularly polarized.

[35]If $\omega - k_\parallel v_\parallel = |\Omega_e|$, then in the reference frame moving with parallel velocity $v_\parallel$, the Doppler shifted wave frequency is equal to the absolute value of the electron cyclotron frequency, i.e., $\omega' \equiv \omega - k_\parallel v_\parallel = |\Omega_e| > 0$. Thus, $\boldsymbol{E}'_{\mathrm{R}}(t,\boldsymbol{x}'=\boldsymbol{x}'_0)$ rotates in the anticlockwise direction, which matches the gyrations of the electrons.

PUZZLE: Sometimes the "left-handed" polarized waves can also resonant with electrons, whose resonance condition is $\omega - k_\parallel v_\parallel + |\Omega_e| = 0$. This sounds curious —— how come a "left-handed" polarized wave could resonant with the "right-handed" gyrating electrons? In fact, polarizations are only concerned with the gyro-direction of the electric fields at a *fixed space point*. In particular, when we say that a wave is "left-handed" polarized, what we mean is that its electric field can be expressed as, at $\boldsymbol{x} = \boldsymbol{x}_0$,

$$\boldsymbol{E}(t,\boldsymbol{x}=\boldsymbol{x}_0) = \hat{\boldsymbol{e}}_{\perp 1}\big|\hat{E}\big|\cos(\omega t - \varphi_0) - \hat{\boldsymbol{e}}_{\perp 2}\big|\hat{E}\big|\sin(\omega t - \varphi_0) \text{ with } \omega > 0; \tag{3.5.21}$$

i.e., we are implicitly assuming (???) $\omega > 0$ when we talk about polarizations. Then in the reference frame moving with parallel velocity $v_\parallel$, the Doppler shifted wave frequency $\omega' \equiv \omega - k_\parallel v_\parallel = -|\Omega_e| < 0$ changes sign and, thus, the wave polarization reverses to a "right-handed" one, matching the gyrations of the electrons. The conclusion is intuitive: in the reference frame moving with parallel velocity $v_\parallel$, only the right-handed polarized waves can resonant with electrons.

From Eq. (3.5.22), the whistler dispersion relation, in this case, reduces to

$$1-\frac{c^2k^2}{\omega^2}-\frac{\omega_{\mathrm{pe}}^2}{\omega(\omega-|\Omega_e|)}\left(1+\frac{k^2v_{\mathrm{t}\perp e}^2}{2\omega(\omega-|\Omega_e|)}\right)=0, \tag{3.6.2}$$

i.e.,

$$\omega^2(\omega-|\Omega_e|)^2-\omega_{\mathrm{pe}}^2\omega(\omega-|\Omega_e|)-c^2k^2(\omega-|\Omega_e|)^2-\frac{1}{2}\omega_{\mathrm{pe}}^2k^2v_{\mathrm{t}\perp e}^2=0. \tag{3.6.3}$$

The low-frequency $|\omega/\Omega_e|\ll 1$ dispersion relation writes

$$\omega^2+\frac{\omega_{\mathrm{pe}}^2}{|\Omega_e|}\omega-k^2c^2\left(1+\frac{\omega_{\mathrm{pe}}^2v_{\mathrm{t}\perp e}^2}{2\Omega_e^2c^2}\right)=0 \tag{3.6.4}$$

$$\begin{aligned}\Rightarrow\omega&=\frac{1}{2}\left(-\frac{\omega_{\mathrm{pe}}^2}{|\Omega_e|}\pm\sqrt{\frac{\omega_{\mathrm{pe}}^4}{|\Omega_e|^2}+4k^2c^2\left(1+\frac{\omega_{\mathrm{pe}}^2v_{\mathrm{t}\perp e}^2}{2\Omega_e^2c^2}\right)}\right)\\ &\xlongequal{kc\ll\omega_{\mathrm{pe}}}-\frac{\omega_{\mathrm{pe}}^2}{2|\Omega_e|}\left(1\mp\left(1+\frac{2k^2c^2|\Omega_e|^2}{\omega_{\mathrm{pe}}^4}\left(1+\frac{\omega_{\mathrm{pe}}^2v_{\mathrm{t}\perp e}^2}{2\Omega_e^2c^2}\right)\right)\right);\end{aligned} \tag{3.6.5}$$

since $|\omega/\Omega_e|\ll 1$, we have to choose the upper sign, which gives the warm whistler wave[36]

$$\omega=\left(k^2c^2\frac{|\Omega_e|}{\omega_{\mathrm{pe}}^2}\right)\left(1+\omega_{\mathrm{pe}}^2\frac{v_{\mathrm{t}\perp e}^2}{2}\Omega_e^2c^2\right)\xlongequal{T_e=0}k^2c^2\frac{|\Omega_e|}{\omega_{\mathrm{pe}}^2}. \tag{3.6.6}$$

That Eq. (3.6.3) exhibits instabilities can be easily seen by taking the $k\to+\infty$ limit, where

$$\omega-|\Omega_e|=\mathrm{i}\omega_{\mathrm{pe}}\frac{v_{\mathrm{t}\perp e}}{\sqrt{2}c}. \tag{3.6.7}$$

We note also that, from Eq. (3.6.3) or Eq. (3.6.7), it is clear that this whistler instability may be regarded as the extension of the Weibel instability (whose dispersion relation is given by Eq. (2.5.14)) to a magnetized plasma.

Next, we consider the electrostatic ion loss-cone instability. To suppress the parallel Landau damping, we let $\left|k_\parallel/k_\perp\right|\ll 1$, so that $|\zeta_e|\gg 1$ and $\left|k_\parallel v_{\mathrm{t}i}/\Omega_i\right|\ll 1$; we further assume that electrons are strongly magnetized, i.e, $|k_\perp\rho_{\mathrm{t}e}|\ll 1$. The electron susceptibility is given by

$$\chi_e\approx\frac{\omega_{\mathrm{pe}}^2}{\Omega_e^2}-\frac{\omega_{\mathrm{pe}}^2}{\omega_{\mathrm{r}}^2}\frac{k_\parallel^2}{k^2}; \tag{3.6.8}$$

here, we remark that the first term on the R.H.S. is due to polarization drift[37], and the second due to parallel motion. Meanwhile, for ions, we have, from Eq. (3.2.29), that

[36]Although the whistler wave is an electromagnetic mode, in a high-density plasma where $\omega_{\mathrm{pe}}\gg|\Omega_e|$, its phase speed $\omega/kc=\mathcal{O}\left(kc|\Omega_e|/\omega_{\mathrm{pe}}^2\right)$ is, in general, far less than the speed of light c. Please refer to footnote[33].

[37]Since $\partial\boldsymbol{P}/\partial t=\boldsymbol{J}$, electron polarization current $\boldsymbol{J}_{\mathrm{pe}}=-(en_ec/\Omega_eB_0)\mathrm{d}\boldsymbol{E}_\perp/\mathrm{d}t$ contributes to a polarization $\boldsymbol{P}=-(en_ec/\Omega_eB_0)\boldsymbol{E}_\perp=(n_em_ec^2)/B_0^2\boldsymbol{E}_\perp$; from $\boldsymbol{P}=(1/4\pi)\tilde{\chi}\cdot\boldsymbol{E}$ we then know that the polarization drift contributes to the perpendiculr susceptibility as $\chi_{\mathrm{pe}}=4\pi n_em_ec^2/B_0^2=\omega_{\mathrm{pe}}^2/\Omega_e^2$.

$$\chi_i = -2\pi \frac{\omega_{\mathrm{p}i}^2}{n_{0i}k^2} \int_{\mathbb{R}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp \sum_{n=-\infty}^{+\infty} J_n^2(\lambda_i) \left(\mathcal{P} \frac{1}{k_\parallel v_\parallel - \omega_\mathrm{r} + n\Omega_i} + \mathrm{i}\pi\delta\big(k_\parallel v_\parallel - \omega_\mathrm{r} + n\Omega_i\big) \right) \left(k_\parallel \frac{\partial}{\partial v_\parallel} + \frac{n\Omega_i}{v_\perp} \frac{\partial}{\partial v_\perp} \right) f_{0i}. \tag{3.6.9}$$

Since the instability is due to wave-ion resonance and noting $\left|k_\parallel v_{\mathrm{t}i}/\Omega_i\right| \ll 1$, we expect that the unstable modes occur very near the ion cyclotron harmonics where $|\omega - N\Omega_i| \approx \left|k_\parallel v_{\mathrm{t}i}\right| \ll \Omega_i$ for some N. Thus

$$\chi_i \approx -2\pi \frac{\omega_{\mathrm{p}i}^2}{n_{0i}k^2} \int_{\mathbb{R}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp J_N^2(\lambda_i) \left(\mathcal{P} \frac{1}{k_\parallel v_\parallel - \omega_\mathrm{r} + N\Omega_i} + \mathrm{i}\pi\delta\big(k_\parallel v_\parallel - \omega_\mathrm{r} + N\Omega_i\big) \right) \frac{N\Omega_i}{v_\perp} \frac{\partial f_{0i}}{\partial v_\perp}. \tag{3.6.10}$$

To further simplify the analysis, we shall assume $\omega_\mathrm{r} = N\Omega_i$, then

$$\chi_i \approx -\mathrm{i}2\pi^2 \frac{\omega_{\mathrm{p}i}^2}{n_{0i}k^2} \frac{N\Omega_i}{\left|k_\parallel\right|} \int_0^{+\infty} \mathrm{d}v_\perp\, J_N^2(\lambda_i) \left.\frac{\partial f_{0i}}{\partial v_\perp}\right|_{v_\parallel=0}. \tag{3.6.11}$$

Now since $\partial(1+\chi_e)/\partial\omega_\mathrm{r} > 0$, i.e., the wave is of positive energy, we would have an instability if

$$\int_0^{+\infty} \mathrm{d}v_\perp\, J_N^2(\lambda_i) \left.\frac{\partial f_{0i}}{\partial v_\perp}\right|_{v_\parallel=0} > 0. \tag{3.6.12}$$

For ion loss-cone distributions and ignoring any effect due to the ambi-polar potential, it is reasonable to assume

$$\left.\frac{\partial f_{0i}}{\partial v_\perp}\right|_{v_\parallel=0} > 0 \text{ for } 0 < v_\perp < v_{\perp\mathrm{c}}. \tag{3.6.13}$$

Moreover, for $\lambda \gg N > 1$, we have

$$\begin{gathered} J_N(\lambda) \sim \sqrt{\frac{2}{\pi\lambda}} \cos\left(\lambda - \frac{N\pi}{2} - \frac{\pi}{4}\right) \\ \Rightarrow J_N^2(\lambda) < \frac{1}{\lambda} \text{ decreases fast.} \end{gathered} \tag{3.6.14}$$

Therefore, if

$$\left|\frac{k_\perp v_{\perp\mathrm{c}}}{\Omega_i}\right| \gg N > 1, \tag{3.6.15}$$

then the integral in Eq. (3.6.12) is dominated by the interval $[0, v_{\perp\mathrm{c}}]$ and we would have an instability.

Finally, we remark that the ion loss-cone instability also exist if the ions are unmagnetized; this

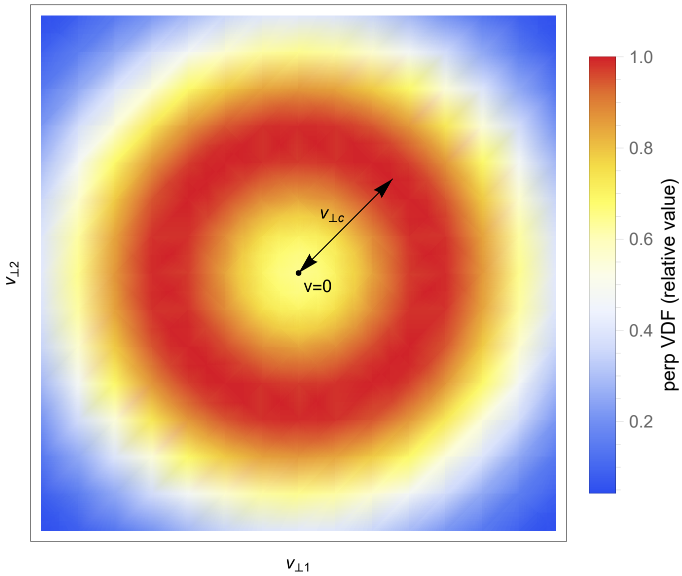


Figure 3.6: Sketch of a loss-cone perpendicular VDF.

is possible only when the phase speed of the wave falls into the valley of the perpendicular ion VDF, i.e., when $|\omega_{\mathrm{r}}/k_\perp| < v_{\perp\mathrm{c}}$. [17]

Mid-Term Take-Home Examination

(1) Consider the electrostatic weak beam-plasma instability for $kv_b = \omega_{\mathrm{pe}}$, i.e., the maximun unstable regime. Estimate the amplitude $\hat{E}$ at which

(i) the linear theory breaks down;

(ii) the unstable (slow) wave begins to trap the cold beam.

Are these two estimatations consistent with each other?

(2) Consider the right-handed circularly polarized electron cyclotron wave for which $\omega < |\omega_e|$. Given $\boldsymbol{B} = \boldsymbol{B}_{\mathrm{R}}(t, \boldsymbol{x}) = \hat{\boldsymbol{B}}_{\mathrm{R}} \exp(\mathrm{i}(kz - \omega t))$ where $\boldsymbol{k} = k\hat{\boldsymbol{e}}_\parallel$, show that

(i) for resonant particles with $v_\parallel = (\omega - |\Omega_e|)/k$, the linearization $|\nabla_{\boldsymbol{v}} f_1| \ll |\nabla_{\boldsymbol{v}} f_0|$ breaks down for

$$t \gtrsim \sqrt{\left|\frac{q}{m} k\hat{E}_{\mathrm{eff}}\right|} \equiv t_{\mathrm{c}}, \quad \hat{E}_{\mathrm{eff}} \equiv \frac{v_\perp}{c}\hat{B}_{\mathrm{R}}; \tag{M.1}$$

(ii) from the single particle picture, t_{c} corresponds to the bounce time of a particle trapped by the $(q/c)\boldsymbol{v}_\perp \times \boldsymbol{B}_{\mathrm{R}}$ force in the parallel direction.

(3) Electrostatic ion cyclotron waves are in the following parameter regime:

$$|\omega| \sim |n\Omega_i| \ll |\Omega_e|, \quad \left|k_\parallel/k_\perp\right| \ll 1, \quad |k_\perp \rho_{\mathrm{t}e}| \ll 1 \text{ and } v_{\mathrm{t}i} \ll \left|\omega/k_\parallel\right| \ll v_{\mathrm{t}e}. \tag{M.2}$$

Plot the $\omega - k_\perp$ dispersion curves.

(4) The following wave equation models the Bohm-Gross wave

$$\left(\delta^2\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial t^2}-\omega_0^2\right)\phi(t,x)=0,\quad \delta^2\ll 1. \tag{M.3}$$

Solve for ϕ subject to the initial conditions

$$\phi(t=0,x)=\mathrm{e}^{-x^2},\quad \frac{\partial\phi}{\partial t}(t=0,x)=0. \tag{M.4}$$

At least, try to establish the $t\to+\infty$ behavior for finite fixed x and the $|x|\to+\infty$ behavior for finite t. Sketch your results.

(5) Consider a one-dimensional unmagnetized plasma with $m_i\to+\infty$. Now we impose an electrostatic potential

$$\varphi_0(x)=\mathrm{e}^{-\alpha x^2},\quad \alpha>0 \tag{M.5}$$

in order to electrostatically confine the plasma.

(i) Solve the unperturbed electron motion.

(ii) Identify the constant of unperturbed motion.

(iii) Use the constant of unperturbed motion to construct an equilibrium distribution f_{0e} which is confined in space, i.e., $f_{0e}\to 0$ as $|x|\to+\infty$.

(iv) Solve, in as much details as you can, the perturbed distribution f_{1e}.
Hint: transform to the action-angle variables, where the action variable is the constant of motion derived in (ii) and angle variable corresponds to the phase of oscillations in the φ_0 potential well.

(6) Give a physical explanation for the velocity-anisotropy-driven right-handed circularly polarized electron cyclotron wave. For this purpose, we shall assume

$$f_{0e}\left(v_\parallel,v_\perp\right)=n_{0e}\delta\left(v_\parallel\right)\frac{1}{\pi}\delta\left(v_\perp^2-v_0^2\right). \tag{M.6}$$

3.7 Linear gyrokinetic equation and the kinetic theory of magnetohydrodynamic waves

Magnetohydrodynamic (MHD) waves have frequencies, typically, much lower than the ion cyclotron frequency. Thus, they are capable of coupling to phenomena with time scales associated with plasma nonuniformities, such as diamagnetic drifts, magnetic trapping, resistive diffusion, etc. We can then expect that MHD waves should play crucial roles in the physics of magnetically confined plasmas. Indeed, in the specific example of tokamaks, one finds ample such examples. For instance, we have the MHD stability analysis, anomalous transport due to drift waves, Alfvén wave heating and magnetic pumping. Since MHD waves are usually derived from the one-fluid description, it will be illuminating to consider these waves using the collisionless Vlasov description. This is one of the purposes of this section. The other purpose is to develop the low-frequency *gyrokinetic* (GK) formalism for uniform plasmas here, which can then be readily extended to nonuniform plasmas as we will do in the next chapter.

We first consider the electrostatic case in order to illustrate the theoretical approach. For low-frequency MHD waves, we adopt the following *maximal* orderings[38] [30]

$$\left|\frac{\omega}{\Omega}\right| = \mathcal{O}(\varepsilon) \ll 1, \quad \left|\frac{\rho_{\mathrm{t}}}{\lambda}\right| = \mathcal{O}(\varepsilon) \ll 1, \quad \underbrace{\left|k_{\parallel}\rho_{\mathrm{t}}\right| = \mathcal{O}(\varepsilon) \ll 1 \text{ and } \left|k_{\perp}\rho_{\mathrm{t}}\right| = \mathcal{O}(1)}_{\left|k_{\parallel}/k_{\perp}\right| \ll 1} \tag{3.7.1}$$

with ε being the smallness parameter. In practice, it is convenient to make the following *book-keeping* [10], [22] substitution

$$\Omega \mapsto \varepsilon^{-1}\Omega, \tag{3.7.2}$$

after which all the quantities appearing explicitly in the equations (e.g., ω, Ω and $k_{\parallel}$) are of order $\mathcal{O}(1)$; this amounts to the small $1/B_0$ expansion [9]. From Eq. (3.2.6) and Eq. (3.2.8), we have

$$f_{1\mathrm{g}} = \frac{q}{m}\varphi_{\mathrm{g}}\left(\frac{\partial}{\partial\mathcal{E}} + \frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}} + G_{\mathrm{g}} \tag{3.7.3}$$

and

$$\begin{aligned}\left(\frac{\partial}{\partial t} + v_{\parallel}\frac{\partial}{\partial X_{\parallel}} - \varepsilon^{-1}\Omega\frac{\partial}{\partial\alpha}\right)G_{\mathrm{g}} &= -\frac{q}{m}\left(\frac{\partial\varphi_{\mathrm{g}}}{\partial t}\frac{\partial}{\partial\mathcal{E}} + \left(\left(\frac{\partial}{\partial t} + v_{\parallel}\frac{\partial}{\partial X_{\parallel}}\right)\varphi_{\mathrm{g}}\right)\frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}} \\ &\equiv \text{R.H.S.}\end{aligned} \tag{3.7.4}$$

We can now carry out a regular perturbative treatment. Letting

$$G_{\mathrm{g}} = G_{\mathrm{g}0} + \varepsilon G_{\mathrm{g}1} + \mathcal{O}(\varepsilon^2), \tag{3.7.5}$$

we have

- at order $\mathcal{O}(\varepsilon^{-1})$, $\partial G_{\mathrm{g}0}/\partial\alpha = 0 \Rightarrow G_{\mathrm{g}0} = G_{\mathrm{g}0}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma)$;
- at order $\mathcal{O}(1)$,

$$\left(\frac{\partial}{\partial t} + v_{\parallel}\frac{\partial}{\partial X_{\parallel}}\right)G_{\mathrm{g}0} - \Omega\frac{\partial G_{\mathrm{g}1}}{\partial\alpha} = \text{R.H.S.} \tag{3.7.6}$$

Imposing the periodicity constraint $G_{\mathrm{g}n}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma, \alpha + 2\pi) = G_{\mathrm{g}n}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma, \alpha)$ to remove the secularity in α, we have, after the gyrophase average,

[38]The following comments are AI generated:
- Physically, $|\omega/\Omega| \ll 1$ implies that the wave frequency is much smaller than the cyclotron frequency, in which case the gyrophase average is invoked; $\left|k_{\parallel}\rho_{\mathrm{t}}\right| \ll 1$ means that the parallel wavelength is much longer than the gyroradius, so that the gyroaverage is self-consistent while retaining the wave-particle resonances $v_{\mathrm{t}} \sim \omega/k_{\parallel}$; finally, $|k_{\perp}\rho_{\mathrm{t}}| \sim 1$ allows the particle to feel the nonuniformaity of the wave fields, so that FLR effects are still preserved.
- “Maximal” means that, under the condition that the gyroaverage remains valid, these orderings are the most relaxed restrictions (e.g., $|k_{\perp}\rho_{\mathrm{t}}| \sim 1$ rather than the drift-kinetic one $|k_{\perp}\rho_{\mathrm{t}}| \to 0$), thereby preserving the physical effects to the maximum extent.

$$\left(\frac{\partial}{\partial t}+v_\parallel\frac{\partial}{\partial X_\parallel}\right)G_{\mathrm{g}0}=\langle\mathrm{R.H.S.}\rangle_0, \tag{3.7.7}$$

where

$$\begin{aligned}\langle\mathrm{R.H.S.}\rangle_0 &\equiv \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}\alpha\,(\mathrm{R.H.S.})\\ &=-\frac{q}{m}\left(\frac{\partial\langle\varphi_{\mathrm{g}}\rangle_0}{\partial t}\frac{\partial}{\partial\mathcal{E}}+\left(\left(\frac{\partial}{\partial t}+v_\parallel\frac{\partial}{\partial X_\parallel}\right)\langle\varphi_{\mathrm{g}}\rangle_0\right)\frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}}.\end{aligned} \tag{3.7.8}$$

Eq. (3.7.7) along with Eq. (3.7.8) is the electrostatic *linear GK equation* valid for uniform magnetized plasmas, in which *particle gyrations are removed while the FLR effects are still preserved* [36].

To go further, we make the Laplace-in-t and Fourier-in-$\boldsymbol{X}$ transforms. Noting Eq. (3.2.22), Eq. (3.7.7) then yields

$$\hat{G}_{\mathrm{g}0}=-\frac{q}{m}\hat{\varphi}J_0(\lambda)\left(\frac{\omega}{\omega-k_\parallel v_\parallel}\frac{\partial}{\partial\mathcal{E}}+\frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}}, \tag{3.7.9}$$

implying (from Eq. (3.7.3) with Eq. (3.2.21))

$$\begin{aligned}\hat{f}_1 &=\frac{q}{m}\hat{\varphi}\left(\frac{\partial}{\partial\mathcal{E}}+\frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}}+\hat{G}_{\mathrm{g}0}\,\mathrm{e}^{\mathrm{i}\lambda\sin\alpha}+\mathcal{O}(\varepsilon)\\ &=\frac{q}{m}\hat{\varphi}\left(\left(1-\frac{\omega}{\omega-k_\parallel v_\parallel}J_0(\lambda)\,\mathrm{e}^{\mathrm{i}\lambda\sin\alpha}\right)\frac{\partial}{\partial\mathcal{E}}+\left(1-J_0(\lambda)\,\mathrm{e}^{\mathrm{i}\lambda\sin\alpha}\right)\frac{1}{B_0}\frac{\partial}{\partial\mu}\right)f_{0\mathrm{g}}+\mathcal{O}(\varepsilon).\end{aligned} \tag{3.7.10}$$

Substituting Eq. (3.7.10) into the Possion's equation $k^2\hat{\varphi}=4\pi\sum_s q_s\int_{\mathbb{R}^3}\mathrm{d}^3v\,\hat{f}_{1s}$, we arrive at the desired linear GK dispersion relation

$$D=1+\sum_s\chi_s=0, \tag{3.7.11}$$

where

$$\chi_s=2\pi\frac{\omega_{\mathrm{p}s}^2}{n_{0s}k^2}\int_{u_{\mathrm{L}}}\mathrm{d}v_\parallel\int_0^{+\infty}\mathrm{d}v_\perp\left(\frac{k_\parallel v_\perp}{\omega-k_\parallel v_\parallel}J_0^2(\lambda_s)\frac{\partial}{\partial v_\parallel}-\left(1-J_0^2(\lambda_s)\right)\frac{\partial}{\partial v_\perp}\right)f_{0s}. \tag{3.7.12}$$

For Maxwellian distributions $f_{0s}(\boldsymbol{v})=\left(n_{0s}/\pi^{3/2}v_{\mathrm{t}s}^3\right)\exp(-v^2/v_{\mathrm{t}s}^2)$, it is then straightforward to carry out the velocity integrals and obtain

$$\chi_s=\frac{1}{k^2\lambda_{\mathrm{D}s}^2}(1+A_0(b_s)\zeta_s Z(\zeta_s)),\quad \zeta_s\equiv\frac{\omega}{k_\parallel v_{\mathrm{t}s}}. \tag{3.7.13}$$

Note that Eq. (3.7.13) doesn't contain the $n\neq 0$ terms as in Eq. (3.2.31), which is a manifestation of the assumption that $\Omega=\mathcal{O}(\varepsilon^{-1})$ is large; the $n=0$ term, nevertheless, preserves the FLR effect.

For the ion acoustic wave (IAW), we assume $v_{ti} \ll \left|\omega/k_\parallel\right| \sim c_s \ll v_{te}$ and $|k_\perp \rho_{te}| \ll 1 \xrightarrow[\omega_{pe}>|\Omega_e|]{\text{high density}} k^2\lambda_{De}^2 \ll 1$ (quasi-neutral). Thus

$$D_r \approx \frac{1}{k^2\lambda_{Di}^2}\left(1 - A_0(b_i)\left(1 + \frac{k_\parallel^2 v_{ti}^2}{2\omega_r^2}\right)\right) + \frac{1}{k^2\lambda_{De}^2} = 0 \tag{3.7.14}$$

$$\Rightarrow \omega_r^2 = k_\parallel^2 c_s^2 \frac{A_0(b_i)}{1 + (T_e/T_i)(1 - A_0(b_i))}. \tag{3.7.15}$$

Noting that $\omega_r^2 \to k_\parallel^2 c_s^2$ as $k_\perp \to 0$, Eq. (3.7.15), thus, corresponds to the IAW with FLR corrections. It is interesting to note, with FLR effects included, that the IAW can now propagate across the magnetic field ($\partial\omega_r/\partial k_\perp \neq 0$) and thus can lead to cross-field energy transport which is absent in the usual MHD description.

At this stage, it is perhaps instructive to recover the more familiar *drift-kinetic* equation [22] by taking the $k_\perp\rho_t \to 0$ limit of the GK equation. In fact, the gyrophase average of Eq. (3.2.5) gives the desired linear drift-kinetic equation in uniform magnetized plasmas for electrostatic waves (noting that $k_\perp\rho_t \to 0 \Rightarrow \boldsymbol{X} \to \boldsymbol{x}$)

$$\left(\frac{\partial}{\partial t} + v_\parallel \frac{\partial}{\partial x_\parallel}\right) f_1 = -\frac{q}{m} E_\parallel \frac{\partial f_0}{\partial v_\parallel}. \tag{3.7.16}$$

We now discuss the generalization of the above theoretical approach to include the full, electromagnetic perturbations. We are, thus, dealing with the *kinetic Alfvén waves* (KAW). We copy the linearized Vlasov equation from Section 3.4:

$$\begin{aligned}\left(-\mathrm{i}\omega + \mathrm{i}k_\parallel v_\parallel - \Omega\frac{\partial}{\partial\alpha}\right)\hat{f}_{1g} = -\frac{q}{m}\Bigg(&\mathrm{i}\frac{\Omega}{k_\perp}\frac{\partial \hat{E}_{g\perp1}}{\partial\alpha}P_1(f_{0g}) + \mathrm{i}v_\perp\frac{\partial\hat{E}_{g\perp2}}{\partial\lambda}P_1(f_{0g}) \\ &+ v_\parallel\hat{E}_{g\parallel}\frac{\partial f_{0g}}{\partial\mathcal{E}} + \mathrm{i}\frac{\Omega}{\omega}v_\parallel\frac{\partial\hat{E}_{g\parallel}}{\partial\alpha}\frac{1}{B_0}\frac{\partial f_{0g}}{\partial\mu}\Bigg),\end{aligned} \tag{3.7.17}$$

where

$$P_1 = \frac{\partial}{\partial\mathcal{E}} + \left(1 - \frac{k_\parallel v_\parallel}{\omega}\right)\frac{1}{B_0}\frac{\partial}{\partial\mu}. \tag{3.7.18}$$

Here, again, we have taken $\hat{\boldsymbol{e}}_\parallel = \boldsymbol{B}_0/B_0$ and $\boldsymbol{k} = k_\perp\hat{\boldsymbol{e}}_{\perp1} + k_\parallel\hat{\boldsymbol{e}}_\parallel$. We now remove the terms involving $\partial\hat{\boldsymbol{E}}_g/\partial\alpha$ on the R.H.S. of Eq. (3.7.17). Letting

$$\hat{f}_{1g} = \mathrm{i}\frac{q}{m}\left(\frac{1}{k_\perp}\hat{E}_{g\perp1}P_1(f_{0g}) + \frac{1}{\omega}v_\parallel\hat{E}_{g\parallel}\frac{1}{B_0}\frac{\partial f_{0g}}{\partial\mu}\right) + \hat{G}_g, \tag{3.7.19}$$

$\hat{G}_g$ then satisfies (applying the orderings given by Eq. (3.7.2))

$$\left(-\mathrm{i}\omega + \mathrm{i}k_\parallel v_\parallel - \varepsilon^{-1}\Omega\frac{\partial}{\partial\alpha}\right)\hat{G}_g = -\frac{q}{m}\left(\frac{\omega - k_\parallel v_\parallel}{k_\perp}\hat{E}_{g\perp1} + \mathrm{i}v_\perp\frac{\partial\hat{E}_{g\perp2}}{\partial\lambda} + v_\parallel\hat{E}_{g\parallel}\right)P_1(f_{0g}) \tag{3.7.20}$$

$$\equiv \text{R.H.S.} \tag{3.7.20}$$

With the perturbative expansion Eq. (3.7.5), we find

- at order $\mathcal{O}(\varepsilon^{-1})$, $\partial\hat{G}_{\mathrm{g}0}/\partial\alpha = 0$;
- at order $\mathcal{O}(1)$, after averaging over the gyrophase,

$$-\mathrm{i}\big(\omega - k_\parallel v_\parallel\big)\hat{G}_{\mathrm{g}0} = -\frac{q}{m}\left(\frac{\omega - k_\parallel v_\parallel}{k_\perp}J_0(\lambda)\hat{E}_{\perp 1} + \mathrm{i}v_\perp J_0'(\lambda)\frac{\partial\hat{E}_{\perp 2}}{\partial\lambda} + v_\parallel J_0(\lambda)\hat{E}_\parallel\right)P_1\big(f_{0\mathrm{g}}\big). \tag{3.7.21}$$

Eq. (3.7.21) is the electromagnetic linear GK equation in uniform magnetized plasmas.

We now examine the specific example where $f_0(\boldsymbol{v}) = \big(n_0/\pi^{3/2}v_{\mathrm{t}}^3\big)\exp(-v^2/v_{\mathrm{t}}^2) \Rightarrow P_1\big(f_{0\mathrm{g}}\big) = -(2/v_{\mathrm{t}}^2)f_0$. The distribution function perturbation is then given by

$$\begin{aligned}
\hat{G}_{\mathrm{g}0} &= \mathrm{i}\frac{q}{T}\left(\frac{J_0(\lambda)}{k_\perp}\hat{E}_{\perp 1} + \mathrm{i}\frac{v_\perp J_0'(\lambda)}{\omega - k_\parallel v_\parallel}\hat{E}_{\perp 2} + \frac{v_\parallel J_0(\lambda)}{\omega - k_\parallel v_\parallel}\hat{E}_\parallel\right)f_0 \\
\Rightarrow \hat{f}_1 &= -\mathrm{i}\frac{q}{T}\frac{1}{k_\perp}\hat{E}_{\perp 1}f_0 + \hat{G}_{\mathrm{g}0}\mathrm{e}^{\mathrm{i}\lambda\sin\alpha} \\
&= -\mathrm{i}\frac{q}{T}\left(\hat{E}_{\perp 1}\frac{1 - J_0(\lambda)\,\mathrm{e}^{\mathrm{i}\lambda\sin\alpha}}{k_\perp} - \mathrm{i}\hat{E}_{\perp 2}\frac{v_\perp J_0'(\lambda)}{\omega - k_\parallel v_\parallel}\mathrm{e}^{\mathrm{i}\lambda\sin\alpha} - \hat{E}_\parallel\frac{v_\parallel J_0(\lambda)}{\omega - k_\parallel v_\parallel}\mathrm{e}^{\mathrm{i}\lambda\sin\alpha}\right)f_0,
\end{aligned} \tag{3.7.22}$$

which can then be substituted into the Maxwell equations and yield the desired linear GK dispersion relation. For low-frequency waves, it is more convenient to use the Possion's equation, which, for $k^2\lambda_{\mathrm{D}e}^2 \ll 1$, is approximated by the quasi-neutrality condition

$$\sum_s \lambda_{\mathrm{D}s}^{-2}\left(\hat{E}_{\perp 1}\frac{1 - A_0(b_s)}{k_\perp} + \mathrm{i}\hat{E}_{\perp 2}\frac{k_\perp v_{\mathrm{t}s}}{2k_\parallel\Omega_s}A_0'(b_s)Z(\zeta_s) + \hat{E}_\parallel\frac{A_0(b_s)}{k_\parallel}(1 + \zeta_s Z(\zeta_s))\right) = 0. \tag{3.7.23}$$

Meanwhile, neglecting the displacement current $(1/c)\partial\boldsymbol{E}/\partial t$, we find, from the $\hat{\boldsymbol{e}}_{\perp 2}$ and $\hat{\boldsymbol{e}}_\parallel$ components of the Ampère's law $\nabla\times\boldsymbol{B} = (4\pi/c)\boldsymbol{J}$, that

$$\begin{aligned}
k^2\hat{E}_{\perp 2} &= \mathrm{i}\frac{4\pi\omega}{c^2}\hat{J}_{\perp 2}, \\
-k_\perp k_\parallel\hat{E}_{\perp 1} + k_\perp^2\hat{E}_\parallel &= \mathrm{i}\frac{4\pi\omega}{c^2}\hat{J}_\parallel.
\end{aligned} \tag{3.7.24}$$

Here, the perturbated current density is given by

$$\begin{aligned}
\hat{\boldsymbol{J}} \approx \hat{\boldsymbol{J}}_0 &= \sum_s q_s\int_{\mathbb{R}^3}\mathrm{d}^3v\,\boldsymbol{v}\,\hat{G}_{\mathrm{g}0s}\,\mathrm{e}^{\mathrm{i}\lambda_s\sin\alpha} \\
&= 2\pi\sum_s q_s\int_{u_{\mathrm{L}}}\mathrm{d}v_\parallel\int_0^{+\infty}\mathrm{d}v_\perp\,v_\perp\hat{G}_{\mathrm{g}0s}\big\langle\boldsymbol{v}\,\mathrm{e}^{\mathrm{i}\lambda_s\sin\alpha}\big\rangle_0,
\end{aligned} \tag{3.7.25}$$

i.e.,

$$\hat{J}_{\perp 1} = o(1) \quad \text{(not adequate, thus the } \hat{\boldsymbol{e}}_{\perp 1} \text{ component of the Ampère's law is not used),} \tag{3.7.26}$$

$$\hat{J}_{\perp 2} \approx 2\pi \sum_s q_s \int_{u_\mathrm{L}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp (-\mathrm{i}v_\perp) J_0'(\lambda_s) \hat{G}_{\mathrm{g}0s}$$
$$= \sum_s \frac{n_{0s} q_s^2}{T_s} \left(\hat{E}_{\perp 1} \frac{v_{\mathrm{t}s}^2}{2\Omega_s} A_0'(b_s) - \mathrm{i}\hat{E}_{\perp 2} \left(\frac{k_\perp^2 v_{\mathrm{t}s}^3}{4k_\parallel \Omega_s^2} A_0''(b_s) + \frac{v_{\mathrm{t}s}}{k_\parallel} \mathrm{e}^{-b_s} I_0'(b_s) \right) Z(\zeta_s) \right.$$
$$\left. - \hat{E}_\parallel \frac{k_\perp v_{\mathrm{t}s}^2}{2k_\parallel \Omega_s} A_0'(b_s)(1 + \zeta_s Z(\zeta_s)) \right), \tag{3.7.26}$$
$$\hat{J}_\parallel = 2\pi \sum_s q_s \int_{u_\mathrm{L}} \mathrm{d}v_\parallel \int_0^{+\infty} \mathrm{d}v_\perp\, v_\perp v_\parallel J_0(\lambda_s) \hat{G}_{\mathrm{g}0s}$$
$$= \sum_s \frac{n_{0s} q_s^2}{T_s} \left(\hat{E}_{\perp 2} \frac{k_\perp v_{\mathrm{t}s}^2}{2k_\parallel \Omega_s} A_0'(b_s)(1 + \zeta_s Z(\zeta(s))) - \mathrm{i}\hat{E}_\parallel \frac{\omega}{k_\parallel^2} A_0(b_s)(1 + \zeta_s Z(\zeta(s))) \right).$$

The quasi-neutrality condition Eq. (3.7.23) and the two components of the Ampère's law Eq. (3.7.24) along with Eq. (3.7.26) completely determine the linear GK normal modes.

To illustrate the physics further, we assume $\left|k_\parallel / k_\perp\right| \ll 1$ and ignore the compressional Alfvén wave (CAW), i.e., take $\hat{E}_{\perp 2} \approx 0$ and $\hat{J}_{\perp 2} \approx 0$. The quasi-neutrality condition Eq. (3.7.23) then becomes

$$\frac{\hat{E}_{\perp 1}}{k_\perp} k_\perp^2 \rho_\mathrm{s}^2 \frac{1 - A_0(b_i)}{b_i} + \frac{\hat{E}_\parallel}{k_\parallel} \left(1 - A_0(b_i) \frac{k_\parallel^2 c_\mathrm{s}^2}{\omega^2} \right) = 0, \quad \rho_\mathrm{s} \equiv \frac{c_\mathrm{s}}{\Omega_i}, \tag{3.7.27}$$

where we have assumed $b_e \ll 1, v_{\mathrm{t}i} \ll \left|\omega / k_\parallel\right| \ll v_{\mathrm{t}e}$ and neglected wave-particle resonances. Meanwhile,

$$\mathrm{i}4\pi \hat{J}_\parallel \approx \frac{\hat{E}_\parallel}{k_\parallel} \frac{1}{\lambda_{\mathrm{D}e}^2} \frac{\omega}{k_\parallel} \left(1 - A_0(b_i) \frac{k_\parallel^2 c_\mathrm{s}^2}{\omega^2} \right), \tag{3.7.28}$$

and the parallel Ampère's law then reduces to (divided by $k_\parallel k_\perp^2$, noting that $\lambda_{\mathrm{D}e}^2 c^2 = v_\mathrm{A}^2 \rho_\mathrm{s}^2$)

$$-\frac{\hat{E}_{\perp 1}}{k_\perp} + \frac{\hat{E}_\parallel}{k_\parallel} = \frac{\hat{E}_\parallel}{k_\parallel} \frac{1}{k_\perp^2 \rho_\mathrm{s}^2} \frac{\omega^2}{k_\parallel^2 v_\mathrm{A}^2} \left(1 - A_0(b_i) \frac{k_\parallel^2 c_\mathrm{s}^2}{\omega^2} \right). \tag{3.7.29}$$

Combining Eq. (3.7.27) and Eq. (3.7.29), we obtain the desired dispersion relation

$$\left(1 - \frac{1 - A_0(b_i)}{b_i} \frac{\omega^2}{k_\parallel^2 v_\mathrm{A}^2} \right) \left(1 - A_0(b_i) \frac{k_\parallel^2 c_\mathrm{s}^2}{\omega^2} \right) = -k_\perp^2 \rho_\mathrm{s}^2 \frac{1 - A_0(b_i)}{b_i}. \tag{3.7.30}$$

Thus, due to the FLR effects, the shear Alfvén wave (SAW) and the IAW are coupled. For $\beta_e = 2c_\mathrm{s}^2 / v_\mathrm{A}^2 \ll 1$, we find, for the slow IAW, that

$$\omega_\mathrm{s}^2 = k_\parallel^2 c_\mathrm{s}^2 \frac{A_0(b_i)}{1 + (T_e/T_i)(1 - A_0(b_i))} \overset{b_i \ll 1}{=\!=} \frac{k_\parallel^2 c_\mathrm{s}^2}{1 + k_\perp^2 \rho_\mathrm{s}^2}, \tag{3.7.31}$$

which is the same as Eq. (3.7.15); on the other hand, for the shear Alfvén wave, we find

$$\omega_{\mathrm{A}}^2 = k_\parallel^2 v_{\mathrm{A}}^2 \left(\frac{b_i}{1 - A_0(b_i)} + k_\perp^2 \rho_{\mathrm{s}}^2 \right) \xlongequal{b_i \ll 1} k_\parallel^2 v_{\mathrm{A}}^2 \left(1 + \left(1 + \frac{3T_i}{4T_e} \right) k_\perp^2 \rho_{\mathrm{s}}^2 \right). \tag{3.7.32}$$

Eq. (3.7.32) is the dispersion relation for the *kinetic shear Alfvén wave* (KSAW). We note that KSAW has finite perpendicular group velocity ($\partial \omega_{\mathrm{A}} / \partial k_\perp \neq 0$, though it still propagates mainly in the parallel direction, which is ensured by the condition $\left| k_\parallel / k_\perp \right| \ll 1$) and plays important roles in shear Alfvén wave heating [37], [38] as well as alpha-particle instabilities [39]. Now, the Landau-damping effects can be included by retaining the imaginary part of the Z function. For $m_e/m_i \lesssim \beta_e \ll 1$, we have $v_{\mathrm{t}i} \ll \left| \omega / k_\parallel \right| \sim v_{\mathrm{A}} \lesssim v_{\mathrm{t}e}$ and, hence, KSAW suffers mainly electron Landau damping[39]

$$\gamma_{\mathrm{A}} = -\frac{\sqrt{\pi}}{2} \frac{v_{\mathrm{A}}}{v_{\mathrm{t}e}} \left(1 + \frac{3T_i}{4T_e} \right) k_\perp^2 \rho_{\mathrm{s}}^2 k_\parallel v_{\mathrm{A}} \exp\left(-\left(\frac{v_{\mathrm{A}}}{v_{\mathrm{t}e}} \right)^2 \left(1 + \left(1 + \frac{3T_i}{4T_e} \right) k_\perp^2 \rho_{\mathrm{s}}^2 \right) \right); \tag{3.7.33}$$

physically, this occurs because, for $k_\perp^2 \rho_{\mathrm{s}}^2 \neq 0$, we have $\hat{E}_\parallel \neq 0$ and particles (especially electrons, which are lighter than ions) can be accelerated (or decelerated) along the background magnetic field line (which is, a neutralizing effect).

Finally, we remark, referring to Eq. (3.7.26), that finite $\hat{E}_{\perp 2}$ can also lead to wave-particle resonance. Since $\hat{E}_{\perp 2} \propto \hat{B}_\parallel$, this resonance, called transit-time magnetic pumping [40], is due to acceleration (or deceleration) of guiding-centers <u>along</u> a varying magnetic field. This a magnetic version of the Landau damping, with the perturbed mirror force $m \, \mathrm{d}v_\parallel / \mathrm{d}t = -\mu \hat{\boldsymbol{e}}_\parallel \cdot \nabla_{\boldsymbol{X}} B$ now playing the role of the perturbed electric force $m \, \mathrm{d}v_\parallel / \mathrm{d}t = q \hat{\boldsymbol{e}}_\parallel \cdot \boldsymbol{E} \big(= -q \hat{\boldsymbol{e}}_\parallel \cdot \nabla_{\boldsymbol{X}} \varphi \big)$ in the (electrostatic) Landau damping.

Homework #6

(1) In this Problem we establish the low-frequency GK equation using the scalar and vector potentials, φ and $\boldsymbol{A}$, such that

$$\boldsymbol{E} = -\nabla \varphi - \frac{1}{c} \frac{\partial \boldsymbol{A}}{\partial t}, \quad \boldsymbol{B} = \nabla \times \boldsymbol{A} \tag{H.6.1}$$

under the Coulomb gauge condition $\nabla \cdot \boldsymbol{A} = 0$.

(i) Show that the linearized Vlasov equation is given by, noting Eq. (3.2.4),

$$\begin{aligned} \hat{\mathcal{L}}_{\mathrm{g}} \hat{f}_{1\mathrm{g}} = -\frac{q}{m} \Bigg(& \Omega \frac{\partial}{\partial \alpha} \left(\hat{\varphi}_{\mathrm{g}} \frac{\partial f_{0\mathrm{g}}}{\partial \mathcal{E}} + \left(\hat{\varphi}_{\mathrm{g}} - \frac{v_\parallel}{c} \hat{A}_{\mathrm{g}\parallel} \right) \frac{1}{B_0} \frac{\partial f_{0\mathrm{g}}}{\partial \mu} \right) \\ & - \mathrm{i} \left(k_\parallel v_\parallel \hat{\varphi}_{\mathrm{g}} - \frac{\omega}{c} \boldsymbol{v} \cdot \hat{\boldsymbol{A}}_{\mathrm{g}} \right) \frac{\partial f_{0\mathrm{g}}}{\partial \mathcal{E}} + \mathrm{i} \frac{\omega - k_\parallel v_\parallel}{c} \boldsymbol{v}_\perp \cdot \hat{\boldsymbol{A}}_{\mathrm{g}} \frac{1}{B_0} \frac{\partial f_{0\mathrm{g}}}{\partial \mu} \Bigg), \end{aligned} \tag{H.6.2}$$

where $\hat{\mathcal{L}}_{\mathrm{g}} = -\mathrm{i}\omega + \mathrm{i} k_\parallel v_\parallel + \Omega \, \partial / \partial \alpha$.

(ii) Letting

[39] CAUTION: The result here, calculated by the retypesetter, is quantitatively different from other references.

$$\hat{f}_{1\mathrm{g}} = \frac{q}{m}\left(\hat{\varphi}_{\mathrm{g}}\frac{\partial}{\partial \mathcal{E}} + \left(\hat{\varphi}_{\mathrm{g}} - \frac{v_\parallel}{c}\hat{A}_{\mathrm{g}\parallel}\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}} + \hat{G}_{\mathrm{g}}, \tag{H.6.3}$$

show that

$$\left(-\mathrm{i}\omega + \mathrm{i}k_\parallel v_\parallel - \Omega\frac{\partial}{\partial \alpha}\right)\hat{G}_{\mathrm{g}} = \mathrm{i}\frac{q}{m}\left(\omega\hat{\psi}_{\mathrm{g}}\frac{\partial}{\partial \mathcal{E}} + \left(\omega - k_\parallel v_\parallel\right)\hat{\psi}_{\mathrm{g}}\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}} \tag{H.6.4}$$

with $\hat{\psi}_{\mathrm{g}} \equiv \hat{\varphi}_{\mathrm{g}} - (\boldsymbol{v}/c)\cdot\hat{\boldsymbol{A}}_{\mathrm{g}}$.

(iii) Performing the gyroaverage, show that the low-frequency response is given by, with $\hat{G}_{\mathrm{g}} = \hat{G}_{\mathrm{g}0} + \hat{G}_{\mathrm{g}1} + \cdots$,

$$\left(\omega - k_\parallel v_\parallel\right)\hat{G}_{\mathrm{g}0} = -\frac{q}{m}\left(\omega\left\langle\hat{\psi}_{\mathrm{g}}\right\rangle_0\frac{\partial}{\partial \mathcal{E}} + \left(\omega - k_\parallel v_\parallel\right)\left\langle\hat{\psi}_{\mathrm{g}}\right\rangle_0\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}}, \tag{H.6.5}$$

where

$$\left\langle\hat{\psi}_{\mathrm{g}}\right\rangle_0 = \left\langle\hat{\varphi}_{\mathrm{g}} - \frac{\boldsymbol{v}}{c}\cdot\hat{\boldsymbol{A}}_{\mathrm{g}}\right\rangle_0 = J_0(\lambda)\left(\hat{\varphi} - \frac{v_\parallel}{c}\hat{A}_\parallel\right) - \frac{v_\perp J_0'(\lambda)}{ck_\perp}\hat{B}_\parallel. \tag{H.6.6}$$

(iv) Show that the Maxwell equations are given by

$$k^2\hat{\varphi} = 8\pi^2\sum_s q_s\sum_\sigma\iint \mathrm{d}\mathcal{E}\,\mathrm{d}\mu\,\frac{B_0}{\left|v_\parallel\right|}\left(\frac{q_s}{m_s}\left(\hat{\varphi}\frac{\partial}{\partial \mathcal{E}} + \left(\hat{\varphi} - \frac{v_\parallel}{c}\hat{A}_\parallel\right)\frac{1}{B_0}\frac{\partial}{\partial \mu}\right) f_{0\mathrm{g}s} + \hat{G}_{\mathrm{g}0s}J_0(\lambda_s)\right),$$

$$\begin{aligned}\left(k^2 - \frac{\omega^2}{c^2}\right)\hat{A}_\parallel = -\frac{\omega}{c}k_\parallel\hat{\varphi} + \frac{8\pi^2}{c}\sum_s q_s\sum_\sigma\iint \mathrm{d}\mathcal{E}\,\mathrm{d}\mu\,\frac{B_0}{\left|v_\parallel\right|}v_\parallel\Bigg(&\frac{q_s}{m_s}\left(\hat{\varphi} - \frac{v_\parallel}{c}\hat{A}_\parallel\right)\frac{1}{B_0}\frac{\partial f_{0\mathrm{g}s}}{\partial \mu}\\ &+\hat{G}_{\mathrm{g}0s}J_0(\lambda_s)\Bigg),\end{aligned} \tag{H.6.7}$$

$$\left(k^2 - \frac{\omega^2}{c^2}\right)\hat{B}_\parallel = \frac{8\pi^2}{c}k_\perp\sum_s q_s\sum_\sigma\iint \mathrm{d}\mathcal{E}\,\mathrm{d}\mu\,\frac{B_0}{\left|v_\parallel\right|}v_\perp\hat{G}_{\mathrm{g}0s}J_0'(\lambda_s).$$

Note that the wave equations involve only three field quantities: φ, $A_\parallel$ and $B_\parallel$.

(2) Either from Problem (1) or your class notes, establish the respective small parameters for

(i) neglecting the compressional Alfvén wave;

(ii) weak coupling between the shear Alfvén wave and the ion acoustic wave.

4 Linear Waves and Instabilities in Nonuniform Plasmas

4.1 Geometric optics and the ray equations

The main references of this section are [41] and [42].

In chapter 1, we have analyzed, using two time and space scales, the propagation of a wave packet in a temporally stationary and spatially homogeneous (uniform) dielectric medium. There, we have found that a *wave packet* (or *quasi-particle*) propagates at the group velocity

$$\boldsymbol{v}_{\mathrm{g}} = \nabla_{\boldsymbol{k}} \Omega(\boldsymbol{k}), \tag{4.1.1}$$

where $\Omega(\boldsymbol{k})$ and $\boldsymbol{k}$ correspond, respectively, to the angular frequency and wavevector of the fast time and space variables. Now, if we relax the restriction and allow the medium to be time- and space-varying, it can be expected that the two-time-and-space-scale approach is still applicable as long as the temporal and spatial variations of the dielectric medium are sufficiently slow when compared with Ω and $\boldsymbol{k}$. This is the essence of the *geometric optics*[40] [43], which deals with the propagation of wave packets in a dielectric medium that is slowly varying in both time and space. The corresponding equations for the wave packet propagation are called the *ray equations*.

Before we get into the detailed analyses, let's consider this subject from the quasi-particle point of view. In this perspective, the wave packet is characterized by its position $\boldsymbol{x}$, momentum $\boldsymbol{k}$ and energy Ω. Thus, the quasi-particle dynamics is completely specified if $\dot{\boldsymbol{x}} \equiv \mathrm{d}\boldsymbol{x}/\mathrm{d}t$, $\dot{\boldsymbol{k}} \equiv \mathrm{d}\boldsymbol{k}/\mathrm{d}t$ and $\dot{\Omega} \equiv \mathrm{d}\Omega/\mathrm{d}t$ are known and those consitute the ray equations. Now, if variations in the medium are sufficiently slow, we then expect that, at least at the lowest order, Eq. (4.1.1) must still hold, i.e.,

$$\dot{\boldsymbol{x}} = (\nabla_{\boldsymbol{k}} \Omega)_{t,\boldsymbol{x}}. \tag{4.1.2}$$

In order to avoid confusion, we have adopted the thermodynamic convention of denoting the independent variables (held constant) in the differentiations as subscripts outside the parentheses. With $\dot{\boldsymbol{x}}$ given, we may move forward and derive the rest of the ray equations.

Let the wave equation be

$$\tilde{D}\left(t, \boldsymbol{x}; \mathrm{i}\frac{\partial}{\partial t}, -\mathrm{i}\nabla\right) \cdot \boldsymbol{E}(t, \boldsymbol{x}) = \boldsymbol{0}. \tag{4.1.3}$$

Adopting the *eikonal* representation such that

[40] This desscription is also known as the Wentzel-Kramers-Brillouin (*WKB* or, along with with Jeffreys, *WKBJ*) or *eikonal* approximation.

$$\boldsymbol{E}(t,\boldsymbol{x}) = \hat{\boldsymbol{E}}\mathrm{e}^{\mathrm{i}S(t,\boldsymbol{x})}, \tag{4.1.4}$$

the local dispersion relation is then given by[41]

$$D(t,\boldsymbol{x};\omega,\boldsymbol{k}) \equiv \det \tilde{D}(t,\boldsymbol{x};\omega,\boldsymbol{k}) = 0, \tag{4.1.10}$$

where[42]

$$\begin{aligned}\omega &:= -\left(\frac{\partial S}{\partial t}\right)_{\boldsymbol{x}} = \omega(t,\boldsymbol{x}),\\ \boldsymbol{k} &:= (\nabla S)_t = \boldsymbol{k}(t,\boldsymbol{x}).\end{aligned} \tag{4.1.12}$$

Eq. (4.1.10), of course, implies[43] two scales, i.e.,

$$\begin{aligned}&\left|\left(\frac{\partial \omega}{\partial t}\right)_{\boldsymbol{x}}\right| \ll \omega^2, \qquad |(\nabla\omega)_t| \ll |\boldsymbol{k}\omega|,\\ &\left|\left(\frac{\partial \boldsymbol{k}}{\partial t}\right)_{\boldsymbol{x}}\right| \ll |\omega\boldsymbol{k}|, \quad \|(\nabla\boldsymbol{k})_t\| \ll \|\boldsymbol{k}\boldsymbol{k}\|.\end{aligned} \tag{4.1.13}$$

From Eq. (4.1.10) we could solve for the on shell conditions, i.e., the dispersion relation for the *real* wave packet propagating in the system

[41] Another approach is as follows [22]: We seek a solution of the form

$$\boldsymbol{E}(t,\boldsymbol{x}) = \hat{\boldsymbol{E}}(\varepsilon t, \varepsilon\boldsymbol{x})\mathrm{e}^{\mathrm{i}S(t,\boldsymbol{x})}, \tag{4.1.5}$$

where $\hat{\boldsymbol{E}}$ is the slow-varying amplitude of the wave and S the fast-varying phase called the *eikonal*. The time derivative and gradient operators acting on $\boldsymbol{E}$ yields

$$\begin{aligned}\frac{\partial \boldsymbol{E}}{\partial t} &= \left(-\mathrm{i}\omega\hat{\boldsymbol{E}} + \frac{\partial \hat{\boldsymbol{E}}}{\partial t}\right)\mathrm{e}^{\mathrm{i}S},\\ \nabla\boldsymbol{E} &= \left(\mathrm{i}\boldsymbol{k}\hat{\boldsymbol{E}} + \nabla\hat{\boldsymbol{E}}\right)\mathrm{e}^{\mathrm{i}S},\end{aligned} \tag{4.1.6}$$

where

$$\begin{aligned}\omega &:= -\left(\frac{\partial S}{\partial t}\right)_{\boldsymbol{x}} = \omega(t,\boldsymbol{x}),\\ \boldsymbol{k} &:= (\nabla S)_t = \boldsymbol{k}(t,\boldsymbol{x}).\end{aligned} \tag{4.1.7}$$

Under the two-scale conditions

$$\frac{\left|\partial\hat{\boldsymbol{E}}/\partial t\right|}{\left|\omega\hat{\boldsymbol{E}}\right|} = \mathcal{O}(\varepsilon) \ll 1, \quad \frac{\left\|\nabla\hat{\boldsymbol{E}}\right\|}{\left\|\boldsymbol{k}\hat{\boldsymbol{E}}\right\|} = \mathcal{O}(\varepsilon) \ll 1, \tag{4.1.8}$$

the lowest-order local dispersion relation has the same form as in a homogeneous plasma, i.e.,

$$D(t,\boldsymbol{x};\omega,\boldsymbol{k}) \equiv \det \tilde{D}(t,\boldsymbol{x};\omega,\boldsymbol{k}) = 0. \tag{4.1.9}$$

[42] Note that

$$\begin{aligned}S(t,\boldsymbol{x}) &= \int_{\boldsymbol{0}}^{\boldsymbol{x}} \mathrm{d}\boldsymbol{x}'\cdot\boldsymbol{k}(t,\boldsymbol{x}') + f(t)\\ &= -\int_0^t \mathrm{d}t'\,\omega(t',\boldsymbol{x}) + g(\boldsymbol{x}).\end{aligned} \tag{4.1.11}$$

[43] "A implies B (A $\Rightarrow$ B)" has the meaning of "B $\subset$ A", i.e., "for A to be true\successful, B is necessary".

$$\omega = \omega_j(\boldsymbol{k}; t, \boldsymbol{x}), \quad j = 1, 2, \cdots \tag{4.1.14}$$

which can be viewed as constraints on the eikonal S.

We now proceed to derive the ray equations for the particular wave packet with $\omega = \Omega(\boldsymbol{k}; t, \boldsymbol{x})$. Noting the exact differentiation condition

$$\begin{aligned}
\left(\frac{\partial \boldsymbol{k}}{\partial t}\right)_{\boldsymbol{x}} &= -(\nabla\omega)_t \\
&\overset{\text{on shell}}{\underset{\omega=\Omega(\boldsymbol{k};t,\boldsymbol{x})}{=\!=\!=}} -(\nabla\Omega)_{t,\boldsymbol{k}} - (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} \cdot (\nabla\boldsymbol{k})_t^{\mathrm{T}} \\
&\overset{(\nabla\times\boldsymbol{k})_t=\boldsymbol{0}}{\underset{\text{i.e., }(\nabla\boldsymbol{k})_t=(\nabla\boldsymbol{k})_t^{\mathrm{T}}}{=\!=\!=}} -(\nabla\Omega)_{t,\boldsymbol{k}} - (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} \cdot (\nabla\boldsymbol{k})_t,
\end{aligned} \tag{4.1.15}$$

we have

$$\left(\frac{\partial \boldsymbol{k}}{\partial t}\right)_{\boldsymbol{x}} + (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} \cdot (\nabla\boldsymbol{k})_t = -(\nabla\Omega)_{t,\boldsymbol{k}}. \tag{4.1.16}$$

This is a first-order linear PDE in $\boldsymbol{k}$; along its *characteristics* or, in this case, the trajectories of the wave packets determined by the ODE[44] $\dot{\boldsymbol{x}} = (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}}$, it reduces to

$$\dot{\boldsymbol{k}} = -(\nabla\Omega)_{t,\boldsymbol{k}}. \tag{4.1.17}$$

For $\dot{\Omega}$, we then find, along the trajectories,

$$\begin{aligned}
\dot{\Omega} &= \left(\frac{\partial\Omega}{\partial t}\right)_{\boldsymbol{x},\boldsymbol{k}} + (\nabla\Omega)_{t,\boldsymbol{k}} \cdot \dot{\boldsymbol{x}} + (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} \cdot \dot{\boldsymbol{k}} \\
&= \left(\frac{\partial\Omega}{\partial t}\right)_{\boldsymbol{x},\boldsymbol{k}} + (\nabla\Omega)_{t,\boldsymbol{k}} \cdot (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} - (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} \cdot (\nabla\Omega)_{t,\boldsymbol{k}} \\
&= \left(\frac{\partial\Omega}{\partial t}\right)_{\boldsymbol{x},\boldsymbol{k}};
\end{aligned} \tag{4.1.18}$$

note that for time-stationary media, we have $(\partial\Omega/\partial t)_{\boldsymbol{x},\boldsymbol{k}} = 0 \Rightarrow \dot{\Omega} = 0$.

Summarizing the results, the ray equations are given by[45]

$$\dot{\boldsymbol{x}} = (\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}}, \tag{4.1.20}$$

[44] This is a proof of the intuitive result Eq. (4.1.2); it is now clear that $\mathrm{d}/\mathrm{d}t$ means the material time derivative along the wave packet's trajectory.

[45] Note that the derivatives of Ω appearing in Eq. (4.1.20) can be calculated from the differentiations of Eq. (4.1.10), i.e.,

$$\begin{aligned}
(\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}} &= -\frac{(\nabla_{\boldsymbol{k}}D)_{t,\boldsymbol{x},\Omega}}{(\partial D/\partial\Omega)_{t,\boldsymbol{x},\boldsymbol{k}}}, \\
(\nabla\Omega)_{t,\boldsymbol{k}} &= -\frac{(\nabla D)_{t,\Omega,\boldsymbol{k}}}{(\partial D/\partial\Omega)_{t,\boldsymbol{x},\boldsymbol{k}}}, \\
\left(\frac{\partial\Omega}{\partial t}\right)_{\boldsymbol{x},\boldsymbol{k}} &= -\frac{(\partial D/\partial t)_{\boldsymbol{x},\Omega,\boldsymbol{k}}}{(\partial D/\partial\Omega)_{t,\boldsymbol{x},\boldsymbol{k}}}.
\end{aligned} \tag{4.1.19}$$

$$\dot{\boldsymbol{k}} = -(\nabla\Omega)_{t,\boldsymbol{k}},$$
$$\dot{\Omega} = \left(\frac{\partial\Omega}{\partial t}\right)_{\boldsymbol{x},\boldsymbol{k}}, \tag{4.1.20}$$

where $\Omega = \Omega(\boldsymbol{k}; t, \boldsymbol{x})$ satisfies the local dispersion relation

$$D(t, \boldsymbol{x}; \Omega, \boldsymbol{k}) = 0. \tag{4.1.21}$$

The above ray equations Eq. (4.1.20) may be regarded as Hamilton's equations of motion for a quasi-particle with Hamiltonian $H = \Omega$ and canonical coordinates $(\boldsymbol{q}, \boldsymbol{p}) = (\boldsymbol{x}, \boldsymbol{k})$. Therefore, if we have an ensemble of wave packets, the corresponding distribution function, $N = N(t, \boldsymbol{x}, \boldsymbol{k})$, would obey the following Liouville's equation

$$\begin{aligned} 0 = \dot{N} &= \frac{\partial N}{\partial t} + \nabla N \cdot \dot{\boldsymbol{x}} + \nabla_{\boldsymbol{k}} N \cdot \dot{\boldsymbol{k}} \\ &= \frac{\partial N}{\partial t} + \nabla_{\boldsymbol{k}}\Omega \cdot \nabla N - \nabla\Omega \cdot \nabla_{\boldsymbol{k}} N, \end{aligned} \tag{4.1.22}$$

which is referred to as the *wave kinetic equation* in plasma physics literature. Eq. (4.1.22) is valid only for a Hermitian medium where there is no wave growth or damping; extensions to include a small anti-Hermitian part will be considered in Section 4.2.

Finally, we remark that geometric optics plays an important role in plasma physics because it serves as the first step toward our understanding of wave dynamics in realistic plasmas. For example, in radio-frequency heating of plasmas, ray tracing is routinely calculated in order to understand how the wave penetrates and, thereby, deposits its energy inside the plasma.

4.2 Derivation of the wave (quasi-particle) kinetic equation

In this section, we generalize Eq. (4.1.22) to include a weak wave growth or dissipation. In another prespective, this discussion can also be regarded as extensions of our analysis in Section 1.3 to nonuniform media.

To simplify the presentation, we shall consider the one-dimensional case and let the wave equation be

$$D\left(t, x; \mathrm{i}\frac{\partial}{\partial t}, -\mathrm{i}\frac{\partial}{\partial x}\right) E(t, x) = 0, \tag{4.2.1}$$

where D is holomorphic in $\partial/\partial t$ and $\partial/\partial x$, which, of course, is physically reasonable. We, furthermore, assume the existence of two time and space scales, with (t_0, x_0) being the fast variables and (t_1, x_1) the slow. Meanwhile, we Fourier transform $E(t, x) \equiv E(t_0, x_0; t_1, x_1)$ in the fast (t_0, x_0) variables by letting

Actually, these formulae are most easily derived by using the properties of the Jacobian Eq. (0.0.10), and noting that $(\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x}}$ is actually $(\nabla_{\boldsymbol{k}}\Omega)_{t,\boldsymbol{x},D}$, etc.

$$E(t_0, x_0; t_1, x_1) = \int_{\mathbb{R}} \frac{\mathrm{d}\omega}{2\pi} \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \hat{E}(\omega, k; t_1, x_1) \mathrm{e}^{\mathrm{i}(kx_0 - \omega t_0)}. \tag{4.2.2}$$

Eq. (4.2.1) then becomes (denoting $D(t_1, x_1; \omega, k) \equiv D$)

$$\begin{aligned}
0 &= D\left(t_1 + t_0, x_1 + x_0; \mathrm{i}\frac{\partial}{\partial t_0} + \mathrm{i}\frac{\partial}{\partial t_1}, -\mathrm{i}\frac{\partial}{\partial x_0} - \mathrm{i}\frac{\partial}{\partial x_1}\right) E(t_0, x_0; t_1, x_1) \\
&\approx \int_{\mathbb{R}} \frac{\mathrm{d}\omega}{2\pi} \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \left(D(t_1, x_1; \omega, k) + t_0 \frac{\partial D}{\partial t_1} + x_0 \frac{\partial D}{\partial x_1} \right. \\
&\qquad \left. + \frac{\partial D}{\partial \omega}\left(\mathrm{i}\frac{\partial}{\partial t_1}\right) + \frac{\partial D}{\partial k}\left(-\mathrm{i}\frac{\partial}{\partial x_1}\right)\right) \hat{E}(\omega, k; t_1, x_1) \mathrm{e}^{\mathrm{i}(kx_0 - \omega t_0)} \\
&\approx \int_{\mathbb{R}} \frac{\mathrm{d}\omega}{2\pi} \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \mathrm{e}^{\mathrm{i}(kx_0 - \omega t_0)} \left((D_\mathrm{r} + \mathrm{i}D_\mathrm{i})\hat{E} - \mathrm{i}\frac{\partial}{\partial \omega}\left(\frac{\partial D_\mathrm{r}}{\partial t_1}\hat{E}\right) + \mathrm{i}\frac{\partial}{\partial k}\left(\frac{\partial D_\mathrm{r}}{\partial x_1}\hat{E}\right) \right. \\
&\qquad \left. + \mathrm{i}\frac{\partial D_\mathrm{r}}{\partial \omega}\frac{\partial \hat{E}}{\partial t_1} - \mathrm{i}\frac{\partial D_\mathrm{r}}{\partial k}\frac{\partial \hat{E}}{\partial x_1} \right),
\end{aligned} \tag{4.2.3}$$

i.e., for each Fourier component,

$$(D_\mathrm{r} + \mathrm{i}D_\mathrm{i})\hat{E} - \mathrm{i}\frac{\partial}{\partial \omega}\left(\frac{\partial D_\mathrm{r}}{\partial t_1}\hat{E}\right) + \mathrm{i}\frac{\partial}{\partial k}\left(\frac{\partial D_\mathrm{r}}{\partial x_1}\hat{E}\right) + \mathrm{i}\frac{\partial D_\mathrm{r}}{\partial \omega}\frac{\partial \hat{E}}{\partial t_1} - \mathrm{i}\frac{\partial D_\mathrm{r}}{\partial k}\frac{\partial \hat{E}}{\partial x_1} = 0. \tag{4.2.4}$$

At zeroth order, we have

$$D_\mathrm{r}\left|\hat{E}\right|^2 = 0. \tag{4.2.5}$$

Let $\omega = \Omega(k; t_1, x_1)$ be a solution to the local Hermitian dispersion relation, i.e.,

$$D_\mathrm{r}(t_1, x_1; \Omega, k) = 0; \tag{4.2.6}$$

since Ω represents the temporal rate of change at the fast time scale, we expect it to be real[46] $\Omega \in \mathbb{R}$. For the particular wave packet with $\omega = \Omega(k; t_1, x_1)$, we could write

$$\left|\hat{E}(\omega, k; t_1, x_1)\right|^2 = P(t_1, x_1, k)\delta(\omega - \Omega(k; t_1, x_1)), \tag{4.2.7}$$

where

$$P(t_1, x_1, k) \equiv \int_{\mathbb{R}} \mathrm{d}\omega \left|\hat{E}(\omega, k; t_1, x_1)\right|^2. \tag{4.2.8}$$

At first order, we have

$$D_\mathrm{i}\hat{E} + \frac{\partial D_\mathrm{r}}{\partial \omega}\frac{\partial \hat{E}}{\partial t_1} - \frac{\partial D_\mathrm{r}}{\partial k}\frac{\partial \hat{E}}{\partial x_1} + \frac{\partial}{\partial k}\left(\frac{\partial D_\mathrm{r}}{\partial x_1}\hat{E}\right) - \frac{\partial}{\partial \omega}\left(\frac{\partial D_\mathrm{r}}{\partial t_1}\hat{E}\right) = 0. \tag{4.2.9}$$

Performing the following operations

$$\hat{E}^* \times \text{Eq. (4.2.9)} + \hat{E} \times \text{Eq. (4.2.9)}^*,$$

[46] This's also the reason why we have used Fourier (instead of Laplace) transform in the fast time variable t_0.

we obtain

$$2D_\mathrm{i}\left|\hat{E}\right|^2+\frac{\partial D_\mathrm{r}}{\partial\omega}\frac{\partial\left|\hat{E}\right|^2}{\partial t_1}-\frac{\partial D_\mathrm{r}}{\partial k}\frac{\partial\left|\hat{E}\right|^2}{\partial x_1}+\frac{\partial}{\partial k}\left(\frac{\partial D_\mathrm{r}}{\partial x_1}\left|\hat{E}\right|^2\right)-\frac{\partial}{\partial\omega}\left(\frac{\partial D_\mathrm{r}}{\partial t_1}\left|\hat{E}\right|^2\right)=0, \quad (4.2.10)$$

which, after substituting Eq. (4.2.7), becomes (denoting $\partial/\partial\Omega\equiv\partial/\partial\omega|_{\omega=\Omega(t_1,x_1,k)}$)

$$\begin{aligned}&\left(2D_\mathrm{i}P+\frac{\partial D_\mathrm{r}}{\partial\Omega}\frac{\partial P}{\partial t_1}-\frac{\partial D_\mathrm{r}}{\partial k}\frac{\partial P}{\partial x_1}+\frac{\partial}{\partial k}\left(\frac{\partial D_\mathrm{r}}{\partial x_1}P\right)-\frac{\partial^2 D_\mathrm{r}}{\partial\Omega\partial t_1}P\right)\delta(\omega-\Omega)\\&+\left(-\frac{\partial\Omega}{\partial t_1}\frac{\partial D_\mathrm{r}}{\partial\omega}P+\frac{\partial\Omega}{\partial x_1}\frac{\partial D_\mathrm{r}}{\partial k}P-\frac{\partial\Omega}{\partial k}\frac{\partial D_\mathrm{r}}{\partial x_1}P-\frac{\partial D_\mathrm{r}}{\partial t_1}P\right)\delta'(\omega-\Omega)=0.\end{aligned} \quad (4.2.11)$$

Integrating the above equation over ω, we find[47]

$$\frac{\mathrm{d}N}{\mathrm{d}t_1}\equiv\frac{\partial N}{\partial t_1}+\frac{\partial\Omega}{\partial k}\frac{\partial N}{\partial x_1}-\frac{\partial\Omega}{\partial x_1}\frac{\partial N}{\partial k}=2\omega_\mathrm{i}N+P\frac{\partial^2 D_\mathrm{r}}{\partial\Omega\partial t_1}-P\frac{\partial^2 D_\mathrm{r}}{\partial k\partial t_1}. \quad (4.2.13)$$

Here,

$$N\equiv P\frac{\partial D_\mathrm{r}}{\partial\Omega}=N(t_1,x_1,k) \quad (4.2.14)$$

has the physical meaning of the quasi-particle (or, sometimes called, *plasmon*) distribution function, $\mathrm{d}/\mathrm{d}t_1$ denotes the material time derivative along the quasi-particle's trajectory, and

$$\omega_\mathrm{i}\equiv-\frac{D_\mathrm{i}}{\partial D_\mathrm{r}/\partial\Omega} \quad (4.2.15)$$

is the linear damping\growth rate. For simplicity, let's assume $\partial^2 D_\mathrm{r}/\partial\omega\partial t_1=\partial^2 D_\mathrm{r}/\partial k\partial x_1=0$, then

$$\frac{\mathrm{d}N}{\mathrm{d}t_1}\equiv\frac{\partial N}{\partial t_1}+\frac{\partial\Omega}{\partial k}\frac{\partial N}{\partial x_1}-\frac{\partial\Omega}{\partial x_1}\frac{\partial N}{\partial k}=2\omega_\mathrm{i}N. \quad (4.2.16)$$

Eq. (4.2.16) is the desired wave kinetic equation including the effect of a small but finite D_i. Thus, with $\omega_\mathrm{i}\neq 0$, the number of quasi-particles is no longer conserved; instead, quasi-particles can be either annihilated or created along the ray path. Eq. (4.2.16) is, therefore, useful in calculating the wave energy deposition with radio-frequency wave heating.

4.3 Cutoff, resonance, and WKB connection formula (boundary layer analysis)

The main references for this section are [44], Chapter 13 of [21] and Chapter 7 of [45].

As indicated by Eq. (4.1.13), the geometric optics is based on the assumption that there exist two scales in both time and spcae. In many cases of practical interest, however, this two-

[47]Noting that the two partial derivatives $\partial/\partial t_1$ and $\partial/\partial\Omega$ do not commute, we have

$$\frac{\partial N}{\partial t_1}=P\frac{\partial}{\partial t_1}\frac{\partial D_\mathrm{r}}{\partial\Omega}=P\frac{\partial^2 D_\mathrm{r}}{\partial\Omega\partial t_1}+P\frac{\partial\Omega}{\partial t_1}\frac{\partial^2 D_\mathrm{r}}{\partial\Omega^2}. \quad (4.2.12)$$

scale assumption is not *uniformly* valid and, in the regions where it breaks down, one has to employ the wave optics, i.e., the full wave equation. This is the subject of the present section.

Since, in confined plasmas, the plasma properties generally evolve much slowlier than the wave phenomena of interest, we shall, for the purpose of this discussion, assume the plasma to be time-stationary but spatially inhomogeneous. The corresponding validity condition for the geometric optics is simply the two-space-scale assumption

$$\|\nabla \boldsymbol{k}\| \ll \|\boldsymbol{k}\boldsymbol{k}\|. \tag{4.3.1}$$

Two cases often occurred in plasma physics such that Eq. (4.3.1) breaks down are *cutoffs* and *resonances* where $|\boldsymbol{k}| \to 0$ and $|\boldsymbol{k}| \to +\infty$, respectively. We shall consider both cases via examples.

(1) Cutoff: As an example of the cutoff, let's consider the following wave equation

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \kappa x\right)\hat{E}(x) = 0. \tag{4.3.2}$$

From the geometric optics, we have $|k(x)| = |\kappa x|^{1/2}$ and, thereby, $|k'(x)| = (1/2)\left|\kappa^{1/2}x^{-1/2}\right|$. Thus, for $|x| \to 0$, the two-scale assumption breaks down

$$|k'(x)| \propto |x|^{-1/2} \gg |k(x)|^2 \propto |x|. \tag{4.3.3}$$

Note that the wave described by Eq. (4.3.2) is propagating for $\kappa x > 0$ and evanescent for $\kappa x < 0$. The wave is *reflected* at $x = 0$ and, in terms of quasi-particles, $x = 0$ is called a *regular turning point*. Just as a reminder to the readers, Eq. (4.3.2) is referred to as the Airy equation and its solutions, called Airy functions, are well tabulated; see Chapter 9 of [2].

(2) Resonance: We now illustrate resonances via the following wave equation

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{\mu}{x}\right)\hat{E}(x) = 0. \tag{4.3.4}$$

From the geometric optics, we have $|k(x)| = \left|\mu^{1/2}x^{-1/2}\right|$ and $|k'(x)| = (1/2)\left|\mu^{1/2}x^{-3/2}\right|$. Thus, for $|x| \to 0$, the two-scale assumption breaks down again

$$|k'(x)| \propto |x|^{-3/2} \gg |k(x)|^2 \propto |x|^{-1}. \tag{4.3.5}$$

In this case, $|k(x)| \to +\infty$ as $|x| \to 0$ and $x = 0$ is called a *singular turning point*.

- Since, quite often, the solution is singular at $x = 0$, $\left|\hat{E}(x)\right|$ tends to become very large near the resonance point (layer) $x = 0$. Thus, in the presence of small but finite dissipaton, we may expect significant wave absorption near $x = 0$ and, thus, resonances play crucial roles in wave heating schemes. This phenomenon is called *resonant absorption*; we shall discuss it in more details in Section 4.4.

- Another feature associated with resonances is that $\lambda(x) = 2\pi/|k(x)| \to 0$ as $|x| \to 0$. We, therefore, expect that those microscopic scales neglected in the physical models should be included near the resonances. This consideration is the basis of the so-called *linear mode conversion* process to be discussed in Section 4.5.

In the above discussions, the full wave solutions are only required near cutoffs or resonances, say, at $x = 0$. Away from $x = 0$, the geometric optics remains valid. By asymptotically matching the full wave solutions[48] with the WKB solutions, we can obtain a relation between the WKB solutions valid for $x > 0$ and those valid for $x < 0$. This relation is called the *WKB connection formula*.

We shall illustrate the derivation of the WKB connection formula for the following case of a second-order wave equation with a regular turning point at $x = 0$:

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2} + f^2(x)\right)\hat{E}(x) = 0, \tag{4.3.6}$$

where

$$f^2(x)\begin{cases} > 0 \text{ for } x > 0 \\ < 0 \text{ for } x < 0 \end{cases}. \tag{4.3.7}$$

- The WKB solutions are written as [45]

$$\hat{E}_{\mathrm{W}}(x) = \begin{cases} C_1|f|^{-1/2}\mathrm{e}^{\mathrm{i}\xi_{\mathrm{W}}} + C_2|f|^{-1/2}\mathrm{e}^{-\mathrm{i}\xi_{\mathrm{W}}} & \text{for } x > 0 \\ D_1|f|^{-1/2}\mathrm{e}^{-\xi_{\mathrm{W}}} + D_2|f|^{-1/2}\mathrm{e}^{\xi_{\mathrm{W}}} & \text{for } x < 0 \end{cases} \tag{4.3.8}$$

  with $\xi_{\mathrm{W}} = \int_0^x \mathrm{d}x'\, |f(x')|$ in either case; note that $\xi_{\mathrm{W}} < 0$ for $x < 0$.
- Near $x = 0$, we could write $f^2(x) = \kappa x + o(x)$ with $\kappa > 0$. Then Eq. (4.3.6) reduces to the Airy eqaution

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}\tilde{x}^2} + \tilde{x}\right)\hat{E}(x) = 0, \quad \tilde{x} \equiv \kappa^{1/3}x, \tag{4.3.9}$$

  whose general solutions can be expressed as[49] [2]

$$\hat{E}_{\mathrm{f}}(x) = \begin{cases} A_1 x^{1/2}J_{1/3}(\xi) + B_1 x^{1/2}J_{-1/3}(\xi) & \text{for } x > 0 \\ A_2|x|^{1/2}I_{1/3}(\xi) + B_2|x|^{1/2}I_{-1/3}(\xi) & \text{for } x < 0 \end{cases} \tag{4.3.10}$$

  with $\xi = (2/3)|\tilde{x}|^{3/2} = (2/3)\kappa^{1/2}|x|^{3/2} > 0$ in either case. Now at $x = 0$, we impose the continuity conditions, $\hat{E}_{\mathrm{f}}\big|_{x=0^+} = \hat{E}_{\mathrm{f}}\big|_{x=0^-}$ and $\mathrm{d}\hat{E}_{\mathrm{f}}/\mathrm{d}x\big|_{x=0^+} = \mathrm{d}\hat{E}_{\mathrm{f}}/\mathrm{d}x\big|_{x=0^-}$, to obtain

$$B_2 = B_1 \text{ and } A_2 = -A_1. \tag{4.3.11}$$

We now perform asymptotic matching of the full solutions $\hat{E}_{\mathrm{f}}$ with the WKB solutions $\hat{E}_{\mathrm{W}}$.

- For $x > 0$, we have, as $x \to +\infty$,

[48] Asymptotic (instead of exact) solutions are enough.
[49] The Airy functions, Ai and Bi, could be expressed in terms of Bessel functions of 1/3 order.

$$\begin{aligned}\hat{E}_{\mathrm{f}}(x) &\sim \sqrt{\frac{2x}{\pi\xi}}\left(A_1\cos\left(\xi-\frac{5\pi}{12}\right)+B_1\cos\left(\xi-\frac{\pi}{12}\right)\right)\\ &=\sqrt{\frac{x}{2\pi\xi}}\left(\left(A_1\mathrm{e}^{-\mathrm{i}5\pi/12}+B_1\mathrm{e}^{-\mathrm{i}\pi/12}\right)\mathrm{e}^{\mathrm{i}\xi}+\left(A_1\mathrm{e}^{\mathrm{i}5\pi/12}+B_1\mathrm{e}^{\mathrm{i}\pi/12}\right)\mathrm{e}^{-\mathrm{i}\xi}\right)\end{aligned}\tag{4.3.12}$$

and, as $x \to 0^+$,

$$\hat{E}_{\mathrm{W}}(x) \sim \sqrt{\frac{2x}{3\xi}}\left(C_1\mathrm{e}^{\mathrm{i}\xi}+C_2\mathrm{e}^{-\mathrm{i}\xi}\right).\tag{4.3.13}$$

Matching the two expressions, we obtain

$$C_1 = A\mathrm{e}^{-\mathrm{i}5\pi/12}+B\mathrm{e}^{-\mathrm{i}\pi/12},\quad C_2 = A\mathrm{e}^{\mathrm{i}5\pi/12}+B\mathrm{e}^{\mathrm{i}\pi/12},\tag{4.3.14}$$

where $A \equiv (1/2)\sqrt{3/\pi}A_1$, $B \equiv (1/2)\sqrt{3/\pi}B_1$.

- Similarly, for $x < 0$, we have,

$$\hat{E}_{\mathrm{f}}(x) \sim \sqrt{\frac{|x|}{2\pi\xi}}\left((A_2+B_2)\mathrm{e}^{\xi}+\left(A_2\mathrm{e}^{-\mathrm{i}5\pi/6}+B_2\mathrm{e}^{-\mathrm{i}\pi/6}\right)\mathrm{e}^{-\xi}\right)\text{ as } x\to-\infty\tag{4.3.15}$$

and (noting $\xi_{\mathrm{W}} < 0$)

$$\hat{E}_{\mathrm{W}}(x) \sim \sqrt{\frac{2|x|}{3\xi}}\left(D_1\mathrm{e}^{\xi}+D_2\mathrm{e}^{-\xi}\right)\text{ as } x\to 0^-.\tag{4.3.16}$$

Matching the two expressions and noting Eq. (4.3.11), we obtain

$$D_1 = -A+B,\quad D_2 = -A\mathrm{e}^{-\mathrm{i}5\pi/6}+B\mathrm{e}^{-\mathrm{i}\pi/6}.\tag{4.3.17}$$

Substituting Eq. (4.3.14) and Eq. (4.3.17) into Eq. (4.3.8), we arrive at the following connection formula for WKB solutions across a regular turning point

$$\begin{aligned}&|f|^{-1/2}\left(\left(A\mathrm{e}^{-\mathrm{i}5\pi/12}+B\mathrm{e}^{-\mathrm{i}\pi/12}\right)\mathrm{e}^{\mathrm{i}\xi_{\mathrm{W}}}+\left(A\mathrm{e}^{\mathrm{i}5\pi/12}+B\mathrm{e}^{\mathrm{i}\pi/12}\right)\mathrm{e}^{-\mathrm{i}\xi_{\mathrm{W}}}\right) &&\text{for } f^2>0\\ \longleftrightarrow\ &|f|^{-1/2}\left((-A+B)\mathrm{e}^{-\xi_{\mathrm{W}}}+\left(-A\mathrm{e}^{-\mathrm{i}5\pi/6}+B\mathrm{e}^{-\mathrm{i}\pi/6}\right)\mathrm{e}^{\xi_{\mathrm{W}}}\right) &&\text{for } f^2<0.\end{aligned}\tag{4.3.18}$$

We remark that, using the same technique, one can also derive a WKB connection formula across a singular turning point [21].

We now shall divert somewhat and make two remarks. First, we note that the connection formula given by Eq. (4.3.18) can be used to derive the *Bohr-Sommerfeld quantization condition* [45], i.e.,

$$\int_{T_1}^{T_2}\mathrm{d}z\, f(z) = \left(n+\frac{1}{2}\right)\pi, n = 0,1,2,\cdots.\tag{4.3.19}$$

Here, we have generalized the wave equation Eq. (4.3.6) to the complex plane where $z = x+\mathrm{i}y$, and $T_1, T_2 \in \mathbb{C}$ are two turning points, i.e., $f(T_1) = f(T_2) = 0$. Referring to Figure 4.1, we see that Eq. (4.3.19) is applicable to solutions which are subdominant in regions S_1 and S_2; that

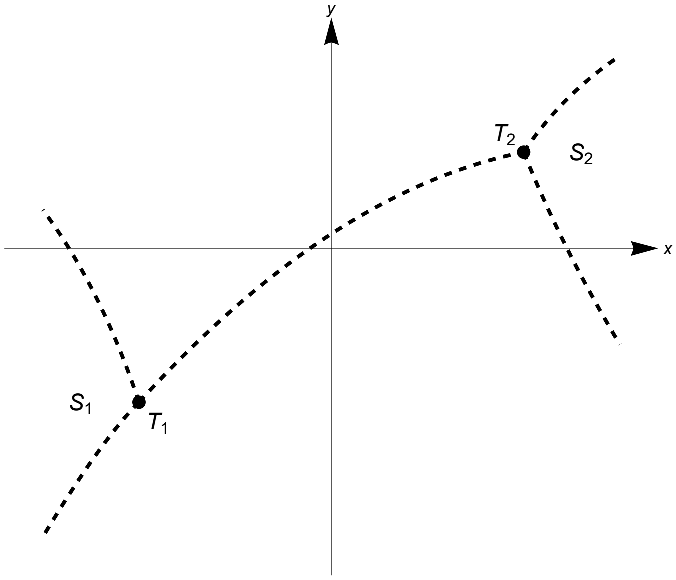


Figure 4.1: The Stokes diagram in the complex plane.

is, solutions in S_1 and S_2 decay away from T_1 and T_2, respectively. Meanwhile, the dashed lines in Figure 4.1 correspond to the *anti-Stokes lines* along which the WKB solutions are purely oscillatory. For a readable account of the WKB method and a more detailed analysis in the complex plane, we refer the readers to the monograph [46]. We note that the WKB technique is useful in solving eigenvalue problems.

The second remark is regarding the derivation of the asymptotic behaviors of the Airy functions (hence, the WKB connection formula) via the *Laplace method* [11], [45], [47], which is often useful in dealing with problems involving asymptotic matchings. Recall the Airy equation

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2} + x\right)\hat{E}(x) = 0. \tag{4.3.20}$$

Let's perform a Laplace-like transform and seek a solution in the form of a contour integral

$$\hat{E}(x) = \int_C \mathrm{d}\omega\, \mathrm{e}^{\mathrm{i}\omega x} p(\omega), \tag{4.3.21}$$

where C is some contour in the complex ω plane and p is a holomorphic function. Then

$$\begin{aligned} \hat{E}''(x) &= -\int_C \mathrm{d}\omega\, \mathrm{e}^{\mathrm{i}\omega x}\omega^2 p(\omega), \\ x\hat{E}(x) &= -\mathrm{i}\int_C \mathrm{d}\omega\, \frac{\partial \mathrm{e}^{\mathrm{i}\omega x}}{\partial \omega} p(\omega) \\ &= -\mathrm{i}\mathrm{e}^{\mathrm{i}\omega x} p(\omega)\big|_{\text{start}}^{\text{end}} + \mathrm{i}\int_C \mathrm{d}\omega\, \mathrm{e}^{\mathrm{i}\omega x} p'(\omega). \end{aligned} \tag{4.3.22}$$

Under the boundary conditions

$$\mathrm{e}^{\mathrm{i}\omega x}p(\omega)\big|_{\omega\in\partial C}=0, \tag{4.3.23}$$

Eq. (4.3.20) reduces to

$$\begin{aligned} &\mathrm{i}p'(\omega)-\omega^2 p(\omega)=0\\ \Rightarrow\ &p(\omega)=A\exp\left(-\mathrm{i}\frac{\omega^3}{3}\right). \end{aligned} \tag{4.3.24}$$

Therefore, we have solved Eq. (4.3.20) in terms of the contour integral

$$\hat{E}(x)=\int_C \mathrm{d}\omega \exp\left(\mathrm{i}x\left(\omega-\frac{\omega^3}{3x}\right)\right). \tag{4.3.25}$$

So far we haven't specified the integration contour C. Since the integrand has no singularities, only the end points of the contour matter; in particular, any closed C would lead to the trivial solution $\hat{E}(x)\equiv 0$. On the other hand, an open C with finite end points would violate the boundary conditions Eq. (4.3.23); hence, both end points of C must be at the complex ∞, and the directions in which the two ends of the contour approach ∞ completely determine the integral. Those directions of approach are controlled by two considerations:

(1) One should approach ∞ along directions in which the integrand decreases rather than increases; for the problem at hand, this allows angles of approach between $\pi/3$ and $2\pi/3$, between π and $4\pi/3$, and between $5\pi/3$ and 2π, that is, within three white sectors in Figure 4.2.

(2) All approaches within the same sector are equivalent; consequently, the two ends of the contour must belong to different sectors.

These two considerations give us three choices of contours —— C_1, C_2 and C_3 —— correspnding to three different solutions $\hat{E}_1(x), \hat{E}_2(x)$ and $\hat{E}_3(x)$. Since, however, the combined contour $C_1+C_2+C_3$ is shrinkable, only two of the three solutions are independent $\hat{E}_1(x)+\hat{E}_2(x)+\hat{E}_3(x)\equiv 0$; this should be expected because Eq. (4.3.20) is a second-order ODE with, therefore, two independent solutions.

With $\hat{E}(x)$ represented by a contour integral, the $|x|\to+\infty$ asymptotic behaviors can then be obtained using the *saddle point method*[50] [1], [16], [47]

[50]The saddle point method is also known as the method of *steepest descent* or *stationary phase*. It estimates the asymptotic behavior of an integral of the following form

$$I(\lambda)=\int_C \mathrm{d}z\,\mathrm{e}^{\lambda S(z)},\quad \lambda\in\mathbb{R} \tag{4.3.26}$$

as $\lambda\to+\infty$. Here, C is a contour in the complex z plane and S is a holomorphic function.

In order to evaluate this integral, we deform C such that it passes through the saddle points of S (where $S'(z)=0$) in the steepest descent directions. The idea of the saddle point method is that, as $\lambda\to+\infty$, the main contribution to the integral comes from the neighborhood of the saddle points. For simplicity, let's assume that S has only one saddle point $z=z_0$; if S has multiple saddle points, we need only to sum the contributions from each. Taylor expanding S about z_0 and noting that the Gaussian profile has width $|z-z_0|\sim 1/\sqrt{\lambda}$, we have

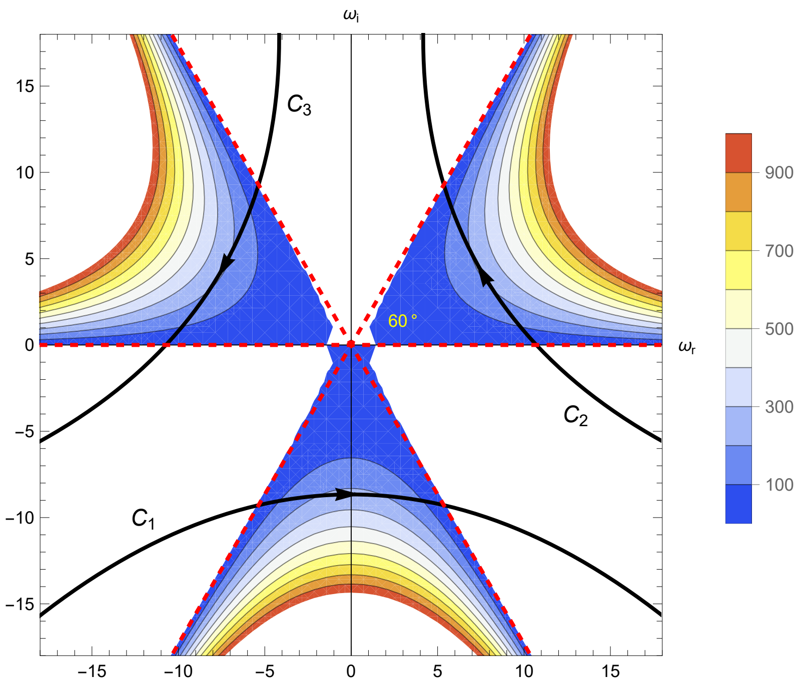


Figure 4.2: Contour plot of the function $\ln\left|\exp(\mathrm{i}(\omega-\omega^3/3))\right|$ in the complex ω plane.

$$\begin{aligned} I(\lambda) &= \int_C \mathrm{d}z \exp\Big(\lambda S(z_0) + \frac{1}{2}\lambda S''(z_0)(z-z_0)^2 + \mathcal{O}\big(\lambda(z-z_0)^3\big) + \mathcal{O}\big(\lambda(z-z_0)^4\big)\Big) \\ &= \mathrm{e}^{\lambda S(z_0)} \int_C \mathrm{d}z \exp\Big(\frac{1}{2}\lambda S''(z_0)(z-z_0)^2\Big) \times \Big(1 + \mathcal{O}\big(\lambda(z-z_0)^3\big) + \mathcal{O}\big(\lambda(z-z_0)^4\big)\Big). \end{aligned} \tag{4.3.27}$$

Now we need to determine the direction of the steepest descent. Let $z - z_0 = \big(\rho/\sqrt{\lambda}\big)\mathrm{e}^{\mathrm{i}\theta}$ and $S''(z_0) = |S''(z_0)|\mathrm{e}^{\mathrm{i}\alpha}$, then the real part of the exponent

$$\mathrm{Re}\Big(\frac{1}{2}\lambda S''(z_0)(z-z_0)^2\Big) = \frac{1}{2}|S''(z_0)|\rho^2 \ \mathrm{Re}\ \mathrm{e}^{\mathrm{i}(2\theta+\alpha)} = \frac{1}{2}|S''(z_0)|\rho^2\cos(2\theta+\alpha) \tag{4.3.28}$$

decreases most rapidly when $\cos(2\theta+\alpha) = -1$, i.e., when $\theta = \pm\pi/2 - \alpha/2 \equiv \theta_0$. Choosing C to pass through z_0 along the direction determined by the proper $\theta = \theta_0$, we find

$$\begin{aligned} I(\lambda) &= \mathrm{e}^{\lambda S(z_0)} \frac{\mathrm{e}^{\mathrm{i}\theta_0}}{\sqrt{\lambda}} \int_{\mathbb{R}} \mathrm{d}\rho \exp\Big(-\frac{1}{2}|S''(z_0)|\rho^2\Big) \times \Big(1 + \mathcal{O}\Big(\frac{1}{\sqrt{\lambda}}\Big)\cancel{\rho^3} + \mathcal{O}\Big(\frac{1}{\lambda}\Big)\rho^4\Big) \\ &= \sqrt{\frac{2\pi}{\lambda|S''(z_0)|}}\mathrm{e}^{\mathrm{i}\theta_0+\lambda S(z_0)}\Big(1 + \mathcal{O}\Big(\frac{1}{\lambda}\Big)\Big). \end{aligned} \tag{4.3.29}$$

In physical situations, one often encounter integrals of the following form

$$I(\lambda) = \int_C \mathrm{d}z\, f(z)\mathrm{e}^{\lambda S(z)} \tag{4.3.30}$$

with f being holomorphic. In this case, we may repeat the above derivation and obtain

$$\begin{aligned} I(\lambda) &= f(z_0)\mathrm{e}^{\lambda S(z_0)} \frac{\mathrm{e}^{\mathrm{i}\theta_0}}{\sqrt{\lambda}} \int_{\mathbb{R}} \mathrm{d}\rho \exp\Big(-\frac{1}{2}|S''(z_0)|\rho^2\Big) \times \Big(1 + \mathcal{O}\Big(\frac{1}{\sqrt{\lambda}}\Big)\rho + \mathcal{O}\Big(\frac{1}{\lambda}\Big)\rho^2\Big)\Big(1 + \mathcal{O}\Big(\frac{1}{\sqrt{\lambda}}\Big)\rho^3 + \mathcal{O}\Big(\frac{1}{\lambda}\Big)\rho^4\Big) \\ &= \sqrt{\frac{2\pi}{\lambda|S''(z_0)|}} f(z_0)\mathrm{e}^{\mathrm{i}\theta_0+\lambda S(z_0)}\Big(1 + \mathcal{O}\Big(\frac{1}{\lambda}\Big)\Big). \end{aligned} \tag{4.3.31}$$

$$\int_C \mathrm{d}z\, \mathrm{e}^{\lambda S(z)} = \sqrt{\frac{2\pi}{\lambda |S''(z_0)|}} \mathrm{e}^{\mathrm{i}\theta_0 + \lambda S(z_0)} \left(1 + \mathcal{O}\left(\frac{1}{\lambda}\right)\right) \text{ for } \lambda \to +\infty, \tag{4.3.32}$$

where S is a holomorphic function, z_0 is the saddle point of S such that $S'(z_0) = 0$ and $\theta_0 = \pm\pi/2 - \alpha/2$ is the inclination angle of the steepest descent direction at $z = z_0$ with $\alpha = \arg(S''(z_0))$ being the argument of $S''(z_0)$. Here we illustrate the derivation of the asymptotic behavior of $\hat{E}(x)$ by adopting the C_1 contour, with $\lambda = |x|$ and

$$S(\omega) = \mathrm{i}(\mathrm{sgn}\ x)\left(\omega - \frac{\omega^3}{3x}\right) \Rightarrow S'(\omega) = \mathrm{i}(\mathrm{sgn}\ x)\left(1 - \frac{\omega^2}{x}\right) \Rightarrow S''(\omega) = -\mathrm{i}2\frac{\omega}{|x|}. \tag{4.3.33}$$

- For $x \to +\infty$, we have two saddle points at $\omega_0 = \pm\sqrt{x}$ with $\theta_0 = \mp\pi/4$, respectively. Thus

$$\begin{aligned}\hat{E}_1(x) &\sim \sqrt{\frac{2\pi}{x(2/\sqrt{x})}}\left(\exp\left(\mathrm{i}\frac{2}{3}x^{3/2} - \mathrm{i}\frac{\pi}{4}\right) + \exp\left(-\mathrm{i}\frac{2}{3}x^{3/2} + \mathrm{i}\frac{\pi}{4}\right)\right) \\ &= \frac{2\sqrt{\pi}}{x^{1/4}}\cos\left(\frac{2}{3}x^{3/2} - \frac{\pi}{4}\right).\end{aligned} \tag{4.3.34}$$

- Similarly, for $x \to -\infty$, we have the saddle points $\omega_0 = \pm\mathrm{i}\sqrt{|x|}$. For the C_1 contour, only the saddle point $\omega_0 = -\mathrm{i}\sqrt{|x|}$ is relevant and, correspondingly, $\theta_0 = 0$. We, thus, derive

$$\hat{E}_1(x) \sim \frac{\sqrt{\pi}}{|x|^{1/4}}\exp\left(-\frac{2}{3}|x|^{3/2}\right). \tag{4.3.35}$$

We note that the connection formula given by Eq. (4.3.34) and Eq. (4.3.35) agrees with that given by Eq. (4.3.18) for the case $A = B$. The case with $A \neq B$ can be derived by adopting the C_2 or C_3 integration contour.

Homework #7

(1) Use the connection formula for the regular turning point to verify the Bohr-Sommerfeld quantization condition Eq. (4.3.19).

(2) Consider the Weber equation

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}z^2} + \mathcal{E} - z^2\right)\phi(z) = 0 \tag{H.7.1}$$

with ϕ being finite (bounded).

  (i) Show that the eigenvalues derived from the Bohr-Sommerfeld quantization condition Eq. (4.3.19) are exact. Answer: $\mathcal{E} = 2n + 1, n \in \mathbb{Z}_{\geq 0}$.

  (ii) Draw the asymptotic anti-Stokes lines.

(3) Using the Laplace method and taking the C_2 or C_3 integration contour, re-derive the connection formula for the regular turning point such that the solution in the evanescent region is <u>divergent</u>.

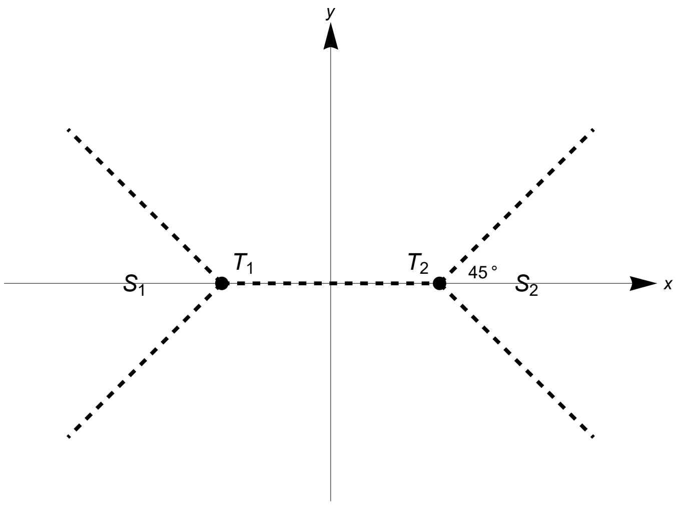


Figure 4.3: Asymptotic anti-Stokes lines for the Weber equation Eq. (H.7.1).

4.4 Tonks-Dattner resonance and resonant absorption

In this section, we shall illustrate some physics associated with resonances via the example of the *Tonks-Dattner resonance*. In this model, we consider a cold, unmagnetized, inhomogeneous plasma located between two capacitor plates which are driven externally by the following electric field

$$\boldsymbol{E}_{\text{ext}}(t) = \hat{\boldsymbol{E}}_0 \mathrm{e}^{-\mathrm{i}\omega_0 t}. \tag{4.4.1}$$

Here, $\omega_0 \sim \omega_{\text{pe}} \gg \omega_{\text{p}i} > 0$ so that the ion dynamics can be neglected, and $\left|\hat{\boldsymbol{E}}_0\right|$ is taken to be sufficiently small to justify a linear description.

Within this model, the electron dynamics is, thus, described by the Possion's equation and the linearized equations[51] of continuity and motion without the thermal pressure term; that is,

$$\nabla \cdot \boldsymbol{E} = -4\pi e n_e, \tag{4.4.2}$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_{0e}(\boldsymbol{x})\boldsymbol{V}_e) = 0, \tag{4.4.3}$$

and

$$m_e \frac{\partial \boldsymbol{V}_e}{\partial t} = -e\boldsymbol{E}. \tag{4.4.4}$$

It is then easy to derive the wave equation for the incident wave

$$\nabla \cdot \left(\left(\frac{\partial^2}{\partial t^2} + \omega_{\text{pe}}^2(\boldsymbol{x}) \right) \boldsymbol{E} \right) = 0, \quad \omega_{\text{pe}}^2(\boldsymbol{x}) \equiv \frac{4\pi e^2 n_{0e}(\boldsymbol{x})}{m_e}. \tag{4.4.5}$$

Expressing the stationary incident wave as

$$\boldsymbol{E}(t, \boldsymbol{x}) = \hat{\boldsymbol{E}}(\boldsymbol{x}) \mathrm{e}^{-\mathrm{i}\omega_0 t}, \tag{4.4.6}$$

[51] Remember that the subscript 1 indicating perturbation is omitted.

Eq. (4.4.5) yields

$$\nabla \cdot \Big(\epsilon(\boldsymbol{x})\hat{\boldsymbol{E}}(\boldsymbol{x})\Big) = 0, \tag{4.4.7}$$

where

$$\epsilon(\boldsymbol{x}) = 1 - \frac{\omega_{\mathrm{pe}}^2(\boldsymbol{x})}{\omega_0^2} \tag{4.4.8}$$

is the dielectric function.

For simplicity and without loss of generality, let's consider the one-dimensional limit, i.e., $\hat{\boldsymbol{E}} = \hat{\boldsymbol{e}}_x \hat{E}, n_{0e} = n_{0e}(x)$, etc. Accordingly, Eq. (4.4.7) becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\epsilon(x)\hat{E}(x)\Big) = 0, \tag{4.4.9}$$

or, equivalently,

$$\frac{\mathrm{d}\hat{E}(x)}{\mathrm{d}x} + \frac{\epsilon'(x)}{\epsilon(x)}\hat{E}(x) = 0. \tag{4.4.10}$$

From Eq. (4.4.10), we see that the WKB wavevector is given by

$$k(x) = \mathrm{i}\frac{\epsilon'(x)}{\epsilon(x)}. \tag{4.4.11}$$

Thus, if for some $x = x_0$ we have

$$\epsilon(x_0) = 1 - \frac{\omega_{\mathrm{pe}}^2(x_0)}{\omega_0^2} = 0 \tag{4.4.12}$$

<u>and</u>

$$\epsilon'(x_0) \neq 0, \tag{4.4.13}$$

then a resonance occurs at $x = x_0$, i.e.,

$$|k(x)| \to +\infty \text{ as } x \to x_0; \tag{4.4.14}$$

in other words, a *resonance occurs where the external driving frequency is equal to the local electron plasma frequency*. This resonance is between the incident wave and the electron plasma oscillation, and is called the *Tonks-Dattner resonance*. Eq. (4.4.9), of course, can be solved directly to give

$$\hat{E}(x) = \frac{\hat{E}_0}{\epsilon(x)}. \tag{4.4.15}$$

Here, we have imposed the condition that $\hat{E}(x) = \hat{E}_0$ in the vacuum where $\epsilon(x) = 1$. Since $\epsilon(x_0) = 0$, Eq. (4.4.15) clearly shows that $\hat{E}(x)$ is singular at $x = x_0$ and, as indicated in Section 4.3, we expect significant wave absorption near $x = x_0$.

To calculate the *resonance absorption*, we invoke Eq. (1.3.14)[52], which gives

$$\frac{\mathrm{d}P(x)}{\mathrm{d}x} = -\frac{\omega_0}{8\pi}\,\mathrm{Im}\Big(\hat{E}^*(x)\epsilon(x)\hat{E}(x)\Big) = -\frac{\omega_0}{8\pi}\left|\hat{E}_0\right|^2\,\mathrm{Im}\frac{1}{\epsilon^*(x)} \tag{4.4.16}$$

where P is the Poynting flux as defined in Eq. (1.3.16). To go further, let's assume that the dielectric function contains a small but finite anti-Hermitian part due to, e.g., *dissipations*. Thus, replacing ω_0^2 by $\omega_0(\omega_0+\mathrm{i}\nu)$ with $|\nu/\omega_0| \ll 1$, we have

$$\epsilon(x) = 1 - \frac{\omega_{\mathrm{pe}}^2(x)}{\omega_0(\omega_0+\mathrm{i}\nu)} \approx 1 - \frac{\omega_{\mathrm{pe}}^2(x)}{\omega_0^2} + \mathrm{i}\frac{\nu}{\omega_0}\frac{\omega_{\mathrm{pe}}^2(x)}{\omega_0^2}. \tag{4.4.17}$$

Resonance occurs at $x = x_0$ where $\omega_0 = \omega_{\mathrm{pe}}(x_0)$. Since Re $\epsilon'(x_0) \neq 0$, we have, for $x \approx x_0$,

$$\epsilon(x) \approx -\frac{2}{\omega_0}\omega_{\mathrm{pe}}'(x_0)(x-x_0) + \mathrm{i}\frac{\nu}{\omega_0} \equiv \kappa(x - x_0 + \mathrm{i}\eta) \tag{4.4.18}$$

with $\kappa \equiv \epsilon'(x_0) = -2\omega_{\mathrm{pe}}'(x_0)/\omega_0 = -\mathrm{d}\ln n_{0e}(x)/\mathrm{d}x|_{x=x_0}$ being the reciprocal of the spatial scale of inhomogeneity, and $\eta \equiv \nu/\omega_0\kappa$ the characteristic length of dissipation (i.e., dissipative effect is non-negligible for $|x-x_0| = \mathcal{O}(\eta)$). Substituting Eq. (4.4.18) into Eq. (4.4.16), we find

$$\begin{aligned}\lim_{|\nu|\to 0}\frac{\mathrm{d}P(x)}{\mathrm{d}x} &= -\frac{\omega_0}{8\pi}\frac{\left|\hat{E}_0\right|^2}{\kappa}\,\mathrm{Im}\left(\lim_{|\eta|\to 0}\frac{1}{x-x_0-\mathrm{i}\eta}\right)\\ &\xlongequal[\text{formula}]{\text{Plemelj}} -\frac{\omega_0}{8}\frac{\left|\hat{E}_0\right|^2}{\kappa}(\mathrm{sgn}\ \eta)\,\delta(x-x_0),\end{aligned} \tag{4.4.19}$$

i.e., the Poynting flux suffers an abrupt change at $x = x_0$,

$$P(x) = P(x_0 - 0) - \frac{\omega_0}{8}\frac{\left|\hat{E}_0\right|^2}{|\kappa|}(\mathrm{sgn}\ \nu)\,\theta(x-x_0)\ \text{for}\ x \approx x_0 \tag{4.4.20}$$

where θ is the Heaviside step function. For positive dissipations $\nu > 0$,

$$\Delta P = \omega_0\frac{\left|\hat{E}_0\right|^2}{8}|\kappa| \tag{4.4.21}$$

gives the energy flux resonantly absorbed at $x = x_0$; see Figure 4.4.

4.5 Linear mode conversion

As indicated in Section 4.3 and, specifically, Section 4.4 for the example of the Tonks-Dantter resonance, wave perturbations become singular and $|k| \to +\infty$ near resonances. Thus, dynamics with microscopic spatial scales neglected in the origional physical models should, in principle, be included near the resonance layer and this can lead to the so-called *linear mode conversion* process. Here, we shall illustrate this process by pursuing the warm plasma analog of the cold Tonks-Dattner resonances.

[52]Eq. (1.3.14) is derived under the two-scale assumption; here, however, the terms in Eq. (1.3.14) that involve two scales vanish.

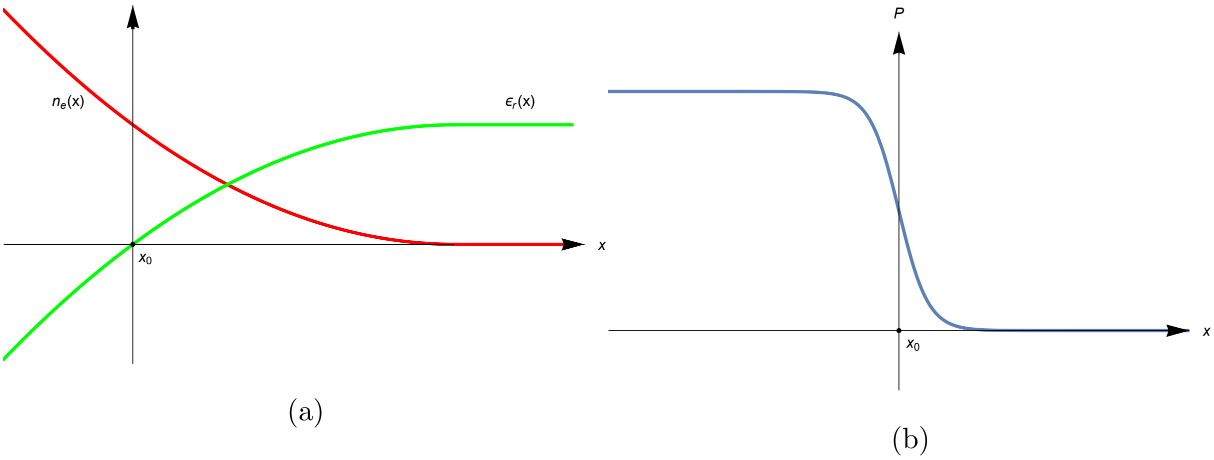


Figure 4.4: Sketches of (a) density n_e, real part of the dielectric function $\epsilon_{\rm r}$ and (b) Poynting flux P v.s. x in the Tonks-Dattner resonance. Here, $x = x_0$ is the resonance point and κ is assumed to be positive for clarity.

In this model where the plasmas are unmagnetized and ions are assumed to be immobile, the only relevant microscopic[53] spatial scale is the electron Debye length, $\lambda_{\rm De}$, which enters the dynamics via the thermal pressure term in the electron equation of motion and the equation of state, i.e.,

$$n_{0e}m_e\frac{\partial \boldsymbol{V}_e}{\partial t} = -\frac{e}{m_e}\boldsymbol{E} - \nabla p_e \tag{4.5.1}$$

and

$$\begin{aligned} &(p_{0e} + p_e)(n_{0e} + n_e)^{-\gamma_e} = \text{const} \\ &\Rightarrow p_e = \gamma_e \frac{p_{0e}}{n_{0e}} n_e = \gamma_e T_e n_e. \end{aligned} \tag{4.5.2}$$

Here, γ_e is the ratio of specific heat for electrons, and T_e is assumed to be constant. Combining with the equation of continuity and the Poisson's equation, i.e., Eq. (4.4.3) and Eq. (4.4.2), we readily derive the following stationary wave equation for the incident wave

$$\nabla\cdot\Big(\big(\overline{\lambda}_{\rm D}^2\nabla^2{+}\epsilon(\boldsymbol{x})\big)\hat{\boldsymbol{E}}(\boldsymbol{x})\Big) = \boldsymbol{0}, \quad \overline{\lambda}_{\rm D}^2 \equiv \frac{\gamma_e}{2}\frac{v_{\rm te}^2}{\omega_0^2}, \tag{4.5.3}$$

which correctly reduces to Eq. (4.4.5) in the cold plasma limit $v_{\rm te} \to 0$.

Again, the physics can be best elucidated in the one-dimensional limit, where Eq. (4.5.3) becomes

$$\left(\overline{\lambda}_{\rm D}^2\frac{{\rm d}^2}{{\rm d}x^2} + \epsilon(x)\right)\hat{E}(x) = \hat{E}_0. \tag{4.5.4}$$

[53]The inhomogeneity scale $|\kappa|^{-1}$ is macroscopic.

Eq. (4.5.4) immediately contrasts the cold plasma Eq. (4.4.15) in that $\hat{E}(x)$ is, in general, no longer singular at $x = x_0$ where $\epsilon(x_0) = 0$ (i.e., $\omega_0 = \omega_{\rm pe}(x_0)$). Physically, the WKB approximation is valid as long as the *inhomogeneity is weak*, i.e., $\left|\kappa\overline{\lambda}_{\rm D}\right| \ll 1$, under which condition a slow-varying wavevector is expected. Furthermore, noting that Eq. (4.5.4) is a second-order nonhomogeneous ODE, whose general solutions can be expressed as linear combinations of two linearly independent solutions of the corresponding homogeneous equation and a particular solution of the nonhomogeneous equation, we shall make the following observations:

(1) The two homogeneous solutions correspond to Bohm-Gross waves which are propagating for $\epsilon(x) > 0$ and evanescent or divergent for $\epsilon(x) < 0$. In fact, away from $x = x_0$ or, more precisely, for $|x - x_0| \gtrsim 1/|\kappa| \gg \overline{\lambda}_{\rm D}$ where $\epsilon(x) = \mathcal{O}(1)$, the WKB wavevector is given by $k_{\rm BG}(x) = \pm\sqrt{\epsilon(x)}/\overline{\lambda}_{\rm D}$ or $\omega_0^2 = \omega_{\rm pe}^2(x) + (\gamma_e/2)k_{\rm BG}^2(x)v_{\rm te}^2$ (Bohm-Gross dispersion relation).

(2) For $|x - x_0| \gtrsim 1/|\kappa| \gg \overline{\lambda}_{\rm D}$, the Bohm-Gross waves are decoupled from the approximate particular solution $\hat{E}_{\rm part}(x) \approx \hat{E}_0/\epsilon(x)$, which is the cold plasma mode (the electrom plasma oscillation) with WKB wavevector $k_{\rm cold}(x) = {\rm i}\epsilon'(x)/\epsilon(x) \approx {\rm i}\kappa/\epsilon(x)$. Alternatively, we may refer to the Bohm-Gross waves as the warm plasma modes. Note that warm plasma modes are short-wavelength modes compared to the cold mode

$$\left|\frac{k_{\rm cold}}{k_{\rm warm}}\right| = \left|\frac{\kappa\overline{\lambda}_{\rm D}}{\epsilon^{3/2}}\right| \ll \left|\epsilon^{-3/2}\right| = \mathcal{O}(1) \text{ for } |x - x_0| \gtrsim \frac{1}{|\kappa|} \gg \overline{\lambda}_{\rm D}. \tag{4.5.5}$$

(3) For $|x - x_0| = \mathcal{O}\left(\overline{\lambda}_{\rm D}\right)$, we expect the warm and cold plasma modes to be strongly coupled. This coupling occurs because as $x \to x_0$, we have $|k_{\rm cold}/k_{\rm warm}| \to +\infty$; thus, we expect that there exists a transition region near $x = x_0$ where the warm and cold plasma modes are strongly coupled $|k_{\rm cold}/k_{\rm warm}| \approx 1$.

To study the effect of couplig in more details, we expand $\epsilon(x) \approx \kappa(x - x_0)$ near $x = x_0$, and assume κ to be positive for clarity. Eq. (4.5.4) then becomes the inhomogeneous Airy equation

$$\left(\frac{{\rm d}^2}{{\rm d}\tilde{x}^2} + \tilde{x}\right)\hat{E}(x) = \frac{\hat{E}_0}{\left(\kappa\overline{\lambda}_{\rm D}\right)^{2/3}}, \quad \tilde{x} \equiv \left(\frac{\kappa}{\overline{\lambda}_{\rm D}^2}\right)^{1/3}(x - x_0), \tag{4.5.6}$$

whose general solution can be expressed as [2]

$$\hat{E}(x) = -\pi\frac{\hat{E}_0}{\left(\kappa\overline{\lambda}_{\rm D}\right)^{2/3}}\left(c_1{\rm Ai}\left(-\frac{x - x_0}{\Delta}\right) + c_2{\rm Bi}\left(-\frac{x - x_0}{\Delta}\right) + {\rm Gi}\left(-\frac{x - x_0}{\Delta}\right)\right). \tag{4.5.7}$$

Here, $\Delta \equiv \left(\overline{\lambda}_{\rm D}^2/\kappa\right)^{1/3} = \overline{\lambda}_{\rm D}\left(\kappa\overline{\lambda}_{\rm D}\right)^{-1/3} \gg \overline{\lambda}_{\rm D}$ is, sometimes called, the *Airy scale length*. To determine the two constants c_1 and c_2, we need to impose two boundary conditions:

(1) There should be no short-wavelength exponentially divergent solution as $x \to -\infty$. This implies $c_2 = 0$.

(2) The other condition concerns with the two Bohm-Gross waves propagating in the $x > x_0$ region with $v_{\rm g} > 0$ and $v_{\rm g} < 0$, respectively. The fact that there is no external source at $x >$

x_0 rules out the possibility of external excitation of the $v_{\mathrm{g}} < 0$ wave. On the other hand, noting that $\left|k_{\mathrm{BG}}(x)\overline{\lambda}_{\mathrm{D}}\right| \to 1$ as $n_{0e}(x) \to 0$, the Bohm-Gross wave suffer strong Landau damping in the low-density region; thus, as the $v_{\mathrm{g}} > 0$ wave propagates from the resonance region near $x = x_0$ outward to the low-density region, it will be strongly damped and the reflected wave is negligible, i.e., the $v_{\mathrm{g}} > 0$ wave is perfectly absorbed at $x = +\infty$. Now both external and internal processes that can give rise to the $v_{\mathrm{g}} < 0$ wave are ruled out, implying that there exist no Bohm-Gross wave with $v_{\mathrm{g}} < 0$ as $x \to +\infty$. This implies [48] $c_1 = \mathrm{i}$.

The corresponding asymptotic behaviors are then given by (see Figure 4.5)

$$\begin{aligned}\hat{E}(x) &= -\pi\frac{\hat{E}_0}{\left(\kappa\overline{\lambda}_{\mathrm{D}}\right)^{2/3}}\left(\mathrm{i}\ \mathrm{Ai}\left(-\frac{x-x_0}{\Delta}\right) + \mathrm{Gi}\left(-\frac{x-x_0}{\Delta}\right)\right)\\ &\sim \begin{cases} -\sqrt{\pi}\frac{\hat{E}_0}{\left(\kappa\overline{\lambda}_{\mathrm{D}}\right)^{2/3}}\left(\frac{\Delta}{x-x_0}\right)^{1/4}\exp\left(\mathrm{i}\left(\frac{2}{3}\left(\frac{x-x_0}{\Delta}\right)^{3/2} + \frac{\pi}{4}\right)\right) + \frac{\hat{E}_0}{\kappa(x-x_0)} & \text{for } x \to +\infty \\ \frac{\hat{E}_0}{\kappa(x-x_0)} + \text{exponentially decay term} & \text{for } x \to -\infty \end{cases}.\end{aligned} \tag{4.5.8}$$

Eq. (4.5.6) and Eq. (4.5.8) indicate that introducing the microscopic dynamics via finite $\overline{\lambda}_{\mathrm{D}}$ removes the singularity at $x = x_0$ even without dissipation; the field amplitude for $|x - x_0| = \mathcal{O}(\Delta)$ is, nonetheless, still significantly amplified $\left|\hat{E}/\hat{E}_0\right| = \mathcal{O}\left(\left|\kappa\overline{\lambda}_{\mathrm{D}}\right|^{-2/3}\right) \gg 1$ — this is, sometimes called, the *Airy amplification (swelling)*. Furthermore, comparing to the cold plasma result, Eq. (4.5.8) contains an additional term corresponding to an Airy-like warm plasma Bohm-Gross wave propagating away from the resonance layer toward the low-density region, which is converted from the homogeneous solution of the incident wave. This process of *exciting short wavelength modes* (here the Bohm-Gross waves) *due to the resonance of long wavelength mdoes* (here the resonance is between the incident wave and the electron plasma oscillation) is called *linear mode conversion* (or *transformation**transition* in certain literature). We remark that, as long as the wave energy carried by the $v_{\mathrm{g}} > 0$ Bohm-Gross wave is perfectly absorbed

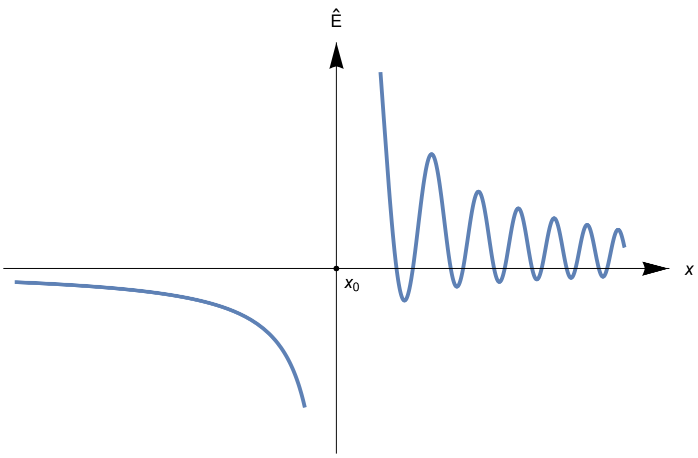


Figure 4.5: Linear mode conversion of Tonks-Dattner model, with no exponentially divergent solution in $x < x_0$ and perfect absorption at $x \to +\infty$.

via electron Landau damping, the rate of resonant energy absorption is the same as that given by the cold plasma theory, i.e., Eq. (4.4.21).

The above example of linear mode conversion originates from thermal effects, where the Debye length $\overline{\lambda}_{\mathrm{D}}$ serves as the microscopic spatial scale (Airy scale length $\Delta = \left(\overline{\lambda}_{\mathrm{D}}^2/\kappa\right)^{1/3}$). In addition, kinetic effects — such as the FLR effect, in which the gyroradius ρ_{t} acts as the microscopic spatial scale (Airy scale length $\Delta = \mathcal{O}\left(\left(\rho_{\mathrm{t}}^2/\kappa\right)^{1/3}\right)$ — can also give rise to linear mode conversion. In magnetized plasmas, for example,

- an incident wave that is resonant with the MHD shear Alfvén wave could be converted to the kinetic shear Alfvén wave [49];
- a resonance with the magnetized electron plasma (Trivelpiece-Gould) wave [34] converts the incident wave to the ion Bernstein wave (IBW) [35], [49].

We now briefly discuss the latter example, which concerns with the linear mode conversion of IBW. For this case, we consider a *slab* plasma in the $x-z$ plane with $\boldsymbol{B}_0 = B_0\hat{\boldsymbol{e}}_z$ and $n_0 = n_0(x)$; without loss of generality, we take $\mathrm{d}n_0/\mathrm{d}x > 0$. Meanwhile, the external driving frequency $\omega_0 = \mathcal{O}(\Omega_i)$; typically, $\omega_0 \lesssim 2\Omega_i$.

- Let's first consider the wave propagation characteristics using the cold plasma model. Here, the corresponding electrostatic wave equation is the following

$$\left(\frac{\partial}{\partial x}\left(\epsilon_{\perp\mathrm{c}}(x)\frac{\partial}{\partial x}\right) + \frac{\partial}{\partial z}\left(\epsilon_{\parallel}(x)\frac{\partial}{\partial z}\right)\right)\phi(x,z) = 0, \tag{4.5.9}$$

where (see Figure 3.1)

$$\epsilon_{\perp\mathrm{c}}(x) = 1 + \frac{\omega_{\mathrm{pe}}^2(x)}{\Omega_e^2} - \frac{\omega_{\mathrm{p}i}^2(x)}{\omega_0^2 - \Omega_i^2}, \tag{4.5.10}$$

$$\epsilon_{\parallel}(x) = 1 - \frac{\omega_{\mathrm{pe}}^2(x)}{\omega_0^2}. \tag{4.5.11}$$

In the eikonal representation, Eq. (4.5.9) then yields

$$k^2(x) = -k_{\parallel}^2\frac{\epsilon_{\parallel}(x)}{\epsilon_{\perp\mathrm{c}}(x)}. \tag{4.5.12}$$

  - For very low densities, we have $\omega_{\mathrm{pe}}^2(x) \ll \Omega_e^2$ and $\omega_{\mathrm{p}i}^2(x) \ll \Omega_i^2$, i.e., $\epsilon_{\perp\mathrm{c}}(x) \approx 1$ and, hence,

$$k^2(x) \approx -k_{\parallel}^2\epsilon_{\parallel}(x). \tag{4.5.13}$$

Noting Eq. (4.5.11), Eq. (4.5.13) predicts the existence of a cutoff at $x = x_{\mathrm{c}}$ where $\omega_{\mathrm{pe}}^2(x_{\mathrm{c}}) = \omega_0^2$. Since for $x \ll x_{\mathrm{c}}$ we have $\epsilon_{\parallel}(x) \approx 1$ and $k^2(x) \approx -k_{\parallel}^2$, the wave has to tunnel into the plasma; this tunneling, however, has little effect because typically $\left|k_{\parallel}x_{\mathrm{c}}\right| \ll 1$. As we move beyong $x = x_{\mathrm{c}}$ such that $\epsilon_{\parallel}(x) \approx -\omega_{\mathrm{pe}}^2(x)/\omega_0^2$, we find the magnetized electron plasma (Trivelpiece-Gould) wave

$$k^2(x) \approx \frac{\omega_{\rm pe}^2(x)}{\omega_0^2} k_\parallel^2. \tag{4.5.14}$$

Moving further into the plasma, we have $\omega_{\rm p}i \sim \omega_0$ and, hence, the possibility of a resonance at $x = x_{\rm r}$ where $\epsilon_{\perp \rm c}(x_{\rm r}) = 0$. Noting that, typically, this resonance is located near the plasma edge and $\omega_0 \lesssim 2\Omega_i$, the resonance condition can be approximated as

$$\omega_{\rm p}^2{}_i \approx \omega_0^2 - \Omega_i \lesssim 3\Omega_i^2. \tag{4.5.15}$$

- We now consider the associated linear mode conversion process. Since $\omega_0 = \mathcal{O}(\Omega_i)$ and $\rho_{{\rm t}i} \gg \rho_{{\rm t}e}$, ion Larmor radius is the relevant microscopic spatial scale. Anticipating the Airy scale length of the order $\mathcal{O}\left(\left(\rho_{{\rm t}i}^2/\kappa\right)^{1/3}\right) \gg \rho_{{\rm t}i}$, we may take the small $\rho_{{\rm t}i}$ expansion. Within the eikonal approximation, we find from Eq. (3.5.7) that

$$\epsilon_\perp(x) \approx \epsilon_{\perp \rm c}(x) - \frac{\omega_{{\rm p}i}^2(x)}{\omega_0^2 - 4\Omega_i^2} k^2(x) \rho_{{\rm t}i}^2. \tag{4.5.16}$$

Near the cold resonance point $x = x_{\rm r}$, we now find

$$\epsilon_\perp(x) \approx \frac{\omega_{{\rm p}i}^2(x)}{4\Omega_i^2 - \omega_0^2} k^2(x) \rho_{{\rm t}i}^2 \tag{4.5.17}$$

and, hence,

$$\begin{aligned} k^2(x) &\approx \frac{\omega_{\rm pe}^2(x)}{\omega_0^2} k_\parallel^2 \frac{4\Omega_i^2 - \omega_0^2}{\omega_{{\rm p}i}^2(x)} \frac{1}{k^2(x)\rho_{{\rm t}i}^2} \\ \Rightarrow k^4(x) &\approx \frac{\omega_{\rm pe}^2(x)}{\omega_{{\rm p}i}^2(x)} \frac{4\Omega_i^2 - \omega_0^2}{\omega_0^2} \frac{k_\parallel^2}{\rho_{{\rm t}i}^2} > 0. \end{aligned} \tag{4.5.18}$$

This shows that, for $\omega_0^2 \lesssim 4\Omega_i^2$, the mode-converted short-wavelength IBW could propagate into the high-density region. This has the desirable feature that the wave energy will be cyclotron damped by ions in the bulk plasma. Figure 4.6 is a sketch of the WKB wavenumber k^2 v.s. x.

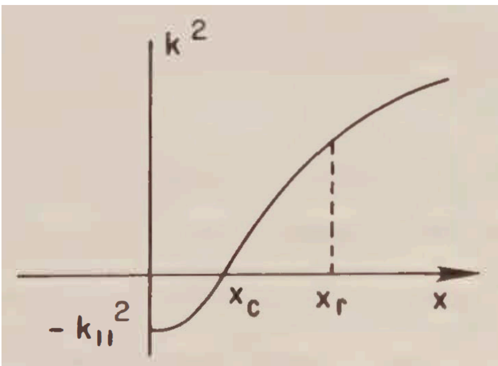


Figure 4.6: Sketch of $k^2 - x$ for the linear mode conversion of IBW.

4.6 Linear gyrokinetic equation in nonuniform plasmas

The main references of this section are [50], [51], [52] and [53].

In this section, we shall generalize the derivation of the linear gyrokinetic (GK) equation in uniform plasmas (Section 3.7) to nonuniform plasmas. The crucial assumption is that the nonuniformities are sufficiently weak such that $\rho_{\mathrm{t}}/L_0 = \mathcal{O}(\varepsilon) \ll 1$ with L_0 being the inhomogeneity scale length and ε the smallness parameter. In practice, it is convenient to make the following *book-keeping* [10], [22] substitution

$$\Omega \mapsto \varepsilon^{-1}\Omega, \tag{4.6.1}$$

after which all the quantities appearing explicitly in the equations (e.g., Ω and $\nabla_{\boldsymbol{X}}$) are of order $\mathcal{O}(1)$.

With $\varepsilon \ll 1$, it can be expected that, to the lowest order, the unperturbed particle orbit corresponds to that of the guiding-center. Thus, we may adopt the same guiding-center transformation as in Section 3.1, which we recall to be

$$\begin{aligned} \boldsymbol{X} &= \boldsymbol{x} + \varepsilon \boldsymbol{v} \times \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}, \\ \boldsymbol{V} &= (\mathcal{E}, \mu, \alpha, \sigma). \end{aligned} \tag{4.6.2}$$

The complexity enters through the spatial dependences of $\boldsymbol{B}_0$ and, hence[54], $\hat{\boldsymbol{e}}_\parallel, \hat{\boldsymbol{e}}_{\perp 1}, \hat{\boldsymbol{e}}_{\perp 2}, v_\parallel, v_\perp$ and α. The corresponding transformation relations of derivatives are then given by[55]

$$\nabla_{\boldsymbol{x}} \mapsto \nabla_{\boldsymbol{X}} + \varepsilon \boldsymbol{\lambda}_{B1} + \boldsymbol{\lambda}_{B2} \tag{4.6.3}$$

and

$$\nabla_{\boldsymbol{v}} \mapsto \boldsymbol{v}\frac{\partial}{\partial \mathcal{E}} + \frac{\boldsymbol{v}_\perp}{B_0}\frac{\partial}{\partial \mu} + \frac{\hat{\boldsymbol{e}}_\alpha}{v_\perp}\frac{\partial}{\partial \alpha} + \varepsilon \frac{\hat{\boldsymbol{e}}_\parallel}{\Omega} \times \nabla_{\boldsymbol{X}}, \tag{4.6.4}$$

where

$$\begin{aligned} \lambda_{B1i} &= \frac{\partial}{\partial x_i}\left(\varepsilon_{jkl} v_k \frac{\left(\hat{\boldsymbol{e}}_\parallel\right)_l}{\Omega}\right)\frac{\partial}{\partial X_j} \\ &\overset{\partial v_k/\partial x_i = 0}{=\!=\!=} \varepsilon_{jkl} v_k \frac{1}{\Omega}\frac{\partial \left(\hat{\boldsymbol{e}}_\parallel\right)_l}{\partial x_i}\frac{\partial}{\partial X_j} - \varepsilon_{jkl} v_k \frac{\left(\hat{\boldsymbol{e}}_\parallel\right)_l}{\Omega}\frac{\partial \ln B_0}{\partial x_i}\frac{\partial}{\partial X_j} \\ &= \varepsilon_{jkl} v_k \frac{1}{\Omega}\left.\frac{\partial \left(\hat{\boldsymbol{e}}_\parallel\right)_l}{\partial x_i}\right|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j} - \varepsilon_{jkl} v_k \left.\frac{\left(\hat{\boldsymbol{e}}_\parallel\right)_l}{\Omega}\frac{\partial \ln B_0}{\partial x_i}\right|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j} + \mathcal{O}(\varepsilon), \end{aligned} \tag{4.6.5}$$

[54] That $\hat{\boldsymbol{e}}_{\perp 1}$ and $\hat{\boldsymbol{e}}_{\perp 2}$ should depend on $\boldsymbol{x}$ is obvious. Now since the velocity (as a phase-space variable) $\boldsymbol{v} = \hat{\boldsymbol{e}}_\parallel v_\parallel + v_\perp(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha)$ is independent of $\boldsymbol{x}$, it then follows that $v_\parallel, v_\perp$ and α must also depend on $\boldsymbol{x}$.

[55] Without loss of generality and for simplicity, let's work in a Cartesian coordinate system $\{x^i \equiv x_i\}$ in $\mathbb{R}^3$, with the basis vectors being $\{\boldsymbol{e}_i \equiv \boldsymbol{e}^i\}$. Then by definition, $\nabla_{\boldsymbol{x}} \coloneqq \boldsymbol{e}_i\partial/\partial x_i$, $\nabla_{\boldsymbol{X}} \coloneqq \boldsymbol{e}_i\partial/\partial X_i$ and $\nabla_{\boldsymbol{v}} \coloneqq \boldsymbol{e}_i\partial/\partial v_i$.

$$\lambda_{B2i} = \frac{\partial\mu}{\partial x_i}\frac{\partial}{\partial\mu} + \frac{\partial\alpha}{\partial x_i}\frac{\partial}{\partial\alpha} \tag{4.6.6}$$

with

$$\begin{aligned}\frac{\partial\mu}{\partial x_i} &= -\frac{v_\perp^2}{2B_0^2}\frac{\partial B_0}{\partial x_i} + \frac{v_\perp}{B_0}\frac{\partial v_\perp}{\partial x_i} \\ &\overset{\partial v^2/\partial x_i=0}{\underset{v_\parallel=\hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{v}}{=\!=\!=}} -\frac{\mu}{B_0}\frac{\partial B_0}{\partial x_i} - \frac{v_\parallel}{B_0}\frac{\partial}{\partial x_i}\left(\hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{v}\right) \\ &\overset{\partial \boldsymbol{v}/\partial x_i=\boldsymbol{0}}{\underset{\partial\hat{\boldsymbol{e}}_\parallel/\partial x_i\cdot\hat{\boldsymbol{e}}_\parallel=0}{=\!=\!=}} -\frac{\mu}{B_0}\frac{\partial B_0}{\partial x_i} - \frac{v_\parallel}{B_0}\frac{\partial\hat{\boldsymbol{e}}_\parallel}{\partial x_i}\cdot\boldsymbol{v}_\perp \\ &= -\frac{\mu}{B_0}\frac{\partial B_0}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}} - \frac{v_\parallel}{B_0}\boldsymbol{v}_\perp\cdot\frac{\partial\hat{\boldsymbol{e}}_\parallel}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}} + \mathcal{O}(\varepsilon)\end{aligned} \tag{4.6.7}$$

and[56]

$$\frac{\partial\alpha}{\partial x_i} = \frac{\partial\hat{\boldsymbol{e}}_{\perp2}}{\partial x_i}\cdot\hat{\boldsymbol{e}}_{\perp1} - \frac{v_\parallel}{v_\perp}\frac{\partial\hat{\boldsymbol{e}}_\parallel}{\partial x_i}\cdot\hat{\boldsymbol{e}}_\alpha. \tag{4.6.10}$$

The unperturbed Vlasov operator then becomes

$$\begin{aligned}\mathcal{L}_0 &= \frac{\partial}{\partial t} + \boldsymbol{v}\cdot\nabla + \Omega\left(\boldsymbol{v}\times\hat{\boldsymbol{e}}_\parallel\right)\cdot\nabla_{\boldsymbol{v}} \\ &\mapsto \mathcal{L}_{\rm g} \equiv \frac{\partial}{\partial t} + v_\parallel\hat{\boldsymbol{e}}_\parallel\cdot\nabla_{\boldsymbol{X}} + \boldsymbol{v}\cdot(\varepsilon\boldsymbol{\lambda}_{B1} + \boldsymbol{\lambda}_{B2}) - \varepsilon^{-1}\Omega\frac{\partial}{\partial\alpha},\end{aligned} \tag{4.6.11}$$

where $\hat{\boldsymbol{e}}_\parallel\cdot\nabla_{\boldsymbol{X}}$ is directional derivative with respective to $\boldsymbol{X}$ along the $\hat{\boldsymbol{e}}_\parallel$ direction (often denoted by $\partial/\partial\hat{\boldsymbol{e}}_\parallel$). Note that generally speaking, $\partial/\partial\hat{\boldsymbol{e}}_\parallel$ is not a derivative with respect to $X_\parallel$; this is because $X_\parallel := \boldsymbol{X}\cdot\hat{\boldsymbol{e}}_\parallel$ is not a coordinate for $\boldsymbol{B}_0 \neq \mathbf{const}$.

Consider the equilibrium distribution function, $f_{0\rm g} = f_{0\rm g}(\boldsymbol{X}, \mathcal{E}, \mu, \alpha, \sigma)$, which obeys

$$\left(v_\parallel\hat{\boldsymbol{e}}_\parallel\cdot\nabla_{\boldsymbol{X}} + \boldsymbol{v}\cdot(\varepsilon\boldsymbol{\lambda}_{B1} + \boldsymbol{\lambda}_{B2}) - \varepsilon^{-1}\Omega\frac{\partial}{\partial\alpha}\right)f_{0\rm g} = 0. \tag{4.6.12}$$

- At order $\mathcal{O}(\varepsilon^{-1})$, we have $\partial f_{0\rm g}/\partial\alpha = 0 \Rightarrow f_{0\rm g} = f_{0\rm g}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma)$. For the sake of simplicity, we shall further assume $f_{0\rm g}$ to be isotropic in velocity space, so that $f_{0\rm g} = f_{0\rm g}(\boldsymbol{X}, \mathcal{E})$.

[56]This follows from

$$\begin{aligned}\boldsymbol{0} = \frac{\partial\boldsymbol{v}}{\partial x_i} &= \frac{\partial}{\partial x_i}\left(\hat{\boldsymbol{e}}_\parallel v_\parallel\right) + \frac{\partial v_\perp}{\partial x_i}(\hat{\boldsymbol{e}}_{\perp1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp2}\sin\alpha) \\ &\quad + v_\perp\frac{\partial\alpha}{\partial x_i}(-\hat{\boldsymbol{e}}_{\perp1}\sin\alpha + \hat{\boldsymbol{e}}_{\perp2}\cos\alpha) + v_\perp\left(\frac{\partial\hat{\boldsymbol{e}}_{\perp1}}{\partial x_i}\cos\alpha + \frac{\partial\hat{\boldsymbol{e}}_{\perp2}}{\partial x_i}\sin\alpha\right)\end{aligned} \tag{4.6.8}$$

and

$$\begin{aligned}&\hat{\boldsymbol{e}}_{\perp\alpha}\cdot\hat{\boldsymbol{e}}_{\perp\beta} = \delta_{\alpha\beta}, \quad \alpha,\beta = 1,2 \\ &\Rightarrow \frac{\partial\hat{\boldsymbol{e}}_{\perp\alpha}}{\partial x_k}\cdot\hat{\boldsymbol{e}}_{\perp\beta} = -\frac{\partial\hat{\boldsymbol{e}}_{\perp\beta}}{\partial x_k}\cdot\hat{\boldsymbol{e}}_{\perp\alpha}.\end{aligned} \tag{4.6.9}$$

- At order $\mathcal{O}(1)$, we obtain $v_\parallel \, \partial f_{0\mathrm{g}}/\partial \hat{\boldsymbol{e}}_\parallel = \Omega \, \partial f_{0\mathrm{g}}^{(1)}/\partial\alpha$, whose gyroaverage yields $\partial f_{0\mathrm{g}}/\partial \hat{\boldsymbol{e}}_\parallel = 0$; note, again, that this does not mean $f_{0\mathrm{g}} = f_{0\mathrm{g}}(\boldsymbol{X}_\perp, \mathcal{E})$ unless $\boldsymbol{B}_0 = \mathbf{const}$.

As for the perturbations, we shall only consider the electrostatic limit. The linearized Vlasov equation is, noting Eq. (3.2.4),

$$\begin{aligned} \mathcal{L}_\mathrm{g} f_{1\mathrm{g}} &= \frac{q}{m}\nabla_{\boldsymbol{x}}\varphi \cdot \left(\boldsymbol{v}\frac{\partial}{\partial\mathcal{E}} + \varepsilon\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega} \times \nabla_{\boldsymbol{X}}\right) f_{0\mathrm{g}} \\ &= \frac{q}{m}(\boldsymbol{v}\cdot\nabla_{\boldsymbol{X}} + \boldsymbol{v}\cdot(\varepsilon\boldsymbol{\lambda}_{B1} + \boldsymbol{\lambda}_{B2}))\varphi_\mathrm{g}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + \varepsilon\frac{q}{m}\nabla_{\boldsymbol{x}}\varphi\cdot\left(\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\nabla_{\boldsymbol{X}} f_{0\mathrm{g}}\right) \\ &= \frac{q}{m}\left(\mathcal{L}_\mathrm{g} - \frac{\partial}{\partial t}\right)\varphi_\mathrm{g}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + \varepsilon\frac{q}{m}\left(\nabla_{\boldsymbol{x}}\varphi\times\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\right)\cdot\nabla_{\boldsymbol{X}} f_{0\mathrm{g}}. \end{aligned} \tag{4.6.13}$$

Letting (recall Eq. (3.2.15))

$$f_{1\mathrm{g}} = \frac{q}{m}\varphi_\mathrm{g}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + h_\mathrm{g}, \tag{4.6.14}$$

we find

$$\begin{aligned} \mathcal{L}_\mathrm{g} h_\mathrm{g} &= \mathcal{L}_\mathrm{g} f_{1\mathrm{g}} - \frac{q}{m}\mathcal{L}_\mathrm{g}\varphi_\mathrm{g}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} \\ &= -\frac{q}{m}\left(\frac{\partial\varphi_\mathrm{g}}{\partial t}\frac{\partial}{\partial\mathcal{E}} - \varepsilon\left(\nabla_{\boldsymbol{x}}\varphi\times\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\right)\cdot\nabla_{\boldsymbol{X}}\right) f_{0\mathrm{g}}. \end{aligned} \tag{4.6.15}$$

Expanding $h_\mathrm{g} = h_{\mathrm{g}0} + \varepsilon h_{\mathrm{g}1} + \mathcal{O}(\varepsilon^2)$, then

- at order $\mathcal{O}(\varepsilon^{-1})$, we have $\partial h_{\mathrm{g}0}/\partial\alpha = 0 \Rightarrow h_{\mathrm{g}0} = h_{\mathrm{g}0}(\boldsymbol{X}, \mathcal{E}, \mu, \sigma)$;
- at order $\mathcal{O}(1)$, but with the $\mathcal{O}(\varepsilon)$ terms correct to their lowest order so as to retain the effects of inhomogeneities, we find

$$\begin{aligned} \mathcal{L}_\mathrm{g} h_{\mathrm{g}0} - \Omega\frac{\partial h_{\mathrm{g}1}}{\partial\alpha} &= -\frac{q}{m}\left(\frac{\partial\varphi_\mathrm{g}}{\partial t}\frac{\partial}{\partial\mathcal{E}} - \varepsilon\left(\nabla_{\boldsymbol{x}}\varphi\times\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\right)\Bigg|_{\boldsymbol{x}=\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\right) f_{0\mathrm{g}} \\ &\equiv \text{R.H.S.} \end{aligned} \tag{4.6.16}$$

  Noting that $\varphi(\boldsymbol{X}) = \varphi(\boldsymbol{x} + \mathcal{O}(\varepsilon)) = \varphi(\boldsymbol{x}) + \mathcal{O}(\varepsilon) = \varphi_\mathrm{g}(\boldsymbol{X}) + \mathcal{O}(\varepsilon)$, we have

$$\nabla_{\boldsymbol{x}}\varphi|_{\boldsymbol{x}=\boldsymbol{X}} = \nabla_{\boldsymbol{X}}\varphi(\boldsymbol{X}) = \nabla_{\boldsymbol{X}}\varphi_\mathrm{g} + \mathcal{O}(\varepsilon), \tag{4.6.17}$$

  and thus

$$\text{R.H.S.} = -\frac{q}{m}\left(\frac{\partial\varphi_\mathrm{g}}{\partial t}\frac{\partial}{\partial\mathcal{E}} - \varepsilon\left(\nabla_{\boldsymbol{X}}\varphi_\mathrm{g}\times\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\Bigg|_{\boldsymbol{x}=\boldsymbol{X}}\right)\cdot\nabla_{\boldsymbol{X}}\right) f_{0\mathrm{g}} + \mathcal{O}(\varepsilon^2). \tag{4.6.18}$$

Averaging over the gyrophase, Eq. (4.6.16) yields, noting that[57] $\langle\boldsymbol{v}\cdot\boldsymbol{\lambda}_{B1}\rangle_0 = \left(u_\parallel\hat{\boldsymbol{b}} + \boldsymbol{v}_\mathrm{d}\right)\cdot\nabla_{\boldsymbol{X}} + \mathcal{O}(\varepsilon)$ and $\langle\boldsymbol{v}\cdot\nabla_{\boldsymbol{x}}\mu\rangle_0 = \mathcal{O}(\varepsilon)$,

[57] Denoting $\hat{\boldsymbol{e}}_\parallel \equiv \hat{\boldsymbol{b}}$, let's prove $\langle\boldsymbol{v}\cdot\boldsymbol{\lambda}_{B1}\rangle_0 = \left(u_\parallel\hat{\boldsymbol{b}} + \boldsymbol{v}_\mathrm{d}\right)\cdot\nabla_{\boldsymbol{X}} + \mathcal{O}(\varepsilon)$ and $\langle\boldsymbol{v}\cdot\nabla_{\boldsymbol{x}}\mu\rangle_0 = \mathcal{O}(\varepsilon)$.

$$\left(\frac{\partial}{\partial t}+\left(\left(v_\parallel+\varepsilon u_\parallel\right)\hat{\boldsymbol{e}}_\parallel+\varepsilon \boldsymbol{v}_{\mathrm{d}}\right)\cdot\nabla_{\boldsymbol{X}}\right)h_{\mathrm{g}0}=\langle\text{R.H.S.}\rangle. \tag{4.6.27}$$

Here, $u_\parallel=(v_\perp^2/2\Omega)\hat{\boldsymbol{e}}_\parallel\cdot\left(\nabla_{\boldsymbol{x}}\times\hat{\boldsymbol{e}}_\parallel\right)\big|_{\boldsymbol{x}=\boldsymbol{X}}$ is the first-order parallel drift (called Baños drift [10], which is unimportant since it is a small correction to a type of motion that already exists at order $\mathcal{O}(1)$), and

$$\boldsymbol{v}_{\mathrm{d}}=\frac{\hat{\boldsymbol{e}}_\parallel}{\Omega}\times\left(\frac{v_\perp^2}{2}\nabla_{\boldsymbol{x}}\ln B_0+v_\parallel^2\hat{\boldsymbol{e}}_\parallel\cdot\nabla_{\boldsymbol{x}}\hat{\boldsymbol{e}}_\parallel\right)\Bigg|_{\boldsymbol{x}=\boldsymbol{X}} \tag{4.6.28}$$

---

(1)
$$\langle\boldsymbol{v}\cdot\boldsymbol{\lambda}_{B1}\rangle_0=\varepsilon_{jkl}\langle v_iv_k\rangle_0\frac{1}{\Omega}\frac{\partial b_l}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}-\varepsilon_{jkl}\langle v_iv_k\rangle_0\frac{b_l}{\Omega}\frac{\partial\ln B_0}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}+\mathcal{O}(\varepsilon). \tag{4.6.19}$$

Noting that

$$\begin{aligned}\boldsymbol{v}&=v_\parallel\hat{\boldsymbol{b}}+\frac{v_\perp}{\sqrt{2}}\sum_{\lambda=\pm}\hat{\boldsymbol{e}}_\lambda\mathrm{e}^{-\mathrm{i}\lambda\alpha}\\ \Rightarrow\langle v_iv_k\rangle_0&=v_\parallel^2b_ib_k+\frac{v_\perp^2}{2}(\delta_{ik}-b_ib_k),\end{aligned} \tag{4.6.20}$$

we have

(i)
$$\begin{aligned}\varepsilon_{jkl}b_ib_k\frac{v_\parallel^2}{\Omega}\frac{\partial b_l}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}&=\frac{v_\parallel^2}{\Omega}\hat{\boldsymbol{b}}\times\left(\hat{\boldsymbol{b}}\cdot\nabla_{\boldsymbol{x}}\hat{\boldsymbol{b}}\right)\big|_{\boldsymbol{x}=\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\\ &=\text{curvature drift}\cdot\nabla_{\boldsymbol{X}};\end{aligned} \tag{4.6.21}$$

(ii)
$$\begin{aligned}\varepsilon_{jkl}\frac{v_\perp^2}{2}(\delta_{ik}-b_ib_k)\frac{1}{\Omega}\frac{\partial b_l}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}&=\frac{v_\perp^2}{2\Omega}\left(\nabla_{\boldsymbol{x}}\times\hat{\boldsymbol{b}}-\hat{\boldsymbol{b}}\times\left(\hat{\boldsymbol{b}}\cdot\nabla_{\boldsymbol{x}}\hat{\boldsymbol{b}}\right)\right)\big|_{\boldsymbol{x}=\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\\ &=\frac{v_\perp^2}{2\Omega}\hat{\boldsymbol{b}}\left(\hat{\boldsymbol{b}}\cdot\left(\nabla_{\boldsymbol{x}}\times\hat{\boldsymbol{b}}\right)\right)\big|_{\boldsymbol{x}=\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\\ &=\text{Baños drift}\cdot\nabla_{\boldsymbol{X}};\end{aligned} \tag{4.6.22}$$

(iii)
$$\begin{aligned}-\varepsilon_{jkl}\frac{v_\perp^2}{2}\delta_{ik}\frac{b_l}{\Omega}\frac{\partial\ln B_0}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}&=\varepsilon_{lij}\frac{v_\perp^2}{2\Omega}b_l\frac{\partial\ln B_0}{\partial x_i}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\frac{\partial}{\partial X_j}\\ &=\frac{v_\perp^2}{2\Omega}\hat{\boldsymbol{b}}\times\nabla_{\boldsymbol{x}}\ln B_0\big|_{\boldsymbol{x}=\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\\ &=\text{gradient drift}\cdot\nabla_{\boldsymbol{X}}.\end{aligned} \tag{4.6.23}$$

(2)
$$\langle\boldsymbol{v}\cdot\nabla_{\boldsymbol{x}}\mu\rangle_0=-\langle\boldsymbol{v}\rangle\cdot\frac{\mu}{B_0}\nabla_{\boldsymbol{x}}B_0\bigg|_{\boldsymbol{x}=\boldsymbol{X}}-\frac{v_\parallel}{B_0}\langle\boldsymbol{v}_\perp\boldsymbol{v}\rangle_0\cdot\cdot\nabla_{\boldsymbol{x}}\hat{\boldsymbol{b}}\big|_{\boldsymbol{x}=\boldsymbol{X}}+\mathcal{O}(\varepsilon). \tag{4.6.24}$$

That the order $\mathcal{O}(1)$ terms cancel out follows from

$$\begin{aligned}&\langle\boldsymbol{v}\rangle=v_\parallel\hat{\boldsymbol{b}},\\ &\langle\boldsymbol{v}_\perp\boldsymbol{v}\rangle_0=\left\langle\boldsymbol{v}_\perp\left(v_\parallel\hat{\boldsymbol{b}}+\boldsymbol{v}_\perp\right)\right\rangle_0=\langle\boldsymbol{v}_\perp\boldsymbol{v}_\perp\rangle=\frac{v_\perp^2}{2}\left(\tilde{g}-\hat{\boldsymbol{b}}\hat{\boldsymbol{b}}\right),\\ &\tilde{g}\cdot\cdot\nabla_{\boldsymbol{x}}\hat{\boldsymbol{b}}=\nabla_{\boldsymbol{x}}\cdot\hat{\boldsymbol{b}}\end{aligned} \tag{4.6.25}$$

and

$$\begin{aligned}0=\nabla_{\boldsymbol{x}}\cdot\boldsymbol{B}_0&=\nabla_{\boldsymbol{x}}\cdot\left(\hat{\boldsymbol{b}}B_0\right)=B_0\nabla_{\boldsymbol{x}}\cdot\hat{\boldsymbol{b}}+\hat{\boldsymbol{b}}\cdot\nabla_{\boldsymbol{x}}B_0\\ \Rightarrow\nabla_{\boldsymbol{x}}\cdot\hat{\boldsymbol{b}}&=-\frac{\hat{\boldsymbol{b}}}{B_0}\cdot\nabla_{\boldsymbol{x}}B_0.\end{aligned} \tag{4.6.26}$$

Direction calculation shows (if the retypesetter does not make any mistake) that the order $\mathcal{O}(\varepsilon)$ terms do not cancel out, so the relation $\langle\boldsymbol{v}\cdot\nabla_{\boldsymbol{x}}\mu\rangle_0=\mathcal{O}(\varepsilon)$ can not be lifted to $\langle\boldsymbol{v}\cdot\nabla_{\boldsymbol{x}}\mu\rangle_0=\mathcal{O}(\varepsilon^2)$.

corresponds to the gradient and curvature drifts of the guiding-center; note that $\boldsymbol{v}_{\mathrm{d}} \perp \hat{\boldsymbol{e}}_{\parallel}\big|_{\boldsymbol{x}=\boldsymbol{X}}$. Setting $\varepsilon = 1$ and ignoring the Baños drift, we arrive at the following user-friendly equation

$$\left(\frac{\partial}{\partial t} + \left(v_{\parallel}\hat{\boldsymbol{e}}_{\parallel} + \boldsymbol{v}_{\mathrm{d}}\right)\cdot\nabla_{\boldsymbol{X}}\right)h_{\mathrm{g}0} = -\frac{q}{m}\left(\frac{\partial\langle\varphi_{\mathrm{g}}\rangle_0}{\partial t}\frac{\partial}{\partial\mathcal{E}} - \nabla_{\boldsymbol{X}}\langle\varphi_{\mathrm{g}}\rangle_0\cdot\left(\frac{\hat{\boldsymbol{e}}_{\parallel}}{\Omega}\bigg|_{\boldsymbol{x}=\boldsymbol{X}}\times\nabla_{\boldsymbol{X}}\right)\right)f_{0\mathrm{g}}, \tag{4.6.29}$$

which is the desired electrostatic linear GK equation valid for nonuniform magnetized plasmas.

Let's go further with the eikonal approximation

$$\begin{aligned} f_{1\mathrm{g}} &= \hat{f}_{1\mathrm{g}}(\boldsymbol{X})\exp\left(\mathrm{i}\left(-\omega t + \int^{\boldsymbol{X}}\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{k}\right)\right),\\ \varphi_{\mathrm{g}} &= \hat{\varphi}_{\mathrm{g}}(\boldsymbol{X})\exp\left(\mathrm{i}\left(-\omega t + \int^{\boldsymbol{X}}\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{k}\right)\right). \end{aligned} \tag{4.6.30}$$

Eq. (4.6.29) then becomes, invoking Eq. (3.2.22),

$$\left(\omega - k_{\parallel}v_{\parallel} - \boldsymbol{k}_{\perp}\cdot\boldsymbol{v}_{\mathrm{d}}\right)\hat{h}_{\mathrm{g}0} = \frac{q}{T}\hat{\varphi}J_0(\lambda)(\hat{\omega} - \omega_*)f_{0\mathrm{g}}, \tag{4.6.31}$$

where $\lambda = k_{\perp}v_{\perp}/\Omega, \hat{\omega} = -(T/m)\,\omega\,\partial/\partial\mathcal{E}, \nabla_{\boldsymbol{X}}f_{0\mathrm{g}} \equiv -\boldsymbol{\kappa}f_{0\mathrm{g}}$ and

$$\omega_* = \boldsymbol{k}_{\perp}\cdot\left(\frac{T}{m\Omega}\boldsymbol{\kappa}\times\hat{\boldsymbol{e}}_{\parallel}\right) = \boldsymbol{k}_{\perp}\cdot\left(\frac{Tc}{qB_0}\boldsymbol{\kappa}\times\hat{\boldsymbol{e}}_{\parallel}\right) \tag{4.6.32}$$

is the Doppler shift due to the diamagnetic-like drift; note that $\boldsymbol{\kappa}$ encodes all the inhomogeneities of the plasma, e.g., density and\or temperature gradients.

4.7 Drift instabilities in eikonal approximation

Here, we adopt the slab-plasma model, with the background magnetic field being uniform and along the z direction $\boldsymbol{B}_0 = B_0\hat{\boldsymbol{e}}_z$, and the density gradient along the x direction; we also assume that there is no temperature gradient[58]. Thus, the local-equilibrium distribution function could be written, in terms of the guiding-center variables[59], $f_{0\mathrm{g}}(X,\mathcal{E}) = n_0(X)\tilde{f}_{\mathrm{M}}(\mathcal{E})$. Here, $\tilde{f}_{\mathrm{M}}(\mathcal{E}) = (m/2\pi T)^{3/2}\exp(-m\mathcal{E}/T) = \left(1/\pi^{3/2}v_{\mathrm{t}}^3\right)\exp(-v^2/v_{\mathrm{t}}^2)$.

The electrostatic linear GK equation writes, under the eikonal approximation Eq. (4.6.30),

$$\left(\omega - k_{\parallel}v_{\parallel}\right)\hat{h}_{\mathrm{g}0} = \frac{q}{T}\hat{\varphi}J_0(\lambda)(\omega - \omega_*)n_0\tilde{f}_{\mathrm{M}}, \tag{4.7.1}$$

where $\omega_* = (Tc/qB_0)k_y\,\mathrm{d}\ln n_0/\mathrm{d}X$. Thus, the perturbed distribution function is given by

$$\hat{f}_1 = \frac{q}{m}\hat{\varphi}\frac{\partial f_{0\mathrm{g}}}{\partial\mathcal{E}} + \hat{h}_{\mathrm{g}0}\mathrm{e}^{\mathrm{i}\lambda\sin\alpha} \tag{4.7.2}$$

[58] Drift instabilities\turbulences caused by temperature gradients, e.g., ion temperature gradient (ITG) and electron temperature gradient (ETG) modes\turbulences, have become important frontiers in tokamak plasma physics research. [49]

[59] Since $f_{0\mathrm{g}}$ depends only on $\boldsymbol{X}_{\perp}$, for n_0 to be dependent on x we must actually have $n_0 = n_0(X)$.

$$= -\frac{q}{T}\hat{\varphi} n_0 \tilde{f}_{\mathrm{M}} \left(1 - J_0(\lambda)\mathrm{e}^{\mathrm{i}\lambda \sin\alpha}\left(1 - \frac{\omega_*}{\omega}\right)\frac{\omega}{\omega - k_\parallel v_\parallel}\right). \tag{4.7.2}$$

For electrons, we assume $b_e \ll 1$ and, hence, ignore the electron Larmor radius

$$\frac{\hat{n}_e}{n_0} = \frac{e}{T_e}\hat{\varphi}\left(1 + \left(1 - \frac{\omega_{*e}}{\omega}\right)\zeta_e Z(\zeta_e)\right); \tag{4.7.3}$$

for ions, we find

$$\frac{\hat{n}_i}{n_0} = -\frac{e}{T_i}\hat{\varphi}\left(1 + \left(1 - \frac{\omega_{*i}}{\omega}\right)A_0(b_i)\zeta_i Z(\zeta_i)\right). \tag{4.7.4}$$

Substituting $\hat{n}_e$ and $\hat{n}_i$ into the quasi-neutrality condition $\hat{n}_e = \hat{n}_i$, we obtain the following linear dispersion relation

$$1 + \left(1 - \frac{\omega_{*e}}{\omega}\right)\zeta_e Z(\zeta_e) + \tau\left(1 + \left(1 - \frac{\omega_{*i}}{\omega}\right)A_0(b_i)\zeta_i Z(\zeta_i)\right) = 0 \tag{4.7.5}$$

or

$$1 + \tau(1 - A_0(b_i)) + \left(1 - \frac{\omega_{*e}}{\omega}\right)\zeta_e Z(\zeta_e) + \frac{\omega_{*e}}{\omega}A_0(b_i)\zeta_i Z(\zeta_i) + \tau A_0(b_i)(1 + \zeta_i Z(\zeta_i)) = 0, \tag{4.7.6}$$

where $\tau := T_e/T_i$ and we have noted $\tau\omega_{*i} = -\omega_{*e}$.

Assuming $|\zeta_i| \gg 1$, Eq. (4.7.6) reduces to

$$1 + \tau(1 - A_0(b_i)) + \left(1 - \frac{\omega_{*e}}{\omega}\right)\zeta_e Z(\zeta_e) - \frac{\omega_{*e}}{\omega}A_0(b_i) - \frac{k_\parallel^2 c_{\mathrm{s}}^2}{\omega^2}A_0(b_i) = 0. \tag{4.7.7}$$

Eq. (4.7.7) shows that in nonuniform plasmas, the electrostatic ion acoustic wave is modified by the diamagnetic drift. To see that such a modification can lead to an instability, we further assume $|\zeta_e| \ll 1$ and[60] $\omega \sim \omega_{*e} \gg \left|k_\parallel c_{\mathrm{s}}\right|$, and write $\omega = \omega_{\mathrm{r}} + \mathrm{i}\gamma$ with $|\gamma/\omega_{\mathrm{r}}| \ll 1$, then an instability arises if the frequency is down shifted due to finite ion Larmor radius $\omega_{\mathrm{r}} < \omega_{*e}$

$$(\omega_{\mathrm{r}}^2 + \mathrm{i}2\gamma\omega_{\mathrm{r}})(1 + \tau(1 - A_0(b_i))) + \mathrm{i}\sqrt{\pi}\omega_{\mathrm{r}}(\omega_{\mathrm{r}} - \omega_{*e})\zeta_e - \omega_{\mathrm{r}}\omega_{*e}A_0(b_i) = 0 \tag{4.7.8}$$

$$\begin{aligned}
&\Rightarrow \omega_{\mathrm{r}} = \omega_{*e}\frac{A_0(b_i)}{1 + \tau(1 - A_0(b_i))} \overset{b_i \ll 1}{=\!=} \frac{\omega_{*e}}{1 + k_\perp^2\rho_{\mathrm{s}}^2},\\
&\frac{\gamma}{\omega_{*e}} = \frac{\sqrt{\pi}}{2}\zeta_e\frac{(1 + \tau)(1 - A_0(b_i))}{(1 + \tau(1 - A_0(b_i)))^2} \underset{\tau \gg 1}{\overset{b_i \ll 1}{=\!=}} \frac{\sqrt{\pi}}{2}\zeta_e\left(\frac{k_\perp\rho_{\mathrm{s}}}{1 + k_\perp^2\rho_{\mathrm{s}}^2}\right)^2.
\end{aligned} \tag{4.7.9}$$

Note that the instability predicted by Eq. (4.7.9) requires only the presence of density gradient, finite ion Larmor radius and electron Landau damping, which are universial ingredients in magnetically confined plasmas. This instability, therefore, is often referred to as the *universal drift instability.*

[60]Without loss of generality, we take $\omega_{*e} > 0$ or, more specifically, $k_y > 0$ and $\kappa = -\mathrm{d}\ln n_0/\mathrm{d}X > 0$.

4.8 Convective damping of drift waves in a sheared magnetic field: an eigenmode analysis

The reference of this section is [54].

It often occurs that stability analyses via eikonal approximation break down due to the presence of turning points and, hence, the associated wave reflection. In such cases, the stability analysis take the form of eigenvalue problems with the frequencies, ω, being the eigenvalues. We shall illustrate this kind of eigenmode stability analysis using the well-known example the shear damping of drift waves, i.e., the convective damping of drift waves in a magnetic field with finite shear.

Again, we adopt the slab-plasma model considered in Section 4.7. This time, however, the background magnetic field is sheared

$$\boldsymbol{B}_{\rm bg} = B_0\left(\hat{\boldsymbol{e}}_z + \hat{\boldsymbol{e}}_y \frac{x}{L_{\rm sh}}\right), \tag{4.8.1}$$

where $L_{\rm sh}$ is the shear scale length and $|x/L_{\rm sh}| \ll 1$ for our purpose. Considering perturbations of the form

$$\varphi(t, \boldsymbol{x}) = \hat{\varphi}(x)\mathrm{e}^{\mathrm{i}(k_y y - \omega t)}, \quad k_y > 0, \tag{4.8.2}$$

we then have $k_z = 0, \boldsymbol{k} = \hat{\boldsymbol{e}}_x k_x(x) + \hat{\boldsymbol{e}}_y k_y = -\mathrm{i}\hat{\boldsymbol{e}}_x \partial/\partial x + \hat{\boldsymbol{e}}_y k_y$ and, hence, $k_\parallel = \boldsymbol{k}\cdot\boldsymbol{B}_{\rm bg}/B_{\rm bg} \approx k_y x/L_{\rm sh} \equiv k_\parallel' x$ with $\mathcal{O}\big((x/L_{\rm sh})^2\big)$ corrections ignored. Since $k_\parallel = 0$ at $x = 0$, the plane $x = 0$ is called the mode-rational surface.

To further simplify the analysis, we shall assume that ions are sufficiently cold such that $T_e \gg T_i, b_i \ll 1$ and $|\zeta_i| \gg 1$; we also suppress wave-electron resonance by letting $\zeta_e \to 0$ so that we may concentrate on the shear-induced convective damping. Under these assumptions, the corresponding eigenmode equation can be readily obtained from Eq. (4.7.7); in fact, with the $\zeta_e Z(\zeta_e)$ term neglected and noting that

$$\begin{aligned} k_\perp^2 &= -\frac{\partial^2}{\partial x^2} + k_y^2 - k_\parallel^2 \approx -\frac{\partial^2}{\partial x^2} + k_y^2 \\ \Rightarrow b_i &= \frac{1}{2}k_y^2\rho_{\rm ti}^2 - \frac{1}{2}\rho_{\rm ti}^2\frac{\partial^2}{\partial x^2} \\ \Rightarrow A_0(b_i) &\approx A_0\left(\frac{1}{2}k_y^2\rho_{\rm ti}^2\right) - \frac{1}{2}\rho_{\rm ti}^2 A_0'(0)\frac{\partial^2}{\partial x^2} \approx 1 - \frac{1}{2}k_y^2\rho_{\rm ti}^2 + \frac{1}{2}\rho_{\rm ti}^2\frac{\partial^2}{\partial x^2}, \end{aligned} \tag{4.8.3}$$

we find that the eigenmodes are described by a Weber equation[61]

[61] The general form of the Weber equation is $\phi''(z) + (\mathcal{E} - z^2)\phi(z) = 0$, where $\mathcal{E}$ is a constant; bounded solutions exist only for $\mathcal{E} = 2n+1, n \in \mathbb{Z}_{\geq 0}$. [11] Recall that the time-independent Schrödinger equation for the harmonic oscillator is a Weber equation; here, however, the eigenmode equation corresponds to a particle in a parabolic potential hill or, sometimes called, anti-well (see Figure 4.7).

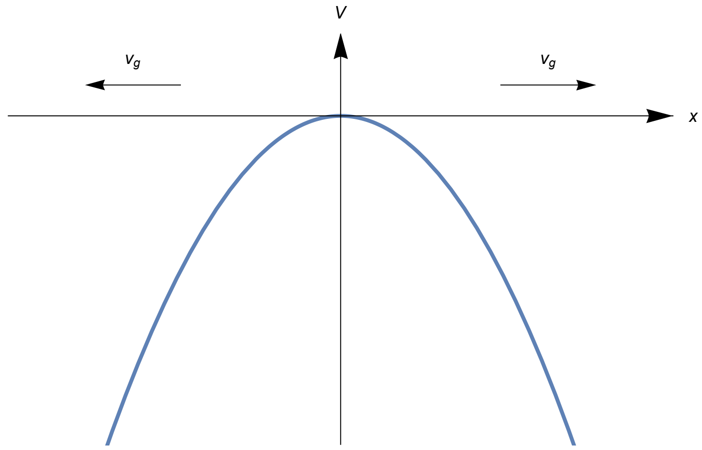


Figure 4.7: Sketch of the "anti" potential well Eq. (4.8.6) for $\omega \in \mathbb{R}$. Here, the wave energy leaks away from the system.

$$\left(\rho_s^2 \frac{\mathrm{d}^2}{\mathrm{d}x^2} + E - V(x)\right)\hat{\varphi}(x) = 0, \tag{4.8.4}$$

where

$$E = \frac{\omega_{*e}}{\omega} - 1 - k_y^2 \rho_s^2 \tag{4.8.5}$$

and

$$V(x) = -\frac{k_\parallel'^2 c_s^2}{\omega^2} x^2. \tag{4.8.6}$$

Now, as an eigenvalue problem, we need to specify the boundary conditions. The proper way to determine the boundary conditions is then to regard the eigenmode analysis as the time-asymptotic limit of an initial-value problem. In this respect, we must apply the following causality requirement

$$\lim_{|x| \to +\infty} \hat{\varphi}(x) = 0 \text{ for } \mathrm{Im}\ \omega > 0. \tag{4.8.7}$$

As $|x| \to +\infty$, the $V(x)$ term dominates and we have wave propagation, i.e., waves can propagate either into or away from the system; the corresponding WKB solutions are

$$\hat{\varphi}(x) \propto \exp\left(\pm \mathrm{i} \frac{k_\parallel' c_s}{2\omega\rho_s} x^2\right). \tag{4.8.8}$$

Imposing the causality condition Eq. (4.8.7) then rejects the solution in Eq. (4.8.8) with the plus sign; the proper boundary condition is, thus, the bounded condition

$$\hat{\varphi}(x) \propto \exp\left(-\mathrm{i} \frac{k_\parallel' c_s}{2\omega\rho_s} x^2\right) \text{ as } |x| \to +\infty. \tag{4.8.9}$$

We now examine the physical implication of the boundary condition Eq. (4.8.9) to wave propagation. In terms of WKB solutions of the form $\exp\big(\mathrm{i}\big(-\omega t + \int^x \mathrm{d}x\, k_x(x)\big)\big)$, Eq. (4.8.9) amounts to

$$k_x = -\frac{k_\parallel' c_\mathrm{s}}{\omega \rho_\mathrm{s}} x \tag{4.8.10}$$

or

$$\omega = -\frac{k_\parallel' c_\mathrm{s}}{k_x \rho_\mathrm{s}} x, \tag{4.8.11}$$

which implies

$$v_{\mathrm{g}x} = \frac{\partial \omega}{\partial k_x} = \frac{k_\parallel' c_\mathrm{s}}{k_x^2 \rho_\mathrm{s}} |x| \ \mathrm{sgn}\ x \begin{cases} > 0 \text{ for } x > 0 \\ < 0 \text{ for } x < 0 \end{cases}; \tag{4.8.12}$$

that is, the wave energy is leaking away from the system (see Figure 4.7). Looking from the initial-value problem point of view and, in the absence of any instability source feeding the waves, any initial perturbation will decay in time due to the convective wave energy leakage. This constitutes the physical picture of the shear damping.

In the following, we perform the eigenmode analysis to demonstrate this shear damping mechanism. From the asymptotic solution Eq. (4.8.9), we know that the asymptotic anti-Stokes lines are given by

$$2 \ \mathrm{Arg}\ x = \mathrm{Arg}\ \omega + k\pi, \quad k \in \mathbb{Z} \tag{4.8.13}$$

or, choosing the particular branch $\arg \omega \in (0, \pi)$ in synchronism with $\mathrm{Im}\ \omega > 0$,

$$\mathrm{Arg}\ x = \begin{cases} (\arg \omega)/2 - \pi/2,\ (\arg \omega)/2 & \text{for } \mathrm{Re}\ x > 0 \\ (\arg \omega)/2 + \pi/2,\ (\arg \omega)/2 + \pi & \text{for } \mathrm{Re}\ x < 0 \end{cases}. \tag{4.8.14}$$

Furthermore, from the causality boundary condition Eq. (4.8.9) we konw that the solution must decay along the real x axis and, hence, be subdominant in regions S_1 and S_2.

In order to simplify the eigenmode equation, we let $x = \alpha t$ and choose α such that

$$\frac{\rho_\mathrm{s}^2}{\alpha^2} = -\frac{k_\parallel'^2 c_\mathrm{s}^2}{\omega^2}\alpha^2 \Rightarrow \alpha^2 = \frac{|\omega| \rho_\mathrm{s}}{k_\parallel' c_\mathrm{s}} \mathrm{e}^{\mathrm{i}(\arg \omega \pm \pi/2)} \Rightarrow \alpha = \sqrt{\frac{|\omega| \rho_\mathrm{s}}{k_\parallel' c_\mathrm{s}}} \mathrm{e}^{\mathrm{i}((\arg \omega)/2 \pm \pi/4)}, \tag{4.8.15}$$

then the eigenmode equation Eq. (4.8.4) writes

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \frac{E\alpha^2}{\rho_\mathrm{s}^2} - t^2 \right) \hat{\varphi}(x) = 0. \tag{4.8.16}$$

With $\arg \omega \in (0, \pi)$, we have $(\arg \omega)/2 - \pi/4 \in (-\pi/4, \pi/4)$ and $(\arg \omega)/2 + \pi/4 \in (\pi/4, 3\pi/4)$. Now the real t axis (defined by the equation $\mathrm{Im}\ t = 0$) should lie inside the subdominant regions S_1 and S_2, so we must choose

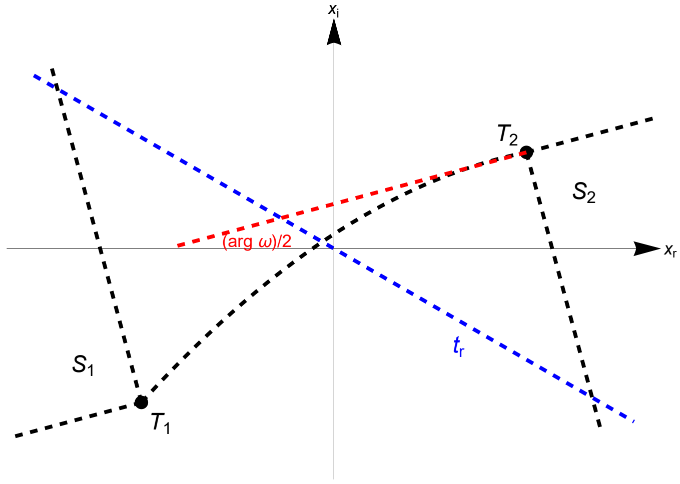


Figure 4.8: Asymptotic anti-Stokes lines of Eq. (4.8.9) and the real t axis on which Im $t = 0 \Rightarrow$ $x_{\mathrm{i}} = x_{\mathrm{r}} \tan((\arg\omega)/2 - \pi/4)$.

$$\alpha = \sqrt{\frac{|\omega|\rho_{\mathrm{s}}}{k'_{\parallel} c_{\mathrm{s}}}} \mathrm{e}^{\mathrm{i}((\arg\omega)/2 - \pi/4)}. \tag{4.8.17}$$

Solutions of Eq. (4.8.16) that satisfies Eq. (4.8.9), i.e., the bounded solutions, exist only for

$$\frac{E\alpha^2}{\rho_{\mathrm{s}}^2} = 2n + 1, \quad n \in \mathbb{Z}_{\geq 0}, \tag{4.8.18}$$

from which we could solve for the dispersion relation

$$\frac{\omega_n}{\omega_{*e}} = \frac{1}{1 + k_y^2\rho_{\mathrm{s}}^2} - \mathrm{i}\frac{2n+1}{1 + k_y^2\rho_{\mathrm{s}}^2}\frac{1}{\kappa L_{\mathrm{sh}}}, \quad n \in \mathbb{Z}_{\geq 0}. \tag{4.8.19}$$

Hence, the *discrete* eigenmodes are indeed damped by the finite magnetic shear; recall (Section 4.7) that in an unsheared background magnetic field, the drift waves possess a continuous frequency spectrum and are generally unstable.

Finally, we remark that our analysis is valid even though the obtained ω has Im $\omega < 0$. This is because, from the argument of analytic continuation, the subdominant regions bounded by the asymptotic anti-Stokes lines remain to be subdominant as ω is continuated from the upper half ω plane to the lower half one; in other words, the boundary condition Eq. (4.8.9) is solely determined by the causality condition Eq. (4.8.7) and remains unchanged throughout analytic continuating.

5 Topics in Nonlinear Plasma Theory

5.1 Quasi-linear theory

In previous chapters, we have shown, via linear theories, that there exist various mechanisms to either excite or damp waves in plasmas; in other words, plasmas as dielectric media may either emit or absorb wave packets (quasi-particles). Considering the plasma *quasi-particle system* as a whole, we would expect some types of energy and momentum conservation and, therefore, that the plasma will react to the emission and\or absorption process. This subject is, of course, beyond the scope of linear theory and is called, correspondingly, *quasi-linear* theory, which possesses the following characteristics:

(1) It deals with effects of perturbed (microscopic) fluctuations, either spontaneously or externally excited, on "equilibrium" (macroscopic) quantities.

(2) The perturbed fluctuations are taken to be of sufficiently small amplitudes, such that perturbative expansions are justified.

(3) It ignores nonlinear mode-coupling processes, e.g., parameteric decays.

(4) In reaction to the perturbations, the "equilibrium" quantities evolve slowly (adiabatically) in time.

In these lectures, we shall only consider collisionless, infinite, uniform plasmas and, thus, the corresponding macroscopic physical quantitity is the VDF. For simplicity, we further limit to the one-dimensional electrostatic (ES1D) case [55]; generalizations to higher-dimensional, magnetized and electromagnetic cases are conceptually straightforward; in fact, we could also generalize the underlying principles to nonuniform plasmas. We, therefore, can expect quasi-linear evolution of "equilibrium" pressure and\or current profiles and, in this sense, quasi-linear theories have important applications in understanding anomalous transport processes and\or nonlinear saturation of kink\tearing modes.

Now, the ES1d Vlasov equation reads

$$\left(\frac{\partial}{\partial t}+v\frac{\partial}{\partial x}+\varepsilon\frac{q}{m}E\frac{\partial}{\partial v}\right)f=0, \tag{5.1.1}$$

where $E=E(t,x)$ denotes the (real) perturbed electric field, and ε is the smallness parameter ordering the perturbations. We, thus, may separate f into its "equilibrium" and perturbed parts

$$f=\langle f\rangle_x(t,v)+\varepsilon\delta f(t,x,v), \tag{5.1.2}$$

where $\langle f\rangle_x := \int_{\mathbb{R}} \mathrm{d}x\, f$ denotes the average over x. Spatial-averaging Eq. (5.1.1), we obtain, noting $\langle E\rangle_x=0$,

$$\frac{\partial\langle f\rangle_x}{\partial t} = -\varepsilon^2 \frac{q}{m}\frac{\partial}{\partial v}\langle E\delta f\rangle_x; \tag{5.1.3}$$

$\langle f\rangle_x$, thus, indeed evolves slowly in time. In expanding the nonlinear Vlasov equation to order $\mathcal{O}(\varepsilon^2)$, we shall distinguish between f_2, the second-order perturbation in $\langle f\rangle_x$, whose *size* is of order $\mathcal{O}(\varepsilon^2)$, and f_0, the part of f independent of E, whose *evolution* is of order $\mathcal{O}(\varepsilon^2)$, i.e.,

$$\langle f\rangle_x(t,v) = f_0(\varepsilon^2 t, v) + \varepsilon^2 f_2(t,v). \tag{5.1.4}$$

Subtracting Eq. (5.1.3) from Eq. (5.1.1), we have

$$\left(\frac{\partial}{\partial t} + v\frac{\partial}{\partial x}\right)\delta f + \frac{q}{m}E\frac{\partial\langle f\rangle_x}{\partial v} + \varepsilon\frac{q}{m}\frac{\partial}{\partial v}\left(E\delta f - \langle E\delta f\rangle_x\right) = 0; \tag{5.1.5}$$

to order $\mathcal{O}(1)$ we find, denoting $\delta f \equiv f_1$ here and $f_0' \equiv \partial f_0/\partial v$,

$$\begin{gathered}\left(\frac{\partial}{\partial t} + v\frac{\partial}{\partial x}\right)f_1 + \frac{q}{m}Ef_0' = 0\\ \Rightarrow f_1(t,x,v) = f_{10}(x-vt,v) - \frac{q}{m}\int_0^t \mathrm{d}\tau\, E(t-\tau, x-v\tau) f_0'(\varepsilon^2(t-\tau),v)\end{gathered} \tag{5.1.6}$$

with $f_{10}(x,v) \equiv f_1(0,x,v)$. We look for solutions of the following form

$$E(t,x) = \hat{E}_k(\varepsilon t)\mathrm{e}^{\mathrm{i}(kx-\omega_k t-\phi_k)}, \quad \hat{E}_k \in \mathbb{R} \tag{5.1.7}$$

with ω_k and ϕ_k being constants[62]. Then Possion's equation

$$\frac{\partial E}{\partial x} = 4\pi\sum_s q_s\int_{\mathbb{R}}\mathrm{d}v\, f_{1s} \tag{5.1.8}$$

becomes, droping the initial value term by appealing to the phase mixing argument (Section 2.2)[63] for $t \gg 1/k\Delta v$,

$$\mathrm{i}k\hat{E}_k(\varepsilon t) = -4\pi\sum_s\frac{q_s^2}{m_s}\int_0^t \mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_k\tau}\hat{E}_k(\varepsilon(t-\tau))\int_{\mathbb{R}}\mathrm{d}v\,\mathrm{e}^{-\mathrm{i}kv\tau}f_{0s}'(\varepsilon^2(t-\tau),v)). \tag{5.1.9}$$

Again by phase mixing, the velocity integral in Eq. (5.1.9) decays with τ in a time scale $1/k\Delta v$; if this is short compared to the evolution time of f_0 (of order $\mathcal{O}(1/\varepsilon^2)$, due to diffusion) and of $\hat{E}_k$ (of order $\mathcal{O}(1/\varepsilon)$, due to wave growth or decay), we may Taylor-expand[64] f_0 and $\hat{E}_k$:

$$\begin{aligned} f_0(\varepsilon^2(t-\tau),v) &= f_0(\varepsilon^2 t, v) + \mathcal{O}(\varepsilon^2),\\ \hat{E}_k(\varepsilon(t-\tau)) &= \hat{E}_k(\varepsilon t) - \varepsilon\tau\left.\frac{\mathrm{d}\hat{E}_k(t')}{\mathrm{d}t'}\right|_{t'=\varepsilon t} + \mathcal{O}(\varepsilon^2).\end{aligned} \tag{5.1.10}$$

[62]Variation of ω_k or ϕ_k with time corresponds to frequency shift, and is of importance for the nonlinear problem [56], but not for the quasi-linear one.

[63]The phase mixing argument is crucial because we are interested in the case where the normal modes may be weakly damped (i.e., $\gamma_k < 0$ and $|\gamma_k| \ll |k\Delta v|$). If, however, the normal modes are unstable, then the normal modes dominate and, hence, any initial perturbation can be safely neglected; as a sconsequence, the results here can be derived in a much simpler fashion.

[64]Corrections are required for nonlinear problems.

Further, we may replace the upper limit of the τ integral by infinity. Eq. (5.1.9) then reads,

$$\begin{aligned} \mathrm{i}k\hat{E}_k(\varepsilon t) &= -\sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}} \int_{\mathbb{R}} \mathrm{d}v\, f_{0s}'(\varepsilon^2 t, v)) \int_0^{+\infty} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}(\omega_k - kv)\tau} \left(\hat{E}_k(\varepsilon t) - \varepsilon\tau \frac{\mathrm{d}\hat{E}_k(t')}{\mathrm{d}t'}\bigg|_{t'=\varepsilon t} \right) \\ &= -\hat{E}_k(\varepsilon t) \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}} \int_{\mathbb{R}} \mathrm{d}v\, f_{0s}'(\varepsilon^2 t, v)) \left(1 + \mathrm{i}\varepsilon\gamma_k(\varepsilon t)\frac{\partial}{\partial\omega_k}\right) \int_0^{+\infty} \mathrm{d}\tau\, \mathrm{e}^{(\mathrm{i}(\omega_k - kv)\tau} \end{aligned} \tag{5.1.11}$$

where $\gamma_k(\varepsilon t) \equiv \frac{\mathrm{d}\hat{E}_k(t')}{\mathrm{d}t'}/\hat{E}_k(t')\Big|_{t'=\varepsilon t}$. Using the causality condition Im $\omega_k = 0^+$, we find

$$-\mathrm{i}\int_0^{+\infty} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}(\omega_k - kv + \mathrm{i}0^+)\tau} = \frac{1}{\omega_k - kv + \mathrm{i}0^+} = \mathcal{P}\frac{1}{\omega_k - kv} - \mathrm{i}\pi\delta(\omega_k - kv), \tag{5.1.12}$$

Eq. (5.1.11) then gives the quasi-linear dispersion relation

$$\epsilon_{\mathrm{r}k} + \mathrm{i}\left(\epsilon_{\mathrm{i}k} + \varepsilon\gamma_k \frac{\partial\epsilon_{\mathrm{r}k}}{\partial\omega_k}\right) = 0, \tag{5.1.13}$$

where

$$\epsilon_{\mathrm{r}k} = \epsilon_{\mathrm{r}k}(\varepsilon^2 t) = 1 + \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k} \mathcal{P}\int_{\mathbb{R}} \mathrm{d}v\, \frac{f_{0s}'(\varepsilon^2 t, v)}{\omega_k - kv} \tag{5.1.14}$$

and

$$\begin{aligned} \epsilon_{\mathrm{i}k} = \epsilon_{\mathrm{i}k}(\varepsilon^2 t) &= -\pi \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k} \int_{\mathbb{R}} \mathrm{d}v\, f_{0s}'(\varepsilon^2 t, v)\delta(\omega_k - kv) \\ &= -\pi \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}k|k|} \frac{\partial f_{0s}(\varepsilon^2 t, v)}{\partial v}\bigg|_{v=\omega_k/k}. \end{aligned} \tag{5.1.15}$$

Note that Eq. (5.1.13) is formally identical to the standard linear dispersion relation, but evolves slowly in time since the "equilibrium" VDF f_0 does; thus the name *quasi-linear*.

With the linear perturbed responses determined, we now proceed to calculate the quasi-linear time evolution of $\langle f\rangle_x$ given by Eq. (5.1.3). To this end, we need to calculate $\langle Ef_1\rangle_x = \langle E^* f_1\rangle_x$. Here we generalize Eq. (5.1.7) to

$$E(t,x) = \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \hat{E}(\varepsilon t, k)\mathrm{e}^{\mathrm{i}(kx - \omega(k)t - \phi(k))}, \quad \hat{E} \in \mathbb{R} \tag{5.1.16}$$

with $\hat{E}(k) = \hat{E}(-k), \omega(-k) = -\omega(k)$ and $\phi(-k) = -\phi(k)$. Using Eq. (5.1.6) and the formula $\left\langle \mathrm{e}^{\mathrm{i}kx}\right\rangle_x = 2\pi\delta(k)$, we find

$$\begin{aligned} \langle E^* f_1\rangle_x(t,v) = \int_{\mathbb{R}} \frac{\mathrm{d}k'}{2\pi} \hat{E}(\varepsilon t, k')\mathrm{e}^{\mathrm{i}(\omega(k')t + \phi(k'))} \Bigg(&\int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \hat{f}_{10}(k,v)\mathrm{e}^{-\mathrm{i}kvt} \left\langle \mathrm{e}^{\mathrm{i}(k-k')x}\right\rangle_x \\ -\frac{q}{m}\int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \int_0^t \mathrm{d}\tau\, \hat{E}(\varepsilon(t-\tau), k)\mathrm{e}^{-\mathrm{i}kv\tau}\mathrm{e}^{-\mathrm{i}\omega(k)(t-\tau)}\mathrm{e}^{-\mathrm{i}\phi(k)} &f_0'(\varepsilon^2(t-\tau), v)\left\langle \mathrm{e}^{\mathrm{i}(k-k')x}\right\rangle_x \Bigg) \end{aligned} \tag{5.1.17}$$

$$
\begin{aligned}
&= \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \hat{E}(\varepsilon t, k) \hat{f}_{10}(k, v) \mathrm{e}^{\mathrm{i}(\omega(k)-kv)t} \mathrm{e}^{\mathrm{i}\phi(k)} \\
&\quad - \frac{q}{m} \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \int_0^t \mathrm{d}\tau\, \hat{E}(\varepsilon t, k) \hat{E}(\varepsilon(t-\tau), k) f_0'(\varepsilon^2(t-\tau), v) \mathrm{e}^{\mathrm{i}(\omega(k)-kv)\tau}.
\end{aligned} \tag{5.1.17}
$$

Now, if the perturbations, $\hat{E}$ and $\hat{f}_{10}$, are sufficiently broadband in the k spectrum, e.g., let's assume that they peak at $k = k_0$ and have a width Δk, we can then argue (just like the phase mixing mechanism) that the first term on the R.H.S. decays on a time scale $t_0 \equiv \left|\Delta k\left(\mathrm{d}\omega(k)/\mathrm{d}k|_{k=k_0} - v\right)\right|$ which is formally of order $\mathcal{O}(1)$. We can, therefore, again drop the the initial value term. On the other hand, the k integral in the second term also decays on a time scale $\tau \sim t_0$; if this is short compared to the evolution time of $\hat{E}(k)$, we may again use Eq. (5.1.10) and extend the upper limit of the τ integral to infinity. Thus[65]

$$
\begin{aligned}
\langle E^* f_1 \rangle_x (t, v) &= -\frac{q}{m} f_0'(\varepsilon^2 t, v) \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \int_0^{+\infty} \mathrm{d}\tau\, \hat{E}(\varepsilon t, k) \left(\hat{E}(\varepsilon t, k) - \varepsilon\tau \frac{\partial \hat{E}(t', k)}{\partial t'}\bigg|_{t'=\varepsilon t} \right) \mathrm{e}^{\mathrm{i}(\omega(k)-kv)\tau} \\
&= -\frac{q}{m} f_0'(\varepsilon^2 t, v) \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \int_0^{+\infty} \mathrm{d}\tau \left(\left|\hat{E}(\varepsilon t, k)\right|^2 - \mathrm{i}\frac{\varepsilon}{2} \frac{\partial \left|\hat{E}(t', k)\right|^2}{\partial t'}\Bigg|_{t'=\varepsilon t} \frac{1}{k}\frac{\partial}{\partial v} \right) \mathrm{e}^{\mathrm{i}(\omega(k)-kv)\tau}
\end{aligned} \tag{5.1.18}
$$

or, using Eq. (5.1.12) and noting that $\langle E^* f_1 \rangle_x$ is real,

$$
\begin{aligned}
\langle E^* f_1 \rangle_x (t, v) &= \mathrm{Re}\ \text{Eq. (5.1.18)} \\
&= -\frac{q}{m} f_0'(\varepsilon^2 t, v) \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} \Bigg(\pi \left|\hat{E}(\varepsilon t, k)\right|^2 \delta(\omega(k) - kv) \\
&\qquad + \frac{1}{2} \frac{\partial \left|\hat{E}(\varepsilon t, k)\right|^2}{\partial t} \frac{1}{k}\frac{\partial}{\partial v} \mathcal{P}\frac{1}{\omega(k) - kv} \Bigg) \\
&\equiv -\frac{q}{m} I(\varepsilon t, v) f_0'(\varepsilon^2 t, v),
\end{aligned} \tag{5.1.19}
$$

where, defining the wave spectrum energy density[66] $\mathcal{E}(\varepsilon t, k) \equiv \left|\hat{E}(\varepsilon t, k)\right|^2 / 8\pi$,

$$
I(\varepsilon t, v) = I_0(\varepsilon t, v) + \frac{\partial I_1(\varepsilon t, v)}{\partial t}, \tag{5.1.20}
$$

$$
\begin{aligned}
I_0(\varepsilon t, v) &= \pi \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} 8\pi \mathcal{E}(\varepsilon t, k) \delta(\omega(k) - kv) \\
&\overset{\omega(k_v) - k_v v = 0}{=\!=\!=} 4\pi \int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E}(\varepsilon t, k) \frac{\delta(k - k_v)}{|\mathrm{d}\omega(k)/\mathrm{d}k - v|}
\end{aligned} \tag{5.1.21}
$$

---

[65]The derivative $\partial/\partial\omega(k)$ is well-defined, because it can also be represented as $-(1/k)\partial/\partial v$.

[66]The key focus of weakly nonlinear wave theory is to study the evolution of $\mathcal{E}$. This is described by the wave kinetic equation, which includes all the nonlinear mode-couplings and wave-particle interactions. [57] Here we are simply content with the quasi-linear result $\partial\mathcal{E}/\partial t = 2\gamma\mathcal{E}$.

$$= 4\pi \frac{\mathcal{E}(\varepsilon t, k)}{|\mathrm{d}\omega(k)/\mathrm{d}k - v|}\bigg|_{k=k_v} \tag{5.1.21}$$

and

$$\begin{aligned} I_1(\varepsilon t, v) &= \frac{1}{2}\mathcal{P}\int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} 8\pi \mathcal{E}(\varepsilon t, k) \frac{1}{k}\frac{\partial}{\partial v}\frac{1}{\omega(k) - kv} \\ &= 2\mathcal{P}\int_{\mathbb{R}} \mathrm{d}k\, \frac{\mathcal{E}(\varepsilon t, k)}{(\omega(k) - kv)^2} \end{aligned} \tag{5.1.22}$$

Eq. (5.1.3) then writes

$$\frac{\partial f_0(\varepsilon^2 t, v)}{\partial t} + \varepsilon^2 \frac{\partial f_2(t, x, v)}{\partial t} = \varepsilon^2 \frac{q^2}{m^2}\frac{\partial}{\partial v}\left(I(\varepsilon t, v) \frac{\partial f_0(\varepsilon^2 t, v)}{\partial v} \right) \tag{5.1.23}$$

or, assocaiting I_0 with $\partial f_0/\partial t$ and $\partial I_1/\partial t$ with $\partial f_2/\partial t$ [67], and setting $\varepsilon = 1$,

$$\frac{\partial f_0}{\partial t} = \frac{q^2}{m^2}\frac{\partial}{\partial v}\left(I_0 \frac{\partial f_0}{\partial v}\right) \equiv \frac{\partial}{\partial v}\left(D_\mathrm{r} \frac{\partial f_0}{\partial v}\right), \tag{5.1.24}$$

$$f_2 = \frac{q^2}{m^2}\frac{\partial}{\partial v}\left(I_1 \frac{\partial f_0}{\partial v}\right), \tag{5.1.25}$$

where

$$D_\mathrm{r} = D_\mathrm{r}(t, v) \equiv \frac{q^2}{m^2} I_0(t, v) = \frac{\omega_\mathrm{p}^2}{m n_0} \frac{\mathcal{E}(t, k)}{|\mathrm{d}\omega(k)/\mathrm{d}k - v|}\bigg|_{k=k_v} \geq 0. \tag{5.1.26}$$

Since only resonant waves with $k = k_v$ contribute to $I_0(v)$, Eq. (5.1.24) describes the redistribution of the resonant particles due to resonant interactions, which, from the form of the equation, is a diffusion in velocity-space with $D_\mathrm{r}(v)$ being the diffusion coefficient. Meanwhile, f_2 corresponds to the quasi-linear deformation of non-resonant particles which do jiggle under the influence of waves; these particles retain memory of the waves and contribute to energy and momentum conservation.

We now establish some conservation properties.

(1) Number of particles:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}} \mathrm{d}v\, \langle f\rangle_x = \int_{\mathbb{R}} \mathrm{d}v\, \frac{\partial \langle f\rangle_x}{\partial t} = -\frac{q}{m}\int_{\mathbb{R}} \mathrm{d}v\, \frac{\partial}{\partial v}\langle E f_1\rangle_x = 0. \tag{5.1.27}$$

(2) Momentum: That

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s v \langle f_s\rangle_x = 0 \tag{5.1.28}$$

follows from

[67] This corresponds to a particular choice of f_0 and f_2, which is consistent with Eq. (5.1.4).

$$
\begin{aligned}
\sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s v \frac{\partial f_{2s}}{\partial t} &= \sum_s \frac{q_s^2}{m_s} \int_{\mathbb{R}} \mathrm{d}v\, v \frac{\partial}{\partial v}\left(\frac{\partial I_1}{\partial t}\frac{\partial f_{0s}}{\partial v}\right) \\
&= -\sum_s \frac{q_s^2}{m_s} \int_{\mathbb{R}} \mathrm{d}v\, \frac{\partial I_1}{\partial t}\frac{\partial f_{0s}}{\partial v} \\
&= 2\sum_s \frac{q_s^2}{m_s}\mathcal{P}\int_{\mathbb{R}} \mathrm{d}v\, f_{0s}' \int_{\mathbb{R}} \mathrm{d}k\, \frac{\partial \mathcal{E}}{\partial t}\frac{\partial}{\partial\omega(k)}\frac{1}{\omega(k)-kv} \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \gamma\mathcal{E}\,\frac{\partial}{\partial\omega(k)}\sum_s \frac{4\pi q_s^2}{m_s}\mathcal{P}\int_{\mathbb{R}} \mathrm{d}v\, \frac{f_{0s}'}{\omega(k)-kv} \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \gamma\mathcal{E}k\frac{\partial\epsilon_{\mathrm{r}}(k)}{\partial\omega(k)} \\
&\overset{\text{dispersion}}{\underset{\text{relation}}{=\!=\!=}} -\frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E}k\epsilon_{\mathrm{i}}(k) \\
&= \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}}\int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E}\int_{\mathbb{R}} \mathrm{d}v\, f_{0s}'\delta(\omega(k)-kv)
\end{aligned}
\tag{5.1.29}
$$

and

$$
\begin{aligned}
\sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s v \frac{\partial f_{0s}}{\partial t} &= \sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s v \frac{\partial}{\partial v}\left(D_{\mathrm{r}s}\frac{\partial f_{0s}}{\partial v}\right) \\
&= -\sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s D_{\mathrm{r}s}\frac{\partial f_{0s}}{\partial v} \\
&= -\sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}}\int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E}\int_{\mathbb{R}} \mathrm{d}v\, f_{0s}'\delta(\omega(k)-kv).
\end{aligned}
\tag{5.1.30}
$$

(3) Energy: That

$$
\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_s \int_{\mathbb{R}} \mathrm{d}v\, \frac{1}{2}m_s v^2 \langle f_s\rangle_x + \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi}\mathcal{E}\right) = 0 \tag{5.1.31}
$$

follows from

$$
\begin{aligned}
\sum_s \int_{\mathbb{R}} \mathrm{d}v\, \frac{1}{2}m_s v^2 \frac{\partial f_2}{\partial t} &= \sum_s \frac{q_s^2}{m_s}\int_{\mathbb{R}} \mathrm{d}v\, \frac{1}{2}v^2\frac{\partial}{\partial v}\left(\frac{\partial I_1}{\partial t}\frac{\partial f_{0s}}{\partial v}\right) \\
&= -\sum_s \frac{q_s^2}{m_s}\int_{\mathbb{R}} \mathrm{d}v\, v\frac{\partial I_1}{\partial t}\frac{\partial f_{0s}}{\partial v} \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \gamma\mathcal{E}\,\frac{\partial}{\partial\omega(k)}\sum_s \frac{4\pi q_s^2}{m_s}\mathcal{P}\int_{\mathbb{R}} \mathrm{d}v\, \frac{vf_{0s}'}{\omega(k)-kv} \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \gamma\mathcal{E}\frac{1}{k}\,\frac{\partial}{\partial\omega(k)}\left(\omega(k)\sum_s \frac{4\pi q_s^2}{m_s}\mathcal{P}\int_{\mathbb{R}} \mathrm{d}v\, \frac{f_{0s}'}{\omega(k)-kv}\right)
\end{aligned}
\tag{5.1.32}
$$

$$\begin{aligned}
&= \frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \gamma\mathcal{E}\left(\epsilon_{\mathrm{r}}(k) - 1 + \omega(k)\frac{\partial \epsilon_{\mathrm{r}}(k)}{\partial \omega(k)}\right) \\
&\overset{\text{dispersion}}{\underset{\text{relation}}{=\!=}} -\frac{1}{\pi}\int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E}\omega(k)\epsilon_{\mathrm{i}}(k) - \int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi} 2\gamma\mathcal{E} \\
&= \sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}} \int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E} \int_{\mathbb{R}} \mathrm{d}v\, v f_{0s}' \delta(\omega(k) - kv) - \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}} \frac{\mathrm{d}k}{2\pi}\mathcal{E}
\end{aligned} \tag{5.1.32}$$

and

$$\begin{aligned}
\sum_s \int_{\mathbb{R}} \mathrm{d}v\, \frac{1}{2} m_s v^2 \frac{\partial f_{0s}}{\partial t} &= \sum_s \int_{\mathbb{R}} \mathrm{d}v\, \frac{1}{2} m_s v^2 \frac{\partial}{\partial v}\left(D_{\mathrm{r}s}\frac{\partial f_{0s}}{\partial v}\right) \\
&= -\sum_s \int_{\mathbb{R}} \mathrm{d}v\, m_s v D_{\mathrm{r}s} \frac{\partial f_{0s}}{\partial v} \\
&= -\sum_s \frac{\omega_{\mathrm{p}s}^2}{n_{0s}} \int_{\mathbb{R}} \mathrm{d}k\, \mathcal{E} \int_{\mathbb{R}} \mathrm{d}v\, v f_{0s}' \delta(\omega(k) - kv).
\end{aligned} \tag{5.1.33}$$

Finally, let's examine the time-asymptotic behavior of f_0. Multiplying Eq. (5.1.24) by f_0 and integrating over v, we obtain, noting $D_{\mathrm{r}} \geq 0$,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}} \mathrm{d}v\, f_0^2 = \int_{\mathbb{R}} \mathrm{d}v\, f_0 \frac{\partial}{\partial v}\left(D_{\mathrm{r}}\frac{\partial f_0}{\partial v}\right) = -\int_{\mathbb{R}} \mathrm{d}v\, D_{\mathrm{r}}\left(\frac{\partial f_0}{\partial v}\right)^2 \leq 0. \tag{5.1.34}$$

However, with f_0 being a physical quantity, we must have

$$\int_{\mathbb{R}} \mathrm{d}v\, f_0^2 > 0 \text{ for all } t. \tag{5.1.35}$$

Eq. (5.1.34) and Eq. (5.1.35) then uniquely impose the following time-asymptotic condition

$$\lim_{t\to+\infty}\int_{\mathbb{R}} \mathrm{d}v\, D_{\mathrm{r}}\left(\frac{\partial f_0}{\partial v}\right)^2 = 0. \tag{5.1.36}$$

Now, in the cases of damped perturbations, this is easily satisfied since $\mathcal{E}(t) \to 0 \Rightarrow D_{\mathrm{r}}(t) \to 0$ as $t \to +\infty$. In the cases of unstable or undamped (e.g., externally driven) perturbations, however, we have $D_{\mathrm{r}}(t, v) > 0$ as $t \to +\infty$ for the resonant particles ($v = v_{\mathrm{r}} = \omega(k)/k$ for some k). Eq. (5.1.36) then indicates

$$\lim_{t\to+\infty} \left.\frac{\partial f_0}{\partial v}\right|_{v=v_{\mathrm{r}}} = 0, \tag{5.1.37}$$

i.e., the VDF becomes flattened at the resonant velocities. This is the well-known *quasi-linear flattening* of the resonant-particle VDF, which is, of course, a characteristic feature of the diffusion process — indeed, such flattening also occurs in damped cases. Therefore, within the framework of quasi-linear theory, an initial perturbation does not blow up (as it would in linear theory) but instead *saturates*, that is, $\dot{\hat{E}} \to 0$ as $t \to +\infty$. We remark that this flattening effect is absent in linear theory, where the wave amplitudes are assumed to be infinitesimal.

5.2 Oscillating-center and ponderomotive force

In this section, we shall consider the long-time single-particle motion in a *weakly nonuniform but fast oscillating* electromagnetic field. Here, the meaning of "long-time" is relative to the periods of the wave oscillations. Thus, as may be expected, the analysis employs two time scales and involves the phase averaging of the fast time scales. The average of the corresponding quasi-linear corrections yields a force that acts on the *oscillating-center*. [22]

Consider a spatially inhomogeneous electromagnetic field oscillating at frequency $\omega = \omega_0$

$$\boldsymbol{E}(t, \boldsymbol{x}) = \frac{1}{2}\Big(\hat{\boldsymbol{E}}(\varepsilon t, \boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega_0 t} + \text{c.c.}\Big). \tag{5.2.1}$$

The particles respond to this field through Newton's equations of motion

$$\dot{\boldsymbol{x}} = \boldsymbol{v}, \tag{5.2.2}$$

$$\dot{\boldsymbol{v}} = \frac{1}{2}\frac{q}{m}\left(\hat{\boldsymbol{E}}(\varepsilon t, \boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega_0 t} + \frac{\boldsymbol{v}}{c} \times \hat{\boldsymbol{B}}(\varepsilon t, \boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega_0 t} + \text{c.c.}\right) + \frac{q}{m}\frac{\boldsymbol{v}}{c} \times \boldsymbol{B}_0, \tag{5.2.3}$$

where $\hat{\boldsymbol{B}} = -\mathrm{i}(c/\omega_0)\nabla \times \hat{\boldsymbol{E}}$. Here, we shall make the following crucial assumption

$$\left\|\boldsymbol{v}\nabla\hat{\boldsymbol{E}}\right\| \ll \left|\omega_0\hat{\boldsymbol{E}}\right|, \tag{5.2.4}$$

which means that

(1) the wave magnetic force is smaller than the wave electric force by one order in the expansion parameter;

(2) the phase speed of the wave is high $v \ll \omega_0/k$, so that the number of resonant particles is negligible compared to that of non-resonant particles or, equivalently, that the wave growth or damping is weak;

(3) the distance a particle travels during one wave period, $2\pi v/\omega_0$, is much less than the wavelength $2\pi/k$, so that the amplitude $\hat{\boldsymbol{E}}$ experienced by the particle is approximately constant during a period of oscillation (adiabaticity).

We now apply the method of averaging. We make the substitution $t \mapsto \tau$ in oscillatory terms (i.e., τ is the t variable for the fast oscillation) and seek a change of variables (oscillating-center transformation) of the form

$$\begin{aligned} \boldsymbol{x}(t) &= \boldsymbol{X}(t) + \boldsymbol{\rho}(\boldsymbol{X}, t, \tau), \\ \boldsymbol{v}(t) &= \boldsymbol{V}(t) + \boldsymbol{u}(\boldsymbol{X}, t, \tau), \end{aligned} \tag{5.2.5}$$

where $\boldsymbol{\rho}$ and $\boldsymbol{u}$ are periodic functions of τ with vanishing means $\langle\cdots\rangle = (\omega_0/2\pi)\int_0^{2\pi/\omega_0} \mathrm{d}\tau\,(\cdots)$.

- The kinematic equation Eq. (5.2.2) gives, to all orders,

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{V} \tag{5.2.6}$$

and, to the lowest order,

$$\frac{\partial \boldsymbol{\rho}}{\partial \tau} = \boldsymbol{u}. \tag{5.2.7}$$

- The momentum equation Eq. (5.2.3) is, to the lowest order,

$$\frac{\partial \boldsymbol{u}}{\partial \tau} = \frac{1}{2}\frac{q}{m}\left(\hat{\boldsymbol{E}}(\varepsilon t, \boldsymbol{X})\mathrm{e}^{-\mathrm{i}\omega_0\tau} + \text{c.c.}\right) + \frac{q}{mc}\boldsymbol{u} \times \boldsymbol{B}_0, \tag{5.2.8}$$

from which we could solve for $\boldsymbol{u}$ and, through Eq. (5.2.7), $\boldsymbol{\rho}$. The acceleration of the oscillating-center is caused by the quasi-linear force

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t} &= \frac{1}{2}\frac{q}{m}\left\langle \boldsymbol{\rho}(\tau)\cdot\left(\nabla_{\boldsymbol{X}}\hat{\boldsymbol{E}}\mathrm{e}^{-\mathrm{i}\omega_0\tau} + \text{c.c.}\right) + \frac{\boldsymbol{u}(\tau)}{\omega_0} \times \left(-\mathrm{i}\nabla_{\boldsymbol{X}} \times \hat{\boldsymbol{E}}\mathrm{e}^{-\mathrm{i}\omega_0\tau} + \text{c.c.}\right)\right\rangle \\ &\quad + \frac{q}{m}\frac{\boldsymbol{V}}{c} \times \boldsymbol{B}_0.\end{aligned} \tag{5.2.9}$$

Here, we have used the Faraday's law $\hat{\boldsymbol{B}} = -\mathrm{i}(c/\omega_0)\nabla \times \hat{\boldsymbol{E}}$, noted that $\left\langle \boldsymbol{V} \times \hat{\boldsymbol{B}}\mathrm{e}^{-\mathrm{i}\omega_0\tau}\right\rangle = \boldsymbol{0}$, and denoted $\hat{\boldsymbol{E}} \equiv \hat{\boldsymbol{E}}(\varepsilon t, \boldsymbol{X})$.

Consider the example of an unmagnetized plasma. With $\boldsymbol{B}_0 = \boldsymbol{0}$, we find

$$\begin{aligned}\boldsymbol{u} &= \frac{1}{2}\frac{q}{m\omega_0}\left(\mathrm{i}\hat{\boldsymbol{E}}\mathrm{e}^{-\mathrm{i}\omega_0\tau} - \mathrm{i}\hat{\boldsymbol{E}}^*\mathrm{e}^{\mathrm{i}\omega_0\tau}\right), \\ \boldsymbol{\rho} &= -\frac{1}{2}\frac{q}{m\omega_0^2}\left(\hat{\boldsymbol{E}}\mathrm{e}^{-\mathrm{i}\omega_0\tau} + \hat{\boldsymbol{E}}^*\mathrm{e}^{\mathrm{i}\omega_0\tau}\right)\end{aligned} \tag{5.2.10}$$

and

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t} &= -\frac{1}{4}\left(\frac{q}{m\omega_0}\right)^2\left(\hat{\boldsymbol{E}}^*\cdot\nabla_{\boldsymbol{X}}\hat{\boldsymbol{E}} + \hat{\boldsymbol{E}}^* \times \left(\nabla_{\boldsymbol{X}}\hat{\boldsymbol{E}}\right) + \text{c.c.}\right) \\ &= -\frac{1}{4}\left(\frac{q}{m\omega_0}\right)^2\left(\nabla_{\boldsymbol{X}}\hat{\boldsymbol{E}}^*\cdot\hat{\boldsymbol{E}} + \nabla_{\boldsymbol{X}}\hat{\boldsymbol{E}}\cdot\hat{\boldsymbol{E}}^*\right);\end{aligned} \tag{5.2.11}$$

that is,

$$\ddot{\boldsymbol{X}} = -\frac{1}{4}\left(\frac{q}{m\omega_0}\right)^2\nabla_{\boldsymbol{X}}\left|\hat{\boldsymbol{E}}\right|^2. \tag{5.2.12}$$

Eq. (5.2.12) is the equation of motion for the oscillating-center, and the effective force

$$\boldsymbol{F}_{\text{pond}} = -\frac{1}{4}\frac{q^2}{m\omega_0^2}\nabla_{\boldsymbol{X}}\left|\hat{\boldsymbol{E}}\right|^2 \tag{5.2.13}$$

is referred to as the *ponderomotive force*; in plasma physics literature, we quite often use the terminology *ponderomotive potential*, defined by

$$\varphi_{\rm pond} = \frac{1}{4}\frac{q}{m\omega_0^2}\left|\hat{\boldsymbol{E}}\right|^2. \tag{5.2.14}$$

The total energy of the oscillating-center

$$\mathcal{E}_{\rm oc} = \frac{1}{2}m|\boldsymbol{V}|^2 + q\varphi_{\rm pond} \tag{5.2.15}$$

is conserved by Eq. (5.2.12). Noting that the ponderomotive potential energy is equal to the average kinetic energy of the oscillatory motion

$$q\varphi_{\rm pond} = \left\langle \frac{1}{2}m|\boldsymbol{u}|^2 \right\rangle, \tag{5.2.16}$$

it is then clear that the force acting on the oscillating-center originates in a transfer of energy from the oscillatory motion to the average motion. Finally, we remark that the ponderomotive force is independent of the sign of charge, so that both ions and electrons can be confined in the same potential well.

5.3 Parametric instabilities

The reference for this section is [58].

In our final topic, we shall briefly discuss the parameteric instabilities, which have abundant applications in both radio-frequency wave heating and nonlinear saturation of instabilities. Specifically, parametric instabilities correspond to waves driven unstable by other waves via nonlinear mode-coupling processses. The prominent example is the nonlinear three-wave interaction. Let the three waves be denoted as $\Omega_0 \equiv (\omega_0, \boldsymbol{k}_0), \Omega_1 \equiv (\omega_1, \boldsymbol{k}_1)$ and $\Omega_{\rm s} \equiv (\omega_{\rm s}, \boldsymbol{k}_{\rm s})$. In order that the nonlinear mode-coupling can occur, we need to satisfy the following $(\omega, \boldsymbol{k})$ matching conditions:

$$\begin{aligned} \omega_0 &= \omega_1 + \omega_{\rm s}, \\ \boldsymbol{k}_0 &= \boldsymbol{k}_1 + \boldsymbol{k}_{\rm s}. \end{aligned} \tag{5.3.1}$$

Here, we take Ω_0 to be the driving (pump) wave, while Ω_1 and $\Omega_{\rm s}$ are the decay (daughter) waves; without loss of generality, we shall assume $\omega_0, \omega_1, \omega_{\rm s} > 0$.

Now, in the three-wave parameteric decay instabilities, there are two types of decay processes. One is the resonant interaction, where both Ω_1 and $\Omega_{\rm s}$ are normal modes of the plasma; the other is the non-resonant interaction, where either Ω_1 or $\Omega_{\rm s}$ is a *virtual* (quasi, i.e., non-normal) mode due to, e.g., heavy damping. We shall clarify the above concepts with ES1D, where Ω_0 and Ω_1 are the Bohm-Gross waves (BGWs) and $\Omega_{\rm s}$ is the ion acoustic wave (IAW). Thus, in terms of the quasi-particle jargon, we shall consider the plasmon-plasmon-phonon interaction. In this case, the resonance interaction then corresponds to a plasmon at Ω_0 decaying into a plasmon at Ω_1 and a phonon at $\Omega_{\rm s}$; meanwhile, the matching conditions given by Eq. (5.3.1) can be interpreted as the conservation of energy and momentum. If, howvere,

$T_e \approx T_i$ such that the phonon is heavily ion Landau damped within the linear time scale, then only the plasmons (the Ω_0 and Ω_1 modes) exist in the nonlinear time scale. In this case, $\Omega_{\rm s}$ is said to be a virtual mode, and the non-resonant decay process may be interpreted as an induced scattering of a plasmon at Ω_0 by thermal ions (i.e., $\omega_{\rm s} \approx k_{\rm s} v_{{\rm t}i}$) into another plasmon at Ω_1. Since the nonlinear wave-particle resonance condition is

$$\omega_0 - \omega_1 \approx |k_0 - k_1| v_{{\rm t}i}, \tag{5.3.2}$$

this decay process is also called the *nonlinear ion Landau damping.*

We now proceed with the detailed calculations. First, we assume $|\omega_0/k_0|, |\omega_1/k_1| \gg v_{{\rm t}e}$ such that the validity of fluid description for the high-frequency BGWs is justified. Noting that $\omega_0, \omega_1 \gg \omega_{\rm s}$, we can decompose the fluid quantities into their high-frequency and low-frequency components, e.g., $n_e = n_{\rm he} + n_{\rm se}$. The corresponding high-frequency electron equations of continuity and motion are then given by

$$\frac{\partial n_{\rm he}}{\partial t} + \frac{\partial}{\partial x}(n_{0e} V_{\rm he} + n_{\rm se} V_{\rm he} + n_{\rm he} V_{\rm se}) = 0 \tag{5.3.3}$$

and

$$\frac{\partial V_{\rm he}}{\partial t} + \left(V_{\rm he} \frac{\partial V_{\rm se}}{\partial x} + V_{\rm se} \frac{\partial V_{\rm he}}{\partial x} \right) = -\frac{e}{m_e} E_{\rm h} - \frac{v_{\rm te}^2}{n_{0e}} \frac{\partial n_{\rm he}}{\partial x}, \quad v_{\rm te} \equiv \sqrt{\frac{\gamma_e T_e}{m_e}}. \tag{5.3.4}$$

Making the perturbative expansion

$$V_{\rm he} = V_{\rm he}^{(1)} + V_{\rm he}^{(2)} + \cdots \tag{5.3.5}$$

and so on, and noting that

$$\left| n_{\rm he}^{(1)} \right| = \mathcal{O}\left(\left| \frac{k_{\rm h}}{\omega_{\rm h}} V_{\rm he}^{(1)} n_{0e} \right| \right) \tag{5.3.6}$$

and

$$\left| n_{\rm se}^{(1)} \right| = \mathcal{O}\left(\left| \frac{k_{\rm s}}{\omega_{\rm s}} V_{\rm se}^{(1)} n_{0e} \right| \right), \tag{5.3.7}$$

it is then straightforward to show that

$$\frac{\left| n_{\rm he}^{(1)} V_{\rm se}^{(1)} \right|}{\left| n_{\rm se}^{(1)} V_{\rm he}^{(1)} \right|} = \mathcal{O}\left(\frac{\omega_{\rm s}}{\omega_{\rm h}} \right) \ll 1; \tag{5.3.8}$$

that is, the dominant low-frequency modification of the high-frequency dynamics is via its density perturbation, $n_{\rm se}^{(1)}$. Eq. (5.3.3) then simplify to

$$\frac{\partial n_{\rm he}^{(1)}}{\partial t} + \frac{\partial}{\partial x}\left(n_{0e} V_{\rm he}^{(1)} + n_{\rm se}^{(1)} V_{\rm he}^{(1)} \right) = 0. \tag{5.3.9}$$

Also, since $|\omega_0/k_0|, |\omega_1/k_1| \gg v_{\rm te}$, we can neglect the convective nonlinear term in the electron equation of motion, Eq. (5.3.4), as

$$\frac{\partial V_{\rm he}^{(1)}}{\partial t} = -\frac{e}{m_e}E_{\rm h}^{(1)} - \frac{v_{\rm te}^2}{n_{0e}}\frac{\partial n_{\rm he}^{(1)}}{\partial x}. \tag{5.3.10}$$

As to the low-frequency dynamics, we need to adopt a kinetic Vlasov description in order to include the induced scattering process. The concept of ponderomotive force, therefore, becomes rather useful here. The low-frequency electron Vlasov equation is then

$$\left(\frac{\partial}{\partial t} + u\frac{\partial}{\partial x}\right)f_{\rm se} - \frac{e}{m_e}\left(-\frac{\partial \varphi_{\rm s}}{\partial x} - \frac{\partial \varphi_{\rm pe}}{\partial x}\right)\frac{\partial f_{0e}}{\partial u} = 0, \tag{5.3.11}$$

where

$$\varphi_{\rm pe} = -\frac{e}{m_e\omega_0^2}\left\langle |E_{\rm h}|^2 \right\rangle_{\rm s} \tag{5.3.12}$$

is the ponderomotive potential for the electrons and $\langle\cdots\rangle_{\rm s} := (1/T)\int_0^T {\rm d}t\,(\cdots)$ with $2\pi/\omega_{\rm s} \ll T \ll 2\pi/\omega_0$ averages the various quantities over high-frequency oscillations (low-frequency oscillations with $\omega \approx \omega_{\rm s}$ are preserved). On the other hand, since $m_e/m_i \ll 1$, ion nonlinearities are small and negligible. Substituting $\hat{f}_{\rm se}$ and $\hat{f}_{{\rm s}i}$ into the low-frequency Possion's equation, we obtain

$$\epsilon_{\rm s}\hat{\varphi}_{\rm s} \equiv (1 + \chi_{\rm se} + \chi_{{\rm s}i})\hat{\varphi}_{\rm s} = -\chi_{\rm se}\hat{\varphi}_{\rm pe}, \tag{5.3.13}$$

where $\chi_{{\rm s}s}$ is the low-frequency *linear* susceptibility for species s (Eq. (2.1.29)); we also calculate $\hat{n}_{\rm se}$ as

$$4\pi e\hat{n}_{\rm se} := 4\pi e\int_{\mathbb{R}^3} {\rm d}^3v\,\hat{f}_{\rm se} = k_{\rm s}^2\chi_{\rm se}\big(\hat{\varphi}_{\rm s} + \hat{\varphi}_{\rm pe}\big) = k_{\rm s}^2\chi_{\rm se}\frac{1+\chi_{{\rm s}i}}{\epsilon_{\rm s}}\hat{\varphi}_{\rm pe}. \tag{5.3.14}$$

Eq. (5.3.9), Eq. (5.3.10), Eq. (5.3.14), Eq. (5.3.12) plus the high-frequency Possion's equation, thus, provide a complete set of equations, from which we can derive a modified "nonlinear" dispersion relation and, hence, parametric growth rates.

To achieve these purposes, we let[68]

$$E_{\rm h} = \frac{1}{2}\Big(\hat{E}_0{\rm e}^{{\rm i}(k_0x-\omega_0t)} + \hat{E}_1{\rm e}^{{\rm i}(k_1x-\omega_1t)} + {\rm c.c.}\Big). \tag{5.3.15}$$

The $-\Omega_1$ components of Eq. (5.3.9) and Eq. (5.3.10) are then

$$\omega_1\hat{n}_{1e}^* - k_1n_{0e}\hat{V}_{1e}^* - k_1\hat{n}_{\rm se}\hat{V}_{0e}^* = 0 \tag{5.3.16}$$

and

$$\omega_1\hat{V}_{1e}^* = {\rm i}\frac{e}{m_e}\hat{E}_1^* + \frac{v_{\rm te}^2}{n_{0e}}k_1\hat{n}_{1e}^*. \tag{5.3.17}$$

[68]Question: Will "c.c." make ω_1 to be ω_1^*?

Using the $-\Omega_1$ component of the high-frequency Possion's equation, i.e., $-\mathrm{i}k_1\hat{E}_1^* = -4\pi e\hat{n}_{1e}^*$, Eq. (5.3.16) and Eq. (5.3.17) further reduce to

$$\mathrm{i}k_1\omega_1\epsilon_\mathrm{h}(-\Omega_1)\hat{E}_1^* = 4\pi e k_1\hat{n}_\mathrm{se}\hat{V}_{0e}^*, \tag{5.3.18}$$

where

$$\epsilon_\mathrm{h}(\Omega) = 1 - \frac{\omega_\mathrm{pe}^2 + k^2 v_\mathrm{te}^2}{\omega^2}. \tag{5.3.19}$$

Meanwhile, noting that

$$\hat{V}_{0e}^* \approx \mathrm{i}\frac{e}{m_e\omega_0}\hat{E}_0^* \tag{5.3.20}$$

and

$$\begin{aligned}\varphi_\mathrm{pe} &= -\frac{e}{m_e\omega_0^2}\left\langle |E_\mathrm{h}|^2\right\rangle_\mathrm{s} = -\frac{e}{4m_e\omega_0^2}\left(\hat{E}_0\hat{E}_1^*\mathrm{e}^{\mathrm{i}(k_\mathrm{s}x-\omega_\mathrm{s}t)} + \mathrm{c.c.}\right)\\ &\Rightarrow \hat{\varphi}_\mathrm{pe} = -\frac{e\hat{E}_0}{4m_e\omega_0^2}\hat{E}_1^*,\end{aligned} \tag{5.3.21}$$

we have Eq. (5.3.18) and Eq. (5.3.14) as

$$\epsilon_\mathrm{h}(-\Omega_1)\hat{E}_1^* = 4\pi e\hat{n}_\mathrm{se}\frac{e\hat{E}_0^*}{4m_e\omega_0^2} \tag{5.3.22}$$

and

$$4\pi e\hat{n}_\mathrm{se} = -k_\mathrm{s}^2\chi_\mathrm{se}\frac{1+\chi_{\mathrm{s}i}}{\epsilon_\mathrm{s}}\frac{e\hat{E}_0}{4m_e\omega_0^2}\hat{E}_1^*. \tag{5.3.23}$$

Eq. (5.3.22) and Eq. (5.3.23) are the desired two coupled equaions, which, when combined, yield the following parametric dispersion relation

$$\epsilon_\mathrm{h}(-\Omega_1)\epsilon_\mathrm{s} = -k_\mathrm{s}^2\chi_\mathrm{se}(1+\chi_{\mathrm{s}i})\left|\frac{e\hat{E}_0}{4m_e\omega_0^2}\right|^2. \tag{5.3.24}$$

Assuming that Ω_1 is a normal mode, i.e., $\epsilon_\mathrm{hr}(\Omega_{1\mathrm{r}}) = 0$, and denoting $\omega_\mathrm{s} = \omega_\mathrm{sr} + \mathrm{i}\gamma \Rightarrow -\omega_1 = \omega_\mathrm{s} - \omega_0 = \omega_\mathrm{sr} - \omega_0 + \mathrm{i}\gamma \equiv -\omega_{1\mathrm{r}} + \mathrm{i}\gamma$, Eq. (5.3.24) becomes, noting[69] $\epsilon(-\Omega_\mathrm{r}) = \epsilon_\mathrm{r}(\Omega_\mathrm{r}) - \mathrm{i}\epsilon_\mathrm{i}(\Omega_\mathrm{r})$,

$$-\mathrm{i}\left(\gamma\frac{\partial\epsilon_\mathrm{hr}}{\partial\omega_{1\mathrm{r}}} + \epsilon_\mathrm{hi}\right)\epsilon_\mathrm{s} = -k_\mathrm{s}^2\chi_\mathrm{se}(1+\chi_{\mathrm{s}i})\left|\frac{e\hat{E}_0}{4m_e\omega_0^2}\right|^2. \tag{5.3.25}$$

- For the resonant decay, we have $\epsilon_\mathrm{sr}(\Omega_\mathrm{sr}) = 0$. Eq. (5.3.25) then reduces to

$$\left(\gamma\frac{\partial\epsilon_\mathrm{hr}}{\partial\omega_{1\mathrm{r}}} + \epsilon_\mathrm{hi}\right)\left(\gamma\frac{\partial\epsilon_\mathrm{sr}}{\partial\omega_\mathrm{sr}} + \epsilon_\mathrm{si}\right) \approx k_\mathrm{s}^2\chi_\mathrm{se}^2\left|\frac{e\hat{E}_0}{4m_e\omega_0^2}\right|^2 \approx \frac{1}{k_\mathrm{s}^2\lambda_\mathrm{De}^2}\left|\frac{e\hat{E}_0}{4m_e\omega_0^2\lambda_\mathrm{De}}\right|^2. \tag{5.3.26}$$

[69] In the framework of fluid description, $\epsilon_\mathrm{hi} = 0$; we will, however, keep the ϵ_hi term so as to formally include the kinetic effects.

Setting $\gamma = 0$ in Eq. (5.3.26), we obtain the following threshold condition

$$\left|\frac{e\hat{E}_0^{\text{th}}}{4m_e\omega_0^2\lambda_{\text{De}}}\right|^2 = \epsilon_{\text{hi}}\epsilon_{\text{si}}k_{\text{s}}^2\lambda_{\text{De}}^2. \tag{5.3.27}$$

For $|E_0| \gg \left|\hat{E}_0^{\text{th}}\right|$, we have a parameteric instability with growth rate

$$\gamma \approx \frac{1}{k_{\text{s}}\lambda_{\text{De}}}\left|\frac{e\hat{E}_0}{4m_e\omega_0^2\lambda_{\text{De}}}\right|\left(\frac{\partial\epsilon_{\text{hr}}}{\partial\omega_{1\text{r}}}\frac{\partial\epsilon_{\text{sr}}}{\partial\omega_{\text{sr}}}\right)^{-1/2}. \tag{5.3.28}$$

- In the case of non-resonant decay, $|\epsilon_{\text{si}}| \sim |\epsilon_{\text{sr}}|$. The imaginary part of Eq. (5.3.25) then yields, assuming ion Landau damping dominates,

$$\begin{aligned}\gamma\frac{\partial\epsilon_{\text{hr}}}{\partial\omega_{1\text{r}}} + \epsilon_{\text{hi}} &= -\left|k_{\text{s}}\frac{e\hat{E}_0}{4m_e\omega_0^2}\right|^2\chi_{se}\ \text{Re}\left(\text{i}\frac{1+\chi_{si}}{\epsilon_{\text{s}}}\right)\\ &= \left|k_{\text{s}}\frac{e\hat{E}_0}{4m_e\omega_0^2}\right|^2\frac{\chi_{\text{se}}}{|\epsilon_{\text{s}}|^2}\ \text{Im}((1+\chi_{si})\epsilon_{\text{s}}^*)\\ &\approx \left|\frac{e\hat{E}_0}{4m_e\omega_0^2\lambda_{\text{De}}}\right|^2\frac{1}{k_{\text{s}}^2\lambda_{\text{De}}^2}\frac{\text{Im}\ \chi_{\text{s}i}}{|\epsilon_{\text{s}}|^2}.\end{aligned} \tag{5.3.29}$$

Eq. (5.3.29) with $\gamma = 0$ indicates that the corresponding threshold condition is sololy set by the high-frequency dissipation ϵ_{hi}, contrary to the resonant case Eq. (5.3.27) where the low-frequency dissipation ϵ_{si} is also involved. Furthermore, since $\gamma \propto \text{Im}\ \chi_{\text{s}i}$, the instability mechanism is indeed associated with the ion Landau damping of the low-frequency virtual mode Ω_{s} and, hence, the growth rate γ is maximized when ion Landau damping is maximized, i.e., when $\omega_{\text{sr}} = \omega_0 - \omega_{1\text{r}} \approx |k_{\text{s}}|v_{\text{t}i}$; noting that $k_{\text{s}} = k_0 - k_1$ and $|\omega_0| \approx |\omega_1| \gg |k_0,k_1|v_{\text{t}i}$, the decay is most easily maximal for an induced *backward* scattering where $k_0k_1 < 0$ so that $|k_{\text{s}}| \approx 2|k_0|$ is not too small.

Final Examination

(1) Consider the linear ideal MHD waves in a uniformly magnetized plasma with $\boldsymbol{B}_0 = B_0\hat{\boldsymbol{e}}_z$.

(i) Derive the dispersion relation.

(ii) Show that there exist seven branches of normal modes; that is, for a given $\boldsymbol{k}$, we have seven solutions of ω. Hint: two fast magnetosonic waves, two slow magnetosonic waves, two shear Alfvén waves, and one trivial $\omega = 0$ entropy mode.

(iii) For the seven branches of normal modes, give their corresponding eigenvectors in terms of the seven variables (perturbed quantities) $p, \rho, \boldsymbol{V}, \boldsymbol{B}$.

(2) Let's consider resistive ($\boldsymbol{E} + (\boldsymbol{V}/c)\times\boldsymbol{B}_0 = \eta\boldsymbol{J}$) shear Alfvén waves in a uniform plasma with a sheared background magnetic field $\boldsymbol{B}_{\text{bg}} = B_0(\hat{\boldsymbol{e}}_z + \hat{\boldsymbol{e}}_y x/L_{\text{sh}})$, where $|x/L_{\text{sh}}| \ll 1$ for our purpose. We shall assume, furthermore, that the plasma is cold and incompressible ($\nabla\cdot$

$\boldsymbol{V}=0$) so that the compressional Alfvén waves and the ion acoustic waves are suppressed. Take perturbations of the form

$$\boldsymbol{V}(t,\boldsymbol{x})=\hat{\boldsymbol{V}}(x)\mathrm{e}^{\mathrm{i}(k_y y-\omega t)}, \tag{F.1}$$

i.e., $k_z=0$, hence $k_\parallel=\boldsymbol{k}\cdot\boldsymbol{B}_{\mathrm{bg}}/B_{\mathrm{bg}}\approx\left(k_y/L_{\mathrm{sh}}\right)x$.

(i) Derive the corresponding differential equation in the x space.

(ii) Fourier transforming from the x space to the k space, derive the following differential equation in the k space[70]

$$\left(\frac{\mathrm{d}}{\mathrm{d}k}\left(\frac{1+k^2}{1+\mathrm{i}\eta(1+k^2)/\omega}\frac{\mathrm{d}}{\mathrm{d}k}\right)+\omega^2(1+k^2)\right)\phi(k)=0. \tag{F.2}$$

(iii) Use the Nyquist technique to prove that the wave is stable.

(3) Consider a cold, uniformly magnetized plasma with $\boldsymbol{B}_0=B_0\hat{\boldsymbol{e}}_z$. The dielectric tensor is given in Figure 3.1.

(i) Show, for parallel propagation (i.e., $\boldsymbol{k}=k\hat{\boldsymbol{e}}_z$), that the dispersion relation reduces to one right-handed circularly polarized wave and one left-handed circularly polarized wave.

(ii) Derive the low-frequency limits where $|\omega/\Omega|\to 0$ of both waves.

(4) Lower-hybrid resonance and mode conversion.

Consider a *slab* plasma in the $x-z$ plane. Here, $\boldsymbol{B}_0=B_0\hat{\boldsymbol{e}}_z$ and x is the inhomogeneous direction, i.e., $n_0=n_0(x)$.

For lower-hybrid waves, we have $\Omega_i\ll\omega\sim\omega_{\mathrm{p}i}$ (at plasma core) $\ll|\Omega_e|$, $\left|k_\parallel/k_\perp\right|\lesssim\sqrt{m_e/m_i}$ where $k_\perp=k_x$ since $k_y=0$, $|k_\perp\rho_{\mathrm{te}}|\ll 1$ and $\left|\omega/k_\parallel\right|\gg v_{\mathrm{te}},v_{\mathrm{t}i}$.

(i) Use the cold plasma model to obtain the following wave equation

$$\left(\frac{\partial}{\partial x}\left(\epsilon_{\perp\mathrm{c}}(x)\frac{\partial}{\partial x}\right)+\frac{\partial}{\partial z}\left(\epsilon_\parallel(x)\frac{\partial}{\partial z}\right)\right)\phi(x,z)=0, \tag{F.3}$$

where (see Figure 3.1)

$$\epsilon_{\perp\mathrm{c}}(x)=1+\frac{\omega_{\mathrm{pe}}^2(x)}{\Omega_e^2}-\frac{\omega_{\mathrm{p}i}^2(x)}{\omega_0^2}, \tag{F.4}$$

$$\epsilon_\parallel(x)=1-\frac{\omega_{\mathrm{pe}}^2(x)}{\omega_0^2}, \tag{F.5}$$

and ω_0 is the driving frequency of the radio frequency (RF) source. Eq. (F.4) shows that the existence of a resonance at $x=x_{\mathrm{r}}$ where $\epsilon_{\perp\mathrm{c}}(x_{\mathrm{r}})=0$.

(ii) As $x\to x_{\mathrm{r}}$, $|k_x|\to+\infty$. Since $\rho_{\mathrm{t}i}^2\gg\rho_{\mathrm{te}}^2$, the first microscopic effect entering is the finite ion Larmor radius (FILR) correction. Assuming f_{0i} to be Maxwellian and using the ion gyrokinetic equation, obtain this correction in the small $\rho_{\mathrm{t}i}$ limit as

[70] This equation is displayed as it was in the original text; note, however, that the dimensionality is wrong.

$$\begin{aligned}\epsilon_\perp(x) &= 1 + \frac{\omega_{\mathrm{pe}}^2(x)}{\Omega_e^2} - \frac{\omega_{\mathrm{p}i}^2(x)}{\omega_0^2}A_0(b_i) \\ &\approx \epsilon_{\perp\mathrm{c}}(x) - \frac{\omega_{\mathrm{p}i}^2(x)}{\omega_0^2}k_x^2\rho_{\mathrm{t}i}^2.\end{aligned} \tag{F.6}$$

Hint: use Eq. (3.7.13).

(iii) With $\partial/\partial z = \mathrm{i}k_z$ well-defined by the RF launcher, plot the $k_x^2 - x$ diagram both for the cold plasma case and with the FILR correction.

(5) Parameteric instability at lower-hybrid frequency.

Consider the resonant decay of a pump lower-hybrid wave (LHW) $\Omega_0 \equiv (\omega_0, \boldsymbol{k}_0)$ to a daughter LHW $\Omega_1 \equiv (\omega_1, \boldsymbol{k}_1)$ and a daughter slow ion acoustic wave (IAW) $\Omega_{\mathrm{s}} \equiv (\omega_{\mathrm{s}}, \boldsymbol{k}_{\mathrm{s}})$. All waves are taken to be electrostatic. Furthermore, we have $\omega_0 = \omega_1 + \omega_{\mathrm{s}}, \omega_0 \approx \omega_1 \gg \omega_{\mathrm{s}}$,

$$\omega_{\left\{\substack{0\\1}\right.} = \omega_{\mathrm{LH}}\left(1 + \frac{m_i}{m_e}\frac{k_\parallel^2}{k_\perp^2}\right)^{1/2}_{\left\{\substack{0\\1}\right.} \tag{F.7}$$

and

$$\omega_{\mathrm{s}} \approx k_{\mathrm{s}\parallel}c_{\mathrm{s}}. \tag{F.8}$$

Here, in Eq. (F.8), we have assumed $|k_{\mathrm{s}\perp}\rho_{\mathrm{s}}| \ll 1$ or $\omega_{\mathrm{s}} \ll \Omega_i$. For further clarification, let's take $\boldsymbol{k}_0 = \hat{\boldsymbol{e}}_x k_{0\perp} + \hat{\boldsymbol{e}}_z k_{0\parallel}, \boldsymbol{k}_1 = \hat{\boldsymbol{e}}_x k_{1x} + \hat{\boldsymbol{e}}_y k_{1y} + \hat{\boldsymbol{e}}_z k_{1\parallel}$ and $|k_\perp \rho_{\mathrm{t}}| \ll 1$.

(i) Calculate the ponderomotive force due to the electrostatic LHWs on the electrons and ions, respectively. Noting that $\omega_{\mathrm{LH}} \sim \omega_{\mathrm{p}i} \gg \Omega_i$, it can be shown that electron nonlinearities dominate over ion nonlinearities. In this calculation, since the slow time scale ω_{s}^{-1} is much longer than Ω_i^{-1}, the long-time dynamics of both electrons and ions are <u>along</u> $\boldsymbol{B}_0$ only.

(ii) Obtain the parametric dispersion relation and the corresponding growth rate.

(6) Oscillating-center transformation for parameteric instabiity.

Similar to the guiding-center transformation used in treating magnetized plasmas, the existence of oscillating-centers suggests that an analogous transformation should be useful in treating parameteric (i.e., nonlinear mode-coupling) processes. We shall illustrate this point by considering unmagnetized plasmas in a <u>dipole</u> (i.e., spatially uniform) pump

$$\boldsymbol{E}_0(t, \boldsymbol{x}) = \hat{\boldsymbol{E}}_0 \cos\omega_0 t. \tag{F.9}$$

The Vlasov equation is

$$\left(\frac{\partial}{\partial t} + \boldsymbol{v}\cdot\nabla + \frac{q}{m}(\boldsymbol{E}_0 + \boldsymbol{E})\cdot\nabla_{\boldsymbol{v}}\right)f = 0. \tag{F.10}$$

The desired transformation[71] is defined as

[71]A comment: This transformation technique can be straightforwardly generalized to magnetized plasmas.

$$\boldsymbol{V} := \boldsymbol{v} + \boldsymbol{u}, \quad \boldsymbol{X} := \boldsymbol{x} + \boldsymbol{\rho}, \tag{F.11}$$

where

$$\dot{\boldsymbol{u}} = \frac{q}{m}\boldsymbol{E}_0, \quad \dot{\boldsymbol{\rho}} = \boldsymbol{u}. \tag{F.12}$$

(i) Show that the linearized Vlasov equation transforms to

$$\left(\frac{\partial}{\partial t} + \boldsymbol{V}\cdot\nabla_{\boldsymbol{X}}\right) f_1^{\mathrm{oc}} = -\frac{q}{m}\boldsymbol{E}^{\mathrm{oc}}\cdot\nabla_{\boldsymbol{V}} f_0^{\mathrm{oc}}. \tag{F.13}$$

Here, the various quantities with superscript "oc" are expressed in terms of the oscillating-center coordinates $(\boldsymbol{X}, \boldsymbol{V})$.

(ii) Following the analogy with the guiding-center treatment, try to show that Eq. (F.11), Eq. (F.12), Eq. (F.13) and the Possion's equation lead to the following coupled set of equations

$$\begin{aligned} (1+\chi_{e,n})E_n &= -\chi_{e,n}\sum_{p=-\infty}^{+\infty} J_{n-p}(\mu)I_p, \\ (1+\chi_{i,n})I_n &= -\chi_{i,n}\sum_{p=-\infty}^{+\infty} J_{p-n}(\mu)E_p. \end{aligned} \tag{F.14}$$

Here, $n \in \mathbb{Z}$, $\chi_{s,n} \equiv \chi_s(\omega + n\omega_0, \boldsymbol{k})$ is the linear susceptibility of species s, and $J_n(\mu)$ is the Bessel function of order n with argument $\mu \equiv |\boldsymbol{k}\cdot(\boldsymbol{\rho}_e - \boldsymbol{\rho}_i)|$. Note that in the no-pump limit $\mu \to 0$, we do recover the usual dispersion relation $1 + \chi_{e,n} + \chi_{i,n} = 0$. Eq. (F.14) reveals an interesting feature: there is no parameteric process if $m_i \to +\infty$, i.e., if $\chi_i \to 0$.

(iii) Examine Eq. (F.14) in the weak-pump limit $\mu \ll 1$. Can you recover the parameteric dispersion relation for the

$$\text{plasmon} \to \text{plasmon} + \text{phonon} \tag{F.15}$$

decay discussed in class? For this purpose, take $\omega = \omega_{\mathrm{s}}, \omega + \omega_0 = \omega_1$ and $\boldsymbol{k}_{\mathrm{s}} = \boldsymbol{k}_1$.

A Method of Averaging and Guiding-Center Motion

In this chapter we use SI units.

A.1 Scale separation and the method of averaging

The main reference of this section is [22].

In many dynamical problems, the motion consists of a rapid oscillation superposed upon a slow drift. For such problems, the most efficient approach is to describe the slow evolution in terms of the average values of the fast-oscillating dynamical variables.

Consider the ODE

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x, t, \alpha), \tag{A.1.1}$$

where the "force" f is a periodic function of its last argument with period 2π, and $\alpha = \varepsilon^{-1}t, \varepsilon \ll 1$. Here, the small parameter ε characterizes the separation between the short oscillation period and the long timescale for the slow evolution of the "position" x .

The basic idea of the averaging method is to make use of this scale separation to treat t and α as distinct independent variables that parameterize the slow drift and the rapid oscillation, respectively, and to look for solutions of the form $x(t, \alpha)$ that are periodic in α .[72] Thus,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\partial x}{\partial t} + \frac{\mathrm{d}\alpha}{\mathrm{d}t}\frac{\partial x}{\partial \alpha} = \frac{\partial x}{\partial t} + \frac{1}{\varepsilon}\frac{\partial x}{\partial \alpha} \tag{A.1.2}$$

and Eq. (A.1.1) is replaced by

$$\frac{\partial x}{\partial t} + \frac{1}{\varepsilon}\frac{\partial x}{\partial \alpha} = f(x, t, \alpha). \tag{A.1.3}$$

A.1.1 The concept of the guiding-center

Let X be the α average of the "position" x, called the "guiding-center position"

$$X(t) := \frac{1}{2\pi}\oint \mathrm{d}\alpha\, x(t, \alpha) \equiv \langle x(t, \alpha)\rangle. \tag{A.1.4}$$

Let's seek a solution of the following form

$$x(t, \alpha) = X(t) + \varepsilon\rho(t, \alpha), \tag{A.1.5}$$

where ρ is a periodic function (whose values are to be determined) of α with vanishing mean: $\langle\rho(t, \alpha)\rangle = 0$. Here ρ encodes the rapid oscillation of x around its average X. We assume that the amplitude of the oscillation is small, so that there is an ε factor in front of ρ.

[72] This is an approximation, since actually α depends on t .

A.1.2 Equations of motion of the guiding-center

We would like to solve for the equation of motion of X order by order in ε. To this end, we substitute the following expansions into Eq. (A.1.3), and solve order by order in ε :

$$\begin{aligned}\rho(t,\alpha) &= \rho_0(t,\alpha) + \varepsilon\rho_1(t,\alpha) + \mathcal{O}(\varepsilon^2),\\ \frac{\mathrm{d}X}{\mathrm{d}t} &= F_0(X,t) + \varepsilon F_1(X,t) + \mathcal{O}(\varepsilon^2).\end{aligned} \tag{A.1.6}$$

To order $\mathcal{O}(\varepsilon)$ the equation of motion becomes

$$\begin{aligned}F_0(X,t) + \varepsilon F_1(X,t) + \varepsilon\frac{\partial\rho_0(t,\alpha)}{\partial t} + \frac{\partial\rho_0(t,\alpha)}{\partial\alpha} + \varepsilon\frac{\partial\rho_1(t,\alpha)}{\partial\alpha}\\ = f(X,t,\alpha) + \varepsilon\rho_0(t,\alpha)\frac{\partial f(X,t,\alpha)}{\partial X} + \mathcal{O}(\varepsilon^2).\end{aligned} \tag{A.1.7}$$

- At order $\mathcal{O}(1)$, we have

$$F_0(X,t) + \frac{\partial\rho_0(t,\alpha)}{\partial\alpha} = f(X,t,\alpha); \tag{A.1.8}$$

  averaging over α gives the solubility condition, we find

$$F_0(X,t) = \langle f(X,t,\alpha)\rangle, \tag{A.1.9}$$

  then

$$\rho_0(t,\beta) = \int^{\beta} \mathrm{d}\gamma\,(f(X,t,\gamma) - \langle f(X,t,\alpha)\rangle. \tag{A.1.10}$$

  Note that ρ_0 (and other higher order terms) is also a function of X, it is a common practice to write $\rho_0(X,t,\alpha)$ to make this dependence explicit.

- At order $\mathcal{O}(\varepsilon)$, we have

$$F_1(X,t) + \frac{\partial\rho_0(X,t,\alpha)}{\partial t} + \frac{\partial\rho_1(X,t,\alpha)}{\partial\alpha} = \rho_0(X,t,\alpha)\frac{\partial f(X,t,\alpha)}{\partial X}, \tag{A.1.11}$$

  whose α average gives the solubility condition

$$F_1(X,t) = \left\langle \rho_0(X,t,\alpha)\frac{\partial f(X,t,\alpha)}{\partial X}\right\rangle. \tag{A.1.12}$$

Thus, to order $\mathcal{O}(\varepsilon)$, the equation of motion for X is written

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \langle f(X,t,\alpha)\rangle + \varepsilon\left\langle \rho_0(X,t,\alpha)\frac{\partial f(X,t,\alpha)}{\partial X}\right\rangle + \mathcal{O}(\varepsilon^2). \tag{A.1.13}$$

Evidently, the slow motion of the "guiding-center position" X is determined to lowest order by the average of the "force", and to next order by the correlation between the oscillation and the spatial gradient of the "force"; note that the gradient of the "force" as well as the "force" itself is evaluated at the "guiding-center position" X rather than the instantaneous "position" x as a consequence of the Taylor expansion about $x = X$.

A.2 Newtonian mechanical approach of guiding-center motion

The main reference of this section is [22].

The aim here is to investigate the motion of a charged particle in a slow-varying electromagnetic field. Here "slow" means that the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ change little in the space scale of one gyroradius ρ and in the time scale of one gyroperiod $1/|\Omega|$:

$$\frac{\rho\|\nabla\ \mathbf{field}\|}{|\mathbf{field}|} \ll 1, \frac{|\partial\ \mathbf{field}/\partial t|}{|\Omega| \times |\mathbf{field}|} \ll 1. \tag{A.2.1}$$

The motion of the charged particle in these fields consists of, intuitively, a rapid oscillation (cyclotron motion around the magnetic field line) and a slow drift (along the magnetic field line or so), as is the case when the fields are constants; if we are only interested in the drift motion, then it is natural to make use of the slow-varying nature to average out the gyration.

To keep track of the orders of the various quantities, let's introduce a *book-keeping* parameter ε, and make the following substitutions:

$$\begin{aligned} \rho &\to \varepsilon\rho, \\ (E, B, \Omega) &\to \varepsilon^{-1}(E, B, \Omega); \end{aligned} \tag{A.2.2}$$

this ε is treated as a small parameter throughout the derivation process[73], but is set to unity in the final answer — that's the meaning of "book-keeping." After these substitutions, all the symbols appearing in the equations represents order $\mathcal{O}(1)$ quantities.

A.2.1 The guiding-center

In order to make use of the method of averaging described above, we write the equations of motion in the first-order differential form

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{v}, \tag{A.2.3}$$

$$\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{q}{m}\frac{1}{\varepsilon}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}). \tag{A.2.4}$$

Based on the knowledge of the solution for the case of constant electromagnetic fields (parallel motion + $E \times B$ drift + gyration), let's seek for a solution of the following form

$$\begin{aligned} \boldsymbol{x}(t,\alpha) &= \boldsymbol{X}(t) + \varepsilon\boldsymbol{\rho}(t,\alpha), \\ \boldsymbol{v}(t,\alpha) &= \boldsymbol{V}(t) + \boldsymbol{w}(t,\alpha). \end{aligned} \tag{A.2.5}$$

Here, α is a new independent variable describing the phase of the gyrating particle, that is, the gyrophase. The functions $\boldsymbol{x}, \boldsymbol{v}, \boldsymbol{\rho}, \boldsymbol{w}$ should be periodic in α with period 2π, and their α averages (i.e., gyro-average, denoted by the bracket $\langle\cdot\rangle$) are

[73]Since the gyration is rapid conceptually, it is resonable to set Ω to be of order $\mathcal{O}(\varepsilon^{-1})$; then the magnetic field should also be of order $\mathcal{O}(\varepsilon^{-1})$. Assume that the electric force is comparable to the magnetic force, then the electric field is also of order $\mathcal{O}(\varepsilon^{-1})$.

$$\begin{aligned} &\boldsymbol{X}(t) = \langle \boldsymbol{x}(t,\alpha) \rangle, \quad \boldsymbol{V}(t) = \langle \boldsymbol{v}(t,\alpha) \rangle, \\ &\langle \boldsymbol{\rho}(t,\alpha) \rangle = \boldsymbol{0}, \qquad \langle \boldsymbol{w}(t,\alpha) \rangle = \boldsymbol{0}. \end{aligned} \tag{A.2.6}$$

The new variables $(\boldsymbol{X}, \boldsymbol{V})$ are free of oscillations along the particle trajectory, and the functions (whose value are to be determined) $\boldsymbol{\rho}$ and $\boldsymbol{w}$ represent the gyration radius and velocity, respectively.

A.2.2 Guiding-center equations of motion

To derive the equations of motion for the guiding-center variables $(\boldsymbol{X}, \boldsymbol{V})$, we substitute the following expansions into the original equations of motion, and solve order by order in ε (noting that $\mathrm{d}\alpha/\mathrm{d}t \approx \Omega = \mathcal{O}(\varepsilon^{-1})$)

$$\begin{aligned} \boldsymbol{V}(t) &= \boldsymbol{V}_0(t) + \varepsilon \boldsymbol{V}_1(t) + \mathcal{O}(\varepsilon^2), \\ \boldsymbol{\rho}(t,\alpha) &= \boldsymbol{\rho}_0(t,\alpha) + \varepsilon \boldsymbol{\rho}_1(t,\alpha) + \mathcal{O}(\varepsilon^2), \\ \boldsymbol{w}(t,\alpha) &= \boldsymbol{w}_0(t,\alpha) + \varepsilon \boldsymbol{w}_1(t,\alpha) + \mathcal{O}(\varepsilon^2), \\ \frac{\mathrm{d}\alpha(t)}{\mathrm{d}t} &= \varepsilon^{-1}\omega_{-1}(t) + \omega_0(t) + \mathcal{O}(\varepsilon); \end{aligned} \tag{A.2.7}$$

the electromagnetic fields experienced by the particle are Taylor-expanded as

$$\begin{aligned} \boldsymbol{E}(\boldsymbol{x}(t,\alpha),t) &= \boldsymbol{E}(\boldsymbol{X}(t),t) + \varepsilon \boldsymbol{\rho}_0(t,\alpha) \cdot \nabla_{\boldsymbol{X}} \boldsymbol{E}(\boldsymbol{X}(t),t) + \mathcal{O}(\varepsilon^2), \\ \boldsymbol{B}(\boldsymbol{x}(t,\alpha),t) &= \boldsymbol{B}(\boldsymbol{X}(t),t) + \varepsilon \boldsymbol{\rho}_0(t,\alpha) \cdot \nabla_{\boldsymbol{X}} \boldsymbol{B}(\boldsymbol{X}(t),t) + \mathcal{O}(\varepsilon^2). \end{aligned} \tag{A.2.8}$$

Let's first look at the kinematic equation Eq. (A.2.3). Substituting Eq. (A.2.6) into Eq. (A.2.3) gives

$$\frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} + \varepsilon \frac{\partial \boldsymbol{\rho}(t,\alpha)}{\partial t} + \varepsilon \frac{\mathrm{d}\alpha(t)}{\mathrm{d}t} \frac{\partial \boldsymbol{\rho}(t,\alpha)}{\partial \alpha} = \boldsymbol{V}(t) + \boldsymbol{w}(t). \tag{A.2.9}$$

The α average is

$$\frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} = \boldsymbol{V}(t) \quad \text{(to all orders in } \varepsilon\text{)}, \tag{A.2.10}$$

which further implies

$$\varepsilon \frac{\partial \boldsymbol{\rho}(t,\alpha)}{\partial t} + \varepsilon \frac{\mathrm{d}\alpha(t)}{\mathrm{d}t} \frac{\partial \boldsymbol{\rho}(t,\alpha)}{\partial \alpha} = \boldsymbol{w}(t,\alpha) \quad \text{(to all orders in } \varepsilon\text{)}. \tag{A.2.11}$$

At order $\mathcal{O}(1)$, Eq. (A.2.11) reduces to

$$\omega_{-1}(t) \frac{\partial \boldsymbol{\rho}_0(t,\alpha)}{\partial \alpha} = \boldsymbol{w}_0(t,\alpha); \tag{A.2.12}$$

that is, once we have solved for $\boldsymbol{w}_0$ as a function of α, then ρ_0 as a function of α follows directly by integration.

Now comes the momentum equation Eq. (A.2.4). Substituting Eq. (A.2.6) into Eq. (A.2.4) gives[74]

$$\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t}+\frac{\partial \boldsymbol{w}}{\partial t}+\frac{\mathrm{d}\alpha}{\mathrm{d}t}\frac{\partial \boldsymbol{w}}{\partial \alpha}=\frac{q}{m}\frac{1}{\varepsilon}(\boldsymbol{E}(\boldsymbol{X}+\varepsilon\boldsymbol{\rho})+(\boldsymbol{V}+\boldsymbol{w})\times\boldsymbol{B}(\boldsymbol{X}+\varepsilon\boldsymbol{\rho})). \tag{A.2.15}$$

At order $\mathcal{O}(\varepsilon^{-1})$, we have

$$\omega_{-1}\frac{\partial \boldsymbol{w}_0}{\partial \alpha}=\frac{q}{m}(\boldsymbol{E}_0+(\boldsymbol{V}_0+\boldsymbol{w}_0)\times\boldsymbol{B}_0); \tag{A.2.16}$$

averaging over α gives the solubility condition

$$\boldsymbol{E}_0+\boldsymbol{V}_0\times\boldsymbol{B}_0=\boldsymbol{0}, \tag{A.2.17}$$

which can only be satisfied if

$$E_{0\parallel}\equiv\hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{E}_0=o(|\boldsymbol{E}_0|),\quad \hat{\boldsymbol{e}}_\parallel:=\frac{\boldsymbol{B}_0}{B_0}, \tag{A.2.18}$$

that is, if $E_{0\parallel}$ is smaller than $|\boldsymbol{E}_0|\approx|\boldsymbol{E}_{0\perp}|$ by at least an order in ε (note that $|\boldsymbol{E}_0|$ is of order $\mathcal{O}(\varepsilon^{-1})$ by assumpution) — a large parallel electric field would invalidate the drift approximation. For this reason, we will then write

$$\boldsymbol{E}_0=\varepsilon E_{0\parallel}\hat{\boldsymbol{e}}_\parallel+\boldsymbol{E}_{0\perp} \tag{A.2.19}$$

so that $E_{0\parallel}$ appearing in the equations are of order $\mathcal{O}(1)$. Solving for $\boldsymbol{V}_0$, we find

$$\boldsymbol{V}_0=u\hat{\boldsymbol{e}}_\parallel+\boldsymbol{v}_{E0}, \tag{A.2.20}$$

where $\boldsymbol{v}_{E0}=\boldsymbol{E}_0\times\boldsymbol{B}_0/B_0^2$ is the $E\times B$ drift velocity (evaluated at the guiding-center position $\boldsymbol{X}$, thus the subscript 0), and $u:=\hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{V}_0$ is the parallel velocity of the guiding-center (or the particle) which is undetermined at this order.
Substituting Eq. (A.2.17) back into Eq. (A.2.16), we see that

$$\omega_{-1}\frac{\partial \boldsymbol{w}_0}{\partial \alpha}=\Omega\boldsymbol{w}_0\times\hat{\boldsymbol{e}}_\parallel,\quad \Omega:=\frac{qB_0}{m}, \tag{A.2.21}$$

whose general solutions are

$$\boldsymbol{w}_0=c\hat{\boldsymbol{e}}_\parallel+w_\perp\left(\hat{\boldsymbol{e}}_{\perp1}\cos\left(\frac{\Omega}{\omega_{-1}}\alpha+\varphi\right)+\hat{\boldsymbol{e}}_{\perp2}\sin\left(\frac{\Omega}{\omega_{-1}}\alpha+\varphi\right)\right), \tag{A.2.22}$$

where $\left\{\boldsymbol{X};\hat{\boldsymbol{e}}_{\perp1},\hat{\boldsymbol{e}}_{\perp2},\hat{\boldsymbol{e}}_\parallel\right\}$ forms an othonormal right-hand frame, and the three "constants" $w_\perp>0$, c and φ have no dependence on α (but they can depend on t). The periodicity of $\boldsymbol{w}_0$ requires

$$\omega_{-1}=\Omega,\quad c=0, \tag{A.2.23}$$

[74] From now on, we will not write the t,α arguments explicitly for brevity, and denote

$$\boldsymbol{E}_0\equiv\boldsymbol{E}(\boldsymbol{X}(t),t),\boldsymbol{B}_0\equiv\boldsymbol{B}(\boldsymbol{X}(t),t). \tag{A.2.13}$$

Note that the ordinary time derivative acts on the field quantities not as partial derivative, but as material derivative, e.g.

$$\frac{\mathrm{d}\boldsymbol{E}_0}{\mathrm{d}t}=\frac{\partial\boldsymbol{E}_0}{\partial t}+\dot{\boldsymbol{X}}\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0. \tag{A.2.14}$$

which in turn implies (absorbing φ into the definition of α)

$$\boldsymbol{w}_0 = w_\perp(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha), \quad w_\perp = |\boldsymbol{w}_0| \equiv w_0. \tag{A.2.24}$$

Note that $w_\perp$ is undetermined at this order. Now Eq. (A.2.12) becomes

$$\frac{\partial \boldsymbol{\rho}_0}{\partial \alpha} = \frac{w_\perp}{\Omega}(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha); \tag{A.2.25}$$

this is easily integrated (periodicity in α requires the integration constant to be zero), and we find

$$\boldsymbol{\rho}_0 = \frac{w_\perp}{\Omega}(\hat{\boldsymbol{e}}_{\perp 1}\sin\alpha - \hat{\boldsymbol{e}}_{\perp 2}\cos\alpha) = \frac{1}{\Omega}\hat{\boldsymbol{e}}_\parallel \times \boldsymbol{w}_0 \tag{A.2.26}$$

or

$$\boldsymbol{w}_0 = \Omega\boldsymbol{\rho}_0 \times \hat{\boldsymbol{e}}_\parallel. \tag{A.2.27}$$

At order $\mathcal{O}(1)$, the momentum equation Eq. (A.2.15) writes

$$\begin{aligned}&\frac{\mathrm{d}\boldsymbol{V}_0}{\mathrm{d}t} + \frac{\partial \boldsymbol{w}_0}{\partial t} + \Omega\frac{\partial \boldsymbol{w}_1}{\partial \alpha} + \omega_0\frac{\partial \boldsymbol{w}_0}{\partial \alpha}\\ &= \frac{q}{m}\Big(E_{0\parallel}\hat{\boldsymbol{e}}_\parallel + \boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0 + \boldsymbol{V}_1\times\boldsymbol{B}_0 + \boldsymbol{V}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0) + \boldsymbol{w}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0) + \boldsymbol{w}_1\times\boldsymbol{B}_0\Big);\end{aligned} \tag{A.2.28}$$

averaging over α gives the solubility condition

$$m\frac{\mathrm{d}\boldsymbol{V}_0}{\mathrm{d}t} = q\Big(E_{0\parallel}\hat{\boldsymbol{e}}_\parallel + \boldsymbol{V}_1\times\boldsymbol{B}_0 + \langle\boldsymbol{w}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0)\rangle\Big). \tag{A.2.29}$$

The calculation of $\langle\boldsymbol{w}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0)\rangle$ is direct:

$$\begin{aligned}\langle\boldsymbol{w}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0)\rangle &= \Omega\Big\langle\Big(\boldsymbol{\rho}_0\times\hat{\boldsymbol{e}}_\parallel\Big)\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0)\Big\rangle\\ &= \Omega\Big(\hat{\boldsymbol{e}}_\parallel\langle\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0\cdot\boldsymbol{\rho}_0\rangle - \langle\boldsymbol{\rho}_0\boldsymbol{\rho}_0\rangle\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0\cdot\hat{\boldsymbol{e}}_\parallel\Big)\\ &= \Omega\Big(\hat{\boldsymbol{e}}_\parallel\langle\boldsymbol{\rho}_0\boldsymbol{\rho}_0\rangle : \nabla_{\boldsymbol{X}}\boldsymbol{B}_0\Big) - \langle\boldsymbol{\rho}_0\boldsymbol{\rho}_0\rangle\cdot\nabla_{\boldsymbol{X}}B_0),\end{aligned} \tag{A.2.30}$$

where the symbol $:$ refers to the double contraction $\tilde{P} : \tilde{Q} = P^{ij}Q_{ij}$. Basically we need only to calculate

$$\begin{aligned}\langle\boldsymbol{\rho}_0\boldsymbol{\rho}_0\rangle &= \left(\frac{w_\perp}{\Omega}\right)^2\langle\hat{\boldsymbol{e}}_{\perp 1}\hat{\boldsymbol{e}}_{\perp 1}\sin^2\alpha + \hat{\boldsymbol{e}}_{\perp 2}\hat{\boldsymbol{e}}_{\perp 2}\cos^2\alpha - (\hat{\boldsymbol{e}}_{\perp 1}\hat{\boldsymbol{e}}_{\perp 2} + \hat{\boldsymbol{e}}_{\perp 2}\hat{\boldsymbol{e}}_{\perp 1})\cos\alpha\sin\alpha\rangle\\ &= \frac{1}{2}\left(\frac{w_\perp}{\Omega}\right)^2(\hat{\boldsymbol{e}}_{\perp 1}\hat{\boldsymbol{e}}_{\perp 1} + \hat{\boldsymbol{e}}_{\perp 2}\hat{\boldsymbol{e}}_{\perp 2})\\ &= \frac{1}{2}\left(\frac{w_\perp}{\Omega}\right)^2\big(\tilde{g} - \hat{\boldsymbol{e}}_\parallel\hat{\boldsymbol{e}}_\parallel\big),\end{aligned} \tag{A.2.31}$$

where $\tilde{g}$ is the metric tensor [3]. Thus,

$$q\langle\boldsymbol{w}_0\times(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{B}_0)\rangle = \frac{qw_\perp^2}{2\Omega}\Big(\hat{\boldsymbol{e}}_\parallel\big(\tilde{g} - \hat{\boldsymbol{e}}_\parallel\hat{\boldsymbol{e}}_\parallel\big) : \nabla_{\boldsymbol{X}}\boldsymbol{B}_0\Big) - \big(\tilde{g} - \hat{\boldsymbol{e}}_\parallel\hat{\boldsymbol{e}}_\parallel\big)\cdot\nabla_{\boldsymbol{X}}B_0) \tag{A.2.32}$$

$$
\begin{aligned}
&= \frac{mw_\perp^2}{2B_0}\big(\hat{\boldsymbol{e}}_\parallel(\nabla_{\boldsymbol{X}}\cdot\boldsymbol{B}_0) - \nabla_{\boldsymbol{X}}B_0 + \text{cancel out}\big) \\
&=: -\mu\nabla_{\boldsymbol{X}}B_0,
\end{aligned}
\tag{A.2.32}
$$

where in the last step the magnetic Gauss's theorem $\nabla_{\boldsymbol{X}}\cdot\boldsymbol{B}_0 = 0$ is used, and

$$
\mu := \frac{mw_\perp^2}{2B_0} = \frac{W_\perp}{B_0} > 0, \quad W_\perp \equiv \frac{1}{2}mw_\perp^2 \tag{A.2.33}
$$

is the "magnetic momentum"; we also define the vector $\boldsymbol{\mu} := -\hat{\boldsymbol{e}}_\parallel\mu$.

- The magnetic momentum:

  From the magnetostatic multiple expansion, we konw that for any current distribution, we can define its magnetic dipole moment

$$
\boldsymbol{m} := \frac{1}{2}\int \mathrm{d}^3x'\, \boldsymbol{x}'\times\boldsymbol{J}(\boldsymbol{x}'). \tag{A.2.34}
$$

  Since $\boldsymbol{B}$ is now of order $\mathcal{O}(\varepsilon^{-1})$, the current density (which is the source of the magnetic field) is also of order $\mathcal{O}(\varepsilon^{-1})$ and we should make the substitution $\boldsymbol{J}\to\varepsilon^{-1}\boldsymbol{J}$ in the above definition.

  Without loss of generality we choose $\boldsymbol{X}$ as our instantaneous origin, then a single particle carrys a current density $\boldsymbol{J}(\boldsymbol{x}') = q(\boldsymbol{V}+\boldsymbol{w})\delta^{(3)}(\boldsymbol{x}'-\varepsilon\boldsymbol{\rho})$. The magnetic dipole momentum is

$$
\boldsymbol{m} = \frac{q}{2}\varepsilon\boldsymbol{\rho}\times\frac{1}{\varepsilon}(\boldsymbol{V}+\boldsymbol{w}), \tag{A.2.35}
$$

  whose α average is the "magnetic momentum" defined above

$$
\boldsymbol{\mu} := \langle\boldsymbol{m}\rangle = \frac{q}{2}\langle\boldsymbol{\rho}_0\times\boldsymbol{w}_0\rangle + \mathcal{O}(\varepsilon) = -\frac{qw_\perp^2}{2\Omega}\hat{\boldsymbol{e}}_\parallel + \mathcal{O}(\varepsilon). \tag{A.2.36}
$$

  Note that the magnetic momentum has nothing to do with the sign of charge the particle carried. The fact that the magnetic momentum is anti-parallel to the magnetic field indicates that the plasmas are diamagnetic.

Finally, the averaged momentum equation at order $\mathcal{O}(1)$ is

$$
m\frac{\mathrm{d}\boldsymbol{V}_0}{\mathrm{d}t} = qE_{0\parallel}\hat{\boldsymbol{e}}_\parallel + q\boldsymbol{V}_1\times\boldsymbol{B}_0 - \mu\nabla_{\boldsymbol{X}}B_0, \tag{A.2.37}
$$

whose parallel component determines the envolution of $u = \hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{V}_0$ (substituting Eq. (A.2.20) and noting that $\hat{\boldsymbol{e}}_\parallel\cdot\mathrm{d}\hat{\boldsymbol{e}}_\parallel/\mathrm{d}t = 0$)

$$
m\frac{\mathrm{d}u}{\mathrm{d}t} = qE_{0\parallel} + \boldsymbol{\mu}\cdot\nabla_{\boldsymbol{X}}B_0 - m\hat{\boldsymbol{e}}_\parallel\cdot\frac{\mathrm{d}\boldsymbol{v}_{E0}}{\mathrm{d}t}. \tag{A.2.38}
$$

Recalling Eq. (A.2.13), the time derivative in $\mathrm{d}\boldsymbol{v}_{E0}/\mathrm{d}t$ acting on the electromagnetic fields is not a partial derivative but the material derivative following the guiding-center motion, that is,

$$
\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t}\cdot\nabla_{\boldsymbol{X}} = \frac{\partial}{\partial t} + \boldsymbol{V}\cdot\nabla_{\boldsymbol{X}} = \frac{\partial}{\partial t} + \boldsymbol{V}_0\cdot\nabla_{\boldsymbol{X}} + \mathcal{O}(\varepsilon). \tag{A.2.39}
$$

The order $\mathcal{O}(\varepsilon)$ perpendicular drift velocity is given by the perpendicular part of Eq. (A.2.37):

$$\boldsymbol{V}_{1\perp} = \frac{1}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \left(\frac{\mathrm{d}\boldsymbol{V}_0}{\mathrm{d}t} + \frac{\mu}{m}\nabla_{\boldsymbol{X}} B_0\right). \tag{A.2.40}$$

The $\mathcal{O}(\varepsilon)$ correction to the parallel velocity u is undetermined to this order.[75] This won't cause a big problem, because it is a small correction to a type of motion (that is, parallel drift) that already exists at $\mathcal{O}(1)$, whereas the $\mathcal{O}(\varepsilon)$ perpendicular drift is a completely new type of motion; in particular, the $\mathcal{O}(\varepsilon)$ perpendicular drift differs fundamentally from the $E \times B$ drift in that it is not the same for all species, and, therefore, cannot be eliminated by transforming to a new reference frame.

A.2.2.1 First-order perpendicular drifts of the guiding-center

Each term in Eq. (A.2.40) has a physical interpretation:

- Magnetic gradient drift:

$$\boldsymbol{v}_{\text{grad}} := \frac{\mu}{m\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \nabla_{\boldsymbol{X}} B_0 = \frac{W_\perp}{qB_0^2}\hat{\boldsymbol{e}}_{\parallel} \times \nabla_{\boldsymbol{X}} B_0. \tag{A.2.41}$$

  This drift is caused by the slight variation of the gyroradius with gyrophase as the charged particle rotates in a nonuniform magnetic field. The gyroradius is reduced on the high-field side of the Larmor orbit, but is increased on the low-field side. The net effect is that the orbit does not quite close on itself and the particle slowly drifts perpendicular to the local gradient of the field strength in the gyration plane.

- Inertial drifts:
  - Magnetic curvature drift:

$$\begin{aligned} \frac{1}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \frac{\mathrm{d}}{\mathrm{d}t}\left(u\hat{\boldsymbol{e}}_{\parallel}\right) &= \frac{u}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \frac{\mathrm{d}\hat{\boldsymbol{e}}_{\parallel}}{\mathrm{d}t} \\ &= \frac{u}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \left(\frac{\partial \hat{\boldsymbol{e}}_{\parallel}}{\partial t} + \boldsymbol{v}_{E0} \cdot \nabla_{\boldsymbol{X}}\hat{\boldsymbol{e}}_{\parallel} + u\hat{\boldsymbol{e}}_{\parallel} \cdot \nabla_{\boldsymbol{X}}\hat{\boldsymbol{e}}_{\parallel}\right). \end{aligned} \tag{A.2.42}$$

    The first two terms represent the acceleration that results when changes in the direction of the magnetic field cause part of the parallel velocity to be converted to perpendicular velocity. In the important limit of stationary magnetic fields and weak electric fields, the previous expression is dominated by the last term

$$\boldsymbol{v}_{\text{curv}} = \frac{u^2}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \left(\hat{\boldsymbol{e}}_{\parallel} \cdot \nabla_{\boldsymbol{X}}\hat{\boldsymbol{e}}_{\parallel}\right) =: \frac{2W_\parallel}{qB_0}\hat{\boldsymbol{e}}_{\parallel} \times \boldsymbol{\kappa}, \quad W_\parallel \equiv \frac{1}{2}mu^2, \tag{A.2.43}$$

    where $\boldsymbol{\kappa} := \hat{\boldsymbol{e}}_{\parallel} \cdot \nabla_{\boldsymbol{X}}\hat{\boldsymbol{e}}_{\parallel}$ is equal to the curvature of the magnetic field line.
  - Polarization drift:

[75] The order $\mathcal{O}(\varepsilon)$ correction to the parallel drift turns out to be $(w_\perp^2/2\Omega)\hat{\boldsymbol{e}}_{\parallel}\hat{\boldsymbol{e}}_{\parallel} \cdot \left(\nabla_{\boldsymbol{X}} \times \hat{\boldsymbol{e}}_{\parallel}\right)$, which is called the *Baños drift*.

$$\boldsymbol{v}_{\text{pola}} := \frac{1}{\Omega}\hat{\boldsymbol{e}}_{\parallel} \times \frac{\mathrm{d}\boldsymbol{v}_{E0}}{\mathrm{d}t} \overset{\text{const } B}{=\!=} \frac{1}{\Omega}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\boldsymbol{E}_{\perp}}{B_0}\right). \tag{A.2.44}$$

We can understand the polarization drift by considering what happens when a gradually increasing electric field is applied to a charged particle at rest. Assume that the magnetic field is a constant $\boldsymbol{B} = B_0\hat{\boldsymbol{e}}_z$, and a perpendicular electric field gradually increases linearly in its magnitude $\boldsymbol{E}(t) = t(E_0/t_0)\hat{\boldsymbol{e}}_y, 0 \leq t \leq t_0$. Initially, the particle is at rest; the particle will continuously gain energy from the electric field and eventually undergo the normal $E \times B$ drift. The energy balance relation

$$\begin{aligned} W_{\text{averaged}} = \frac{1}{2}qE_0\delta = \frac{1}{2}m\left(\frac{E_0}{B_0}\right)^2 = \Delta E \\ \Rightarrow \delta = \frac{E_0}{\Omega B_0} \end{aligned} \tag{A.2.45}$$

indicates that the position of the guiding center is shifted from the initial position by an amount of $\delta\hat{\boldsymbol{e}}_y$. The polarization drift is essentially the time derivative of this displacement. Since ions are shifted in the opposite direction from the electrons, polarization drift gives rise to a current, called the polarization current.

A.2.2.2 Kinetic energy theorem and invariance of the magnetic moment

The evolution of the magnetic momentum, or equivalently the perpendicular kinetic energy, is determined by the energy, obtained by taking the dot product of the momentum equation Eq. (A.2.4) with the velocity:

$$\frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}|\boldsymbol{v}|^2 = \frac{1}{\varepsilon}q\boldsymbol{v}\cdot\boldsymbol{E}. \tag{A.2.46}$$

The α averaged version of the above equation is, to lowest order in ε (that is, $\mathcal{O}(1)$),

$$\frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(u^2 + w_{\perp}^2 + |\boldsymbol{v}_{E0}|^2\right) = q\left(uE_{0\parallel} + \boldsymbol{V}_{1\perp}\cdot\boldsymbol{E}_{0\perp} + \langle\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0\cdot\boldsymbol{w}_0\rangle\right). \tag{A.2.47}$$

Here, $\mathrm{d}/\mathrm{d}t$ is to be understood as the material derivative when acting on the electromagnetic fields. The calculation of the remaining term $\langle\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0\cdot\boldsymbol{w}_0\rangle = \Omega\hat{\boldsymbol{e}}_{\parallel}\cdot\langle(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0)\times\boldsymbol{\rho}_0\rangle$ is straightforward:

$$\begin{aligned} \Omega\hat{\boldsymbol{e}}_{\parallel}\cdot\langle(\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{E}_0)\times\boldsymbol{\rho}_0\rangle &= \Omega\varepsilon^{ijk}\left\langle\rho_0^m\rho_{0j}\right\rangle(\nabla_m E_{0i})b_{0k} \\ &\overset{\text{Eq. (A.2.31)}}{=\!=} \frac{w_{\perp}^2}{2\Omega}\varepsilon^{ijk}\left(\delta_j^m - b_0^m b_{0j}\right)(\nabla_m E_{0i})b_{0k} \\ &= -\frac{\mu}{q}\hat{\boldsymbol{e}}_{\parallel}\cdot(\nabla_{\boldsymbol{X}}\times\boldsymbol{E}_0) = -\frac{\mu}{q}\left(\hat{\boldsymbol{b}}\cdot(\nabla\times\boldsymbol{E})\right)|_{\boldsymbol{x}=\boldsymbol{X}} \\ &\overset{\text{Faraday's law}}{=\!=} \frac{\mu}{q}\left(\hat{\boldsymbol{b}}\cdot\frac{\partial\boldsymbol{B}}{\partial t}\right)|_{\boldsymbol{x}=\boldsymbol{X}} = \frac{\mu}{q}\frac{\partial B_0}{\partial t}. \end{aligned} \tag{A.2.48}$$

Finally, Eq. (A.2.47) writes

$$\frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(u^2+|\boldsymbol{v}_{E0}|^2\right)+\frac{\mathrm{d}}{\mathrm{d}t}(\mu B_0)=q\left(uE_{0\|}+\boldsymbol{V}_{1\perp}\cdot\boldsymbol{E}_{0\perp}\right)+\mu\frac{\partial B_0}{\partial t}. \tag{A.2.49}$$

Eliminating u with Eq. (A.2.38) and $\boldsymbol{V}_{1\perp}$ with Eq. (A.2.40), we arrive at, after a tedious but straightforward calculation,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mu B_0)=\mu\frac{\mathrm{d}B_0}{\mathrm{d}t}\Rightarrow\frac{\mathrm{d}\mu}{\mathrm{d}t}=0; \tag{A.2.50}$$

that is, the *magnetic moment* μ *is conserved* to lowest order in ε. Here, again, we stress that the derivative $\mathrm{d}/\mathrm{d}t$ is to be understood as the material derivative when acting on the electromagnetic fields.

A.2.2.3 Adiabatic invariance and the magnetic moment

The magnetic moment is a conserved quantity related with gyration, which is a periodic motion. It is well-known that for a periodic Hamiltonian system characterized by a slow-varying parameter[76] λ, the *action variable* defined by

$$J:=\frac{1}{2\pi}\oint_{\lambda=\text{ const}}\mathrm{d}\boldsymbol{q}\cdot\boldsymbol{p} \tag{A.2.51}$$

is a conserved quantity, called an *adiabatic invariant.* [59]

For a charged particle in a slow-varying electromagnetic field, the gyration is periodic, and the electromagnetic fields (or gyrofrequency, gyroradius, etc.) serve as the slow-varying parameters in the Hamiltonian. The action variable, i.e., the momentum conjugate to the *angle variable* (here the gyrophase) α, is given by

$$\begin{aligned}
J&:=\frac{1}{2\pi}\oint_{\boldsymbol{E},\boldsymbol{B}=\text{ const}}\mathrm{d}\boldsymbol{q}\cdot\boldsymbol{p}\\
&\overset{\boldsymbol{p}=m\boldsymbol{v}+q\boldsymbol{A}}{\underset{\boldsymbol{q}=\boldsymbol{X}+\boldsymbol{\rho}}{=\!=\!=}}\frac{1}{2\pi}\oint_{\text{gyration orbit}}\mathrm{d}\alpha\,\varepsilon\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\cdot\left(m(\boldsymbol{V}+\boldsymbol{w})+q\frac{1}{\varepsilon}\boldsymbol{A}\right)\\
&=\left\langle\varepsilon\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\cdot\left(m(\boldsymbol{V}+\boldsymbol{w})+q\frac{1}{\varepsilon}\boldsymbol{A}\right)\right\rangle\\
&=\varepsilon\left(m\left\langle\frac{\partial\boldsymbol{\rho}_0}{\partial\alpha}\cdot\boldsymbol{w}_0\right\rangle+q\left\langle\boldsymbol{\rho}_0\cdot\nabla_{\boldsymbol{X}}\boldsymbol{A}_0\cdot\frac{\partial\boldsymbol{\rho}_0}{\partial\alpha}\right\rangle\right)(1+\mathcal{O}(\varepsilon))\\
&\overset{\text{direct calculation}}{\underset{\text{as in Eq. (A.2.48)}}{=\!=\!=}}\varepsilon\left(\frac{mw_\perp^2}{\Omega}+\frac{q}{\Omega}\left(-\frac{\mu}{q}\left(\hat{\boldsymbol{b}}\cdot(\nabla\times\boldsymbol{A})\right)|_{\boldsymbol{x}=\boldsymbol{X}}\right)\right)(1+\mathcal{O}(\varepsilon))\\
&=\varepsilon\frac{m}{q}\mu(1+\mathcal{O}(\varepsilon)).
\end{aligned} \tag{A.2.52}$$

[76] "Slow" means that over a period of motion T the relative change of λ is small, i.e. $T|\mathrm{d}\lambda/\mathrm{d}t/\lambda|\ll1$; we say that λ changes "adiabaticly". Actually all we need is that λ is a parameter that appears in the Hamiltonian, it need not really characterize the system itself, but, for example, could be a parameter characterizing the external fields.

Order	Dimensionless	Fields	Distances	Rates	Velocities
ϵ^{-1}		$\mathbf{B}$, $\mathbf{E}_\perp$		Ω	
1		$E_\parallel$	L	$\mathbf{v}/\mathrm{L}$, $\mathbf{v}_E/\mathrm{L}$, τ^{-1}	$\mathbf{v}$, $\mathbf{v}_E$
ϵ	ρ/L, $(\Omega\tau)^{-1}$		ρ	$\mathbf{v}_\nabla/L$, $\mathbf{v}_{\mathrm{pol}}/\mathrm{L}$, $\mathbf{v}_\kappa/\mathrm{L}$	$\mathbf{v}_\nabla$, $\mathbf{v}_\kappa$, $\mathbf{v}_{\mathrm{pol}}$

Figure 6.1: Guiding-center ordering. [10]

Thus to lowest order, the action variable J is proportional to the magnetic moment μ, and the conservation of μ follows from the adiabatic invariance of J.

A.2.3 Summary of the results

The particle always gyrates around the magnetic field at the local gyrofrequency $\Omega = qB_0/m$. The local perpendicular gyration velocity $w_\perp$ is determined by the requirement that the magnetic moment $\mu = mw_\perp^2/2B_0$ be a constant of the motion, and the gyroradius is equal to $w_\perp/\Omega$. The parallel velocity u is determined by Eq. (A.2.38), and the perpendicular drift velocity is the sum of the $E\times B$ drift velocity $\boldsymbol{v}_{E0}$ and the first-order perpendicular drift velocity $\boldsymbol{V}_{1\perp}$.

The particle's "state", which is defined by the six phase-space variables $(\boldsymbol{x}, \boldsymbol{v})$, can be equally well described by the six guiding-center phase-space variables $(\boldsymbol{X}, u, \mu, \alpha)$. As there are still six variables parametrizing the phase space, there is no loss of information in making the guiding-center transformation $(\boldsymbol{x}, \boldsymbol{v}) \mapsto (\boldsymbol{X}, u, \mu, \alpha)$. It is tempting to use $(J = m/q\mu, \alpha)$ as a pair of canonical coordinates since the action variable J is an adiabatic invariant, which means that the conjugate angle variable α is ignorable (cyclic); see next section.

A.3 Variational approach of guiding-center motion

The main reference of this section is [10].

A.3.1 Phase-space action principle and phase-space Lagrangian

The main reference of this subsection is [60].

Hamilton's principle

$$\delta \int_{t_1}^{t_2} \mathrm{d}t\, L(q^i, \dot{q}^i, t) = 0 \tag{A.3.1}$$

is originally formulated in configuration space, whose *justification is only that it leads to the correct equations of motion*, that is, the Lagrange equations (n is the number of degrees of freedom)

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0, i = 1, 2, \cdots, n. \tag{A.3.2}$$

In the same sprit, we can seek for a phase-space action principle, whose validity is ensured as long as it generates the correct equations of motion (now the canonical equations of Hamilton).

The Hamilton's principle itself gives us some insights on how to construct a phase-space action principle. The first modification is that the integral must be evaluated over the trajectory of the system point in phase space, and the varied paths must be in the neighborhood of this phase space trajectory. In the spirit of the Hamiltonian formulation, both q and p must be treated as independent coordinates to be varied independently, and we require that the varied paths have the same end points in both q and p. Also, the integrand in the action integral must be expressed as a function of both q and p , and their time derivatives[77]. The following "modified Hamilton's princple" is vividly portrayed:

$$\delta \int_{t_1}^{t_2} \mathrm{d}t \, \mathcal{L}(q^i, \dot{q}^i, p_i, t) = 0, \tag{A.3.3}$$

where

$$\mathcal{L}(q^i, \dot{q}^i, p_i, t) \coloneqq p_i \dot{q}^i - H(q^i, p_i, t) \tag{A.3.4}$$

is a phase-space Lagrangian (which is equal *in value* to the original configuration-space Lagrangian) and $H(q^i, p_i, t)$ is the Hamiltonian. The $2n$ Euler-Lagrange equations derived from this variational principle are the canonical equations:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{q}^i} - \frac{\partial \mathcal{L}}{\partial q^i} &= 0 \Rightarrow \dot{p}_i = -\frac{\partial H}{\partial q^i}, \\ \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{p}_i} - \frac{\partial \mathcal{L}}{\partial p_i} &= 0 \Rightarrow \dot{q}^i = \frac{\partial H}{\partial p_i}. \end{aligned} \tag{A.3.5}$$

Thus the validity of the modified Hamilton's principle is justified.

Note that the phase-space Lagrangian is not unique; for example, we can always add a total time derivative of an arbitrary (well-behaved) phase-space function $F(q^i, p_i, t)$ without changing the equations of motion. This freedom is useful in the derivation of the guiding-center Lagrangian [22], since

$$\langle \mathcal{L} - \langle \mathcal{L} \rangle \rangle = 0 \Rightarrow \mathcal{L} - \langle \mathcal{L} \rangle = \Omega \frac{\partial F}{\partial \alpha} = \frac{\mathrm{d}F}{\mathrm{d}t}(1 + \mathcal{O}(\varepsilon)) \tag{A.3.6}$$

allows us to replace $\mathcal{L}$ with its gyroaverage $\langle \mathcal{L} \rangle$.

A.3.1.1 Phase-space Lagrangian of a charged particle in an electromagnetic field

The configuration-space Lagrangian of a charged particle in an electromagnetic field is

$$L(\boldsymbol{x}, \dot{\boldsymbol{x}}, t) = \frac{1}{2} m |\dot{\boldsymbol{x}}|^2 + q \boldsymbol{A}(\boldsymbol{x}, t) \cdot \dot{\boldsymbol{x}} - q \varphi(\boldsymbol{x}, t), \tag{A.3.7}$$

where $\boldsymbol{A}$ is the vector potential and φ the scalar potential; the electromagnetic fields are expressed in terms of these potnetials as

[77]Note that $\dot{q}$ and p are assumed to be independent when calculating the variation; the resultant equations of motion will specify their relation.

$$\boldsymbol{E} = -\nabla\varphi - \frac{\partial \boldsymbol{A}}{\partial t}, \quad \boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{A.3.8}$$

The canonical momentum is

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{x}}} = m\dot{\boldsymbol{x}} + q\boldsymbol{A}, \tag{A.3.9}$$

thus the Hamiltonian is

$$\begin{aligned} H(\boldsymbol{x}, \boldsymbol{p}, t) &= \boldsymbol{p} \cdot \dot{\boldsymbol{x}} - L(\boldsymbol{x}, \dot{\boldsymbol{x}}, t) \\ &= \frac{1}{2}m|\dot{\boldsymbol{x}}|^2 + q\varphi(\boldsymbol{x}, t) \\ &= \frac{|\boldsymbol{p} - q\boldsymbol{A}(\boldsymbol{x}, t)|^2}{2m} + q\varphi(\boldsymbol{x}, t). \end{aligned} \tag{A.3.10}$$

For now we would like to use the non-canonical coordinate[78] $\boldsymbol{v} := (\boldsymbol{p} - q\boldsymbol{A})/m$ instead of the canonical one $\boldsymbol{p}$ to serve as the "momentum", so the non-canonical phase-space Lagrangian we will use is[79]

$$\mathcal{L}(\boldsymbol{x}, \dot{\boldsymbol{x}}, \boldsymbol{v}, t) = (m\boldsymbol{v} + q\boldsymbol{A}(\boldsymbol{x}, t)) \cdot \dot{\boldsymbol{x}} - \frac{1}{2}m|\boldsymbol{v}|^2 - q\varphi(\boldsymbol{x}, t). \tag{A.3.11}$$

A.3.2 Derivation of the guiding-center Lagrangian

Introducting the book-keeping parameter ε, the non-canonical phase-space Lagrangian is written

$$\mathcal{L}(\boldsymbol{x}, \dot{\boldsymbol{x}}, \boldsymbol{v}, t) = (m\boldsymbol{v} + q\varepsilon^{-1}\boldsymbol{A}(\boldsymbol{x}, t)) \cdot \dot{\boldsymbol{x}} - \left(\frac{1}{2}m|\boldsymbol{v}|^2 + q\varepsilon^{-1}\varphi(\boldsymbol{x}, t)\right). \tag{A.3.12}$$

Let's expand the phase-space variables in power series of ε, and solve order by order:

$$\begin{aligned} \boldsymbol{x}(t, \alpha) &= \boldsymbol{X}(t) + \varepsilon\boldsymbol{\rho}(t, \alpha) + \mathcal{O}(\varepsilon^2), \\ \boldsymbol{v}(t, \alpha) &= \boldsymbol{V}(t) + \boldsymbol{w}(t, \alpha) + \mathcal{O}(\varepsilon), \end{aligned} \tag{A.3.13}$$

where all the symbols appearing in the equation represent $\mathcal{O}(1)$ quantities; the time derivative of $\boldsymbol{x}$ equals

$$\dot{\boldsymbol{x}} = \dot{\boldsymbol{X}} + \varepsilon(\varepsilon^{-1}\dot{\alpha})\frac{\partial \boldsymbol{\rho}}{\partial \alpha} + \mathcal{O}(\varepsilon) = \dot{\boldsymbol{X}} + \dot{\alpha}\frac{\partial \boldsymbol{\rho}}{\partial \alpha} + \mathcal{O}(\varepsilon). \tag{A.3.14}$$

To order $\mathcal{O}(1)$, the Lagrangian is

$$\begin{aligned} \mathcal{L} = \varepsilon^{-1}q\left(\boldsymbol{A} \cdot \left(\dot{\boldsymbol{X}} + \varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\right) - \varphi\right) + q\boldsymbol{\rho} \cdot \left(\nabla\boldsymbol{A} \cdot \left(\dot{\boldsymbol{X}} + \varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\right) - \nabla\varphi\right) \\ + m(\boldsymbol{V} + \boldsymbol{w}) \cdot \left(\dot{\boldsymbol{X}} + \varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\right) - \frac{1}{2}m(|\boldsymbol{V}|^2 + |\boldsymbol{w}|^2 + 2\boldsymbol{w} \cdot \boldsymbol{V}) + \mathcal{O}(\varepsilon). \end{aligned} \tag{A.3.15}$$

[78]Clearly, the canonical equations of Hamilton cannot apply to non-canonical coordinates; the evolution of these variables are determined by their Lagrange equations.

[79]Note that $\dot{\boldsymbol{x}}$ and $\boldsymbol{v}$ are assumed to be independent when calculating the variation; the resultant equations of motion will specify that they are equal.

In the first two terms, the time derivative on $\boldsymbol{\rho}$ can be converted through integration by parts to that on the electromagnetic fields, and in this process the Lagrangian is much simpfied[80]:

(1)
$$\boldsymbol{A}\cdot\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}=\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}\cdot\boldsymbol{\rho})-\frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\cdot\boldsymbol{\rho}=\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}\cdot\boldsymbol{\rho})-\left(\frac{\partial\boldsymbol{A}}{\partial t}+\dot{\boldsymbol{X}}\cdot\nabla\boldsymbol{A}\right)\cdot\boldsymbol{\rho};\tag{A.3.16}$$

at this stage, the Lagrangian writes

$$\begin{aligned}\mathcal{L}=&\,\varepsilon^{-1}q\big(\boldsymbol{A}\cdot\dot{\boldsymbol{X}}-\varphi\big)+q\boldsymbol{\rho}\cdot\big(\boldsymbol{E}+\dot{\boldsymbol{X}}\times\boldsymbol{B}\big)+q\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\\&+m(\boldsymbol{V}+\boldsymbol{w})\cdot\left(\dot{\boldsymbol{X}}+\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\right)-\frac{1}{2}m(|\boldsymbol{V}|^2+|\boldsymbol{w}|^2+2\boldsymbol{w}\cdot\boldsymbol{V})+\mathcal{O}(\varepsilon)+\text{total derivative.}\end{aligned}\tag{A.3.17}$$

(2)
$$\begin{aligned}\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}+\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\cdot\nabla\boldsymbol{A}\cdot\boldsymbol{\rho}&=\varepsilon\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\boldsymbol{\rho})-\varepsilon\boldsymbol{\rho}\cdot\frac{\mathrm{d}\nabla\boldsymbol{A}}{\mathrm{d}t}\cdot\boldsymbol{\rho}\\&=\text{total derivative}+\mathcal{O}(\varepsilon),\end{aligned}\tag{A.3.18}$$

thus

$$\begin{aligned}\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}&=\frac{1}{2}\left(\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}-\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\cdot\nabla\boldsymbol{A}\cdot\boldsymbol{\rho}\right)\\&\quad+\frac{1}{2}\left(\boldsymbol{\rho}\cdot\nabla\boldsymbol{A}\cdot\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}+\varepsilon\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}\cdot\nabla\boldsymbol{A}\cdot\boldsymbol{\rho}\right)\\&=\frac{1}{2}\dot{\alpha}\left(\boldsymbol{\rho}\times\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\right)\cdot\boldsymbol{B}+\mathcal{O}(\varepsilon)+\text{total derivative;}\end{aligned}\tag{A.3.19}$$

at this stage, the Lagrangian writes

$$\begin{aligned}\mathcal{L}=&\,\varepsilon^{-1}q\big(\boldsymbol{A}\cdot\dot{\boldsymbol{X}}-\varphi\big)+q\boldsymbol{\rho}\cdot\big(\boldsymbol{E}+\dot{\boldsymbol{X}}\times\boldsymbol{B}\big)+\frac{1}{2}q\dot{\alpha}\left(\boldsymbol{\rho}\times\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\right)\cdot\boldsymbol{B}\\&+m(\boldsymbol{V}+\boldsymbol{w})\cdot\left(\dot{\boldsymbol{X}}+\dot{\alpha}\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\right)-\frac{1}{2}m(|\boldsymbol{V}|^2+|\boldsymbol{w}|^2+2\boldsymbol{w}\cdot\boldsymbol{V})+\mathcal{O}(\varepsilon)+\text{total derivatives.}\end{aligned}\tag{A.3.20}$$

The Lagrange equation of $\boldsymbol{X}$ at order $\mathcal{O}(\varepsilon^{-1})$ gives Eq. (A.2.16), from which we derived

$$\dot{\boldsymbol{X}}_\perp=\boldsymbol{v}_E,\quad\dot{\alpha}\frac{\partial\boldsymbol{w}}{\partial\alpha}=\Omega\dot{\alpha}\frac{\partial\boldsymbol{\rho}}{\partial\alpha}\times\boldsymbol{b};\tag{A.3.21}$$

the Lagrange equation of $\boldsymbol{V}$ gives

$$\dot{\boldsymbol{X}}+\dot{\alpha}\frac{\partial\boldsymbol{\rho}}{\partial\alpha}=\boldsymbol{V}+\boldsymbol{w}\Rightarrow\boldsymbol{V}=\dot{\boldsymbol{X}}=:u\hat{\boldsymbol{e}}_\parallel+\boldsymbol{v}_E,\quad\boldsymbol{w}=\dot{\alpha}\frac{\partial\boldsymbol{\rho}}{\partial\alpha},\tag{A.3.22}$$

where $u:=\hat{\boldsymbol{e}}_\parallel\cdot\dot{\boldsymbol{X}}$. From the above two equations, we could solve for $\boldsymbol{w}$ and $\boldsymbol{\rho}$, and the answer is the same as Eq. (A.2.24) and Eq. (A.2.27)

$$\boldsymbol{w}=w(\hat{\boldsymbol{e}}_{\perp1}\cos\alpha+\hat{\boldsymbol{e}}_{\perp2}\sin\alpha),\quad\boldsymbol{\rho}=\frac{1}{\Omega}\hat{\boldsymbol{e}}_\parallel\times\boldsymbol{w},\quad\dot{\alpha}=\Omega.\tag{A.3.23}$$

[80] Another reason to integrate by parts is to eliminate the gyrophase since we have already known from the previous section that it is ignorable (cyclic).

Now let's substitute $\boldsymbol{V} = u\hat{\boldsymbol{e}}_\parallel + \boldsymbol{v}_E$, $\boldsymbol{w} = w(\hat{\boldsymbol{e}}_{\perp 1}\cos\alpha + \hat{\boldsymbol{e}}_{\perp 2}\sin\alpha)$ and $\boldsymbol{\rho} = \hat{\boldsymbol{e}}_\parallel \times \boldsymbol{w}/\Omega$ into the Lagrangian[81], and then average over α. The result is

$$\begin{aligned}\langle\mathcal{L}\rangle = &\Big(q\varepsilon^{-1}\boldsymbol{A} + m\big(u\hat{\boldsymbol{e}}_\parallel + \boldsymbol{v}_E\big)\Big)\cdot\dot{\boldsymbol{X}} + J\dot{\alpha}\\ &-\Big(\frac{1}{2}m\big(u^2 + |\boldsymbol{v}_E|^2\big) + \mu B + q\varepsilon^{-1}\varphi\Big) + \mathcal{O}(\varepsilon) + \text{total derivatives.}\end{aligned}\tag{A.3.24}$$

Finally, setting $\varepsilon = 1$, we arrive at the non-canonical phase-space guiding-center Lagrangian[82] [61]

$$\begin{aligned}\mathcal{L}_{\text{gc}}\big(\boldsymbol{X}, \dot{\boldsymbol{X}}, \dot{\alpha}, u, J, t\big) = &\Big(q\boldsymbol{A}(\boldsymbol{X},t) + m\big(u\hat{\boldsymbol{e}}_\parallel(\boldsymbol{X},t) + \boldsymbol{v}_E(\boldsymbol{X},t)\big)\Big)\cdot\dot{\boldsymbol{X}}\\ &+J\dot{\alpha} - H_{\text{gc}}(\boldsymbol{X}, u, J, t),\end{aligned}\tag{A.3.25}$$

where

$$H_{\text{gc}}(\boldsymbol{X}, u, J, t) = \frac{1}{2}m\big(u^2 + |\boldsymbol{v}_E(\boldsymbol{X},t)|^2\big) + \mu B(\boldsymbol{X},t) + q\varphi(\boldsymbol{X},t), \quad \mu = \frac{q}{m}J. \tag{A.3.26}$$

A.3.3 Equations of motion through variational method

In the guiding-center Lagrangian, $\boldsymbol{X}$ and α are coordinates, while u and J are momentums[83]. Note that only (J, α) are canonical.

The canonical equations for α and J are

$$\begin{aligned}\dot{\alpha} &= \frac{\partial H_{\text{gc}}}{\partial J} = \Omega,\\ \dot{J} &= -\frac{\partial H_{\text{gc}}}{\partial \alpha} = 0.\end{aligned}\tag{A.3.27}$$

The Lagrange equation for u, $\partial\mathcal{L}_{\text{gc}}/\partial u = 0$, gives

$$u = \hat{\boldsymbol{e}}_\parallel\cdot\dot{\boldsymbol{X}}. \tag{A.3.28}$$

Finally, the Lagrange equation for $\boldsymbol{X}$ is

$$m\dot{u}\hat{\boldsymbol{e}}_\parallel = q\big(\boldsymbol{E}^* + \dot{\boldsymbol{X}}\times\boldsymbol{B}^*\big), \tag{A.3.29}$$

where

$$\boldsymbol{E}^* = -\nabla\varphi^* - \frac{\partial\boldsymbol{A}^*}{\partial t}, \quad \boldsymbol{B}^* = \nabla\times\boldsymbol{A}^* \tag{A.3.30}$$

and

---

[81] Recall that the idea of analytical mechanics is to find the real motion out of all possible motions. This substitution means that we are narrowing the range of what is meant by "possible". Note that we should not substitute $\dot{\alpha} = \Omega$ since α is a coordinate that need to be reserved — the gyrophase locates the particle along the gyration orbit; for a similar reason, in this subsititution u should not be regard as $\hat{\boldsymbol{e}}_\parallel\cdot\dot{\boldsymbol{X}}$, but is simply an order $\mathcal{O}(1)$ quantity which is the parallel component of $\boldsymbol{v}$.

[82] This is the famous Northrop guiding-center (Ngc) Lagrangian; there also exist other forms of Lagrangians in the literature [10].

[83] The canonical momentum conjugate to $\boldsymbol{X}$ is $q\boldsymbol{A} + m\big(u\hat{\boldsymbol{e}}_\parallel + \boldsymbol{v}_E\big)$, so u may be referred to as (and is essentially) the "non-canonical momentum" conjugate to $\boldsymbol{X}$.

$$q\varphi^* = q\varphi + \mu B + \frac{1}{2}m|\boldsymbol{v}_E|^2, \quad q\boldsymbol{A}^* = q\boldsymbol{A} + m\big(u\hat{\boldsymbol{e}}_\parallel + \boldsymbol{v}_E\big). \tag{A.3.31}$$

The dot product of Eq. (A.3.29) with $\boldsymbol{B}^*$ solves u,

$$\dot{u} = \frac{q}{m}\frac{\boldsymbol{B}^*}{B_\parallel^*}\cdot\boldsymbol{E}^*, \tag{A.3.32}$$

and then the perpendicular part of Eq. (A.3.29) solves $\boldsymbol{X}$,

$$\dot{\boldsymbol{X}} = u\frac{\boldsymbol{B}^*}{B_\parallel^*} + \frac{\boldsymbol{E}^*\times\hat{\boldsymbol{e}}_\parallel}{B_\parallel^*}, \tag{A.3.33}$$

where

$$B_\parallel^* := \hat{\boldsymbol{e}}_\parallel\cdot\boldsymbol{B}^* = B + \frac{m}{q}\hat{\boldsymbol{e}}_\parallel\cdot\Big(\nabla\times\big(u\hat{\boldsymbol{e}}_\parallel + \boldsymbol{v}_E\big)\Big). \tag{A.3.34}$$

Note that in the variational approach, the first-order correction to the parallel velocity is automatically included.[84] In fact, all the higher-order terms that are needed to conserve phase-space volume and energy are retained here, whereas the equations derived from Newtonian mechanical approach (see the previous section) conserves neither phase-space volume nor energy.

A.3.3.1 Conservation of energy

The time derivative of the guiding-center Hamilton is (CAUTION: $\partial/\partial t$ does not act on phase-spcae variables such as u)

$$\begin{aligned}\frac{\mathrm{d}H_{\mathrm{gc}}}{\mathrm{d}t} &= \frac{\partial H_{\mathrm{gc}}}{\partial t} + \dot{\boldsymbol{X}}\cdot\nabla H_{\mathrm{gc}} + \dot{u}\frac{\partial H_{\mathrm{gc}}}{\partial u} \\ &= \text{several terms, each has a multiplicative factor of } \frac{\partial\ \textbf{field}}{\partial t},\end{aligned} \tag{A.3.35}$$

which means that the guiding-center energy

$$E_{\mathrm{gc}} = \frac{1}{2}mu^2 + q\varphi^* = \frac{1}{2}m\Big(u^2 + |\boldsymbol{v}_E|^2\Big) + \mu B + q\varphi \tag{A.3.36}$$

is a constant of motion if the electromagnetic fields are time-independent.

A.3.3.2 Conservation of phase-space volume

It is direct to prove the the guiding-center Liouville's theorem [10]

$$\frac{\partial B_\parallel^*}{\partial t} + \nabla\cdot\Big(B_\parallel^*\dot{\boldsymbol{X}}\Big) + \frac{\partial}{\partial u}\Big(B_\parallel^*\dot{u}\Big) = 0. \tag{A.3.37}$$

The reason why Eq. (A.3.37) is referred to as Liouville's theorem is that $\mathcal{J}_{\mathrm{gc}} = qmB_\parallel^*$ (essentially $B_\parallel^*$) is the Jacobian of the guiding-center phase-space transformation, i.e.,

$$\mathrm{d}^3q\,\mathrm{d}^3p = \mathcal{J}_{\mathrm{gc}}\mathrm{d}^3X\,\mathrm{d}\alpha\,\mathrm{d}u\,\mathrm{d}J\,. \tag{A.3.38}$$

[84] TODO: derive the various drifts from the above EoMs.

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